FLUXGATE MAGNETOMETER CALIBRATION FOR

JUICE

JUI–IGEP–TR–0004

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Protocol and Analysis of the

JUICE FGM CALIBRATION for the FM-OB and FM-IB sensor connected to the complete FM ELECTRONICS

Dr. Ingo Richter

Institut für Geophysik und extraterrestrische Physik
Technische Universität Braunschweig
Mendelssohnstraße 3, 38106 Braunschweig
Germany

Contents

1 Introduction

This document describes the ground calibration of the JUICE Fluxgate magnetometer comprising the sensors FM-OB, FM-IB and the complete FM electronics. The calibration was conducted at the Magnetsrode Calibration facility operated by the Institute for Geophysics and extraterrestrial Physics, TU Braunschweig.
The tests were executed from April 14 to April 22, 2021.

Part Identification:

Sensors:	FM-OB, FM-IB
Electronics	FM

Key Personnel:

Richard Baughen	operating FGM, responsible for FGM-OB
Rachel Hudson	operating FGM
Christian Hagen	setup FGM
Irmgard Jernej	setup FGM
Uli Auster	alignment Tests of FGM, responsible for FGM-IB
Ingo Richter	operating MRode Facility

General Setup:

- Sensors initially installed at CoC without thermal box.
- FM Electronics placed in House 2 anteroom.
- EGSE placed in House 2 / anteroom.

Remarks:

- The Analysis has been performed using a nominal conversion factor of 1 nT/V (nT/ADCcounts) used at the CALIBRATOR Software.
- The magnetometer was operated with a sampling rate of 16 vectors per second for the linearity and offset tests.
  128 Hz, 32 Hz, 16 Hz, 1 Hz were used for frequency measurements.
- **All values (e.g Offsets) given in engineering nanotesla (enT) have to be multiplied with the calculated sensitivities to obtain real nanotesla values.**

- Sensor saturation limits: Not checked.
- MRode-Fieldcontrol Software: *FLDCTRL_26_I2C_TCP.vi*
- MRode-Terminal software: *TERMINAL_17.vi*
- MRode-Tempcontrol software: *TEMPCTRL_32_I2C_TCP.vi*
- Analysis software: *CALIBRATOR_73.vi*
- Calibration-data of the JUICE instrument are merged into the CDRS using ASCII-files containing 16 s averaged or 128 Hz FGM data.

2 General Setup Pictures

Figure 1: The FM Electronics in the Anteroom of House 2.

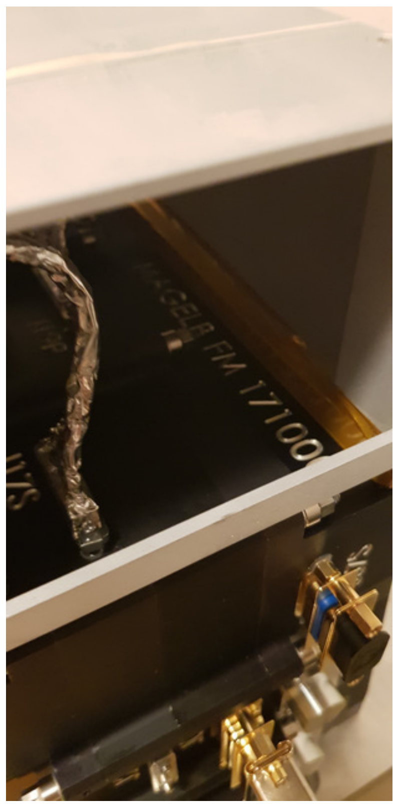

Figure 2: The zoomed FM Board in the Anteroom of House 2.

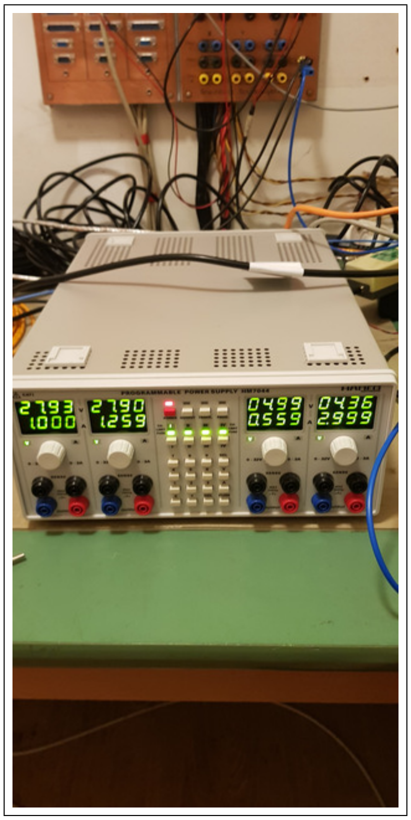

Figure 3: The Power Supply in the Anteroom of House 2.

3 Setup on April 14, 2021

The test equipment has been shipped to MRode on April 14. Sensors and electronics have been installed in H2. System works properly.

4 Measurements on April 15, 2021

4.1 Data

CCD File	Configuration File	Remark
21-04-15\08-20-18.CCR	FREQ400.MAG	
21-04-15\09-03-01.CCR	FREQ400.MAG	
21-04-15\09-49-04.CCR	FREQ400.MAG	
21-04-15\10-33-19.CCR	FREQ400.MAG	
21-04-15\11-17-54.CCR	FREQ400.MAG	
21-04-15\19-23-48.CCR	OFFSET_200.MAG	
21-04-15\19-33-14.CCR	OFFSET_200.MAG	
21-04-15\19-44-03.CCR	LIN60000XYZ.MAG	
21-04-15\20-04-09.CCR	LIN8000XYZ.MAG	

5 Sensor Setup Pictures for Frequency Measurements

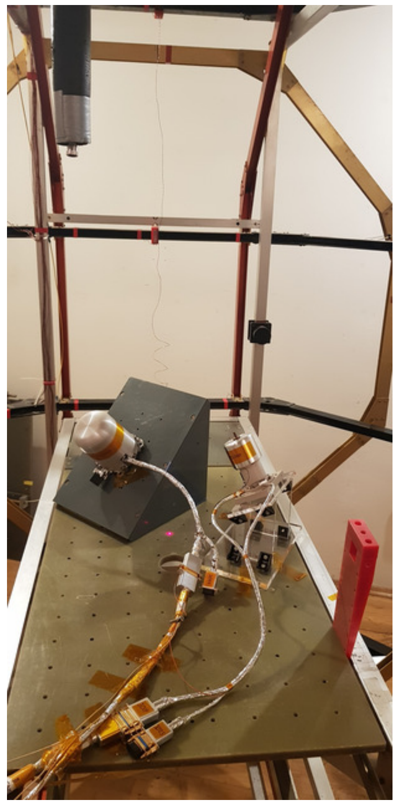

Figure 4: Setup for Frequency Measurements of OB and IB Sensor, view from south.

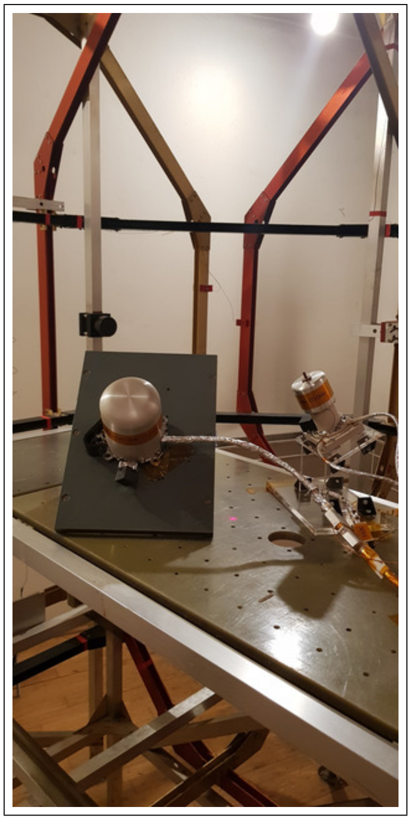

Figure 5: Setup for Frequency Measurements of OB and IB Sensor, view from south-west.

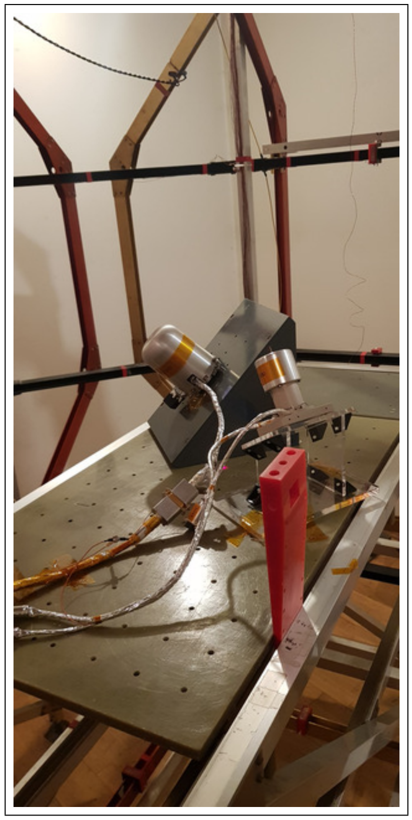

Figure 6: Setup for Frequency Measurements of OB and IB Sensor, view from south-east.

5.1 Calibration Initialisation

Initial reading of the thermistors:

T_{59}	=	16.1°C
$T_{\mathrm{ELEC-IB}}$	=	16.5°C
$T_{\mathrm{FM-IB}}$	=	18°C
$T_{\mathrm{FM-OB}}$	=	21°C
T_{PCU}	=	18°C

The FM sensors were placed diagonal in space left and right of CoC. Both sensors shall be operated in parallel for upcoming frequency measurements. See pictures 4 to 6. The sensors were connected to the electronics in the anteroom of H2. The electronics was powered by an external Power supply, refer to picture 3.

The system was set up, powered and tested successfully.

5.2 AC-Measurements on FM-OB, 128 Hz, no Filters, 40nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.
- 20dB attenuator

5.2.1 Frequency Measurements — SENSOR OB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
`21-04-15\08-20-18.CCR`	`FREQ400.MAG`	

Parameter File: `FREQ_PARAMETER__21-04-15-08-20-18_OB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	128.0013	0.002666
Y	127.9982	0.002784
Z	128.0016	0.003206
Mean Sampling Frequency	128.0004	0.001849

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	13.15
Y	16.37
Z	16.38

Applied Frequencies and Measured Amplitudes

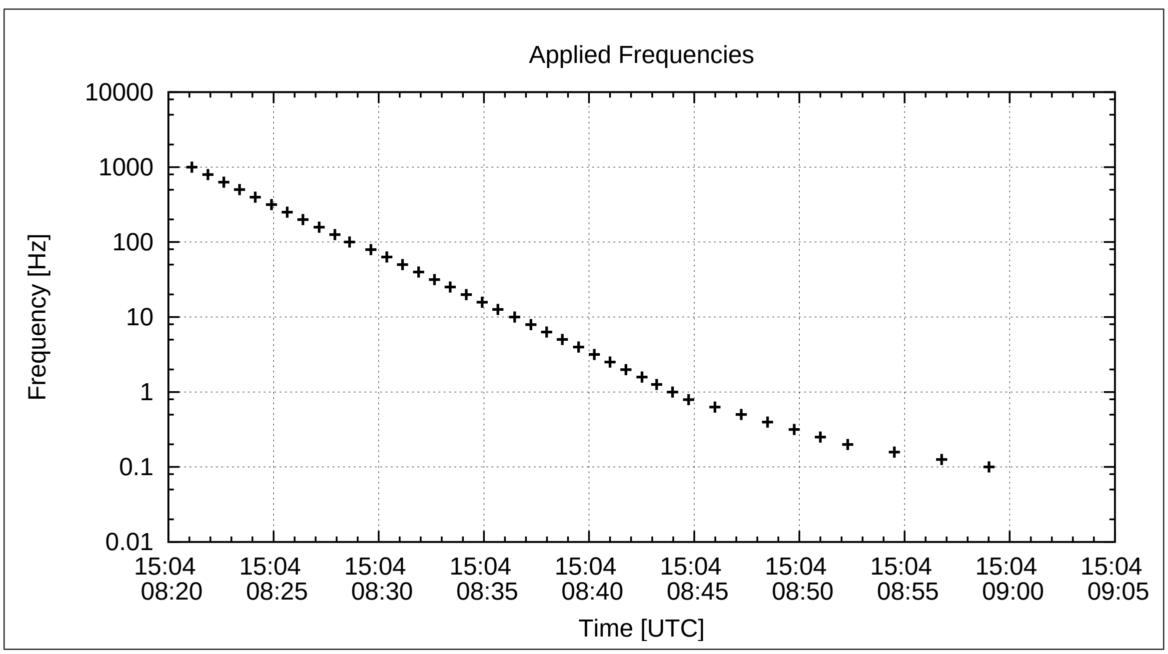

Figure 7: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.00	0.00	0.00
794.000	0.00	0.00	0.00
631.000	0.00	0.00	0.00
501.000	0.01	0.01	0.02
398.000	0.01	0.02	0.03
316.000	0.03	0.04	0.07
251.000	0.01	0.08	0.13
199.000	0.10	0.14	0.20
158.000	0.16	0.24	0.34
126.000	0.26	0.39	0.54
100.000	0.42	0.61	0.83
79.400	0.67	0.97	1.29
63.100	1.04	1.45	1.91
50.100	1.58	2.07	2.71
39.800	2.31	2.90	3.79
31.600	3.27	3.84	5.02
25.100	4.42	4.86	6.34
19.900	5.70	5.86	7.64
15.800	6.94	6.77	8.81
12.600	8.03	7.50	9.76
10.000	8.94	8.08	10.51
7.900	9.63	8.52	11.07
6.310	10.10	8.79	11.44
5.010	10.43	9.00	11.70
3.980	10.65	9.13	11.87
3.160	10.80	9.20	11.98
2.510	10.89	9.26	12.05
1.990	10.94	9.30	12.09
1.580	10.98	9.32	12.12
1.260	11.01	9.32	12.14
1.000	11.01	9.34	12.15
0.790	11.04	9.35	12.16
0.631	11.05	9.36	12.16
0.501	11.05	9.36	12.15
0.398	11.04	9.37	12.16
0.316	11.04	9.37	12.17
0.251	11.03	9.37	12.17
0.199	11.05	9.36	12.17
0.158	11.05	9.37	12.17
0.126	11.05	9.37	12.18
0.100	11.05	9.37	12.17

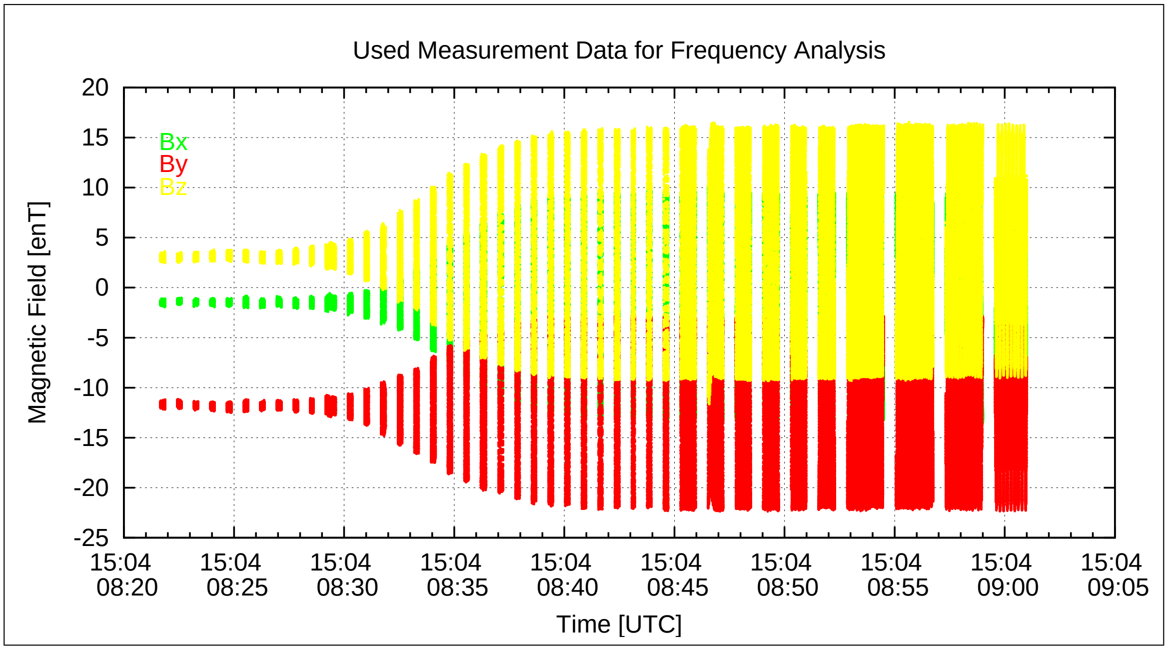

Figure 8: Used packets of Measured data for the Frequency analysis

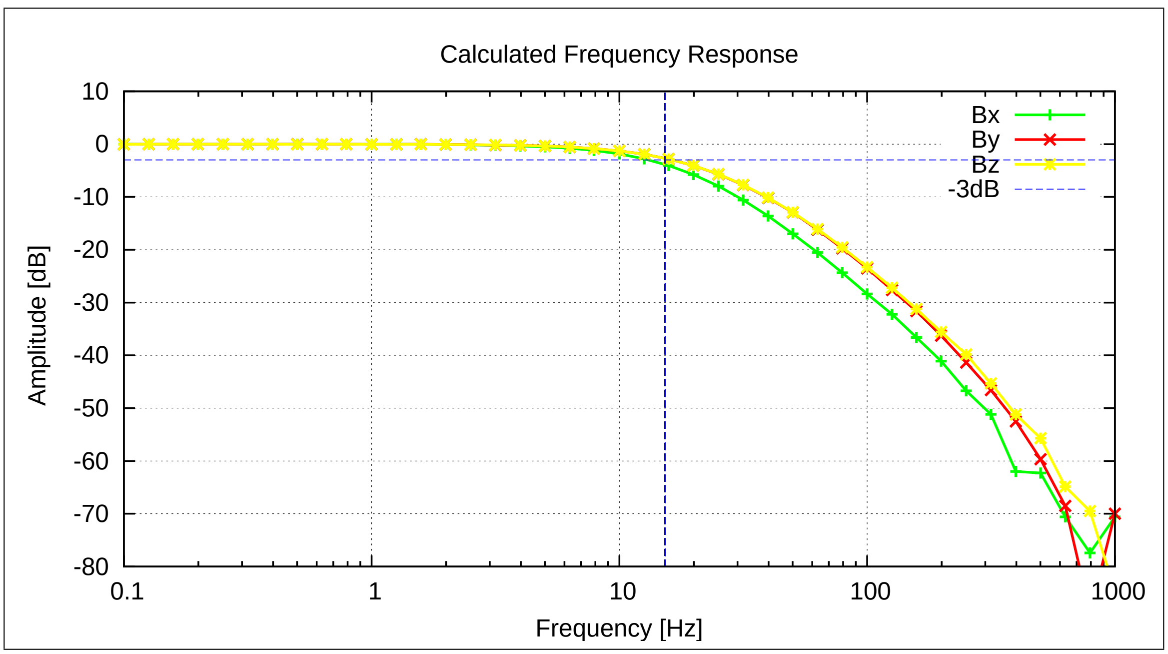

Figure 9: Calculated Frequency Response

5.3 AC-Measurements on FM-IB, 128 Hz, no Filters, 40nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.
- 20dB attenuator

5.3.1 Frequency Measurements — SENSOR IB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
`21-04-15\08-20-18.CCR`	`FREQ400.MAG`	

Parameter File: `FREQ_PARAMETER__21-04-15-08-20-18_IB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	127.9923	0.007549
Y	128.0023	0.001222
Z	127.9976	0.001091
Mean Sampling Frequency	127.9974	0.005004

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	57.79
Y	68.39
Z	31.59

Applied Frequencies and Measured Amplitudes

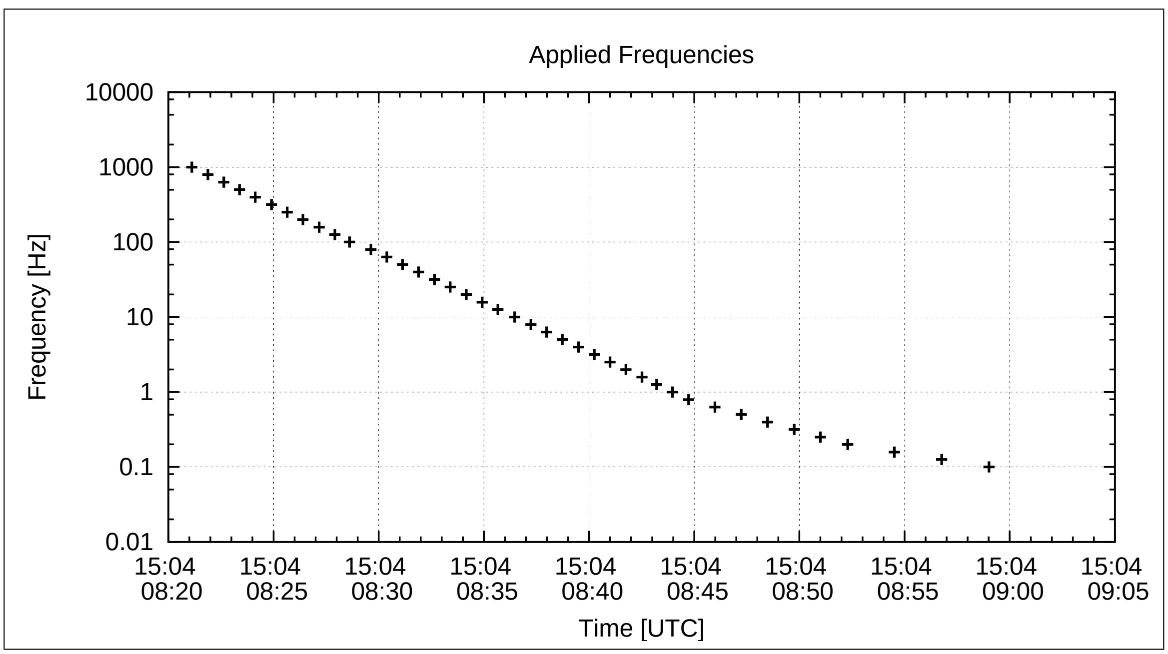

Figure 10: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.00	0.13	0.01
794.000	0.07	0.64	0.15
631.000	0.30	0.93	0.31
501.000	0.59	1.22	0.45
398.000	0.43	0.89	0.34
316.000	0.71	1.57	0.45
251.000	1.04	1.73	0.83
199.000	1.90	3.61	1.19
158.000	0.94	1.69	0.66
126.000	1.14	1.80	0.97
100.000	3.22	5.79	2.35
79.400	5.23	9.27	3.45
63.100	6.58	11.78	4.32
50.100	7.63	13.39	4.95
39.800	8.35	14.47	5.69
31.600	8.88	15.03	6.26
25.100	9.25	15.29	6.78
19.900	9.49	15.39	7.25
15.800	9.64	15.41	7.67
12.600	9.74	15.41	7.99
10.000	9.80	15.39	8.25
7.900	9.83	15.37	8.44
6.310	9.85	15.37	8.56
5.010	9.87	15.36	8.66
3.980	9.87	15.35	8.72
3.160	9.88	15.36	8.75
2.510	9.88	15.36	8.77
1.990	9.89	15.35	8.80
1.580	9.89	15.35	8.81
1.260	9.89	15.36	8.79
1.000	9.88	15.35	8.82
0.790	9.89	15.36	8.81
0.631	9.91	15.36	8.82
0.501	9.90	15.34	8.82
0.398	9.90	15.34	8.83
0.316	9.90	15.35	8.83
0.251	9.89	15.34	8.85
0.199	9.89	15.35	8.82
0.158	9.89	15.35	8.83
0.126	9.90	15.36	8.83
0.100	9.90	15.35	8.83

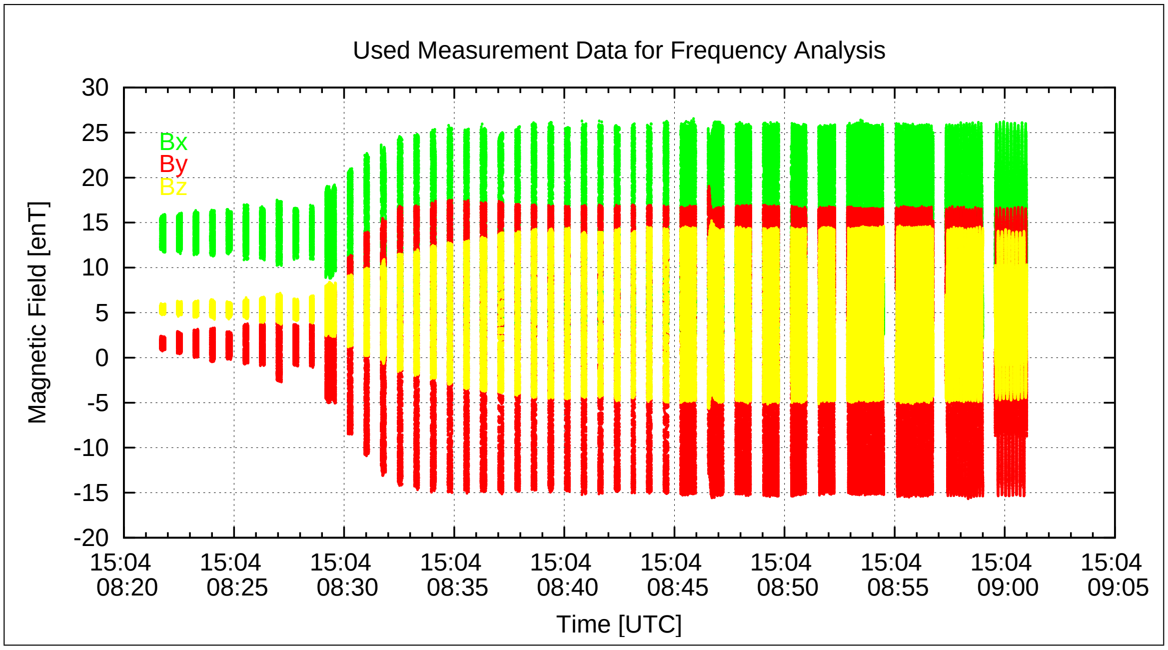

Figure 11: Used packets of Measured data for the Frequency analysis

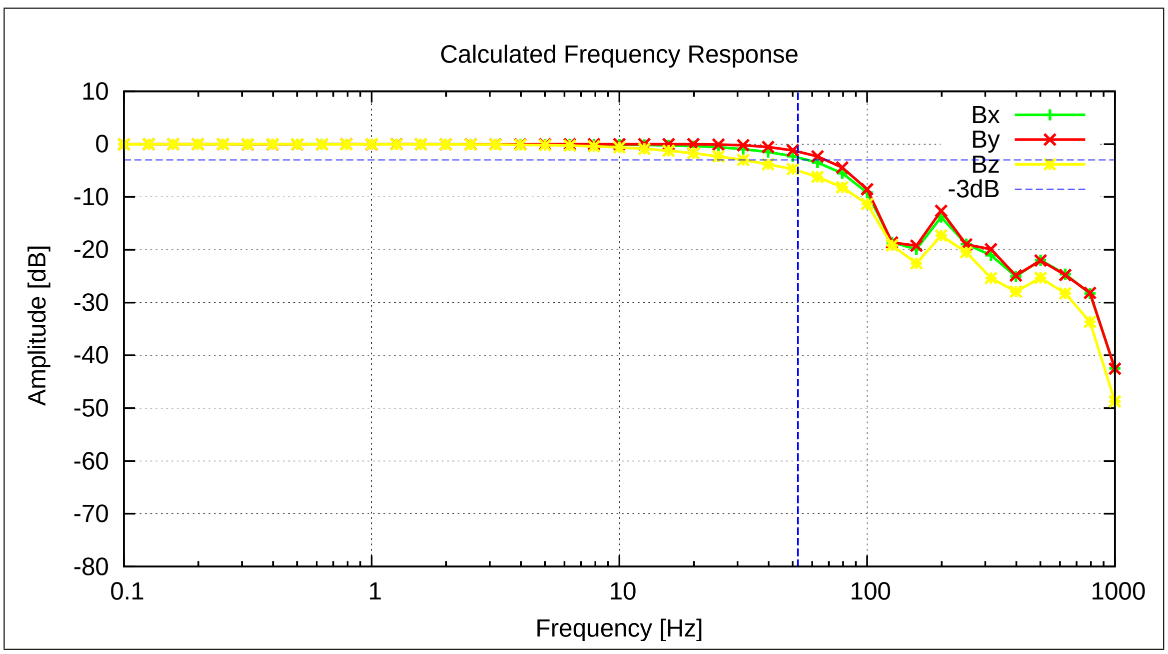

Figure 12: Calculated Frequency Response

5.4 AC-Measurements on FM-OB, 128 Hz, no Filters, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.4.1 Frequency Measurements — SENSOR OB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
21-04-15\09-03-01.CCR	FREQ400.MAG	

Parameter File: FREQ_PARAMETER__21-04-15-09-03-01_OB.FPF

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	127.9995	0.000629
Y	128.0024	0.006287
Z	128.0008	0.000135
Mean Sampling Frequency	128.0009	0.001464

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	13.14
Y	16.37
Z	16.42

Applied Frequencies and Measured Amplitudes

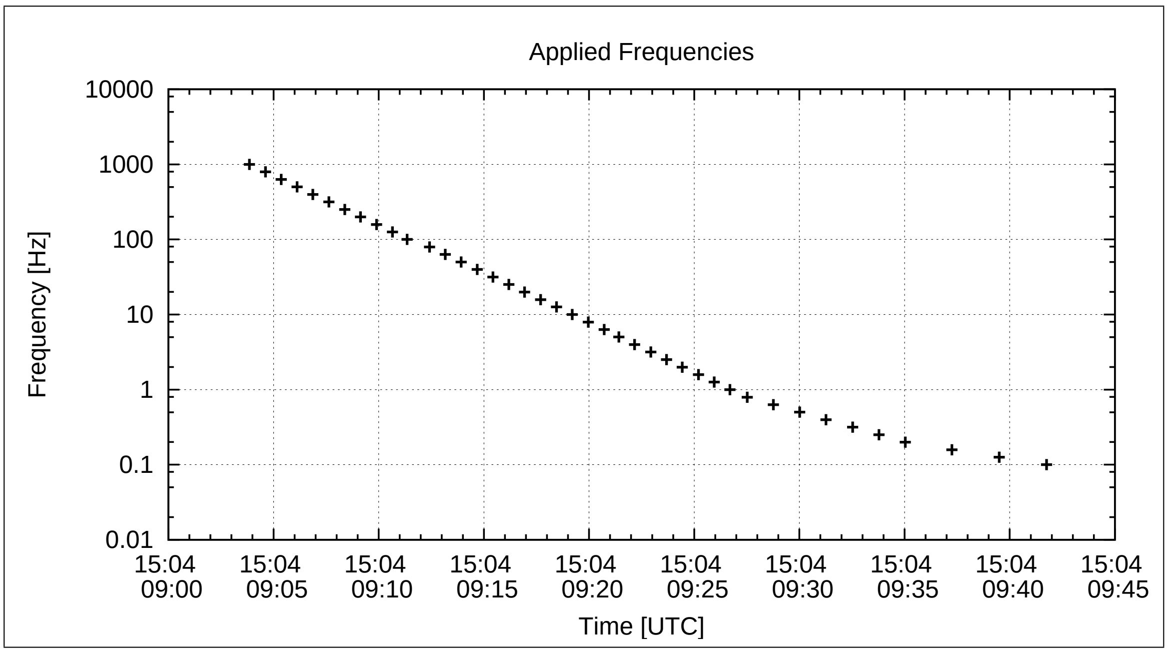

Figure 13: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.00	0.00	0.01
794.000	0.01	0.02	0.04
631.000	0.04	0.04	0.08
501.000	0.08	0.10	0.17
398.000	0.16	0.22	0.34
316.000	0.31	0.44	0.67
251.000	0.56	0.82	1.20
199.000	0.98	1.46	2.05
158.000	1.64	2.46	3.39
126.000	2.66	3.89	5.39
100.000	3.14	4.64	6.22
79.400	6.75	9.76	12.98
63.100	10.40	14.56	19.22
50.100	15.81	21.09	27.75
39.800	23.23	29.24	38.33
31.600	32.94	38.79	50.71
25.100	44.58	49.03	63.97
19.900	57.42	59.19	77.12
15.800	69.99	68.29	88.90
12.600	80.97	75.71	98.51
10.000	90.11	81.57	106.11
7.900	97.10	85.90	111.72
6.310	101.86	88.77	115.44
5.010	105.21	90.77	118.03
3.980	107.43	92.08	119.73
3.160	108.89	92.92	120.84
2.510	109.82	93.48	121.55
1.990	110.43	93.83	122.00
1.580	110.81	94.05	122.28
1.260	111.04	94.18	122.47
1.000	111.20	94.29	122.59
0.790	111.29	94.34	122.66
0.631	111.37	94.38	122.71
0.501	111.41	94.41	122.75
0.398	111.44	94.43	122.76
0.316	111.46	94.43	122.79
0.251	111.47	94.46	122.79
0.199	111.48	94.46	122.79
0.158	111.50	94.48	122.80
0.126	111.49	94.46	122.83
0.100	111.52	94.48	122.82

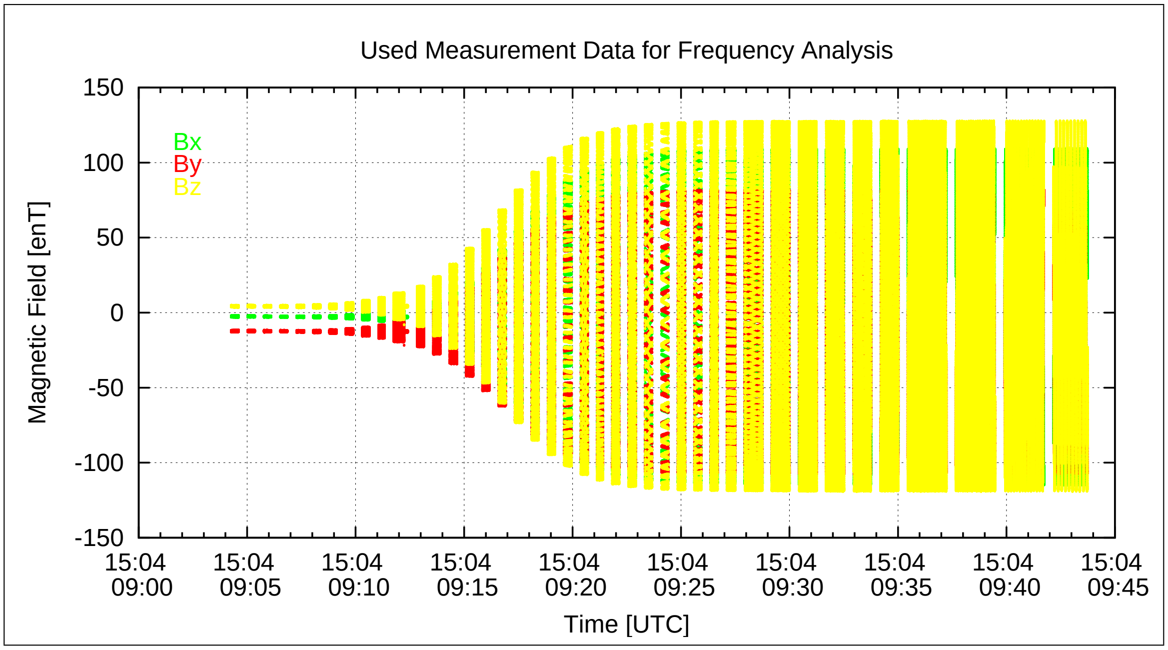

Figure 14: Used packets of Measured data for the Frequency analysis

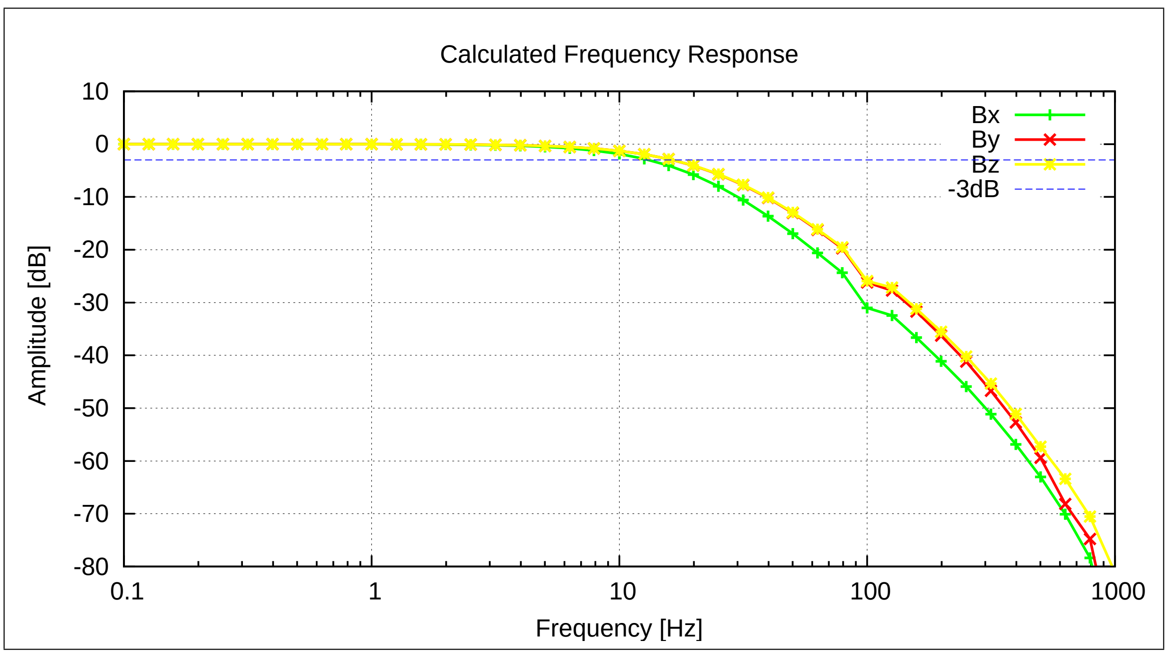

Figure 15: Calculated Frequency Response

5.5 AC-Measurements on FM-IB, 128 Hz, no Filters, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.5.1 Frequency Measurements — SENSOR IB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
`21-04-15\09-03-01.CCR`	`FREQ400.MAG`	

Parameter File: `FREQ_PARAMETER__21-04-15-09-03-01_IB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	128.0003	0.000134
Y	127.9999	0.000194
Z	128.0016	0.000204
Mean Sampling Frequency	128.0006	0.000876

3 dB Corner Frequency:

Component	$f_{3\text{dB}}$ [Hz]
X	57.73
Y	68.35
Z	31.72

Applied Frequencies and Measured Amplitudes

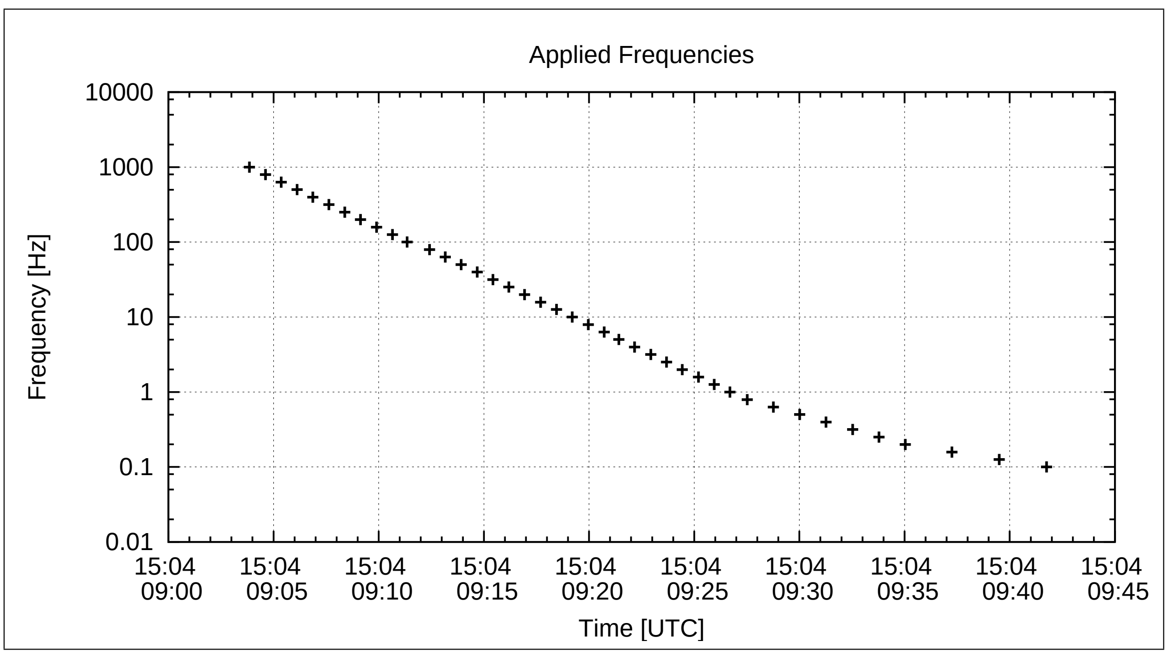

Figure 16: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.82	1.19	0.38
794.000	4.10	6.35	1.94
631.000	5.98	8.83	3.59
501.000	7.95	12.18	4.91
398.000	5.71	8.88	3.57
316.000	8.94	15.69	4.87
251.000	11.32	17.48	8.51
199.000	20.55	36.38	12.19
158.000	10.14	17.00	6.68
126.000	11.78	18.28	9.90
100.000	25.68	42.83	17.81
79.400	53.08	93.50	34.83
63.100	66.70	118.86	43.69
50.100	76.90	135.81	51.14
39.800	84.41	146.07	57.49
31.600	89.75	151.66	63.20
25.100	93.40	154.27	68.43
19.900	95.83	155.25	73.22
15.800	97.35	155.45	77.37
12.600	98.26	155.36	80.69
10.000	98.86	155.22	83.32
7.900	99.20	155.05	85.24
6.310	99.42	154.96	86.52
5.010	99.57	154.90	87.41
3.980	99.65	154.84	87.99
3.160	99.69	154.82	88.37
2.510	99.74	154.80	88.62
1.990	99.76	154.78	88.77
1.580	99.77	154.77	88.88
1.260	99.77	154.77	88.93
1.000	99.79	154.77	88.99
0.790	99.79	154.77	89.01
0.631	99.80	154.78	89.03
0.501	99.81	154.80	89.05
0.398	99.82	154.79	89.06
0.316	99.82	154.80	89.06
0.251	99.84	154.79	89.08
0.199	99.83	154.80	89.07
0.158	99.86	154.80	89.08
0.126	99.83	154.83	89.09
0.100	99.86	154.84	89.09

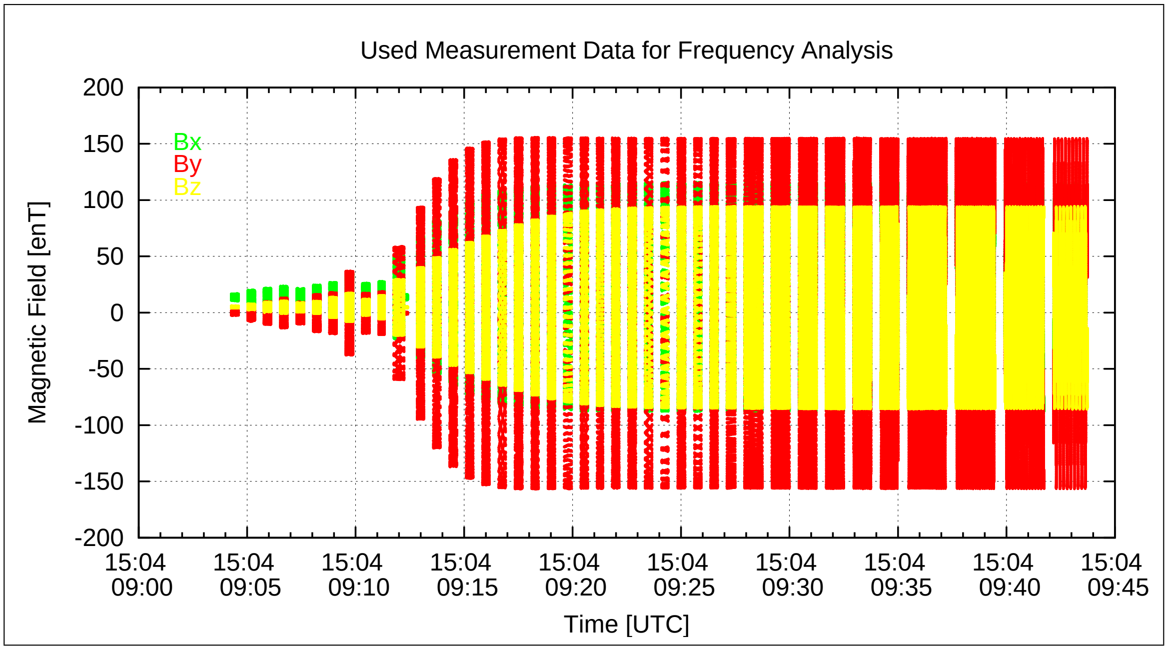

Figure 17: Used packets of Measured data for the Frequency analysis

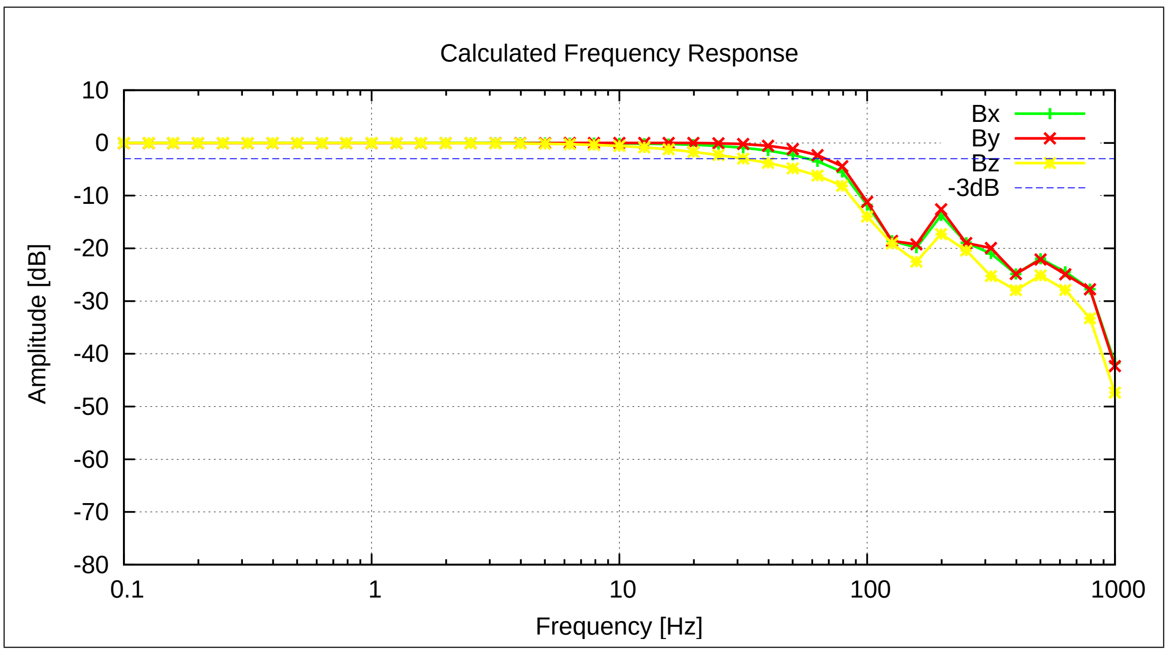

Figure 18: Calculated Frequency Response

5.6 AC-Measurements on FM-OB, 32 Hz, FIR-32Hz Filter, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.6.1 Frequency Measurements — SENSOR OB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
21-04-15\09-49-04.CCR	FREQ400.MAG	

Parameter File: `FREQ_PARAMETER__21-04-15-09-49-04_OB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	31.9987	0.008408
Y	32.0005	0.004748
Z	31.9963	0.011470
Mean Sampling Frequency	31.9985	0.002121

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	12.62
Y	13.08
Z	13.08

Applied Frequencies and Measured Amplitudes

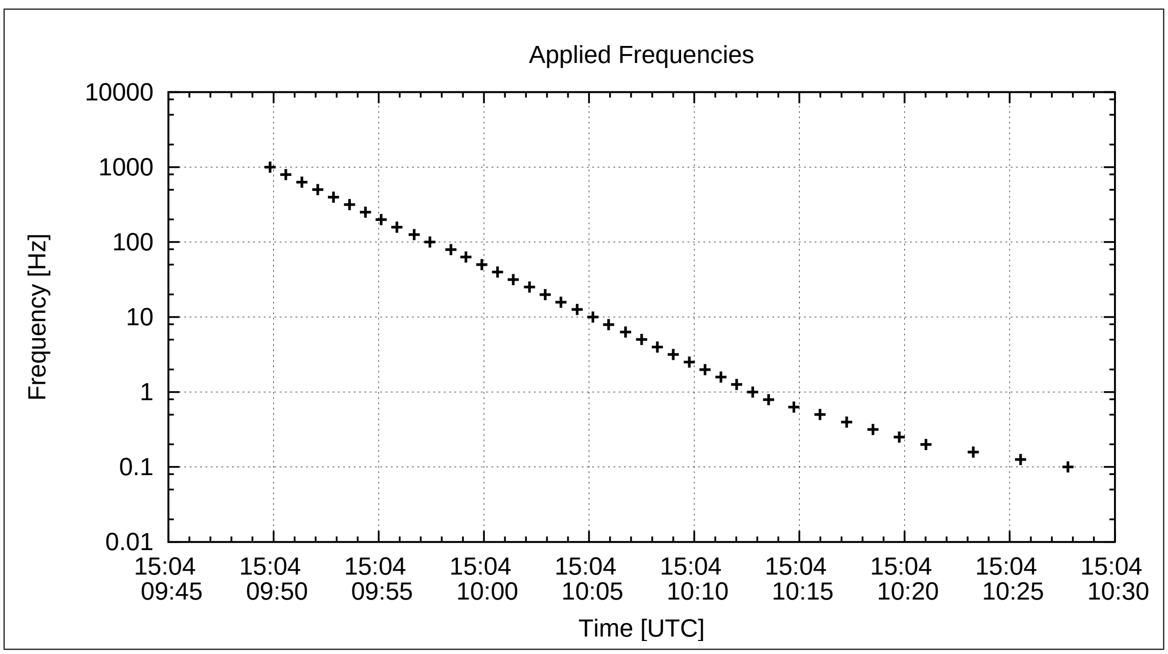

Figure 19: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.01	0.01	0.01
794.000	0.00	0.01	0.00
631.000	0.04	0.05	0.08
501.000	0.08	0.10	0.17
398.000	0.14	0.19	0.30
316.000	0.01	0.01	0.01
251.000	0.56	0.82	1.19
199.000	0.00	0.00	0.00
158.000	0.02	0.03	0.05
126.000	2.67	3.98	5.38
100.000	0.00	0.01	0.00
79.400	0.00	0.00	0.02
63.100	0.02	0.01	0.01
50.100	0.00	0.00	0.00
39.800	0.24	0.29	0.37
31.600	0.03	0.01	0.02
25.100	0.00	0.00	0.00
19.900	0.34	0.35	0.46
15.800	38.23	37.30	48.57
12.600	79.43	74.26	96.64
10.000	90.08	81.55	106.09
7.900	97.10	85.90	111.72
6.310	101.85	88.77	115.44
5.010	105.20	90.77	118.03
3.980	107.43	92.08	119.73
3.160	108.88	92.93	120.83
2.510	109.82	93.47	121.55
1.990	110.42	93.83	122.01
1.580	110.82	94.06	122.29
1.260	111.06	94.18	122.47
1.000	111.21	94.28	122.60
0.790	111.30	94.33	122.66
0.631	111.37	94.37	122.71
0.501	111.41	94.39	122.74
0.398	111.46	94.43	122.77
0.316	111.46	94.43	122.80
0.251	111.48	94.44	122.79
0.199	111.48	94.44	122.81
0.158	111.49	94.45	122.80
0.126	111.51	94.47	122.85
0.100	111.51	94.47	122.83

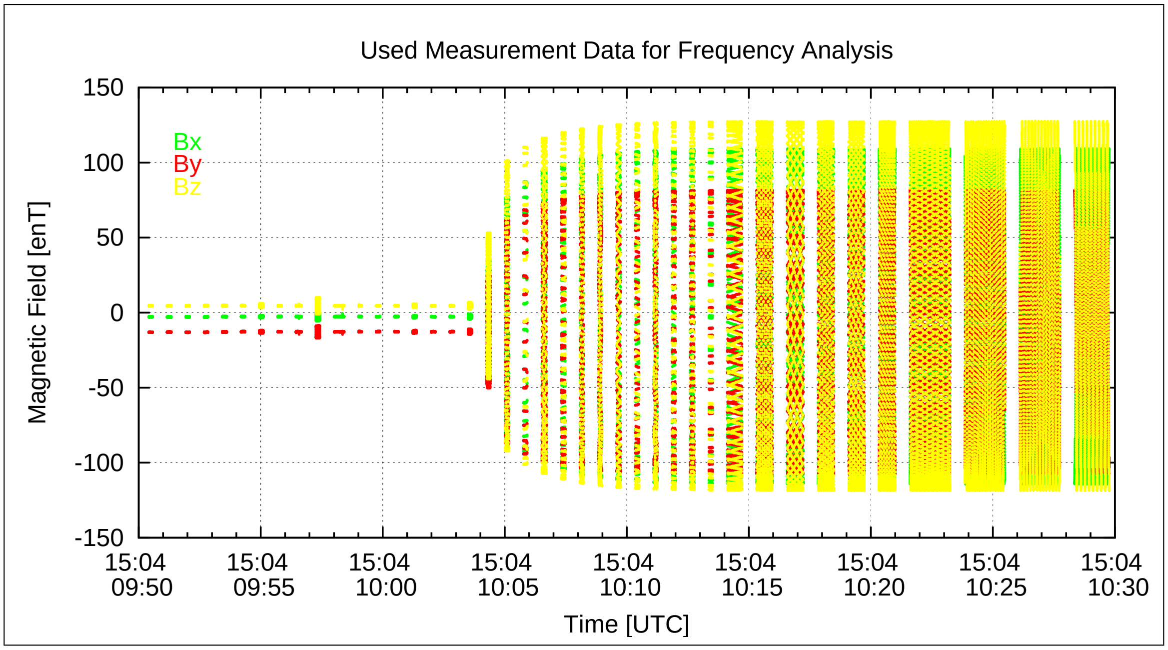

Figure 20: Used packets of Measured data for the Frequency analysis

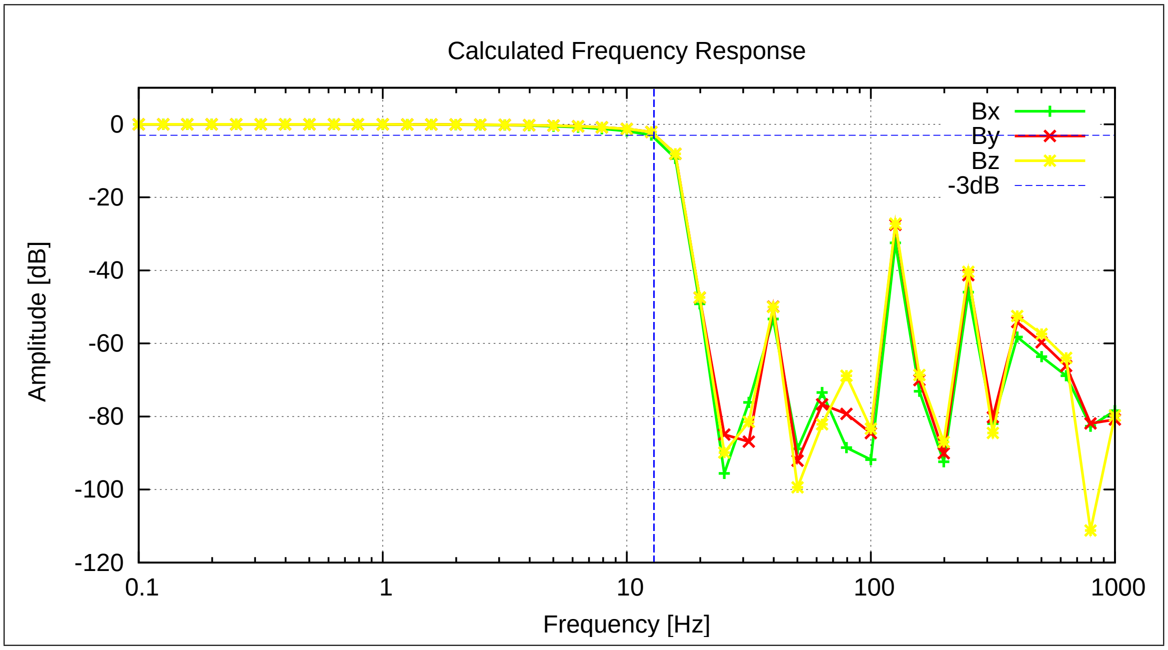

Figure 21: Calculated Frequency Response

5.7 AC-Measurements on FM-IB, 32 Hz, FIR-32Hz Filter, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.7.1 Frequency Measurements — SENSOR IB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
21-04-15\09-49-04.CCR	FREQ400.MAG	

Parameter File: FREQ_PARAMETER__21-04-15-09-49-04_IB.FPF

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	31.9977	0.007336
Y	31.9985	0.005306
Z	32.0002	0.009036
Mean Sampling Frequency	31.9988	0.001243

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	14.27
Y	14.41
Z	13.75

Applied Frequencies and Measured Amplitudes

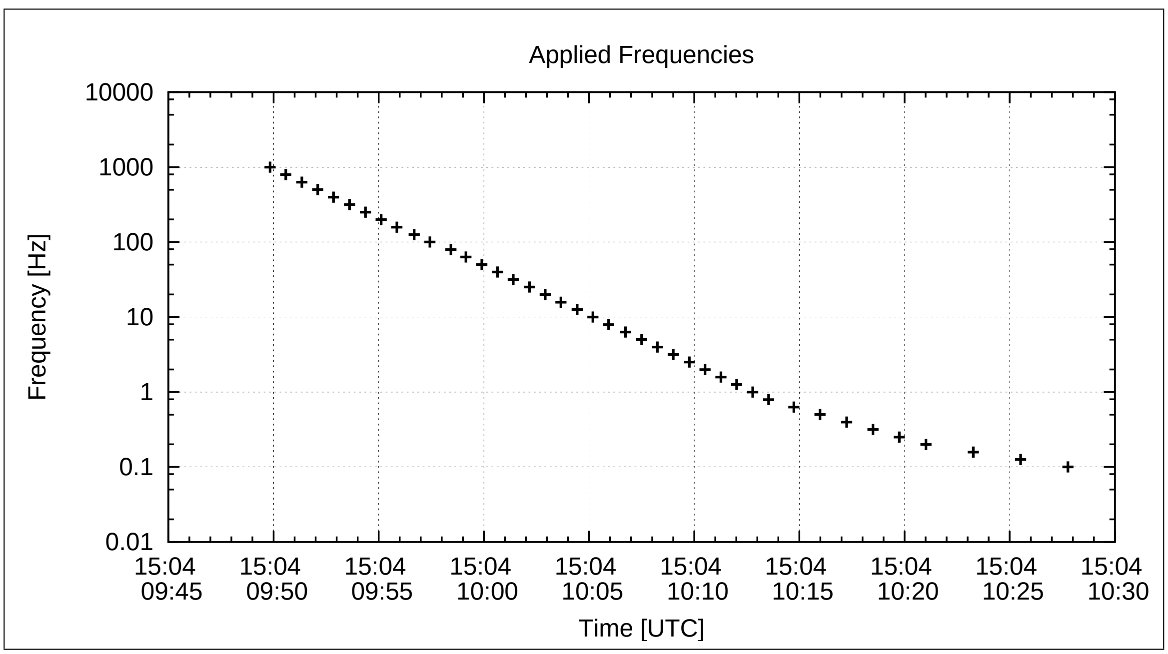

Figure 22: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.09	0.15	0.06
794.000	0.15	0.22	0.08
631.000	6.15	9.35	3.62
501.000	8.08	12.42	4.92
398.000	5.04	7.89	3.15
316.000	0.16	0.25	0.11
251.000	11.32	17.52	8.49
199.000	0.05	0.07	0.03
158.000	0.10	0.11	0.07
126.000	11.78	18.28	9.93
100.000	0.05	0.08	0.04
79.400	0.00	0.06	0.04
63.100	0.05	0.03	0.02
50.100	0.12	0.16	0.09
39.800	0.06	0.09	0.04
31.600	0.08	0.03	0.01
25.100	0.10	0.14	0.08
19.900	0.84	1.33	0.63
15.800	53.18	84.92	42.27
12.600	96.41	152.42	79.18
10.000	98.86	155.21	83.32
7.900	99.21	155.07	85.26
6.310	99.42	154.96	86.54
5.010	99.56	154.89	87.42
3.980	99.65	154.85	88.00
3.160	99.70	154.80	88.38
2.510	99.74	154.79	88.62
1.990	99.76	154.78	88.79
1.580	99.80	154.78	88.89
1.260	99.78	154.79	88.94
1.000	99.80	154.79	88.99
0.790	99.79	154.77	89.01
0.631	99.80	154.78	89.03
0.501	99.81	154.79	89.04
0.398	99.83	154.80	89.05
0.316	99.81	154.80	89.07
0.251	99.83	154.80	89.06
0.199	99.82	154.81	89.07
0.158	99.83	154.81	89.07
0.126	99.84	154.85	89.09
0.100	99.85	154.84	89.09

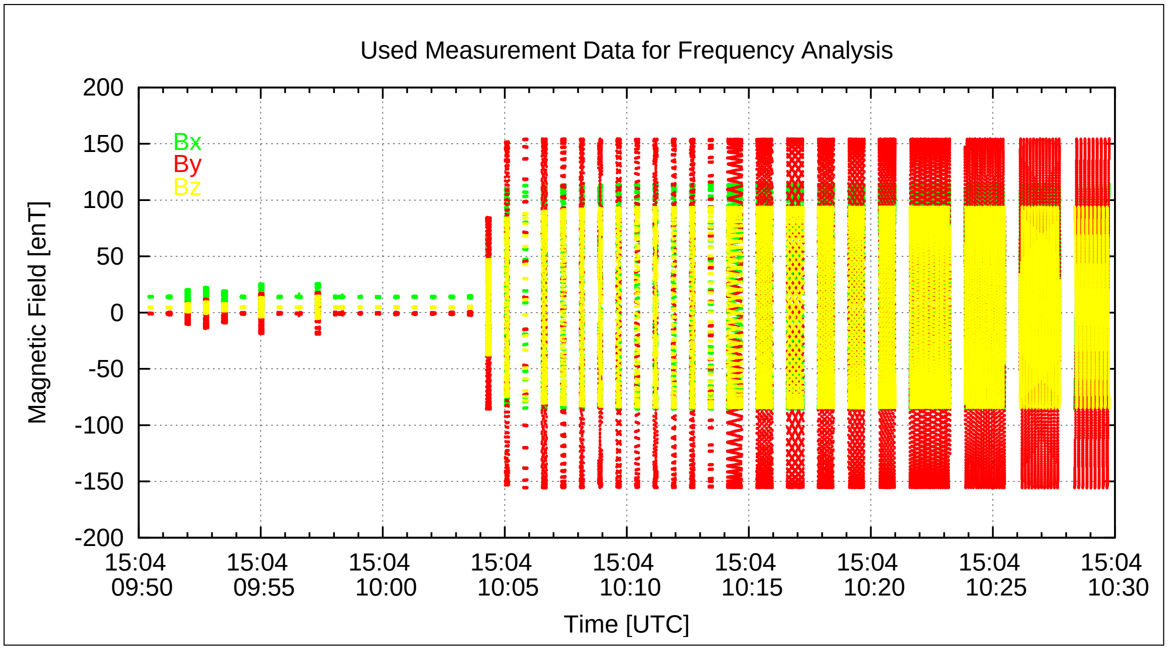

Figure 23: Used packets of Measured data for the Frequency analysis

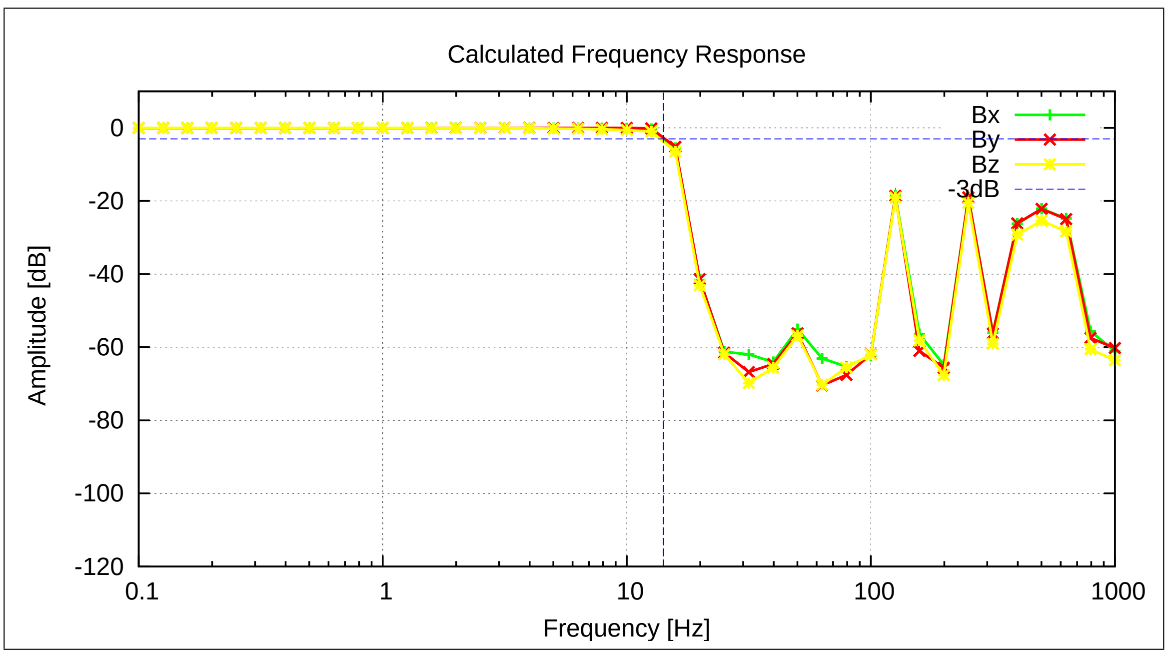

Figure 24: Calculated Frequency Response

5.8 AC-Measurements on FM-OB, 16 Hz, FIR-16Hz Filter, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.8.1 Frequency Measurements — SENSOR OB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
`21-04-15\10-33-19.CCR`	`FREQ400.MAG`	

Parameter File: `FREQ_PARAMETER__21-04-15-10-33-19_OB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	16.0019	0.012674
Y	15.9992	0.011276
Z	16.0021	0.007690
Mean Sampling Frequency	16.0011	0.001602

3 dB Corner Frequency:

Component	f_{3dB} [Hz]
X	6.55
Y	6.63
Z	6.63

Applied Frequencies and Measured Amplitudes

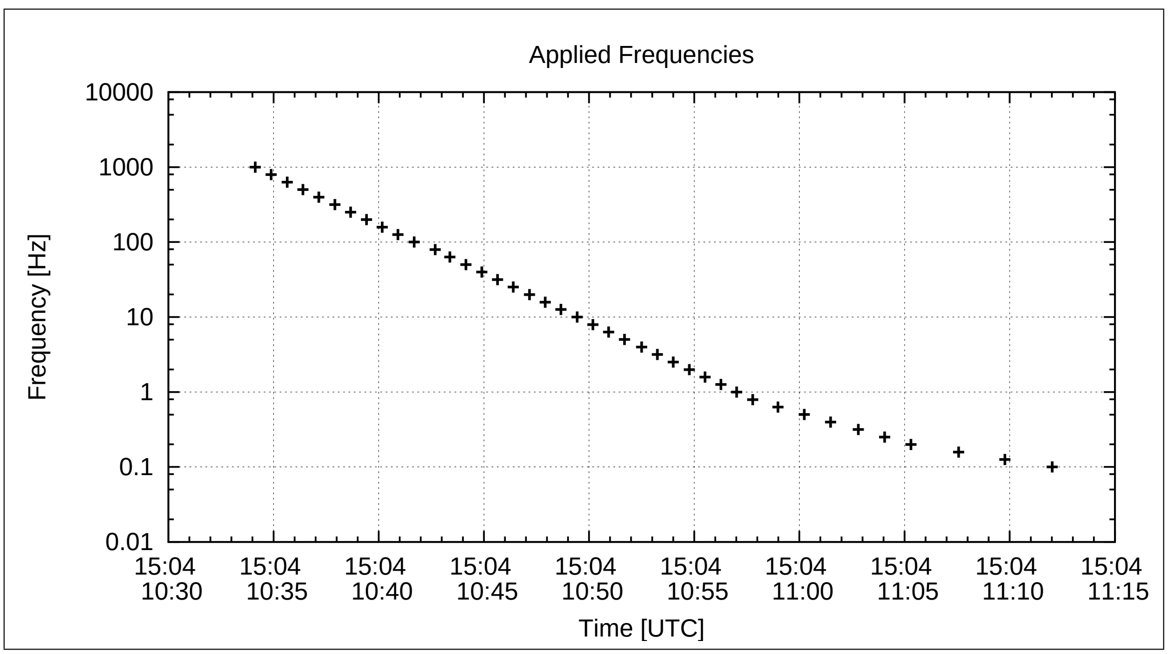

Figure 25: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.00	0.00	0.01
794.000	0.01	0.00	0.01
631.000	0.01	0.02	0.02
501.000	0.00	0.01	0.00
398.000	0.04	0.05	0.13
316.000	0.01	0.00	0.00
251.000	0.55	0.79	1.16
199.000	0.00	0.01	0.00
158.000	0.00	0.01	0.02
126.000	2.67	3.98	5.39
100.000	0.00	0.00	0.01
79.400	0.02	0.03	0.01
63.100	0.00	0.00	0.01
50.100	0.01	0.00	0.00
39.800	0.01	0.02	0.00
31.600	0.04	0.01	0.01
25.100	0.00	0.00	0.00
19.900	0.27	0.28	0.37
15.800	0.02	0.01	0.00
12.600	0.13	0.11	0.15
10.000	10.68	9.67	12.58
7.900	50.81	44.95	58.47
6.310	85.58	74.58	96.99
5.010	101.46	87.52	113.83
3.980	106.79	91.53	119.02
3.160	108.81	92.85	120.74
2.510	109.81	93.48	121.55
1.990	110.43	93.85	122.01
1.580	110.82	94.06	122.29
1.260	111.07	94.19	122.47
1.000	111.20	94.26	122.61
0.790	111.32	94.35	122.67
0.631	111.37	94.39	122.73
0.501	111.42	94.42	122.78
0.398	111.44	94.44	122.78
0.316	111.47	94.45	122.79
0.251	111.50	94.45	122.81
0.199	111.47	94.45	122.81
0.158	111.47	94.43	122.80
0.126	111.46	94.45	122.81
0.100	111.48	94.44	122.78

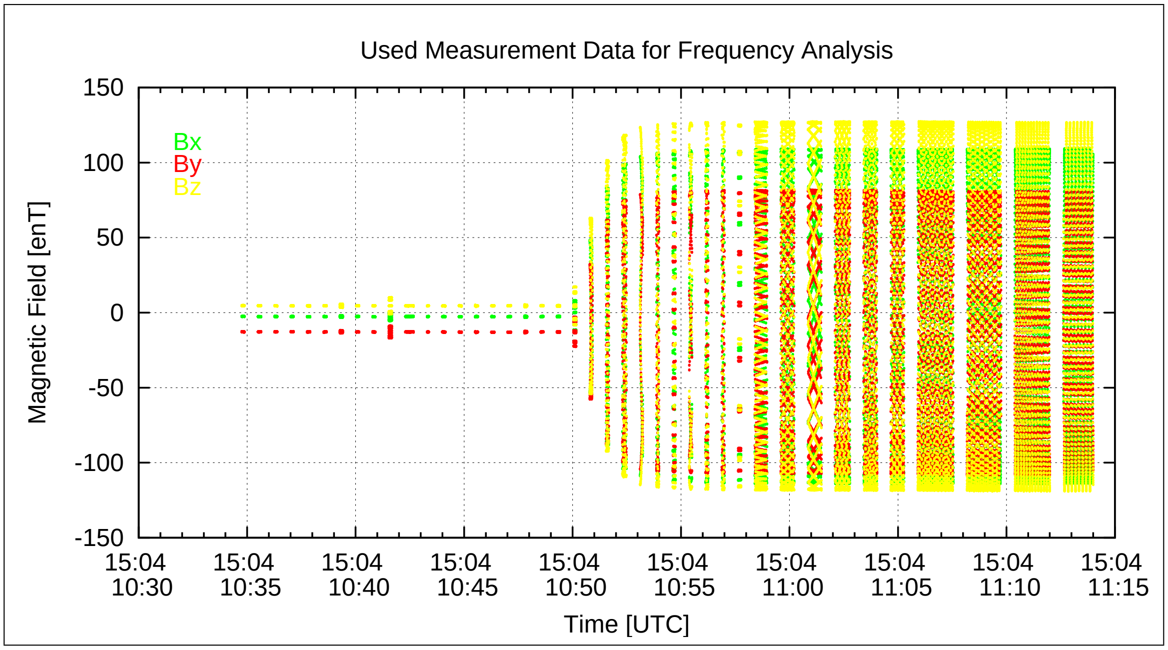

Figure 26: Used packets of Measured data for the Frequency analysis

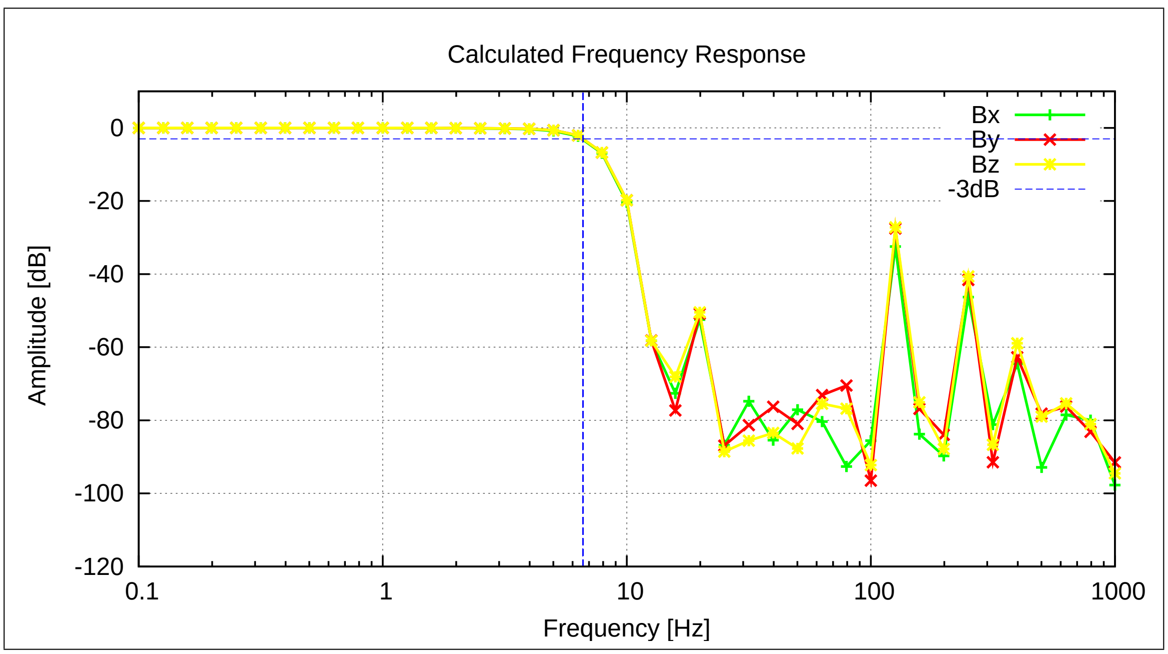

Figure 27: Calculated Frequency Response

5.9 AC-Measurements on FM-IB, 16 Hz, FIR-16Hz Filter, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.9.1 Frequency Measurements — SENSOR IB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
21-04-15\10-33-19.CCR	FREQ400.MAG	

Parameter File: FREQ_PARAMETER__21-04-15-10-33-19_IB.FPF

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	16.0008	0.005174
Y	16.0007	0.003914
Z	16.0018	0.007817
Mean Sampling Frequency	16.0011	0.000616

3 dB Corner Frequency:

Component	$f_{3\text{dB}}$ [Hz]
X	6.80
Y	6.81
Z	6.72

Applied Frequencies and Measured Amplitudes

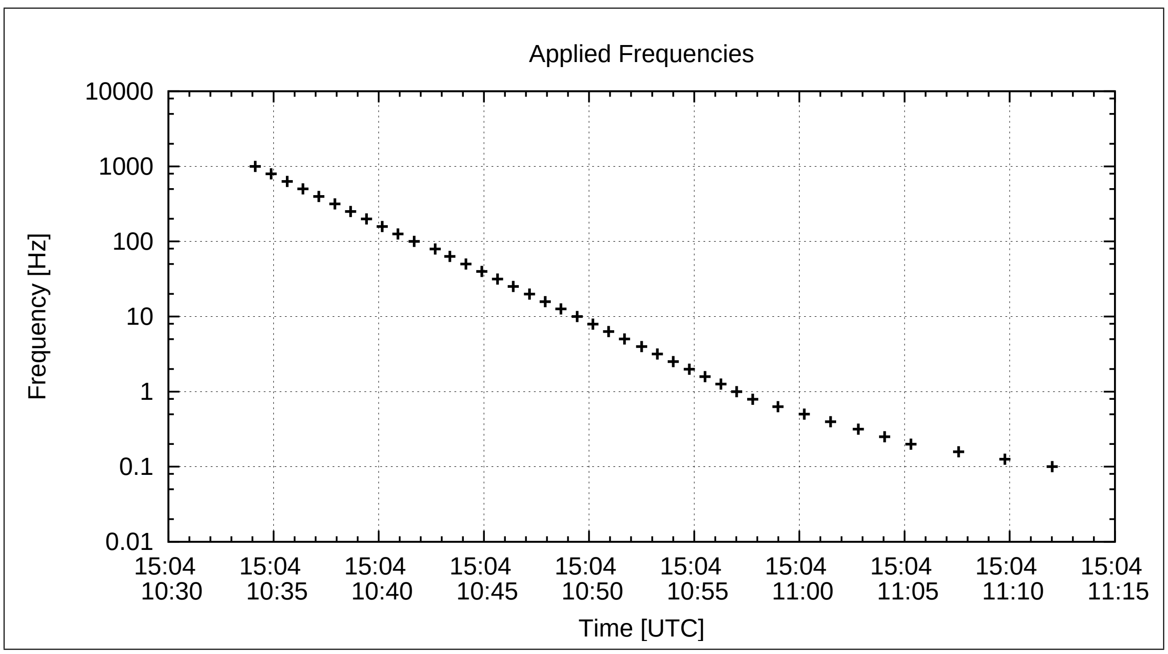

Figure 28: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.02	0.05	0.02
794.000	0.13	0.21	0.07
631.000	1.82	2.76	1.07
501.000	0.33	0.53	0.20
398.000	0.46	0.67	0.31
316.000	0.05	0.07	0.04
251.000	10.90	16.89	8.14
199.000	0.04	0.07	0.03
158.000	0.09	0.13	0.07
126.000	11.76	18.26	9.92
100.000	0.04	0.07	0.06
79.400	0.17	0.20	0.13
63.100	0.05	0.10	0.05
50.100	0.27	0.39	0.22
39.800	0.08	0.14	0.03
31.600	0.03	0.05	0.03
25.100	0.13	0.18	0.10
19.900	0.35	0.49	0.28
15.800	0.05	0.08	0.05
12.600	0.14	0.20	0.12
10.000	11.70	18.24	9.84
7.900	51.90	81.12	44.60
6.310	83.53	130.19	72.70
5.010	96.02	149.38	84.30
3.980	99.05	153.92	87.48
3.160	99.63	154.69	88.30
2.510	99.73	154.79	88.64
1.990	99.77	154.79	88.80
1.580	99.80	154.79	88.88
1.260	99.80	154.79	88.95
1.000	99.78	154.80	88.99
0.790	99.82	154.79	89.02
0.631	99.82	154.79	89.05
0.501	99.82	154.82	89.06
0.398	99.83	154.80	89.08
0.316	99.84	154.81	89.08
0.251	99.84	154.83	89.08
0.199	99.82	154.81	89.08
0.158	99.81	154.80	89.06
0.126	99.82	154.79	89.09
0.100	99.83	154.78	89.05

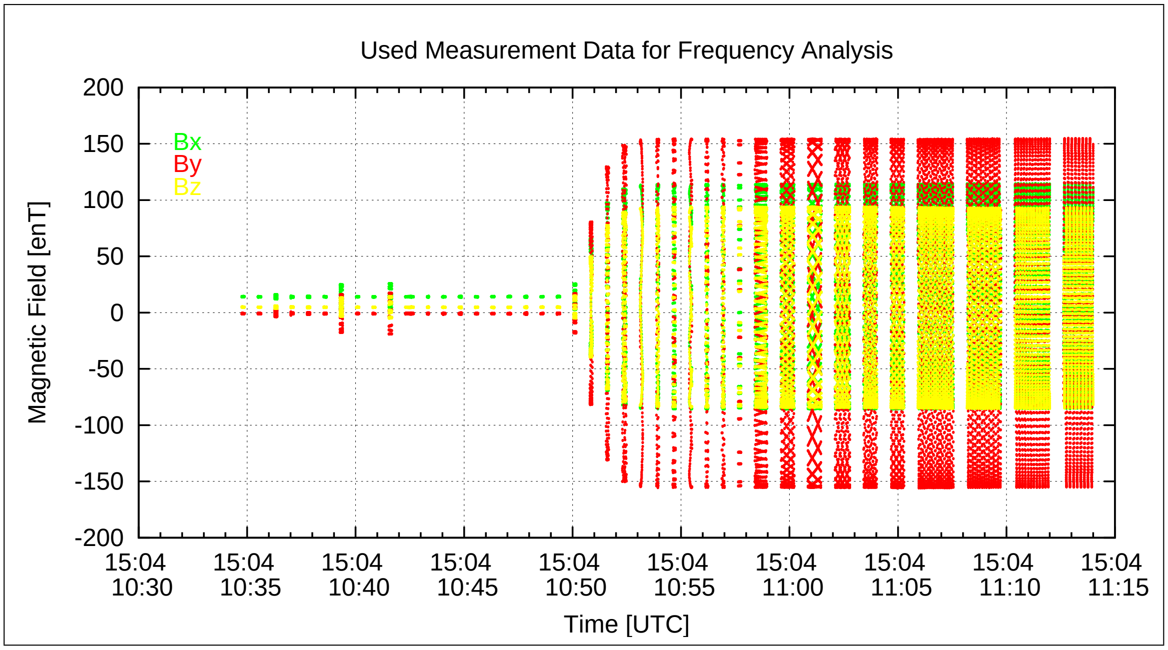

Figure 29: Used packets of Measured data for the Frequency analysis

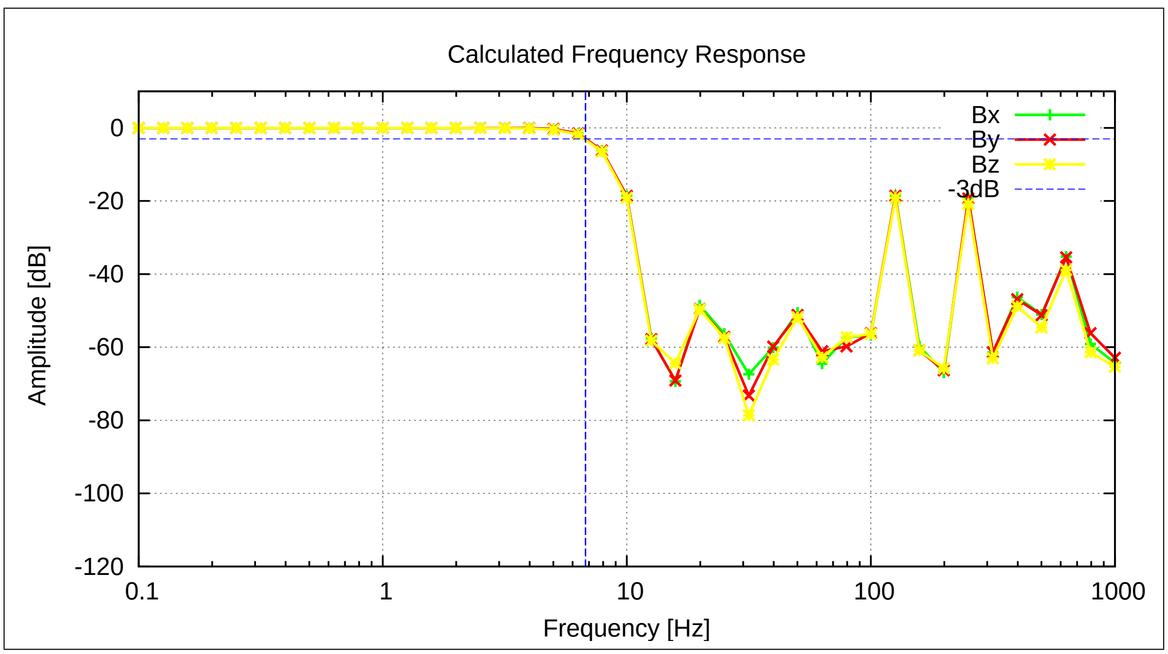

Figure 30: Calculated Frequency Response

5.10 AC-Measurements on FM-OB, 1 Hz, FIR-1Hz Filter, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.10.1 Frequency Measurements — SENSOR OB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
`21-04-15\11-17-54.CCR`	`FREQ400.MAG`	

Parameter File: `FREQ_PARAMETER__21-04-15-11-17-54_OB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	1.0005	0.003261
Y	1.0005	0.003113
Z	1.0000	0.001601
Mean Sampling Frequency	1.0003	0.000302

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	0.54
Y	0.54
Z	0.54

Applied Frequencies and Measured Amplitudes

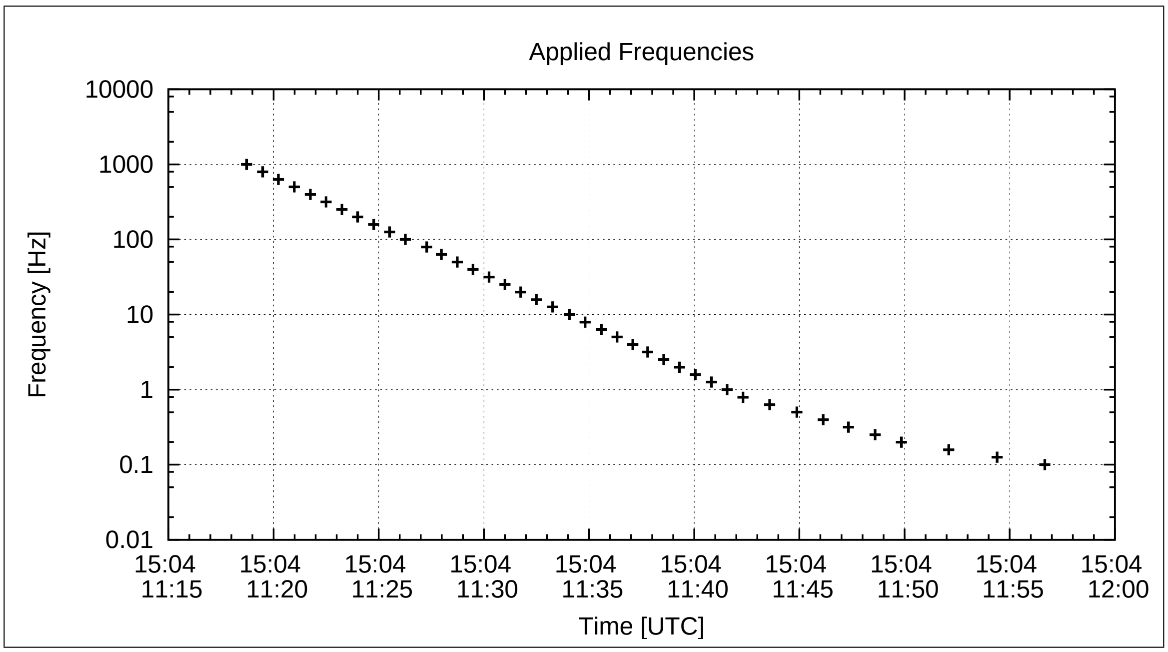

Figure 31: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	1.63	0.44	0.05
794.000	0.07	0.33	0.19
631.000	0.15	0.37	0.07
501.000	0.19	0.48	4.31
398.000	0.04	0.52	0.12
316.000	0.21	0.52	0.18
251.000	4.43	0.57	0.17
199.000	0.28	0.51	0.19
158.000	0.26	0.37	0.18
126.000	0.22	0.40	0.14
100.000	0.37	0.31	0.15
79.400	0.17	0.31	0.22
63.100	0.56	0.43	0.45
50.100	0.51	0.20	0.79
39.800	0.23	0.25	0.48
31.600	0.27	0.42	0.71
25.100	1.09	1.20	0.88
19.900	0.34	0.24	0.19
15.800	0.66	0.60	0.59
12.600	1.61	1.49	1.95
10.000	0.09	0.17	0.22
7.900	1.04	0.91	1.08
6.310	4.10	3.58	4.71
5.010	0.10	0.14	0.12
3.980	0.09	0.27	0.29
3.160	5.24	4.47	5.76
2.510	6.11	6.38	8.28
1.990	0.36	0.23	0.18
1.580	21.54	18.30	23.77
1.260	20.39	17.30	22.47
1.000	0.39	0.35	0.23
0.790	27.45	23.26	30.24
0.631	51.38	43.55	56.64
0.501	50.90	42.31	55.08
0.398	84.50	71.60	93.09
0.316	93.92	79.57	103.46
0.251	100.15	84.86	110.32
0.199	104.21	88.30	114.83
0.158	106.84	90.49	117.70
0.126	108.49	91.91	119.50
0.100	109.58	92.81	120.67

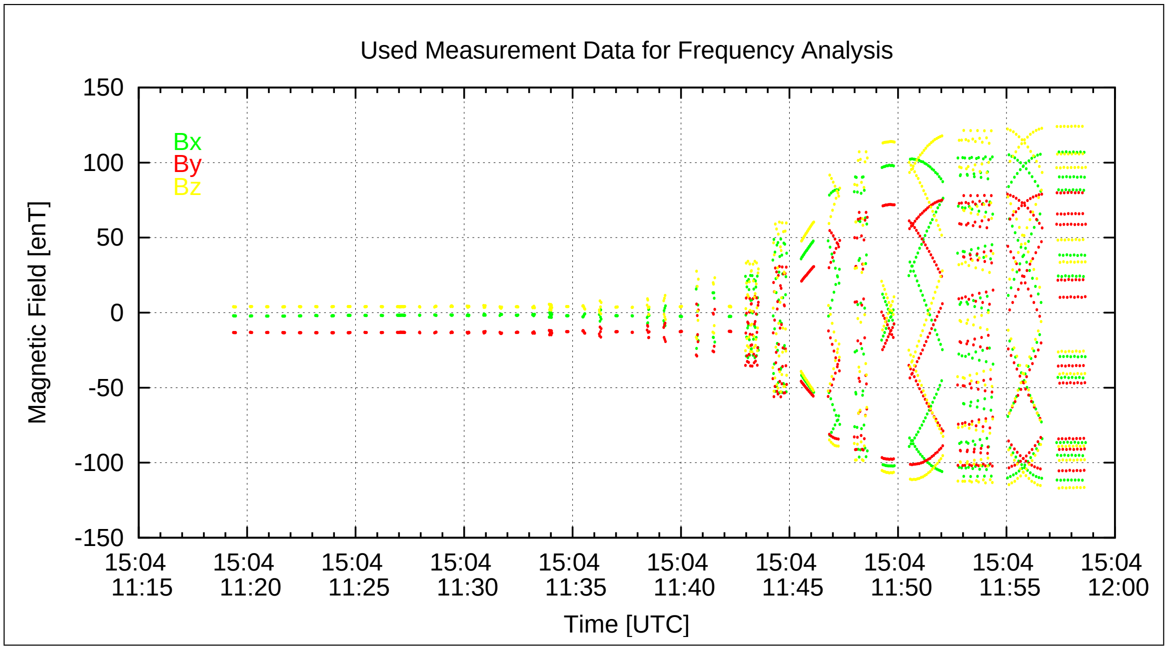

Figure 32: Used packets of Measured data for the Frequency analysis

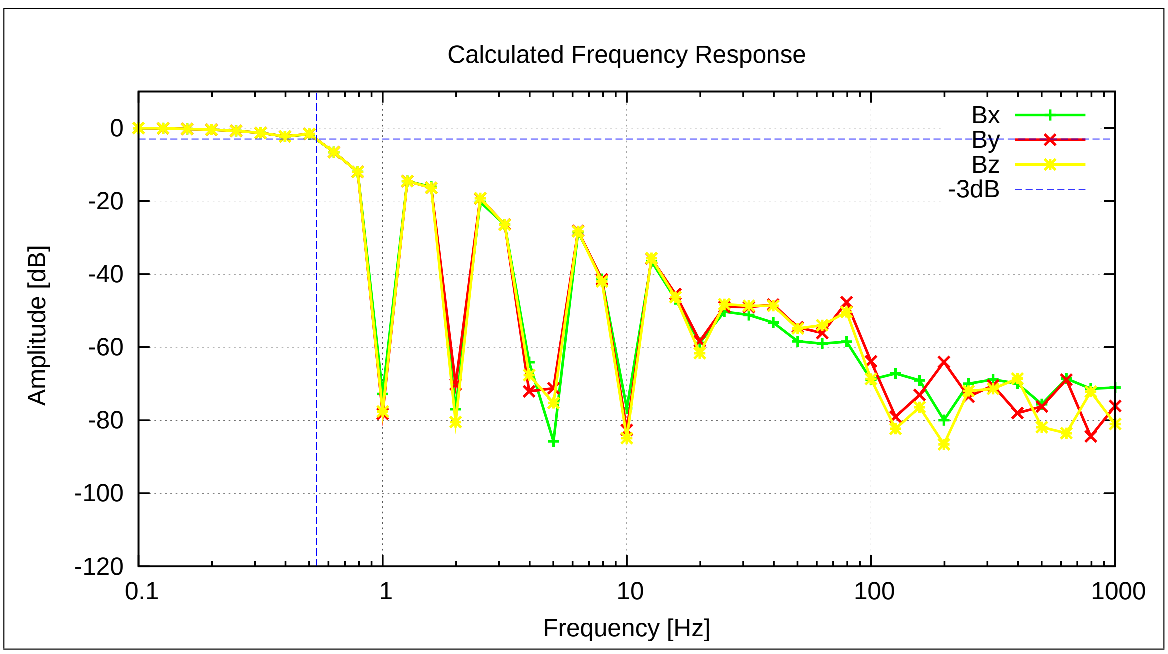

Figure 33: Calculated Frequency Response

5.11 AC-Measurements on FM-IB, 1 Hz, FIR-1Hz Filter, 400nT amplitude

A frequency measurement was performed at 17°C.

Setup:

- Both sensors mounted diagonal roughly at CoC.
- See pictures 4 to 6.
- Field applied on Y_c.

5.11.1 Frequency Measurements — SENSOR IB

This section is dedicated to the frequency behavior of the instrument. The analysis of the performed AC measurements allows to calculate the actual sampling frequency f_s of the instrument and the frequency response (amplitude vs. frequency). The measurements have been performed with the sensor placed at CoC in a diagonal in space orientation. The AC-fields have been applied on the Y_c axis only. Using this setup it can be guaranteed that only one frequency is applied at a time and no beat effects occur.

Used Frequency Measurements:

CCD File	Configuration File	Remark
21-04-15\11-17-54.CCR	FREQ400.MAG	

Parameter File: `FREQ_PARAMETER__21-04-15-11-17-54_IB.FPF`

Calibration Parameter:

Sampling Frequency:

Component	f_s [Hz]	Standard deviation [Hz]
X	1.0005	0.001849
Y	1.0008	0.004141
Z	1.0009	0.006167
Mean Sampling Frequency	1.0007	0.000216

3 dB Corner Frequency:

Component	$f_{3\mathrm{dB}}$ [Hz]
X	0.54
Y	0.54
Z	0.54

Applied Frequencies and Measured Amplitudes

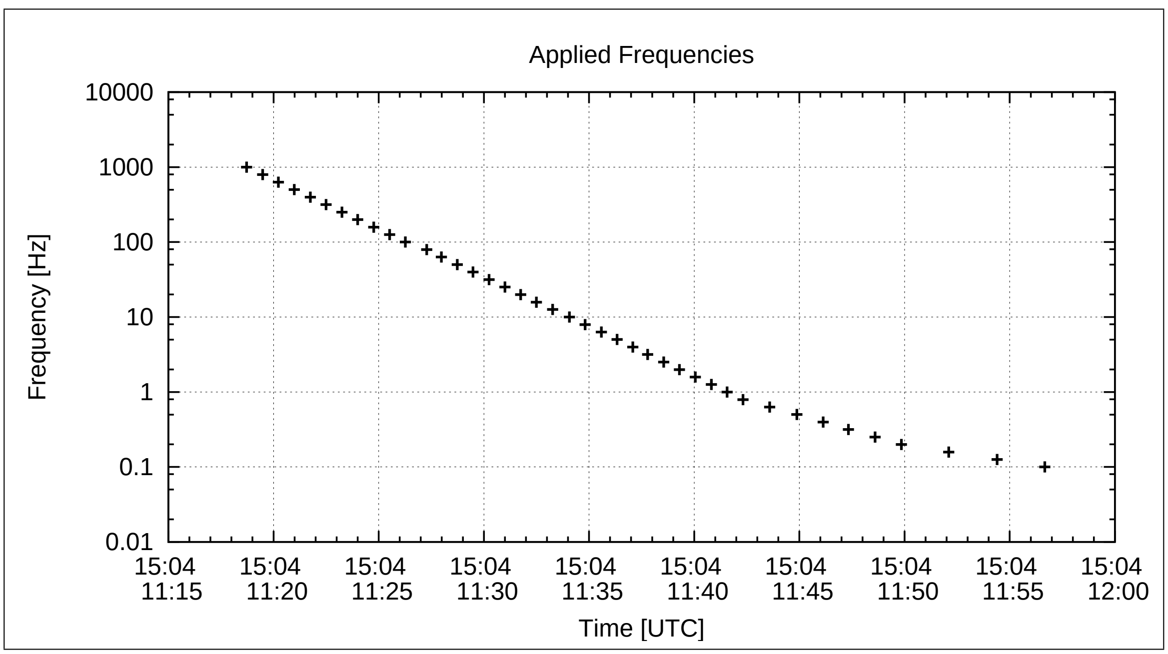

Figure 34: Applied Frequencies for AC Analysis

Applied Frequency [Hz]	Bx [enT]	By [enT]	Bz [enT]
1000.000	0.32	0.17	0.45
794.000	0.31	0.34	0.29
631.000	0.28	0.23	0.39
501.000	0.29	0.23	0.53
398.000	0.44	0.27	0.42
316.000	0.71	0.06	0.31
251.000	0.51	0.28	0.40
199.000	0.95	0.42	0.43
158.000	0.47	0.19	0.16
126.000	0.51	0.06	0.23
100.000	0.81	0.46	0.21
79.400	1.07	1.63	0.67
63.100	0.95	1.96	0.71
50.100	0.51	2.23	0.90
39.800	0.63	1.58	0.59
31.600	0.92	1.86	0.86
25.100	2.03	2.18	1.13
19.900	0.49	0.27	0.10
15.800	0.78	1.04	0.54
12.600	1.89	3.07	1.57
10.000	0.02	0.25	0.32
7.900	1.03	1.46	0.83
6.310	3.98	6.26	3.54
5.010	0.15	0.06	0.09
3.980	0.19	0.25	0.45
3.160	4.84	7.42	4.23
2.510	7.20	11.26	6.67
1.990	0.26	0.27	0.20
1.580	19.40	30.07	17.28
1.260	18.35	28.42	16.30
1.000	0.43	0.33	0.17
0.790	24.62	38.16	21.93
0.631	46.05	71.44	41.11
0.501	46.94	73.01	40.43
0.398	75.70	117.38	67.53
0.316	84.11	130.44	75.05
0.251	89.68	139.08	80.03
0.199	93.31	144.74	83.29
0.158	95.65	148.37	85.36
0.126	97.15	150.63	86.67
0.100	98.11	152.13	87.52

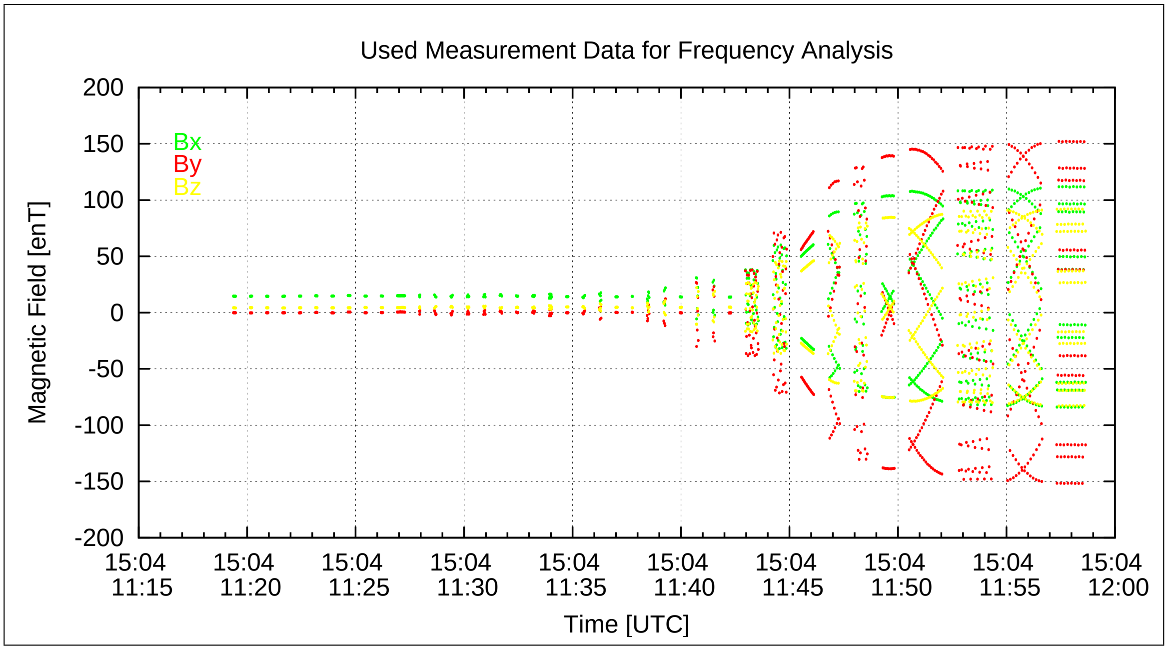

Figure 35: Used packets of Measured data for the Frequency analysis

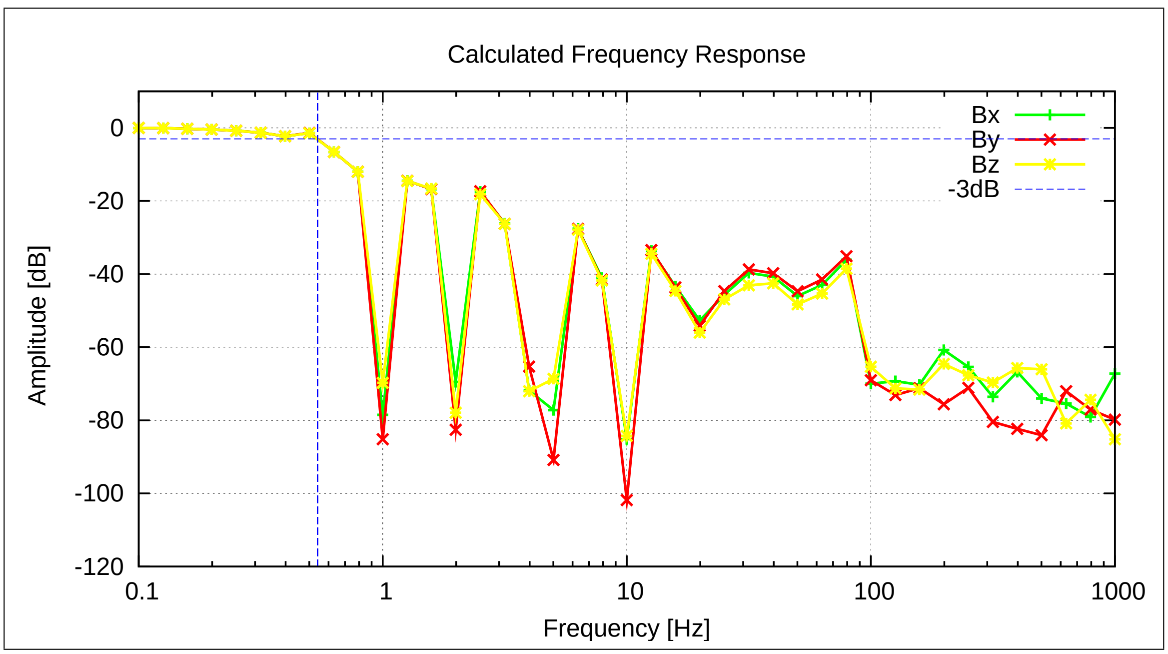

Figure 36: Calculated Frequency Response

6 Sensor Setup Pictures for Mode Checks

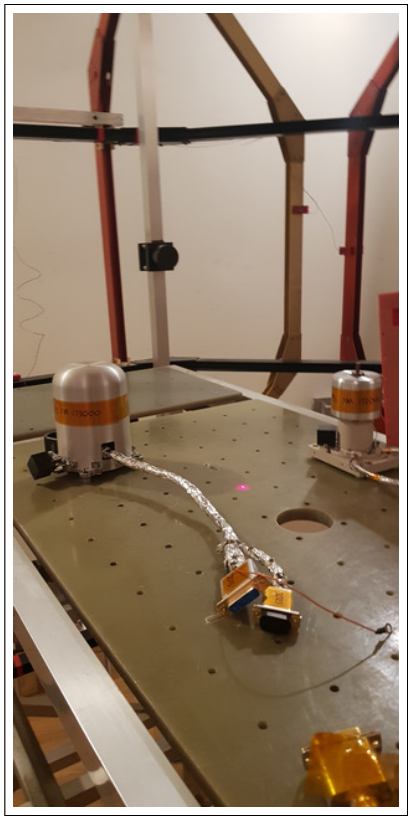

Figure 37: Setup for Mode Checks of OB and IB Sensor, view from south.

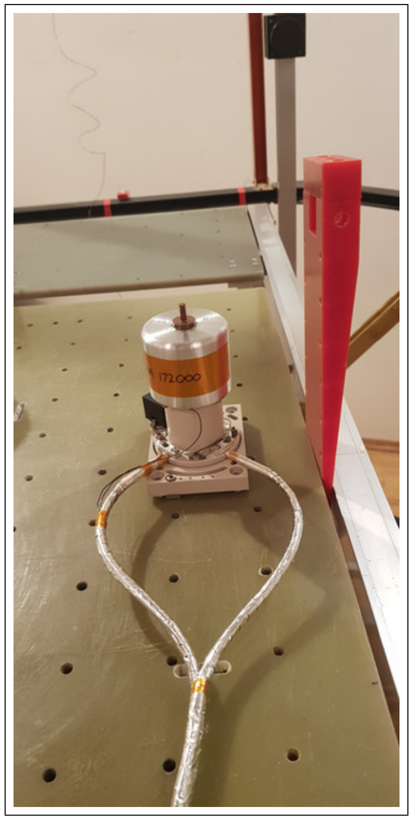

Figure 38: Setup for Mode Checks of OB and IB Sensor, view of IB from south.

Figure 39: Setup for Mode Checks of OB and IB Sensor, view of OB from west.

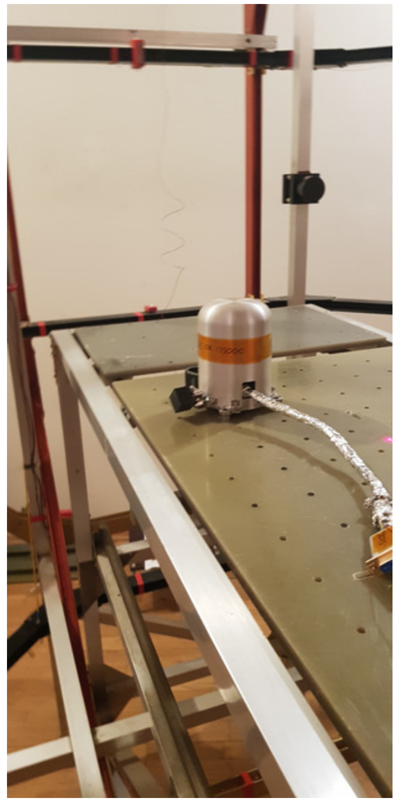

Figure 40: Setup for Mode Checks of OB and IB Sensor, view of OB from south.

6.1 Sensor Rotations, Mode checks: FGM-IB and FGM OB

All sensors were rotated at CoC in 90 deg steps around all axes.

For the IB sensor the rotations were done in

- zerofield conditions
- Field: 10000 nT, 4000 nT, 0 nT

For the OB sensor the rotations were done in

- zerofield conditions
- Field: 6000 nT, 2000 nT, 0 nT
- Field: 40000 nT, 10000 nT, 0 nT

The analysis of these measurements is done by Uli Auster.

6.2 Alignment Check

An extended alignment check was executed on the FGM-OB, FGM-IB and the spare IB sensor. This was done using Uli's special rotating device.

For the IB sensor the rotations were done in the following field conditions.

- Field: 0 nT, 0 nT, -10000 nT
- Field: uncompensated full Earthfield

For the OB sensor the rotations were done in the following configuration:

- Field: uncompensated full Earthfield

The analysis of these measurements is done by Uli Auster/ IC team.

6.3 Sensor Mounting: FGM-OB /Orientation / Alignment Check

After these tests the FGM-OB sensor was mounted in the thermal test box. See pictures 39 to 44.

Figure 41: Setup for Standard Calibration of the OB Sensor, Azimuth=0, Elevation=0, view from south.

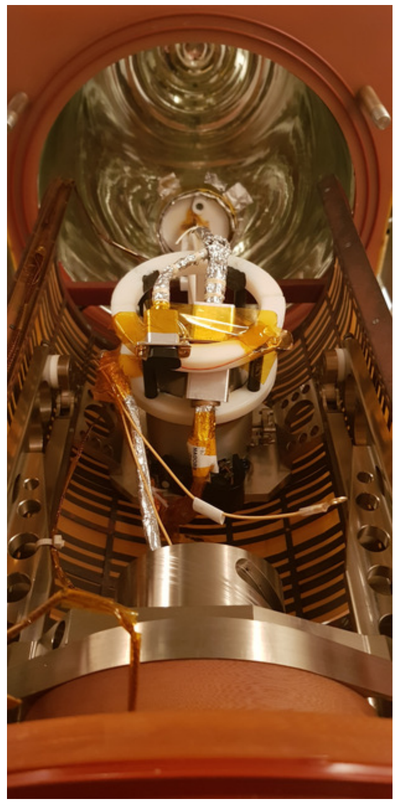

Figure 42: Setup for Standard Calibration of the OB Sensor, Azimuth=0, Elevation=90, view from south.

Figure 43: Setup for Standard Calibration of the OB Sensor, Azimuth=180, Elevation=0, view from south.

Figure 44: Setup for Standard Calibration of the OB Sensor, Azimuth=180, Elevation=90, view from south.

- POSITION 1: Normal Position for Offset Measurements XZ:
  $X_m = +Y_c, Y_m = -Z_c, Z_m = -X_c$
  Elevation=0°. Azimuth= 180°.
- POSITION 2: Turned Position for turned Offset Measurements XZ:
  $X_m = -Y_c, Y_m = -Z_c, Z_m = +X_c$
  Elevation=0°. Azimuth= 0°.
- POSITION 3: Normal Position for Offset Measurements Y:
  $X_m = -Y_c, Y_m = -X_c, Z_m = +Z_c$
  Elevation=90°. Azimuth= 00°.
- POSITION 4: Normal Position for turned Offset Measurements Y:
  $X_m = +Y_c, Y_m = +X_c, Z_m = +Z_c$
  Elevation=0°. Azimuth= 180°.

All linearity measurements are executed in POSITION 1.

Application of separated constant fields in all main axes in standard orientation yielded:

X_m	=	$+Y_\mathrm{c}$
Y_m	=	-Z_c
Z_m	=	-X_c

Thus, this test confirmed that the FM-OB sensor coordinate system is RIGHT-HANDED.

6.4 DC-Analysis at Room Temperature, Range=R0, FGM-OB

6.4.1 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-15\19-23-48.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER__21-04-15-19-23-48_.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	8.28	0.103
Y	-15.19	0.204
Z	-8.55	0.227

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-0.84
Y	6.96
Z	6.36

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

6.4.2 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
21-04-15\19-44-03.CCR	LIN60000XYZ.MAG	

Parameter File: PARAMETER_LIN___21-04-15_19-44-03.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.007521 & +0.053763 & -0.998525 \\ -0.999861 & -0.015247 & +0.006710 \\ -0.014863 & +0.998437 & +0.053870 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 25'51'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 52'25'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 54'43'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: FLDS
Fields applied for 20.0 s
Mean Sensor Temperature: 17.7°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (7.862 , -23.763 , -2.871) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.031	0.000	0.031
Mean	0.099	0.096	0.098
Maximum	0.275	0.270	0.273

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.152459 & +0.018430 & -0.008061 \\ -0.009934 & +1.204981 & +0.064713 \\ -0.000424 & -0.068854 & +1.199516 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.152444 & +0.000000 & +0.000000 \\ +0.007675 & +1.207080 & +0.000000 \\ -0.008691 & -0.003964 & +1.201287 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.152444 & +0.000000 & +0.000000 \\ +0.000000 & +1.207054 & +0.000000 \\ +0.000000 & +0.000000 & +1.201246 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} \quad = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006660 & +1.000022 & +0.000000 \\ -0.007542 & -0.003284 & +1.000034 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\xi_{\text{x,y}} = +90^\circ\ 22'54''$$
$$\xi_{\text{x,z}} = +89^\circ\ 34'9''$$
$$\xi_{\text{y,z}} = +89^\circ\ 48'53''$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	14.480	12.848	10.131
Maximum Error	7.096	6.783	7.026
Minimum Error	-49.895	-45.712	-37.033

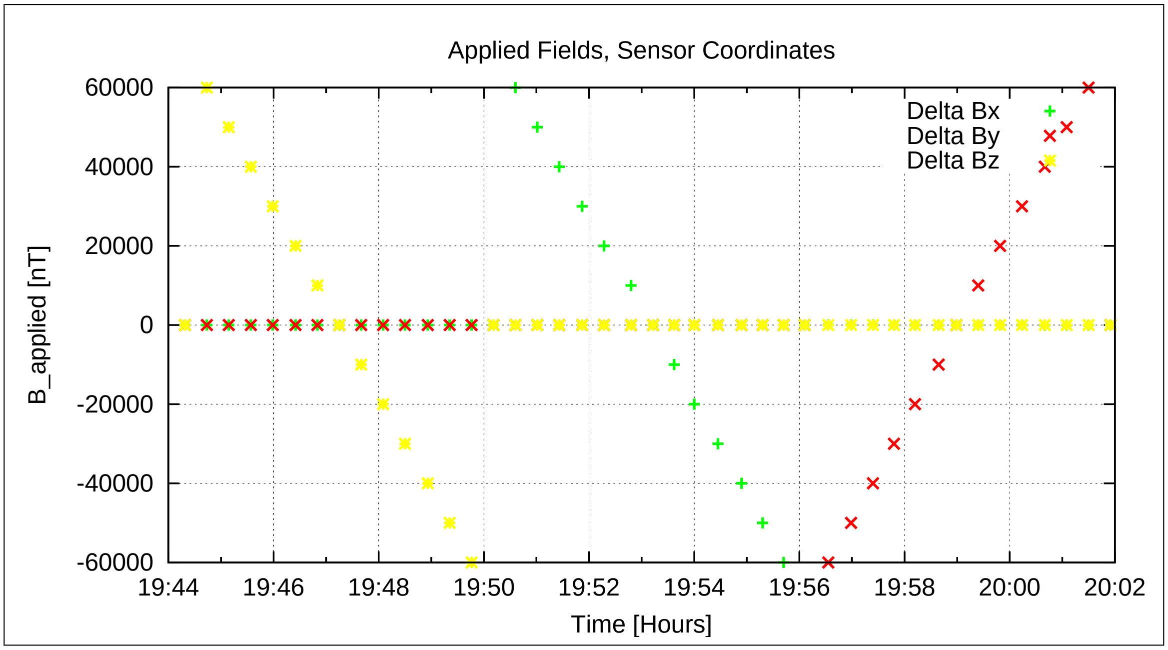

Figure 45: Applied Fields

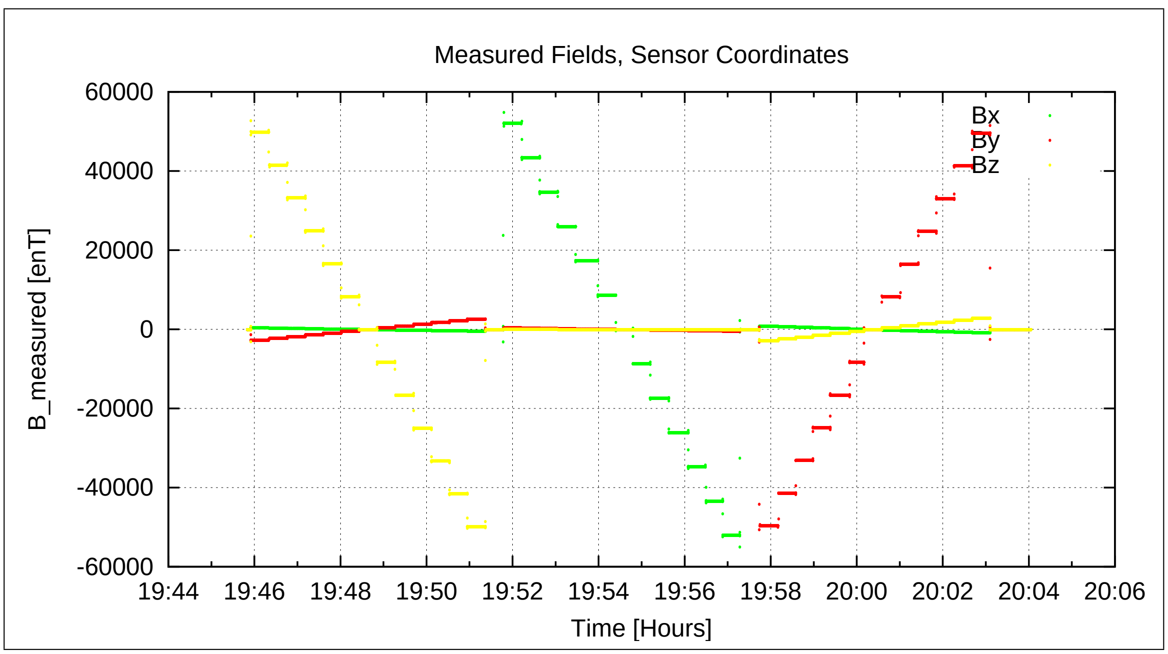

Figure 46: Measured Data

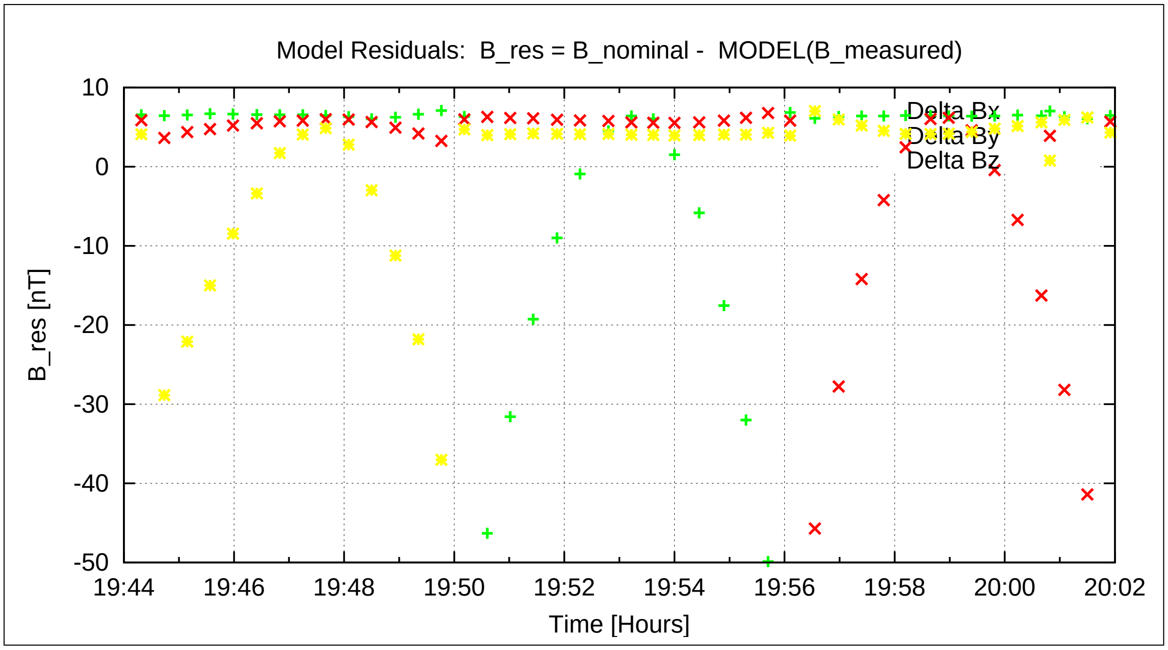

Figure 47: Model Quality - Differences of applied field and modelled measurement data

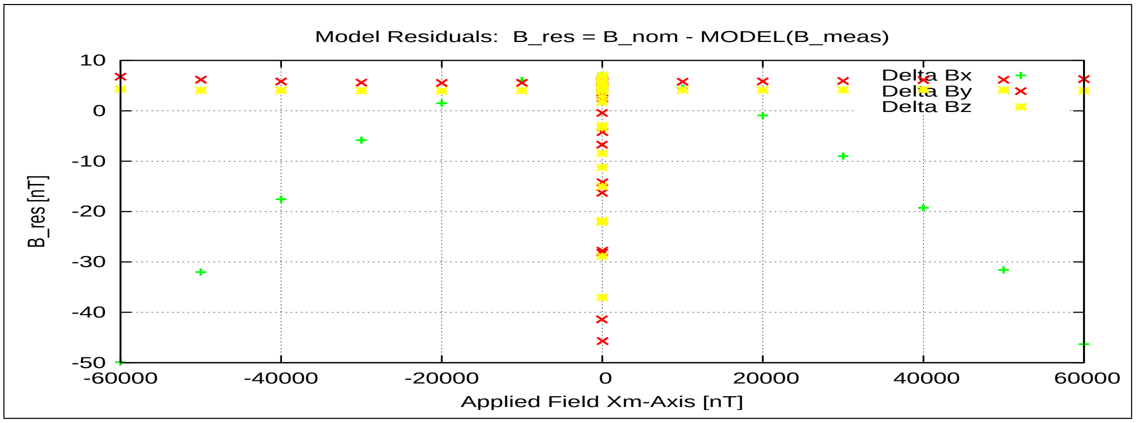

Figure 48: RESIDUALS vs FLD X

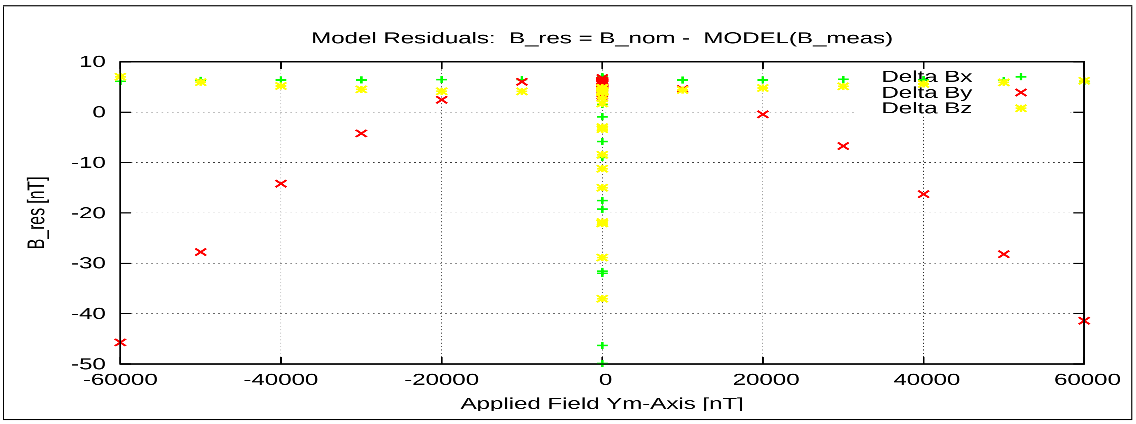

Figure 49: RESIDUALS vs FLD Y

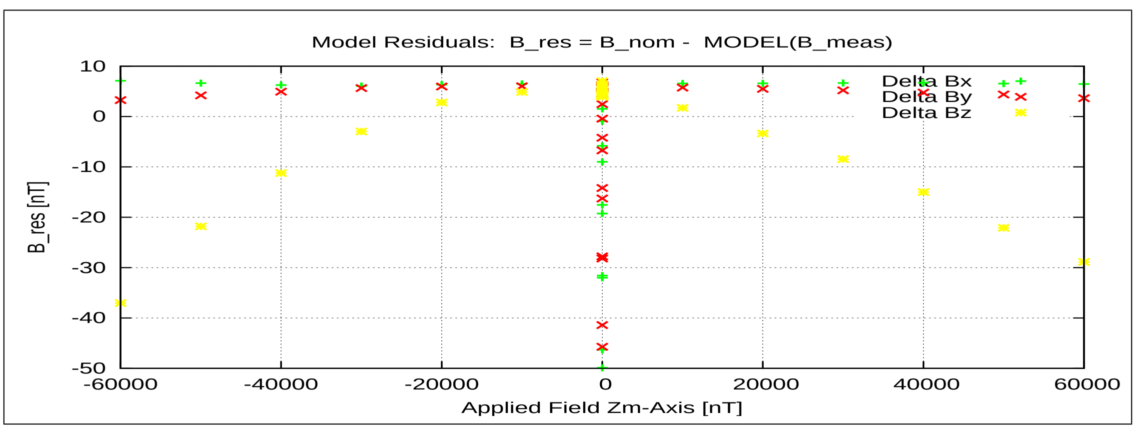

Figure 50: RESIDUALS vs FLD Z

6.5 DC-Analysis at Room Temperature, Range=R1, FGM-OB

6.5.1 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-15\19-33-14.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER__21-04-15-19-33-14_.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	-2.85	0.072
Y	-5.95	0.109
Z	-3.50	0.096

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-1.13
Y	7.89
Z	6.97

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

6.5.2 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
21-04-15\20-04-09.CCR	LIN8000XYZ.MAG	

Parameter File: PARAMETER_LIN___21-04-15_20-04-09.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.007403 & +0.053758 & -0.998527 \\ -0.999862 & -0.015262 & +0.006591 \\ -0.014885 & +0.998437 & +0.053864 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 25'27'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 52'28'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 54'44'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: FLDS
Fields applied for 19.0 s
Mean Sensor Temperature: 17.8°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 0 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (-3.905 , -12.175 , 4.137) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.000	0.000	0.000
Mean	0.042	0.045	0.042
Maximum	0.119	0.107	0.126

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.035407 & +0.016514 & -0.007125 \\ -0.008859 & +1.078626 & +0.058223 \\ -0.000426 & -0.061623 & +1.079329 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.035393 & +0.000000 & +0.000000 \\ +0.006979 & +1.080505 & +0.000000 \\ -0.007727 & -0.003542 & +1.080922 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.035393 & +0.000000 & +0.000000 \\ +0.000000 & +1.080480 & +0.000000 \\ +0.000000 & +0.000000 & +1.080886 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006741 & +1.000023 & +0.000000 \\ -0.007463 & -0.003279 & +1.000033 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +90^\circ\ 23'10'' \\ \xi_{\text{x,z}} &= +89^\circ\ 34'25'' \\ \xi_{\text{y,z}} &= +89^\circ\ 48'54'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.161	0.514	0.217
Maximum Error	0.344	0.988	0.292
Minimum Error	-0.463	-0.963	-0.640

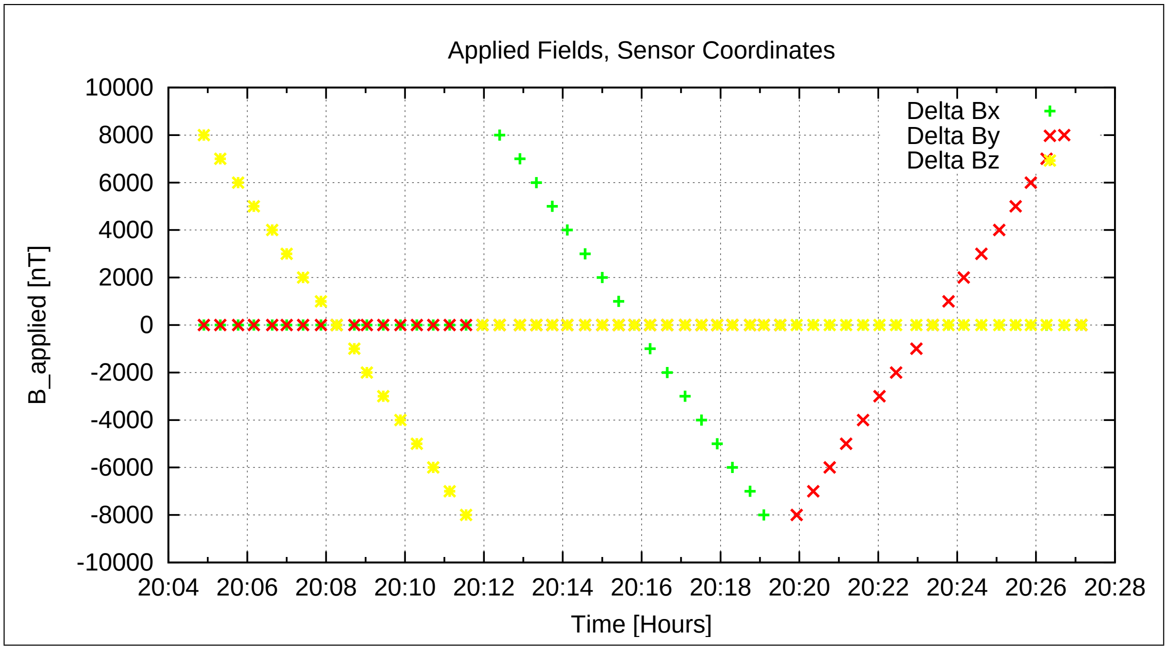

Figure 51: Applied Fields

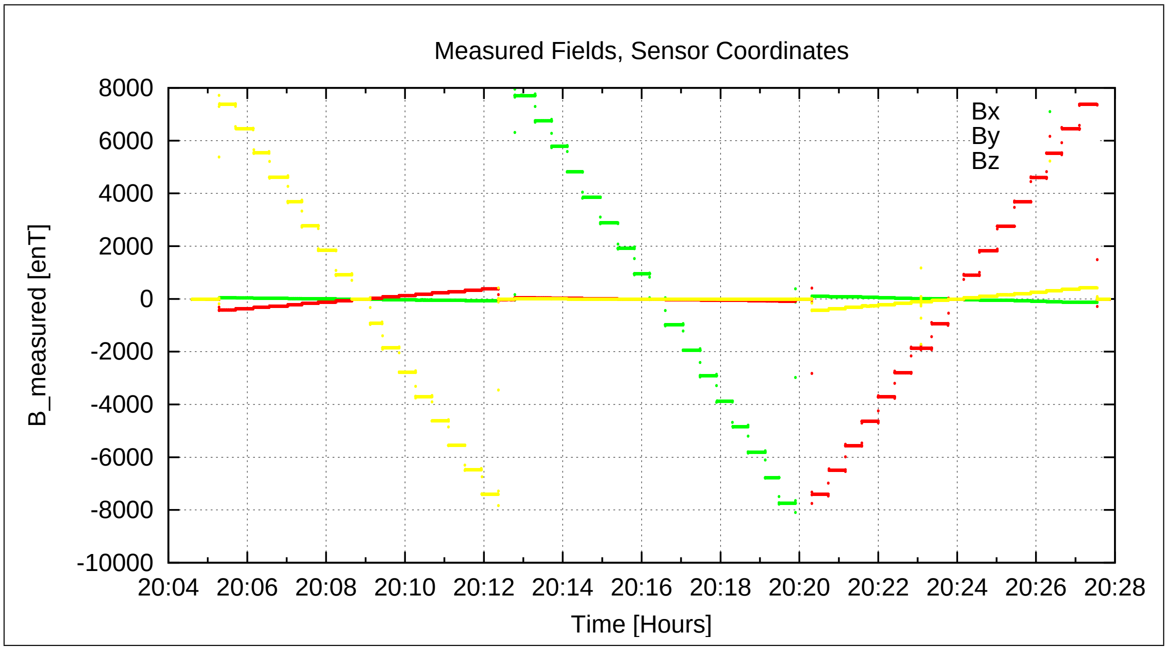

Figure 52: Measured Data

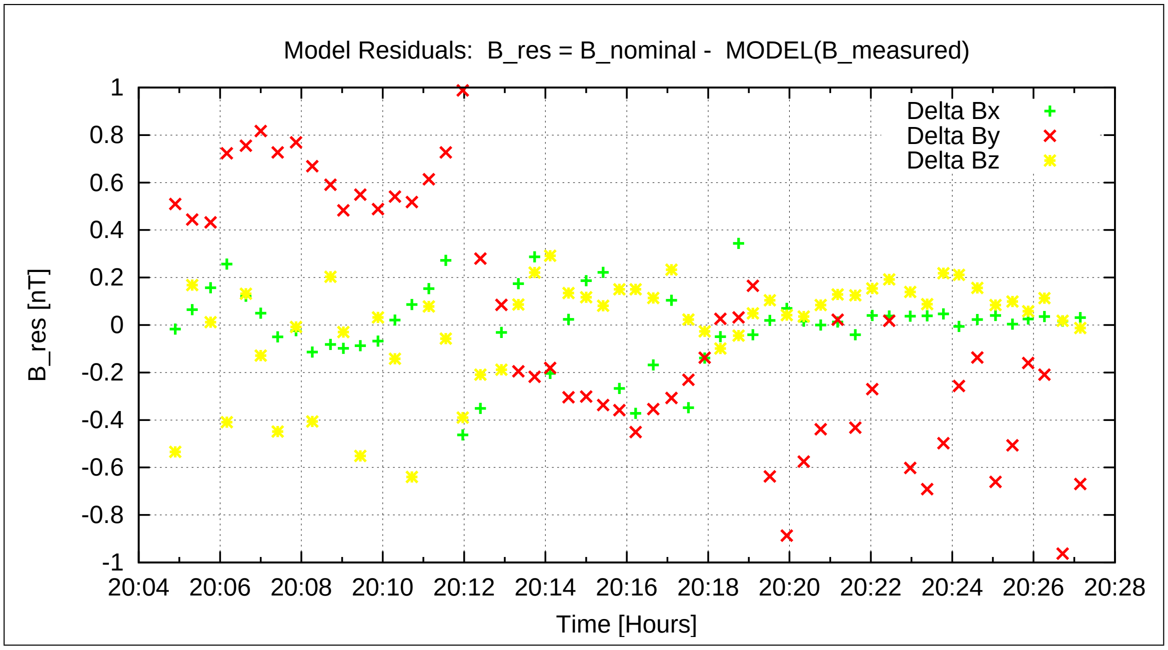

Figure 53: Model Quality - Differences of applied field and modelled measurement data

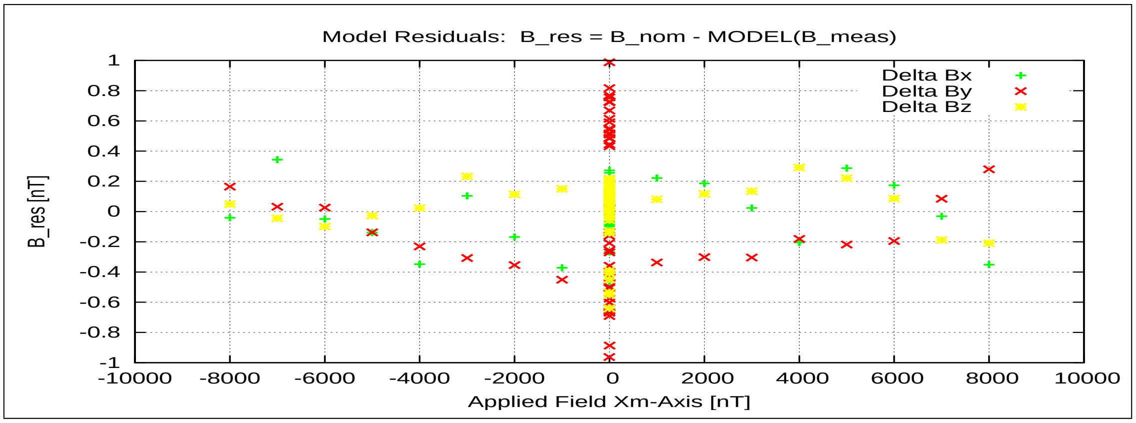

Figure 54: RESIDUALS vs FLD X

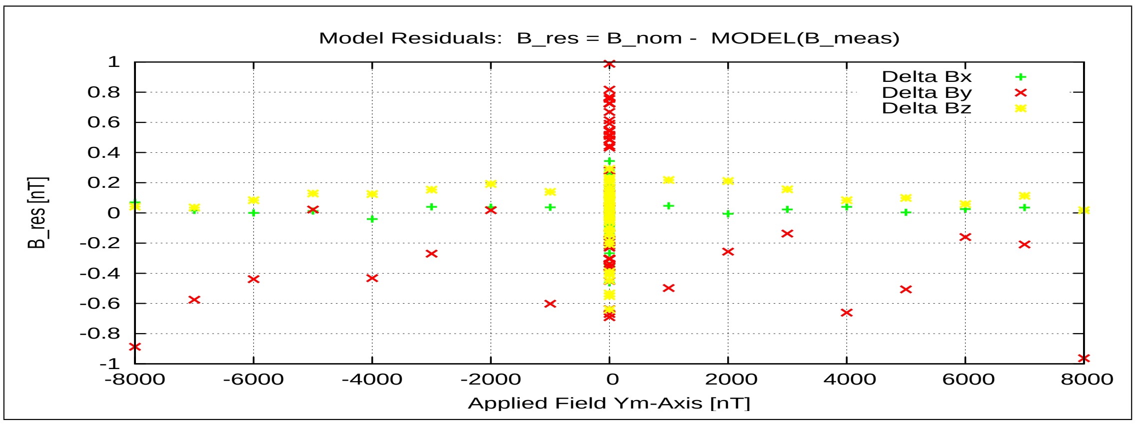

Figure 55: RESIDUALS vs FLD Y

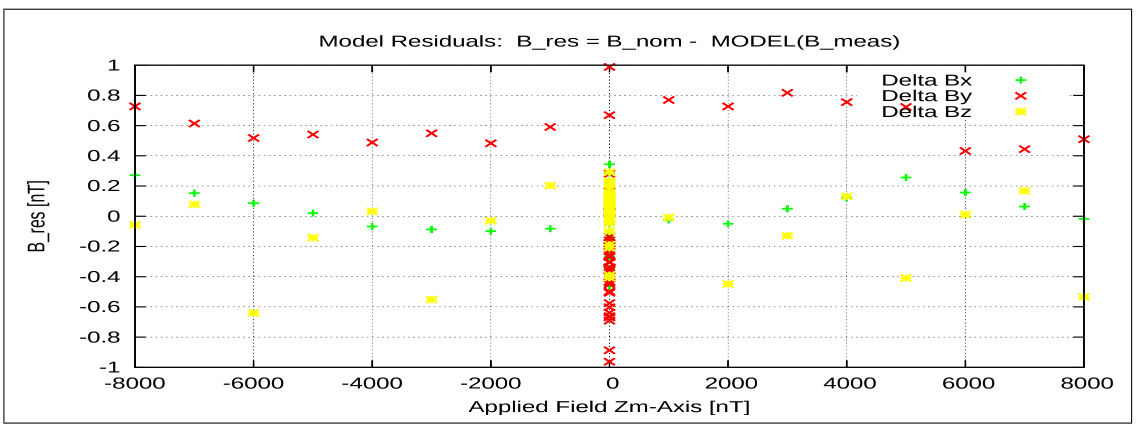

Figure 56: RESIDUALS vs FLD Z

7 Measurements on April 16, 2021

7.1 Data

CCD File	Configuration File	Remark
21-04-16\08-11-05.CCR	LIN60000XYZ.MAG	
21-04-16\09-08-53.CCR	LIN8000XYZ.MAG	
21-04-16\09-37-17.CCR	OFFSET_200.MAG	
21-04-16\13-00-07.CCR	LIN60000XYZ.MAG	
21-04-16\13-23-27.CCR	LIN8000XYZ.MAG	
21-04-16\13-48-45.CCR	OFFSET_200.MAG	
21-04-16\13-57-59.CCR	OFFSET_200.MAG	
21-04-16\14-07-15.CCR	LIN60000XYZ.MAG	
21-04-16\15-50-45.CCR	LIN8000XYZ.MAG	
21-04-16\16-17-42.CCR	OFFSET_200.MAG	
21-04-16\16-24-15.CCR	OFFSET_200.MAG	
21-04-16\16-52-07.CCR	LIN8000XYZ.MAG	
21-04-16\17-15-34.CCR	NULL.MAG	
21-04-16\17-52-51.CCR	LIN8000XYZ.MAG	
21-04-16\18-16-18.CCR	NULL.MAG	
21-04-16\18-53-35.CCR	LIN8000XYZ.MAG	
21-04-16\19-17-02.CCR	NULL.MAG	
21-04-16\19-54-20.CCR	LIN8000XYZ.MAG	
21-04-16\20-17-46.CCR	NULL.MAG	
21-04-16\20-55-04.CCR	LIN8000XYZ.MAG	
21-04-16\21-18-31.CCR	NULL.MAG	
21-04-16\21-55-48.CCR	LIN8000XYZ.MAG	
21-04-16\22-19-16.CCR	NULL.MAG	
21-04-16\22-56-33.CCR	LIN8000XYZ.MAG	
21-04-16\23-20-00.CCR	NULL.MAG	
21-04-16\23-57-18.CCR	LIN8000XYZ.MAG	

Inspection of the system at 0630. Everything is ok. Temperature in the box is about 18 degC. Today a cooling cycle down to $-140°$C with some intermediate temperature levels shall be conducted. The cooling cycle started with a temperature goal of -30 degC. Initial cooling time was set to 20s. System did not need any additional pressurization, as pressure already built up during night. After reaching -30 degC the thermal parameters were set as follows: Thermal Parameter:

- T-COOL-MAX = 5 s
- FLD-ACTIVE = 25 s
- Asynchtronous cooling
- Delay =0

7.2 DC-Analysis at T = -30°C, Range=R0, FGM-OB

7.2.1 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
21-04-16\08-11-05.CCR	LIN60000XYZ.MAG	

Parameter File: PARAMETER_LIN___21-04-16_08-11-05.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.007784 & +0.054841 & -0.998465 \\ -0.999858 & -0.015328 & +0.006953 \\ -0.014923 & +0.998377 & +0.054952 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 26'46'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 52'42'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 50'60'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: FLDS
Fields applied for 22.0 s
Mean Sensor Temperature: -33.3°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (10.118 , -26.124 , 1.433) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.000	0.000	0.000
Mean	0.115	0.100	0.114
Maximum	0.371	0.367	0.415

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.154989 & +0.018566 & -0.008369 \\ -0.010151 & +1.207287 & +0.066143 \\ -0.000129 & -0.070266 & +1.201788 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.154976 & +0.000000 & +0.000000 \\ +0.007577 & +1.209466 & +0.000000 \\ -0.008718 & -0.003944 & +1.203636 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.154976 & +0.000000 & +0.000000 \\ +0.000000 & +1.209440 & +0.000000 \\ +0.000000 & +0.000000 & +1.203595 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006560 & +1.000022 & +0.000000 \\ -0.007548 & -0.003261 & +1.000034 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +90^\circ\ 22'33'' \\ \xi_{\text{x,z}} &= +89^\circ\ 34'7'' \\ \xi_{\text{y,z}} &= +89^\circ\ 48'58'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	13.405	11.861	10.428
Maximum Error	7.311	6.002	7.234
Minimum Error	-46.699	-42.484	-37.555

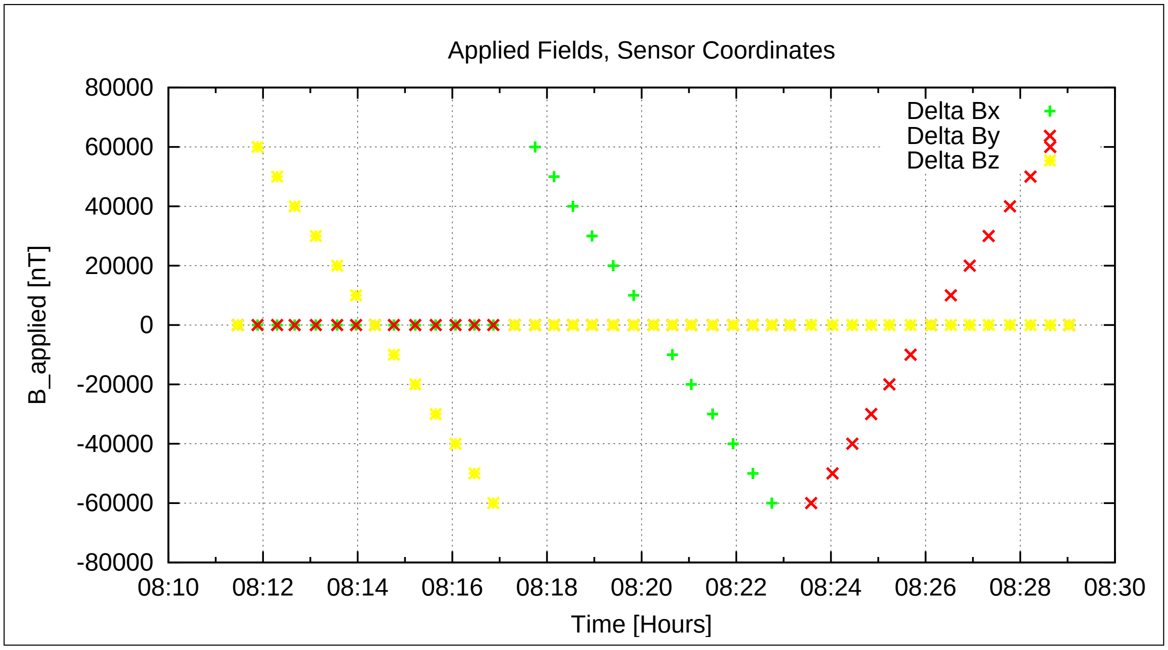

Figure 57: Applied Fields

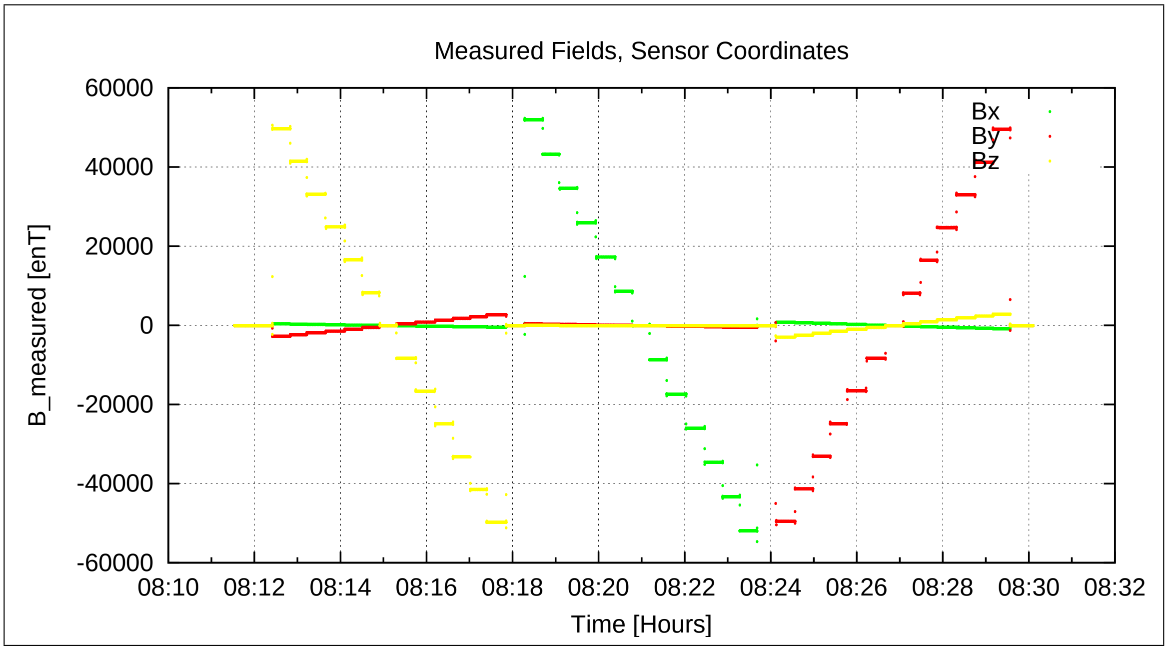

Figure 58: Measured Data

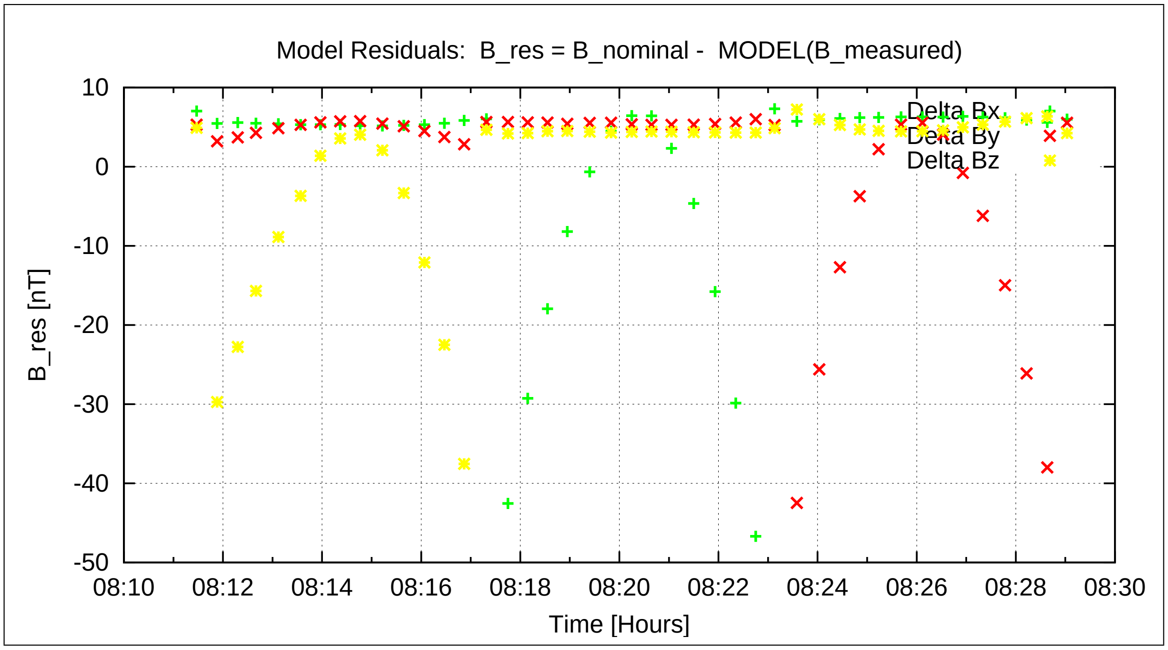

Figure 59: Model Quality - Differences of applied field and modelled measurement data

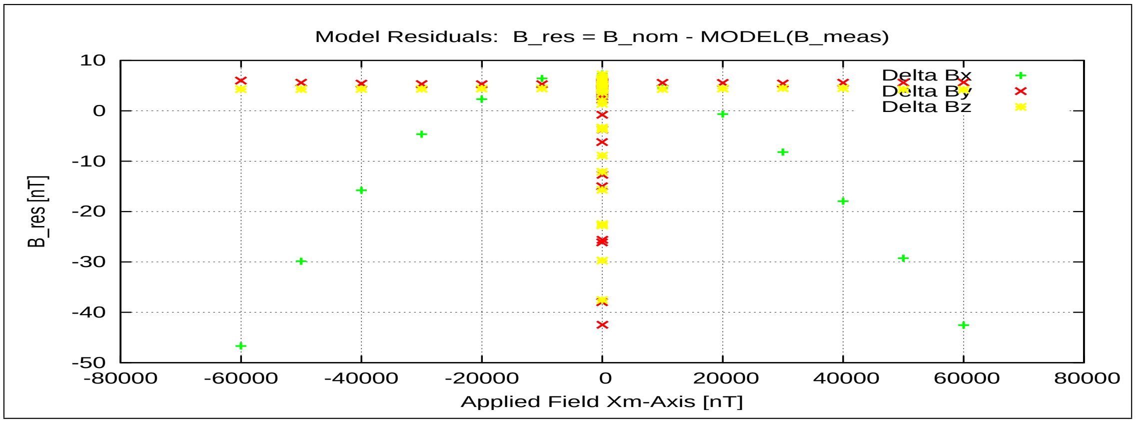

Figure 60: RESIDUALS vs FLD X

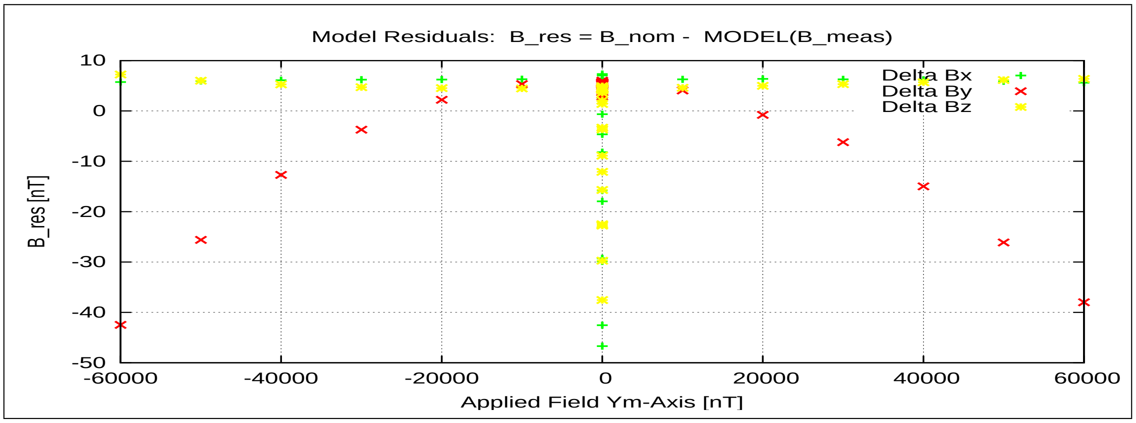

Figure 61: RESIDUALS vs FLD Y

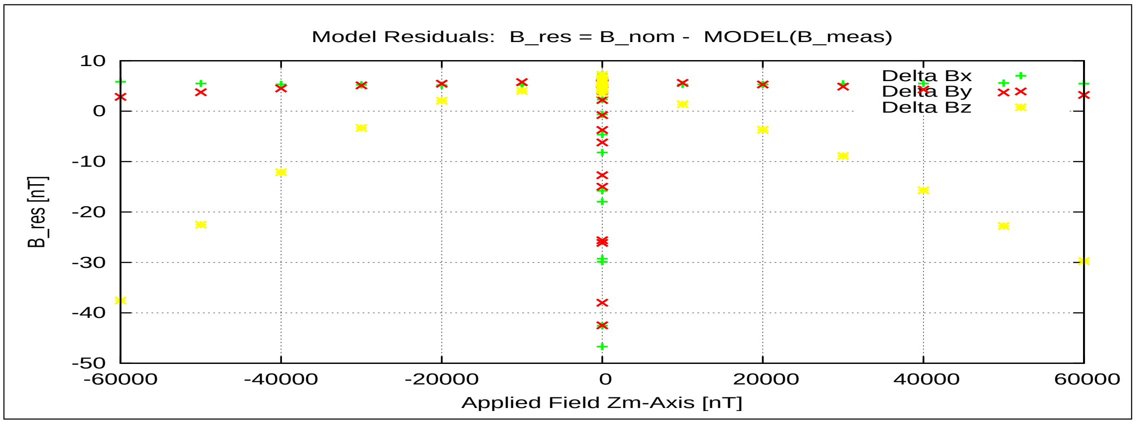

Figure 62: RESIDUALS vs FLD Z

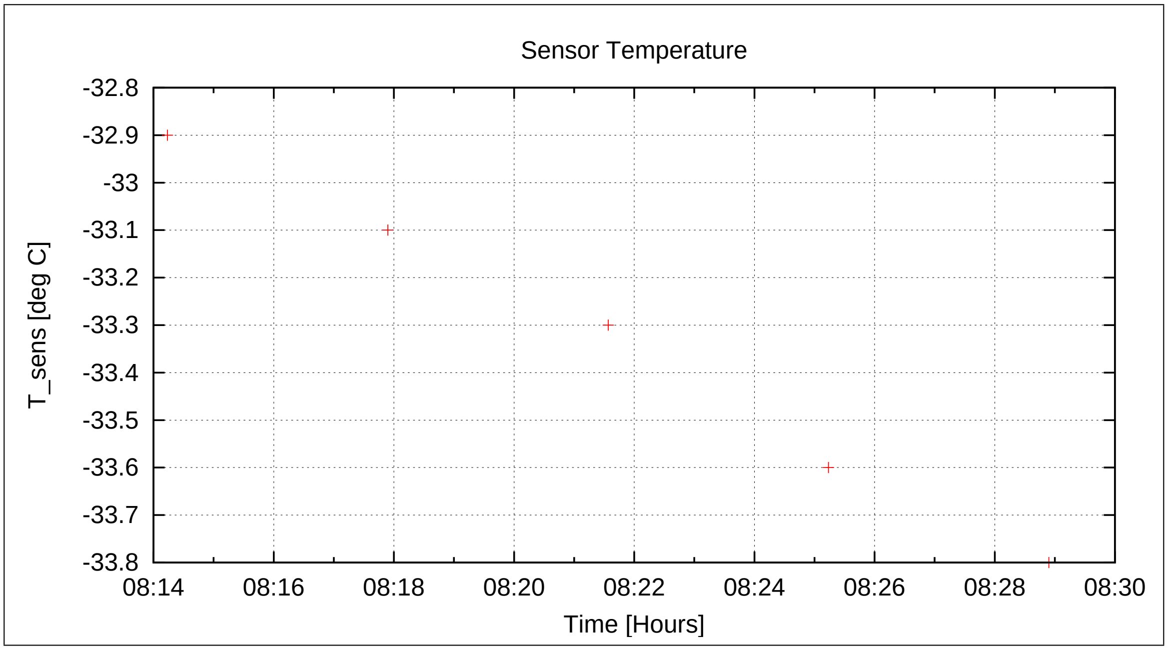

Figure 63: Sensor Temperature

7.3 DC-Analysis at T = $-30°$C, Range=R1, FGM-OB

7.3.1 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
`21-04-16\09-08-53.CCR`	`LIN8000XYZ.MAG`	

Parameter File: `PARAMETER_LIN___21-04-16_09-08-53.CPF`

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.007928 & +0.054920 & -0.998459 \\ -0.999858 & -0.015287 & +0.007098 \\ -0.014873 & +0.998374 & +0.055034 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 27'15'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 52'33'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 50'43'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: FLDS
Fields applied for 23.0 s
Mean Sensor Temperature: -36.7°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (-1.099 , -12.683 , 7.437) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.016	0.018	0.015
Mean	0.043	0.046	0.041
Maximum	0.104	0.109	0.150

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\, \underline{\underline{\rho}}\, \underline{\underline{\omega}}\, \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.036457 & +0.016555 & -0.007677 \\ -0.009121 & +1.079385 & +0.059525 \\ -0.000044 & -0.062884 & +1.079942 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\, \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.036445 & +0.000000 & +0.000000 \\ +0.006740 & +1.081336 & +0.000000 \\ -0.007903 & -0.003502 & +1.081609 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.036445 & +0.000000 & +0.000000 \\ +0.000000 & +1.081313 & +0.000000 \\ +0.000000 & +0.000000 & +1.081572 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} \quad = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006503 & +1.000021 & +0.000000 \\ -0.007625 & -0.003239 & +1.000034 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\xi_{\mathrm{x,y}} = +90^\circ\ 22'21''$$
$$\xi_{\mathrm{x,z}} = +89^\circ\ 33'52''$$
$$\xi_{\mathrm{y,z}} = +89^\circ\ 49'2''$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.466	0.230	0.330
Maximum Error	0.980	0.589	0.493
Minimum Error	-0.686	-0.696	-0.830

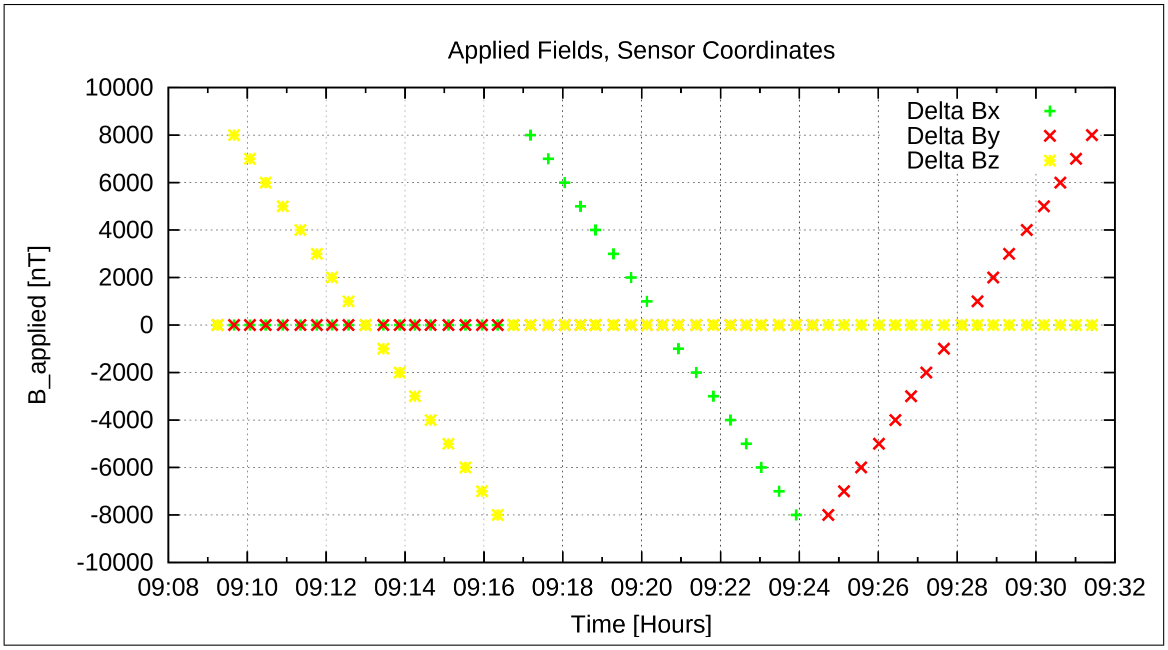

Figure 64: Applied Fields

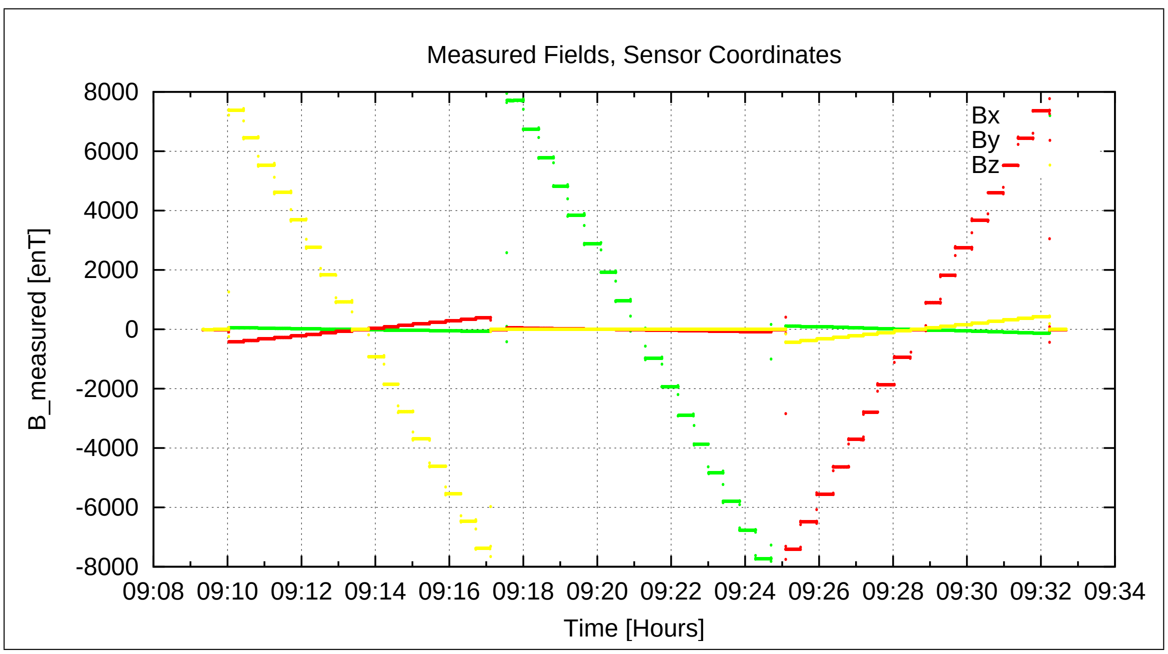

Figure 65: Measured Data

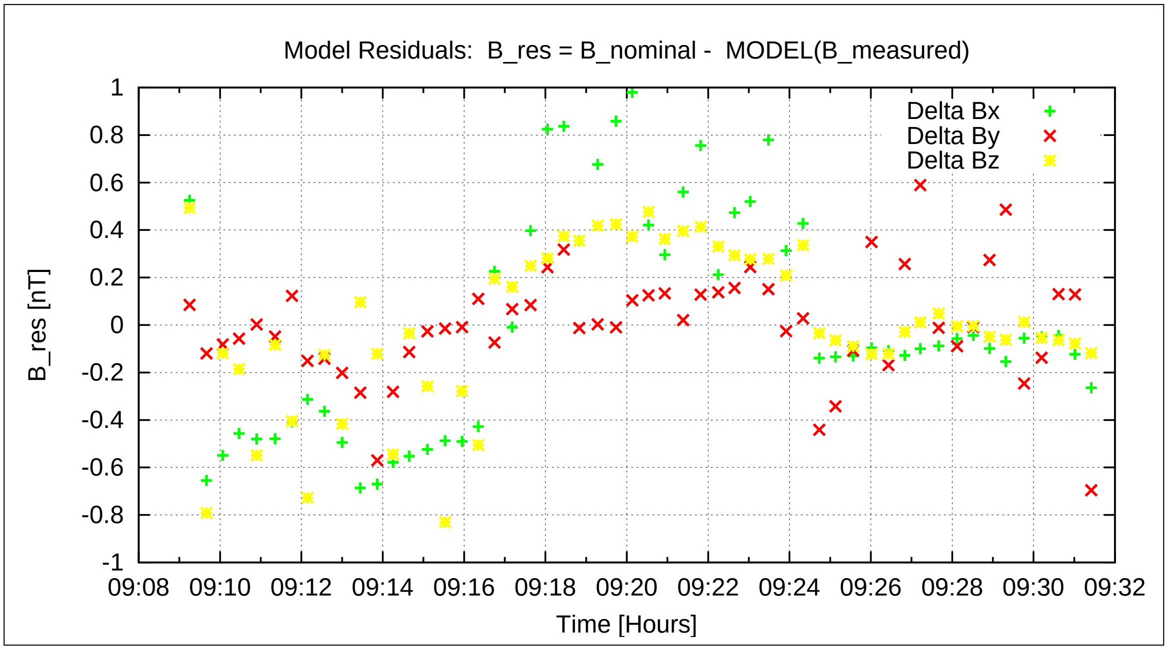

Figure 66: Model Quality - Differences of applied field and modelled measurement data

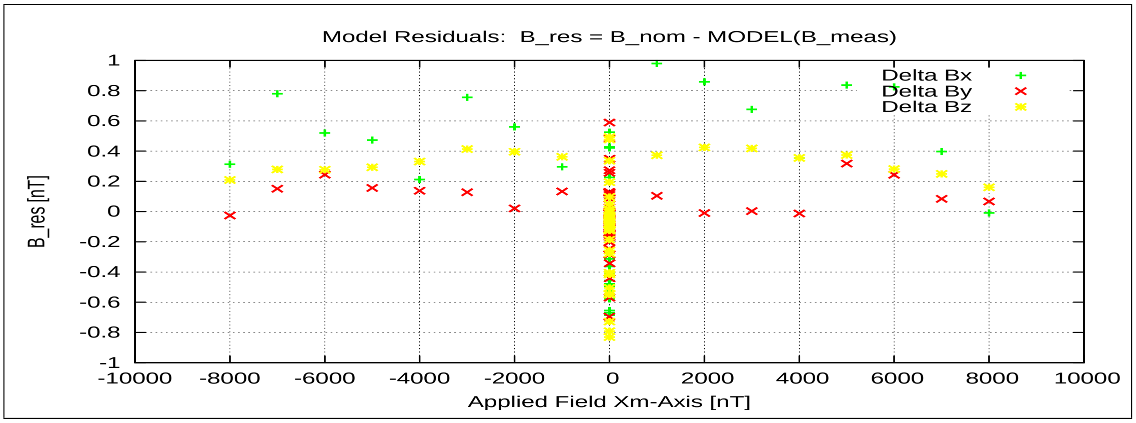

Figure 67: RESIDUALS vs FLD X

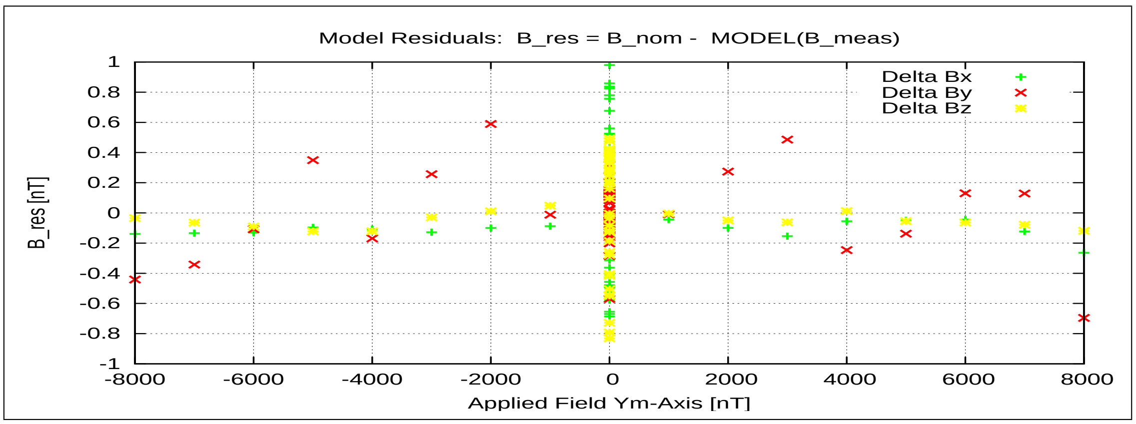

Figure 68: RESIDUALS vs FLD Y

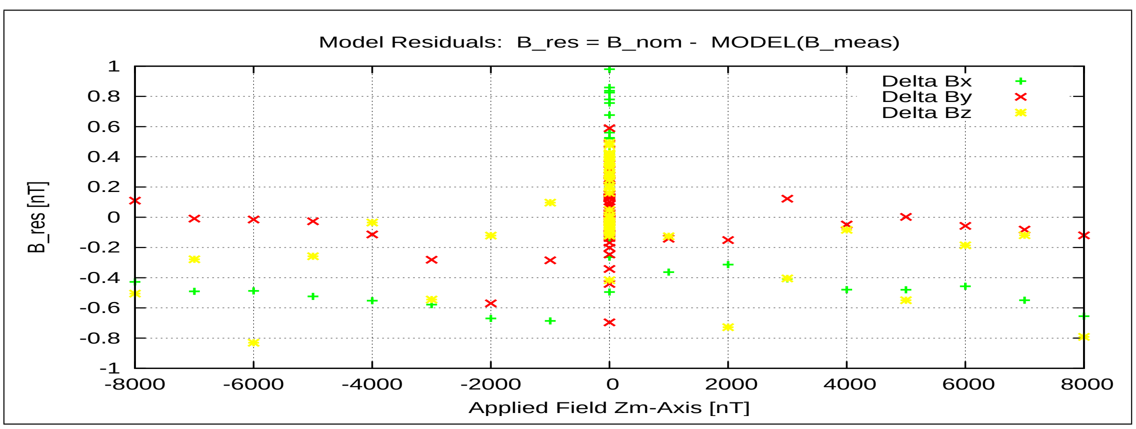

Figure 69: RESIDUALS vs FLD Z

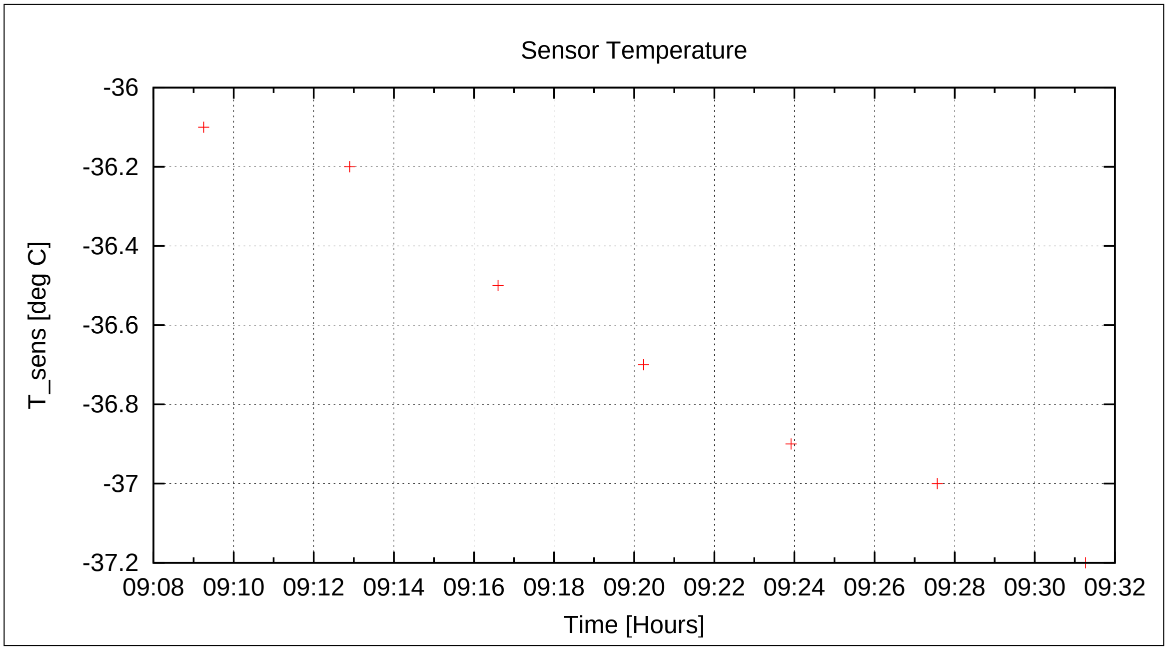

Figure 70: Sensor Temperature

7.3.2 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-16\09-37-17.CCR	OFFSET_200.MAG	

Parameter File: `OFF_PARAMETER__21-04-16-09-37-17_.OPF`

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	-0.01	0.134
Y	-5.42	0.103
Z	-0.95	0.167

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-1.46
Y	9.59
Z	8.51

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

The next temperature goal was -60 degC. Cooling started at 0950.

7.4 Special Frequency Tests, Range=R1, FGM-OB

During cooling down there were some special frequency tests executed under responsibility of IC. We applied for a few minutes

- 500 nT- at 1 Hz sequentially on Xc, Yc, Zc.
- 500 nT- at 5 Hz sequentially on Xc, Yc, Zc.

As the FGM-OB sensor has a huge thermal inertia, cooling of the sensor was very slow. Therefore we lowered the temperature goal to T=-67 deg C at 11:50.

7.5 DC-Analysis at T $= -60°$C, Range=R0, FGM-OB, Nominal Load

7.5.1 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
21-04-16\13-00-07.CCR	LIN60000XYZ.MAG	

Parameter File: `PARAMETER_LIN___21-04-16_13-00-07.CPF`

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.012352 & +0.056804 & -0.998309 \\ -0.999815 & -0.015414 & +0.011494 \\ -0.014735 & +0.998266 & +0.056984 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 42'28'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 52'59'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 43'60'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -67.0°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (12.995 , -24.133 , 2.094) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.028	0.028	0.028
Mean	0.101	0.100	0.101
Maximum	0.281	0.281	0.282

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.155252 & +0.018689 & -0.013835 \\ -0.010159 & +1.207211 & +0.068591 \\ +0.005168 & -0.072679 & +1.201645 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \underline{\underline{\tilde{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.155252 & +0.000000 & +0.000000 \\ +0.007372 & +1.209535 & +0.000000 \\ -0.008698 & -0.003979 & +1.203681 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.155252 & +0.000000 & +0.000000 \\ +0.000000 & +1.209510 & +0.000000 \\ +0.000000 & +0.000000 & +1.203640 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006382 & +1.000020 & +0.000000 \\ -0.007529 & -0.003290 & +1.000034 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +90^\circ\ 21'56'' \\ \xi_{\text{x,z}} &= +89^\circ\ 34'11'' \\ \xi_{\text{y,z}} &= +89^\circ\ 48'51'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	12.224	11.381	9.170
Maximum Error	6.422	5.949	6.719
Minimum Error	-43.316	-40.874	-33.308

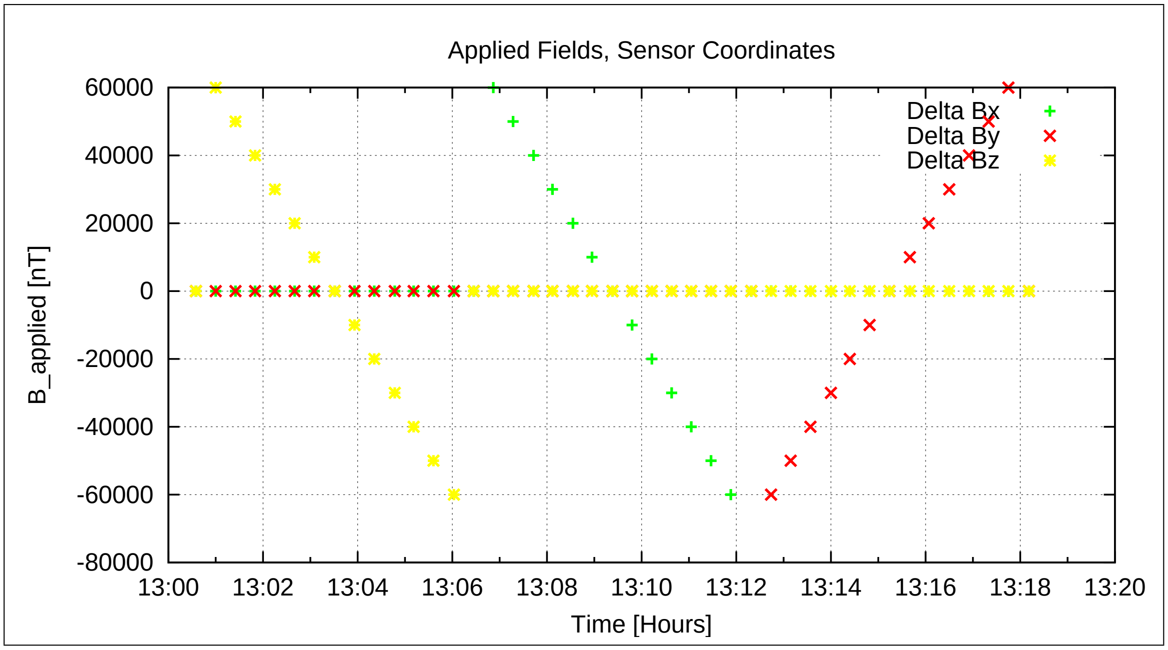

Figure 71: Applied Fields

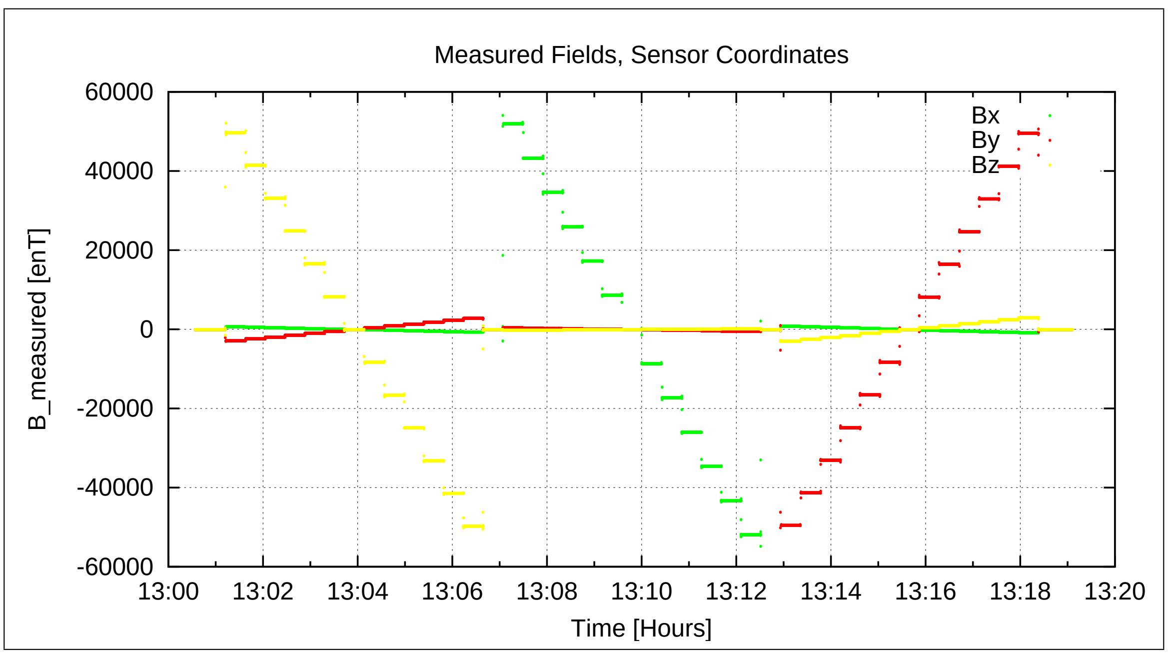

Figure 72: Measured Data

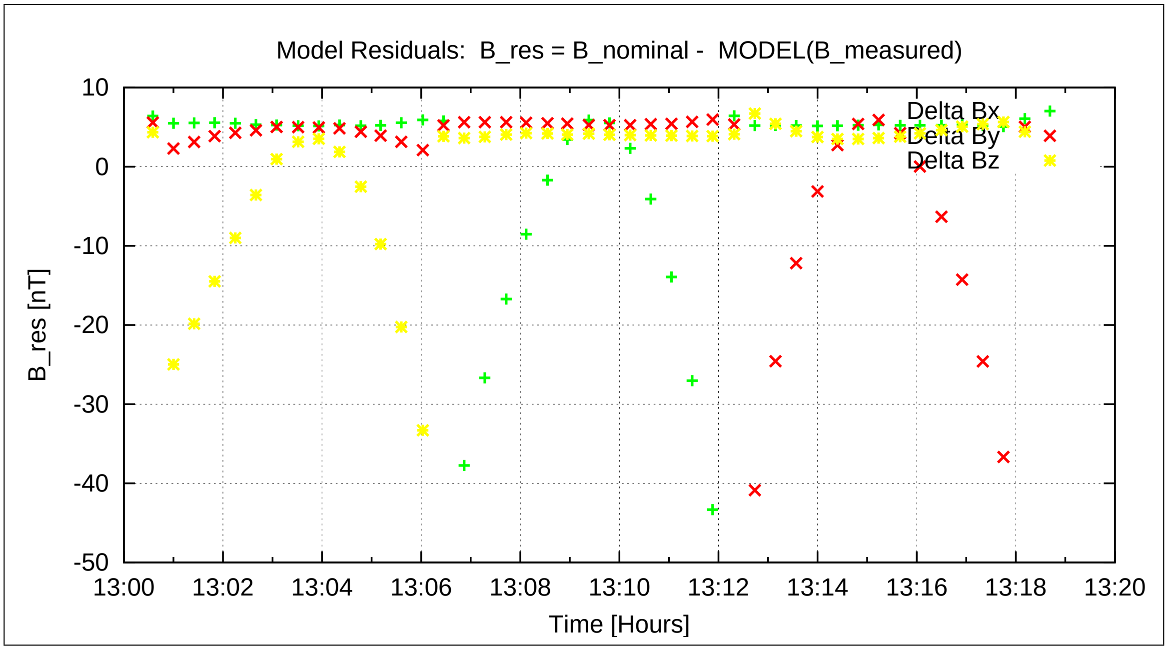

Figure 73: Model Quality - Differences of applied field and modelled measurement data

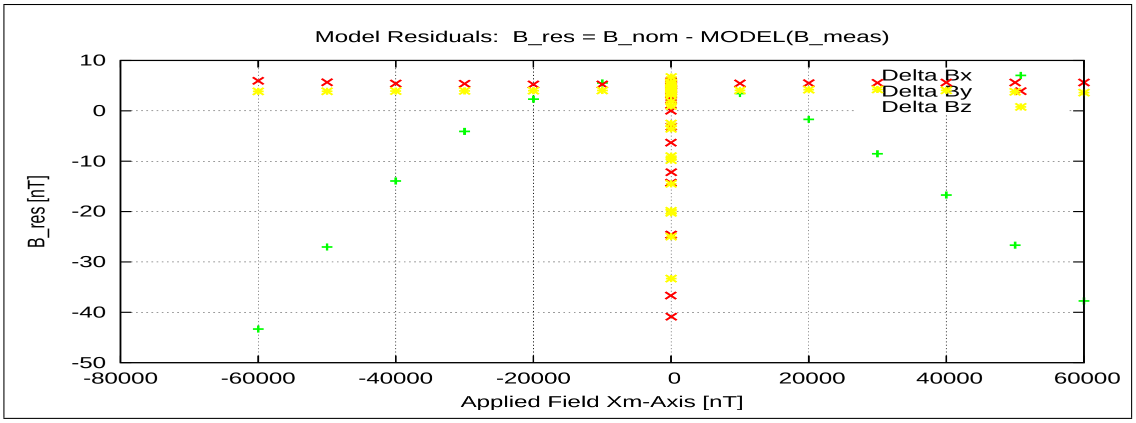

Figure 74: RESIDUALS vs FLD X

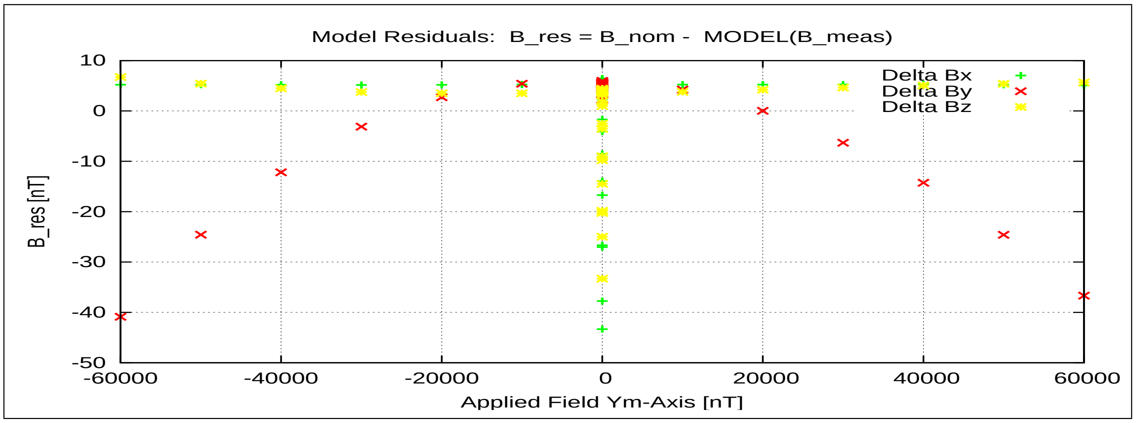

Figure 75: RESIDUALS vs FLD Y

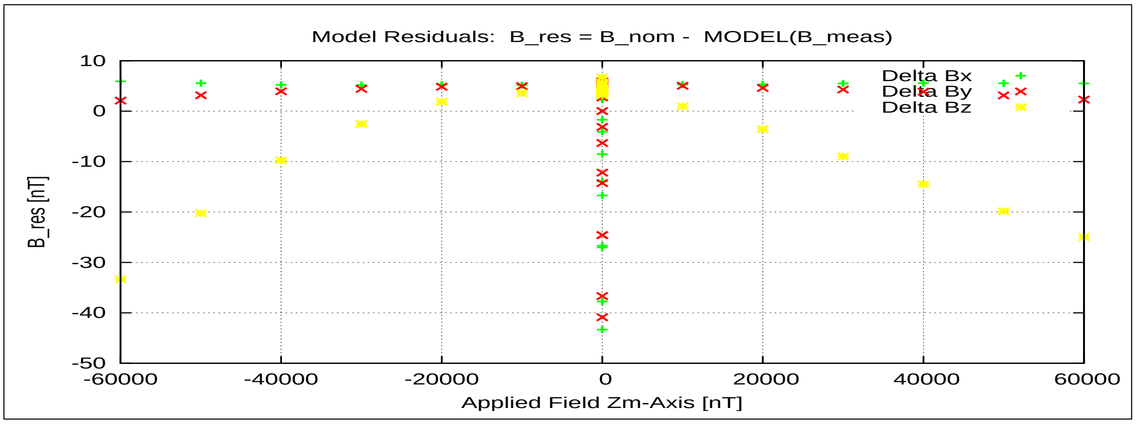

Figure 76: RESIDUALS vs FLD Z

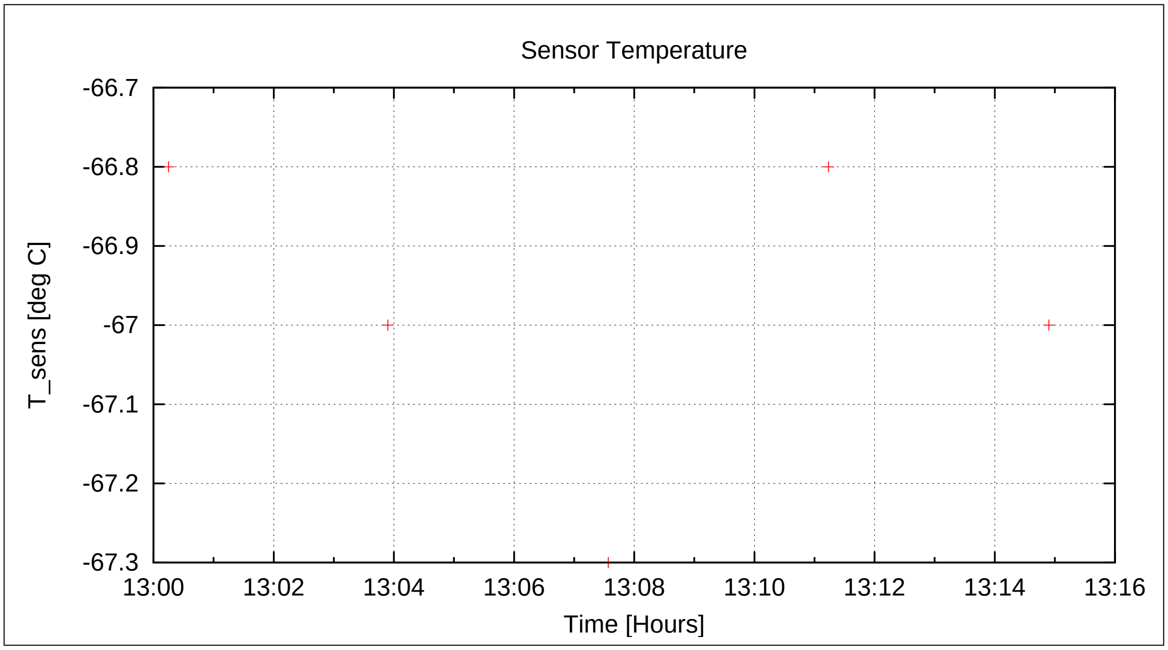

Figure 77: Sensor Temperature

7.6 DC-Analysis at T = -60°C, Range=R1, FGM-OB

7.6.1 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
21-04-16\13-23-27.CCR	LIN8000XYZ.MAG	

Parameter File: PARAMETER_LIN___21-04-16_13-23-27.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.012386 & +0.056860 & -0.998305 \\ -0.999815 & -0.015375 & +0.011529 \\ -0.014694 & +0.998264 & +0.057040 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = & +90^\circ\ 42'35'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = & +90^\circ\ 52'51'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = & +86^\circ\ 43'48'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -66.8°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (0.492 , -12.428 , 7.338) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.022	0.019	0.014
Mean	0.047	0.050	0.045
Maximum	0.260	0.265	0.165

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.037160 & +0.016677 & -0.012477 \\ -0.009117 & +1.079911 & +0.061731 \\ +0.004608 & -0.065091 & +1.080399 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \underline{\underline{\tilde{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.037159 & +0.000000 & +0.000000 \\ +0.006583 & +1.081994 & +0.000000 \\ -0.007877 & -0.003574 & +1.082233 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.037159 & +0.000000 & +0.000000 \\ +0.000000 & +1.081972 & +0.000000 \\ +0.000000 & +0.000000 & +1.082197 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006347 & +1.000020 & +0.000000 \\ -0.007595 & -0.003303 & +1.000034 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +90^\circ\ 21'49'' \\ \xi_{\text{x,z}} &= +89^\circ\ 33'58'' \\ \xi_{\text{y,z}} &= +89^\circ\ 48'49'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.318	0.159	0.299
Maximum Error	0.553	0.265	0.507
Minimum Error	-0.463	-0.510	-0.620

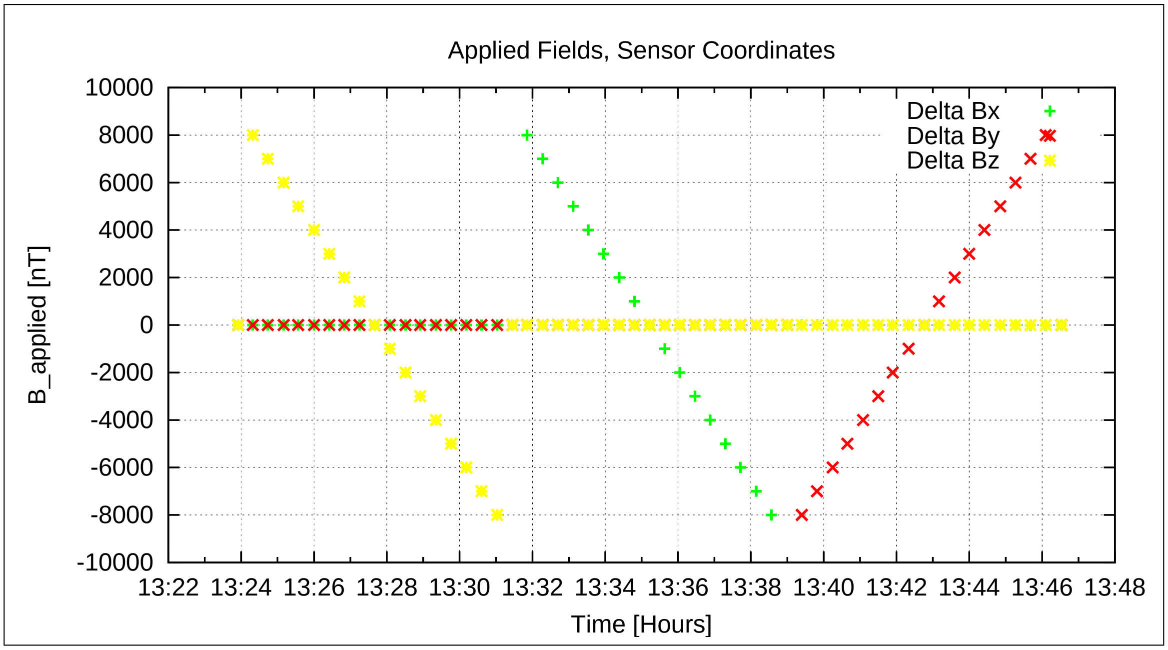

Figure 78: Applied Fields

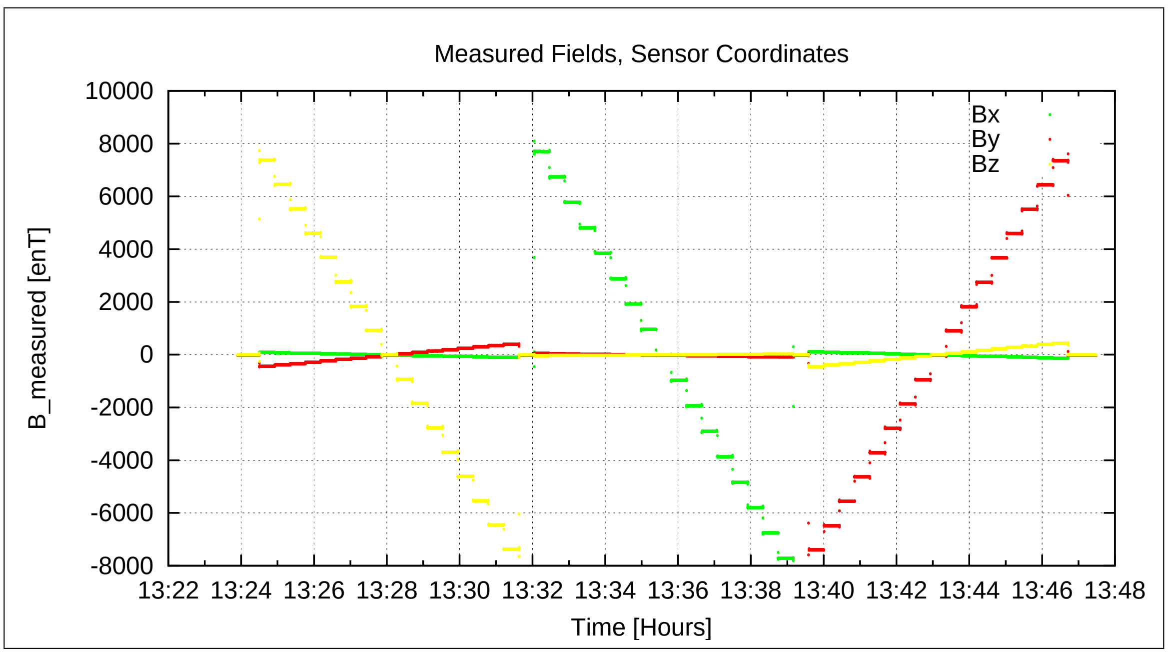

Figure 79: Measured Data

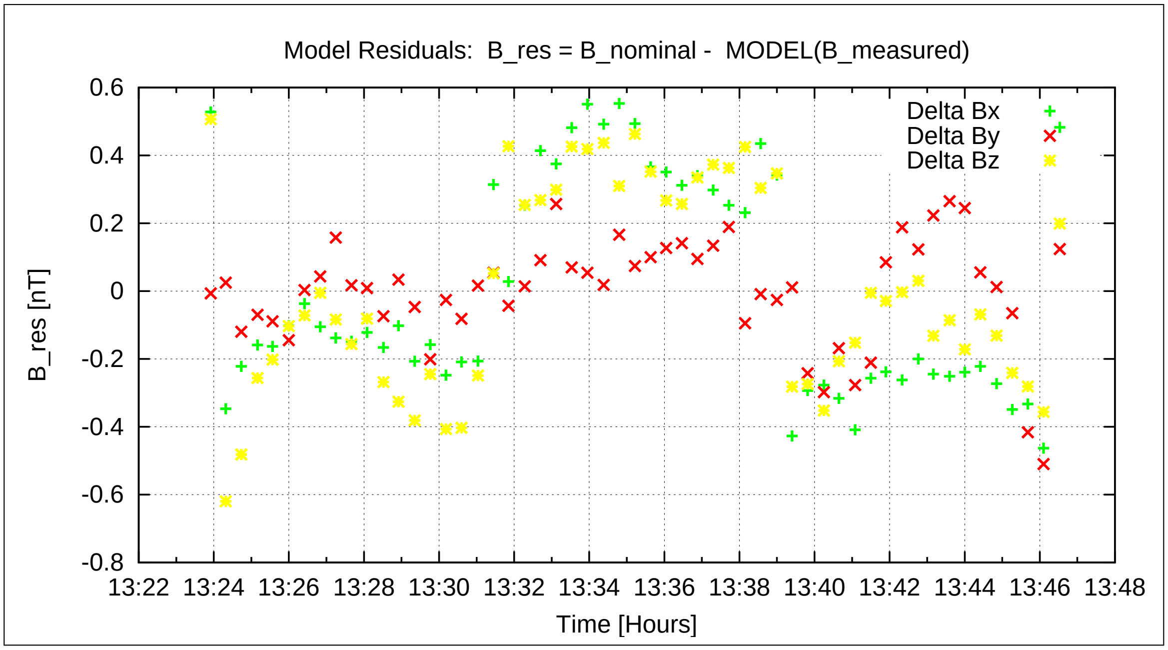

Figure 80: Model Quality - Differences of applied field and modelled measurement data

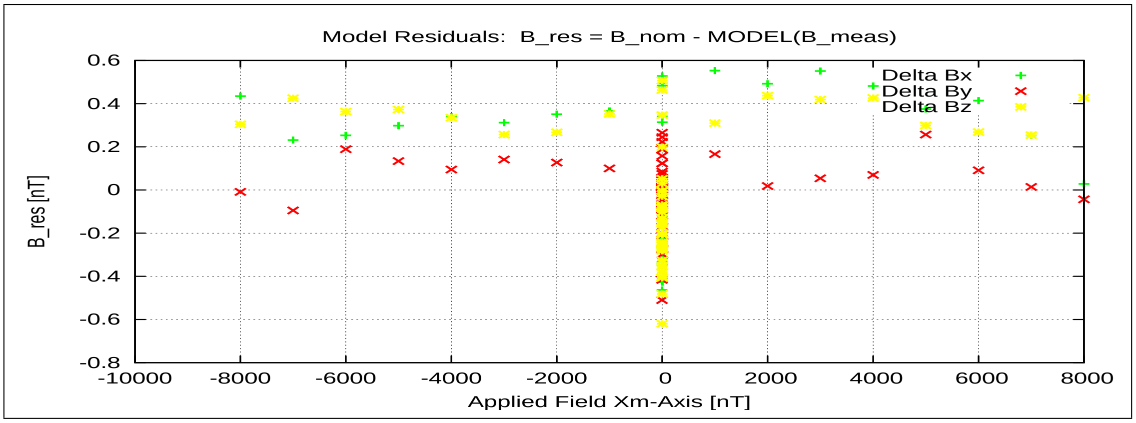

Figure 81: RESIDUALS vs FLD X

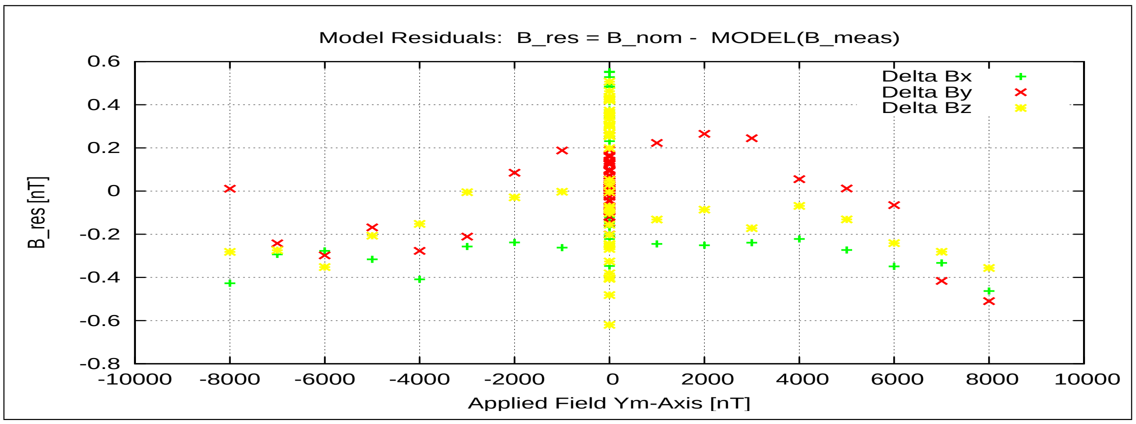

Figure 82: RESIDUALS vs FLD Y

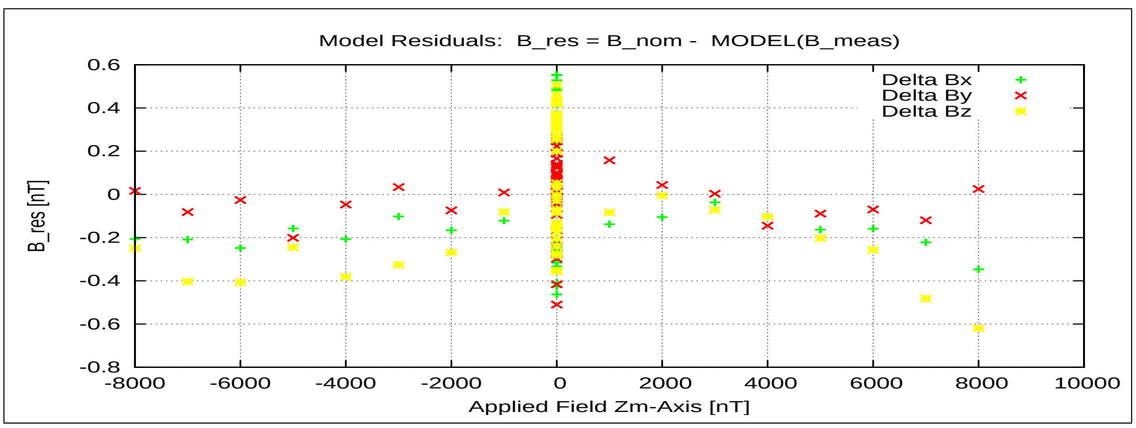

Figure 83: RESIDUALS vs FLD Z

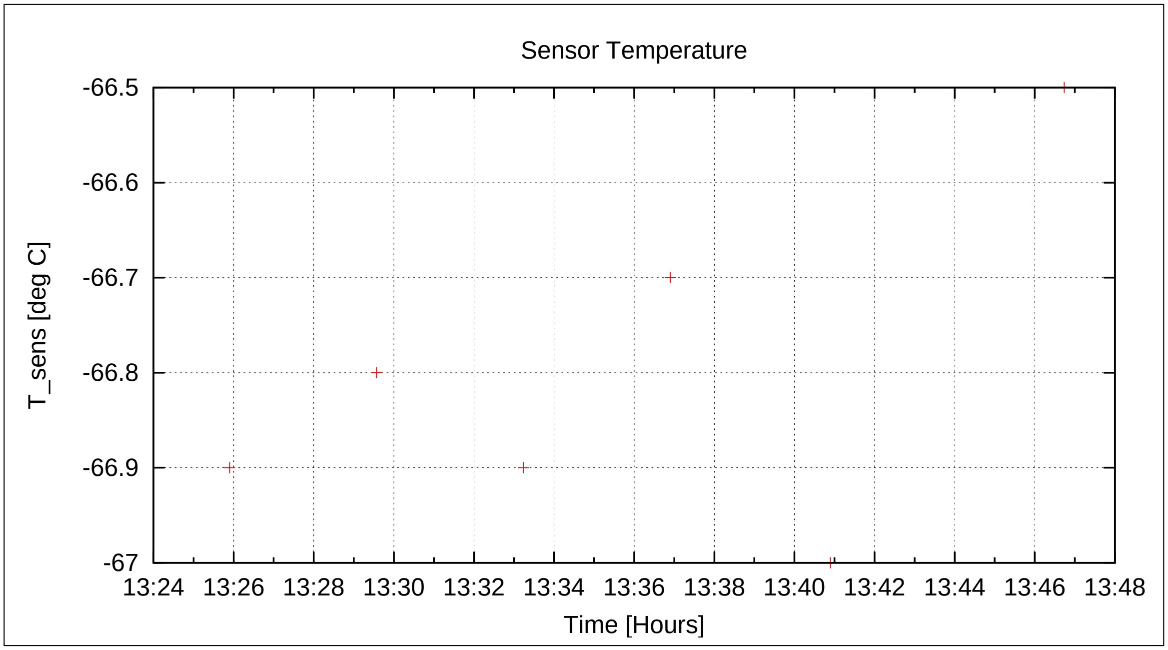

Figure 84: Sensor Temperature

7.6.2 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$

$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-16\13-48-45.CCR	OFFSET_200.MAG	

Parameter File: `OFF_PARAMETER__21-04-16-13-48-45_.OPF`

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	1.27	0.096
Y	-5.37	0.095
Z	-1.54	0.123

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-0.54
Y	9.56
Z	8.44

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

7.7 DC-Analysis at T = $-60°$C, Range=R0, FGM-OB

7.7.1 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\mathrm{off}} = \frac{B_{\mathrm{normal}} + B_{\mathrm{turned}}}{2}$$
$$B_{\mathrm{res}} = \frac{B_{\mathrm{normal}} - B_{\mathrm{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-16\13-57-59.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER__21-04-16-13-57-59_.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	12.70	0.109
Y	-14.48	0.198
Z	-5.51	0.199

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-0.51
Y	8.34
Z	7.70

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

7.8 DC-Analysis at T = $-60°$C, Range=R0, FGM-OB, Reduced Load

7.8.1 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
`21-04-16\14-07-15.CCR`	`LIN60000XYZ.MAG`	

Parameter File: `PARAMETER_LIN___21-04-16_14-07-15.CPF`

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.007777 & +0.057816 & -0.998297 \\ -0.999837 & -0.016709 & +0.006821 \\ -0.016286 & +0.998187 & +0.057937 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 26'44'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 57'27'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 40'43'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: FLDS
Fields applied for 25.0 s
Mean Sensor Temperature: -81.7°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (10.747 , -24.394 , 1.295) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.038	0.028	0.038
Mean	0.114	0.099	0.114
Maximum	0.284	0.281	0.261

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.158747 & +0.020300 & -0.008233 \\ -0.011938 & +1.210848 & +0.069936 \\ -0.000144 & -0.074213 & +1.205055 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.158752 & +0.000000 & +0.000000 \\ +0.007453 & +1.213284 & +0.000000 \\ -0.008739 & -0.004072 & +1.207111 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.158752 & +0.000000 & +0.000000 \\ +0.000000 & +1.213258 & +0.000000 \\ +0.000000 & +0.000000 & +1.207070 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006432 & +1.000021 & +0.000000 \\ -0.007542 & -0.003357 & +1.000034 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +90^\circ\ 22'7'' \\ \xi_{\text{x,z}} &= +89^\circ\ 34'9'' \\ \xi_{\text{y,z}} &= +89^\circ\ 48'38'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	12.792	10.255	9.423
Maximum Error	6.538	5.724	5.372
Minimum Error	-44.683	-36.998	-34.396

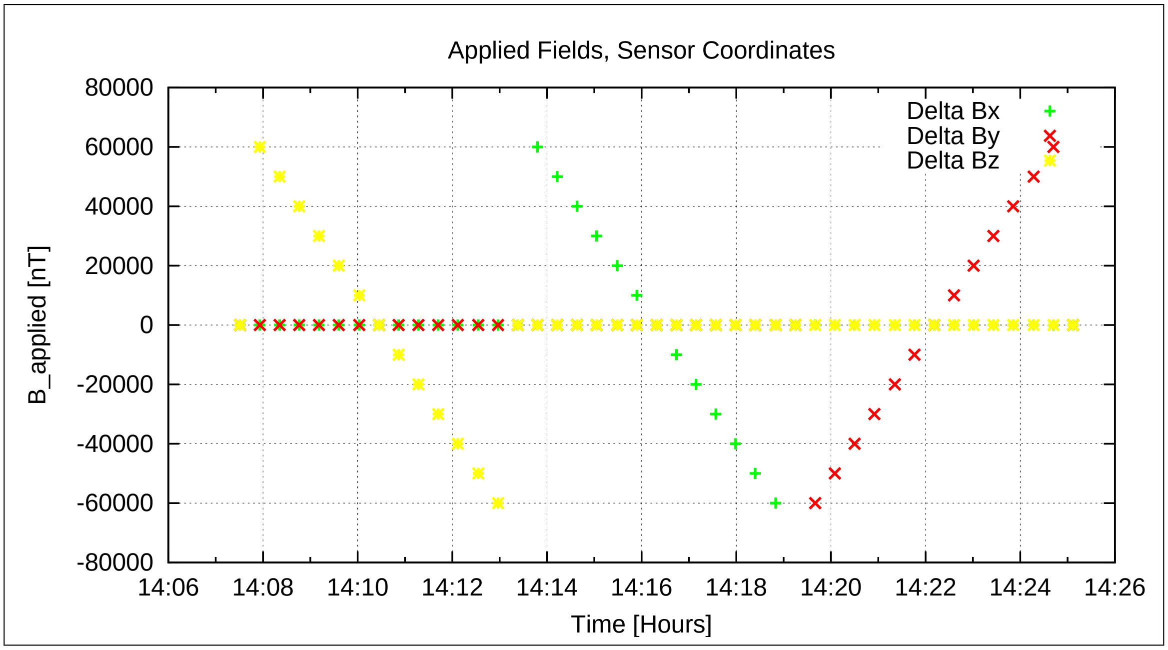

Figure 85: Applied Fields

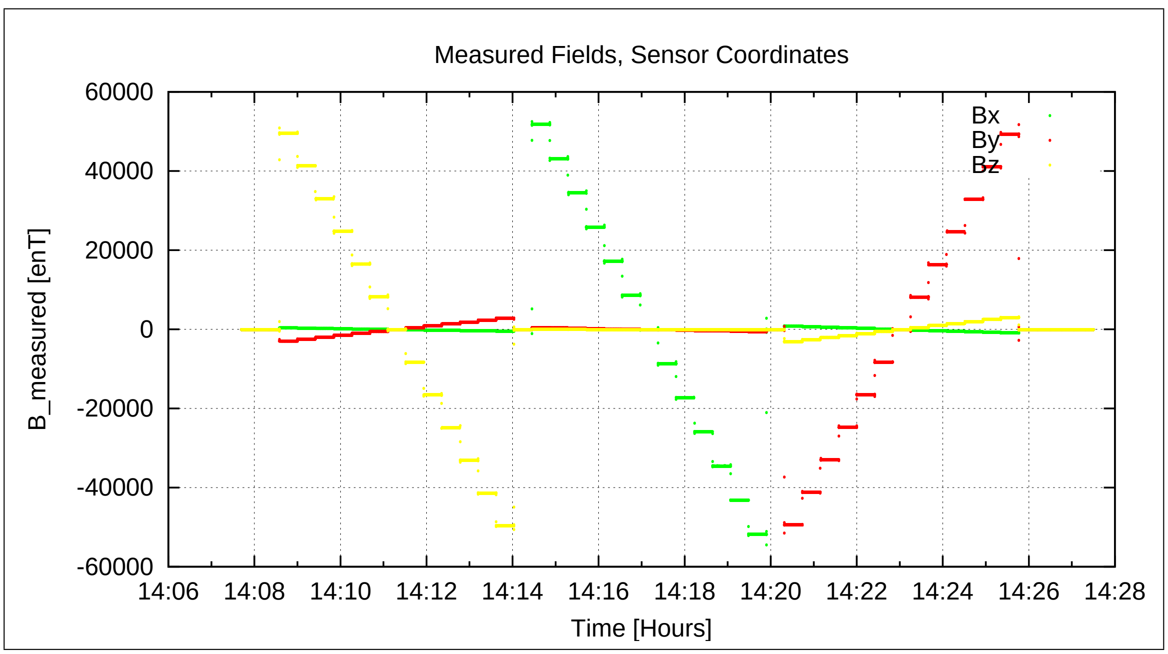

Figure 86: Measured Data

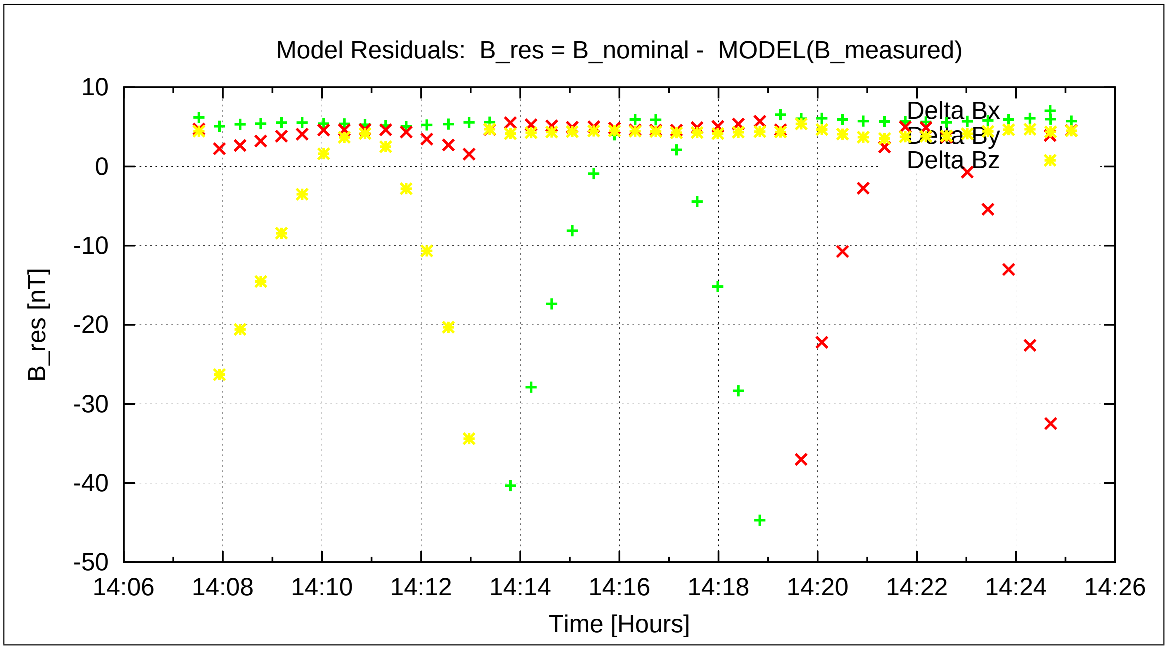

Figure 87: Model Quality - Differences of applied field and modelled measurement data

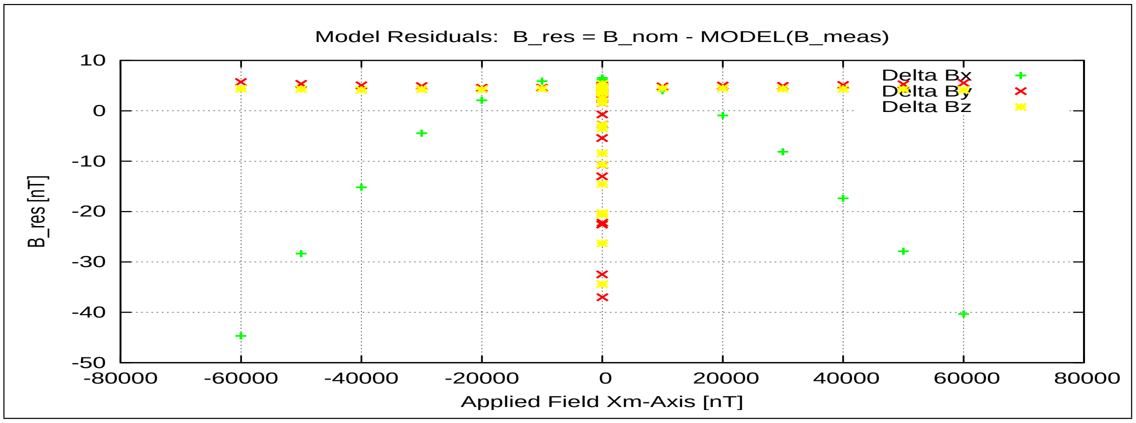

Figure 88: RESIDUALS vs FLD X

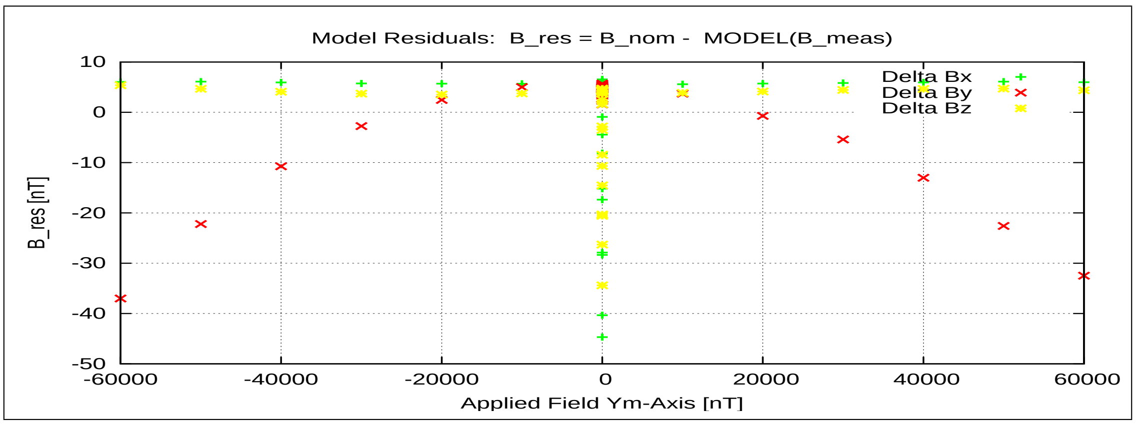

Figure 89: RESIDUALS vs FLD Y

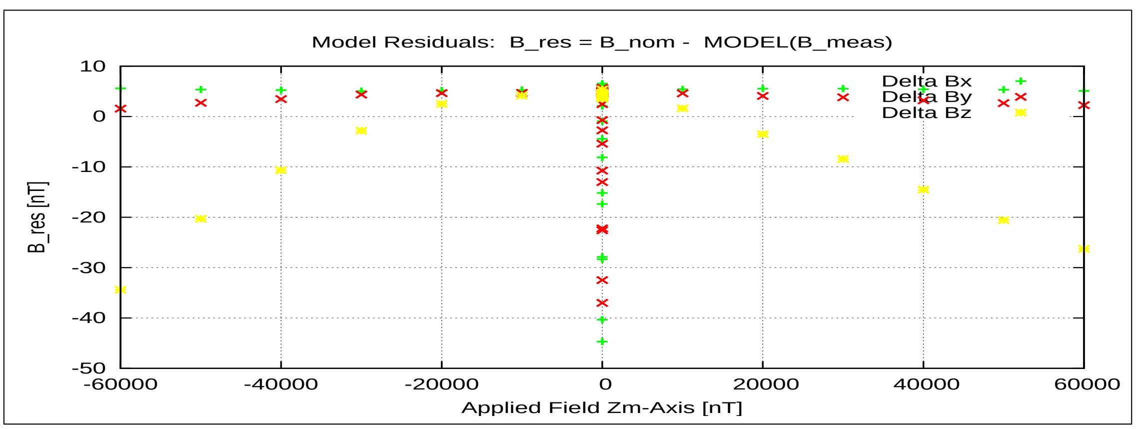

Figure 90: RESIDUALS vs FLD Z

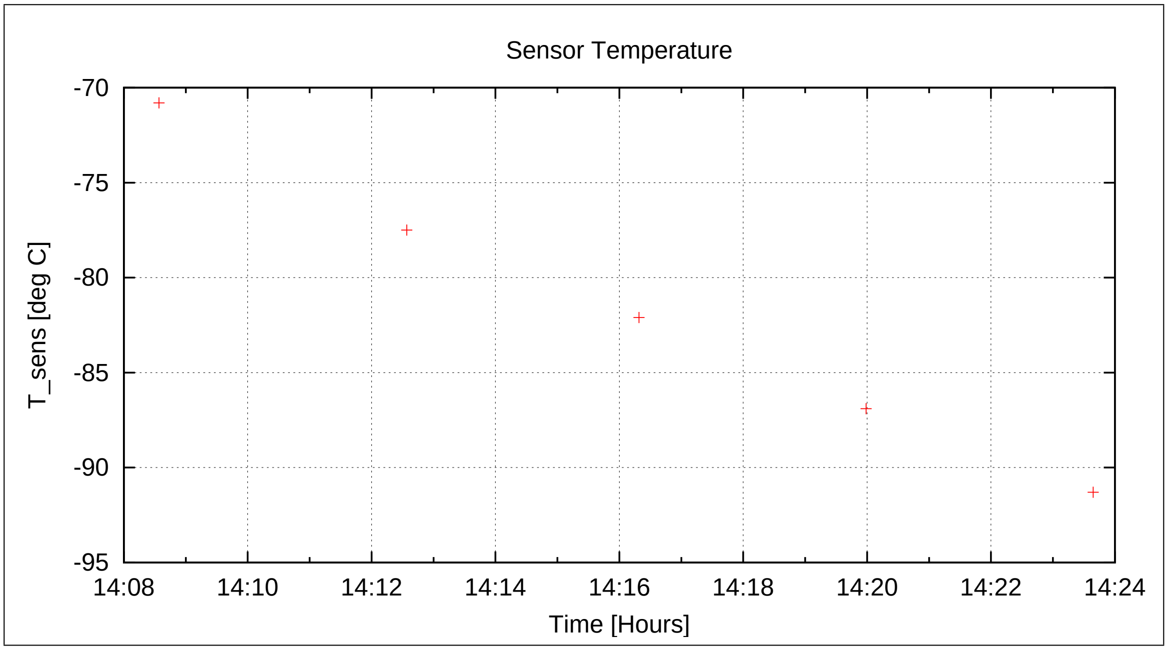

Figure 91: Sensor Temperature

The next temperature goal was -110 degC. Cooling started at 14:46.

7.9 DC-Analysis at T = $-110°$C, Range=R1, FGM-OB

7.9.1 Calibration on 3 Linear Axes

Used Files:

CCD File	Configuration File	Remark
`21-04-16\15-50-45.CCR`	`LIN8000XYZ.MAG`	

Parameter File: `PARAMETER_LIN___21-04-16_15-50-45.CPF`

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.008073 & +0.058528 & -0.998253 \\ -0.999833 & -0.016817 & +0.007099 \\ -0.016372 & +0.998144 & +0.058654 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 27'45'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 57'49'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 38'15'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -109.8°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (-0.318 , -11.667 , 7.908) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.019	0.018	0.019
Mean	0.047	0.052	0.047
Maximum	0.127	0.124	0.141

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\, \underline{\underline{\rho}}\, \underline{\underline{\omega}}\, \underline{\underline{\sigma}} = \begin{pmatrix} +1.037773 & +0.018234 & -0.007687 \\ -0.010835 & +1.080372 & +0.063507 \\ +0.000003 & -0.067232 & +1.080856 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\, \underline{\underline{\sigma}} = \begin{pmatrix} +1.037778 & +0.000000 & +0.000000 \\ +0.006638 & +1.082609 & +0.000000 \\ -0.008000 & -0.003876 & +1.082747 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.037778 & +0.000000 & +0.000000 \\ +0.000000 & +1.082587 & +0.000000 \\ +0.000000 & +0.000000 & +1.082708 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006396 & +1.000020 & +0.000000 \\ -0.007708 & -0.003581 & +1.000036 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +90^\circ\ 21'59'' \\ \xi_{\text{x,z}} &= +89^\circ\ 33'35'' \\ \xi_{\text{y,z}} &= +89^\circ\ 47'52'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.273	0.140	0.222
Maximum Error	0.562	0.360	0.487
Minimum Error	-0.394	-0.270	-0.435

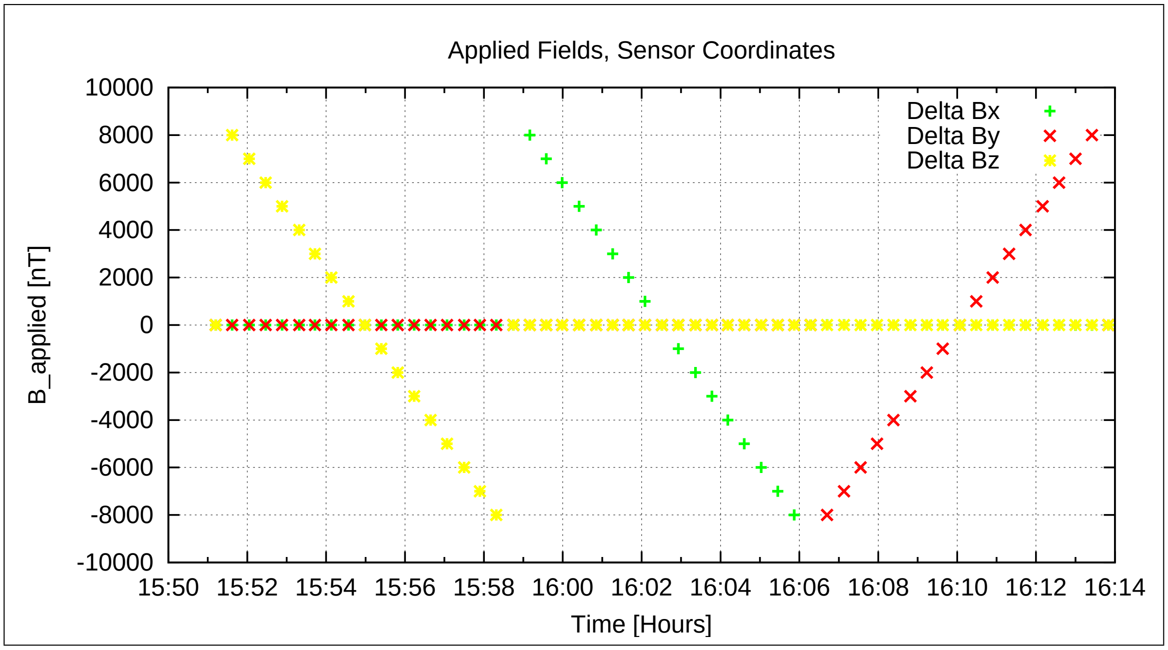

Figure 92: Applied Fields

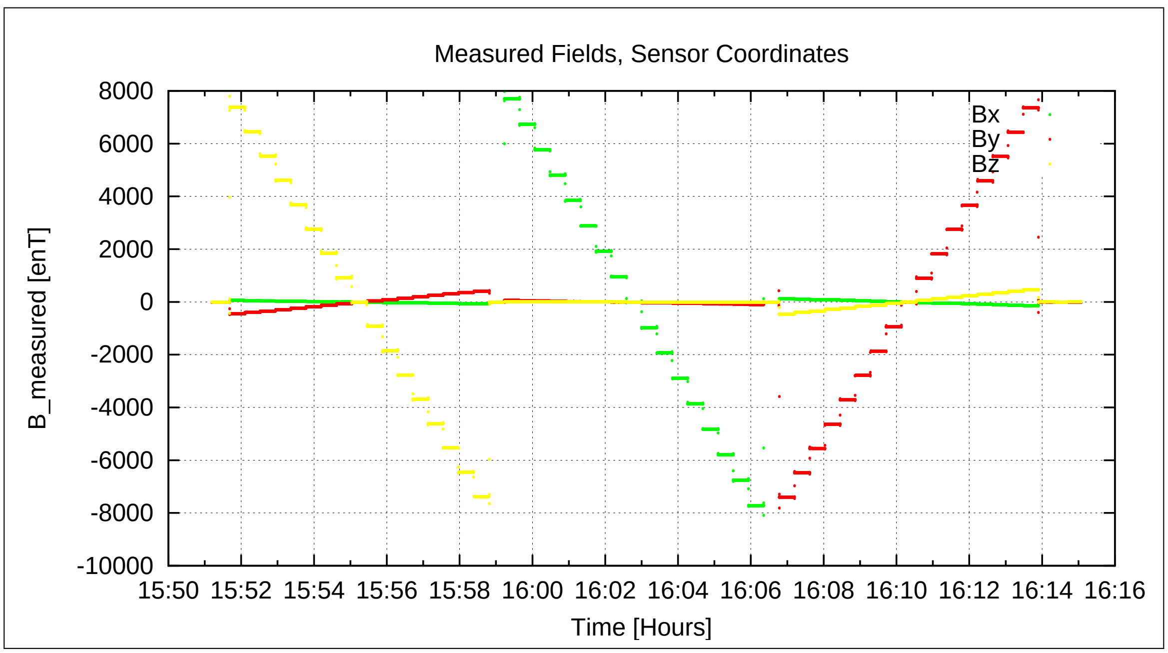

Figure 93: Measured Data

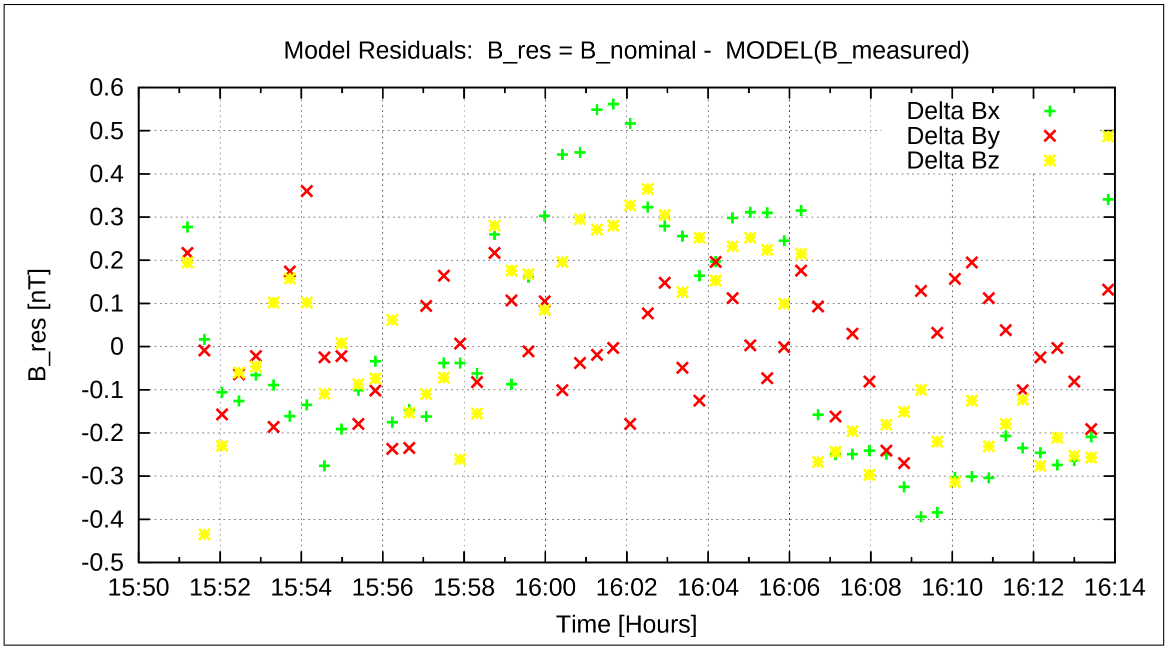

Figure 94: Model Quality - Differences of applied field and modelled measurement data

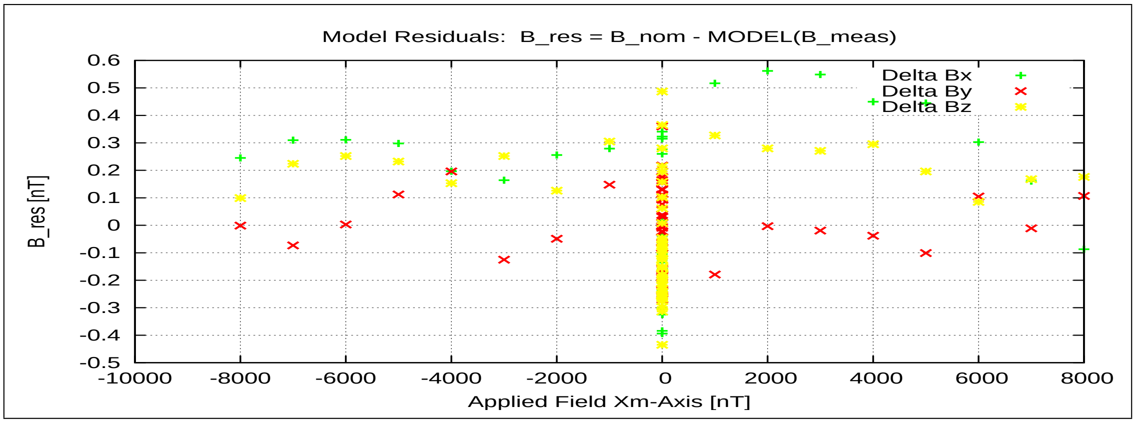

Figure 95: RESIDUALS vs FLD X

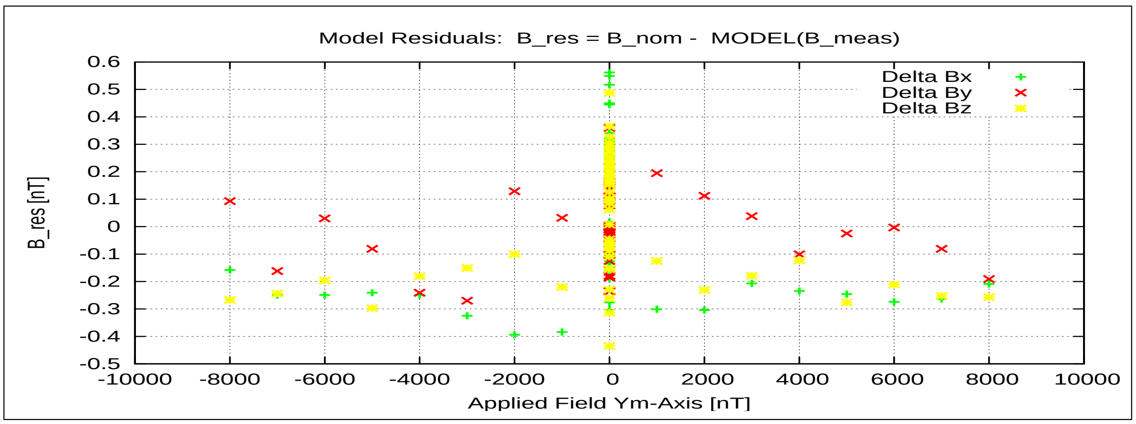

Figure 96: RESIDUALS vs FLD Y

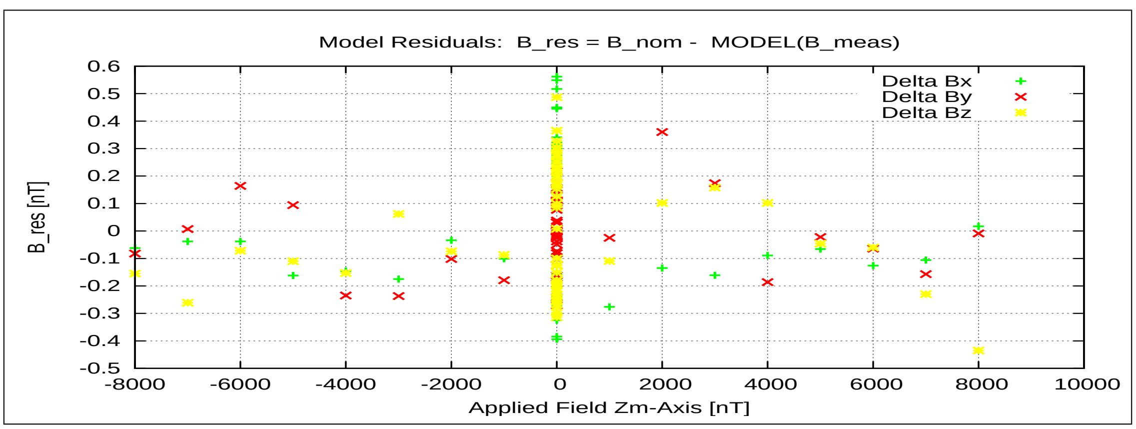

Figure 97: RESIDUALS vs FLD Z

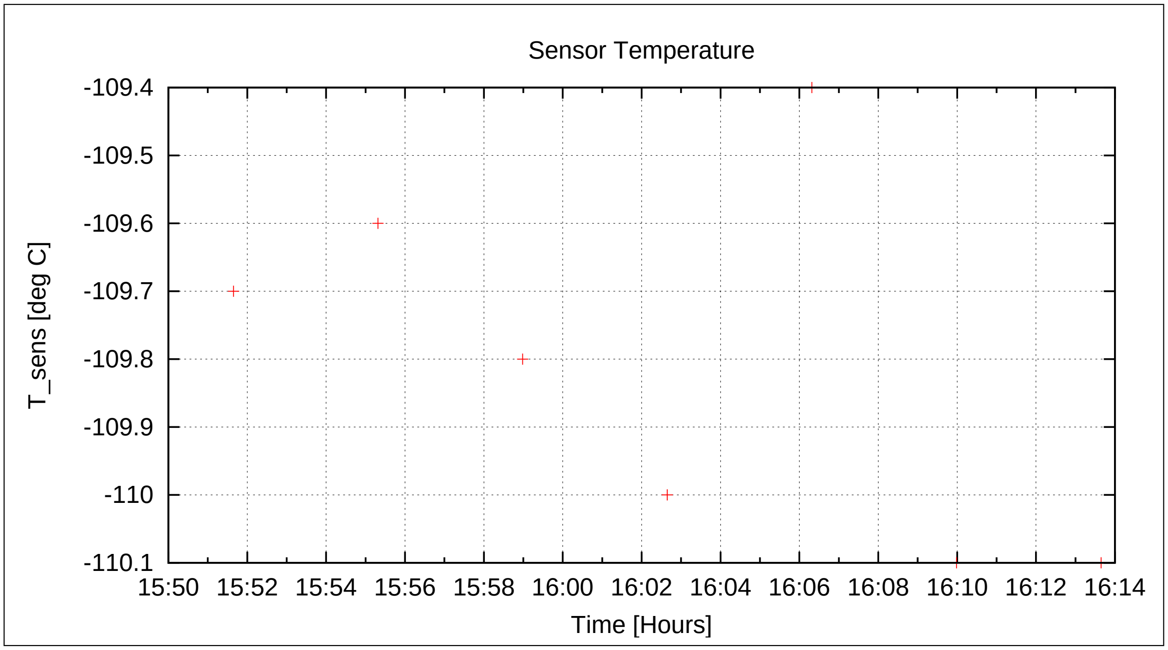

Figure 98: Sensor Temperature

7.9.2 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-16\16-17-42.CCR	OFFSET_200.MAG	

Parameter File: `OFF_PARAMETER__21-04-16-16-17-42_.OPF`

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	1.22	0.094
Y	-5.52	0.115
Z	-0.06	0.113

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-1.08
Y	8.80
Z	7.87

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

7.10 DC-Analysis at T = −110°C, Range=R0, FGM-OB

7.10.1 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-16\16-24-15.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER__21-04-16-16-24-15_.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	13.59	0.109
Y	-14.32	0.250
Z	-1.59	0.128

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-0.38
Y	7.72
Z	8.60

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

The next temperature goal was -140 degC. Cooling started at 16:50. During the night we alternatively performed linearity measurements in R1 (8000 nT max.) and zerofield measurements.

7.11 DC-Analysis at T = $-140°$C, Range=R1, FGM-OB

7.11.1 Calibration on 3 Linear Axes — RANGE R1 — SENSOR OB

Used Files:

CCD File	Configuration File	Remark
21-04-16\18-53-35.CCR	LIN8000XYZ.MAG	

Parameter File: PARAMETER_LIN_R1_OB_21-04-16_18-53-35.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.009024 & +0.063457 & -0.997944 \\ -0.999826 & -0.016883 & +0.007967 \\ -0.016343 & +0.997842 & +0.063598 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 31'1'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +90^\circ\ 58'3'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +86^\circ\ 21'13'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -140.4°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (-1.271 , -9.273 , 5.209) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.018	0.022	0.018
Mean	0.049	0.053	0.047
Maximum	0.137	0.136	0.141

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.038249 & +0.018316 & -0.008630 \\ -0.010858 & +1.080464 & +0.068890 \\ +0.000784 & -0.072616 & +1.080977 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.038253 & +0.000000 & +0.000000 \\ +0.006644 & +1.083049 & +0.000000 \\ -0.008180 & -0.003898 & +1.083205 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.038253 & +0.000000 & +0.000000 \\ +0.000000 & +1.083027 & +0.000000 \\ +0.000000 & +0.000000 & +1.083164 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.006399 & +1.000020 & +0.000000 \\ -0.007879 & -0.003599 & +1.000037 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\xi_{\text{x,y}} = +90^\circ\ 21'60''$$
$$\xi_{\text{x,z}} = +89^\circ\ 32'60''$$
$$\xi_{\text{y,z}} = +89^\circ\ 47'48''$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.230	0.151	0.261
Maximum Error	0.420	0.258	0.392
Minimum Error	-0.398	-0.526	-0.557

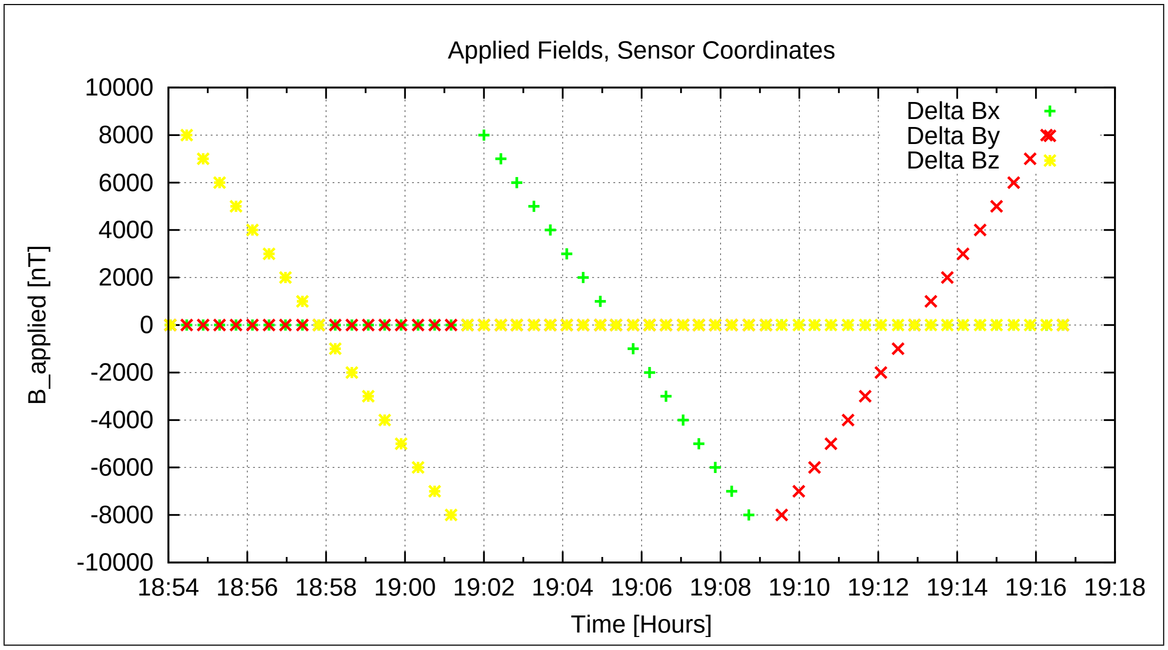

Figure 99: Applied Fields

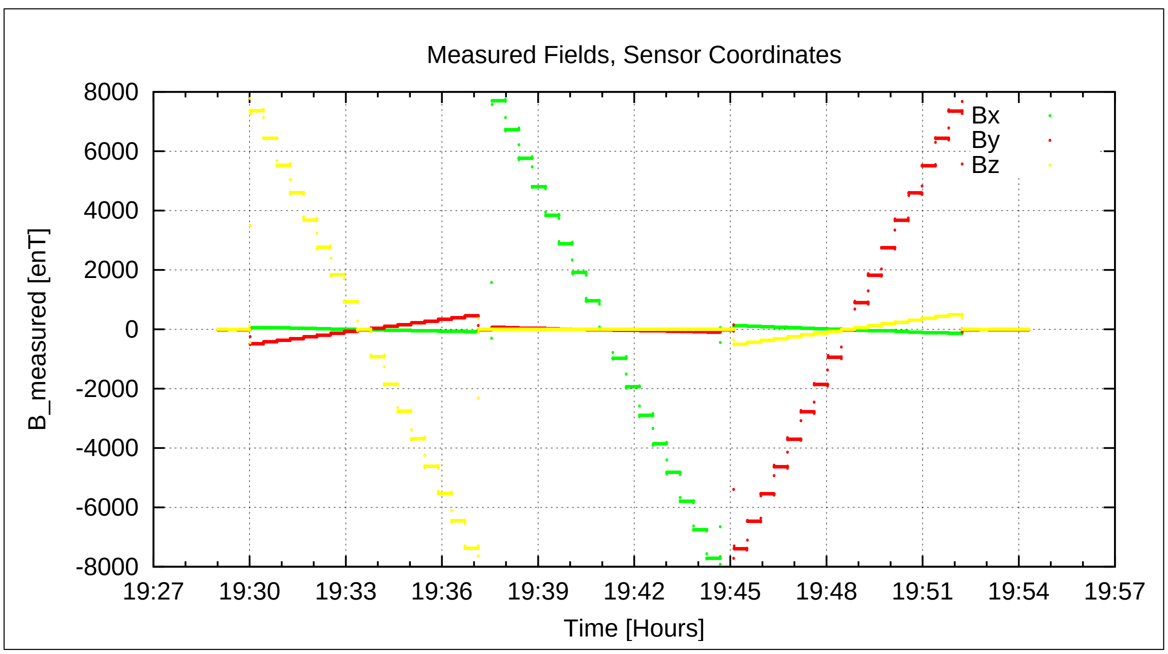

Figure 100: Measured Data

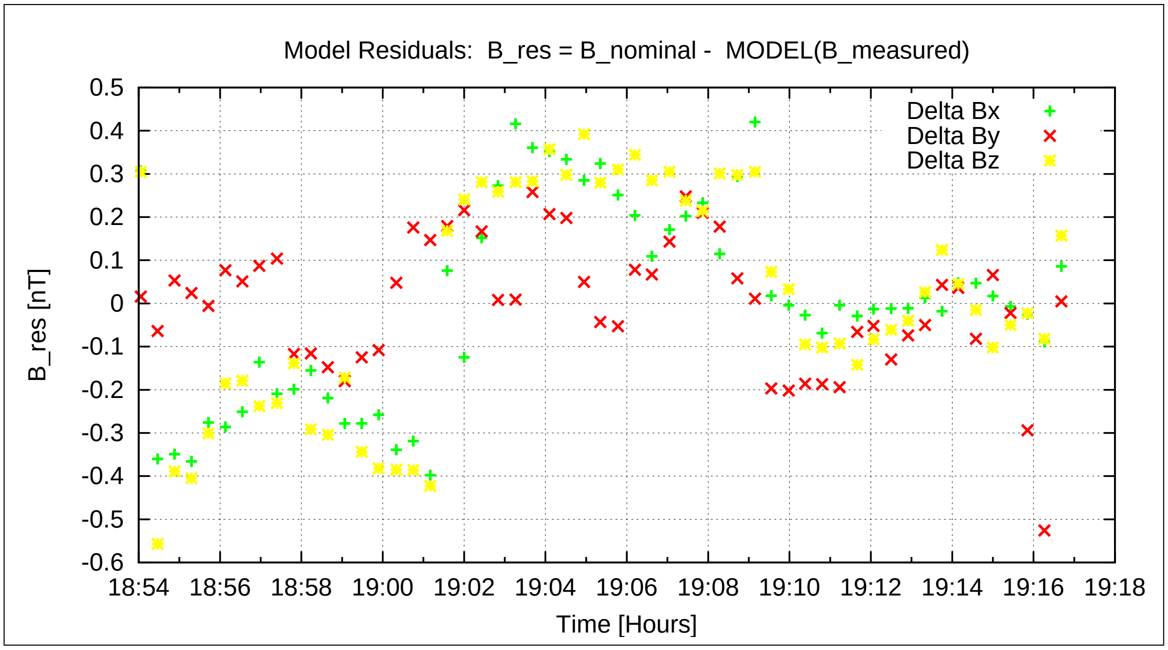

Figure 101: Model Quality - Differences of applied field and modelled measurement data

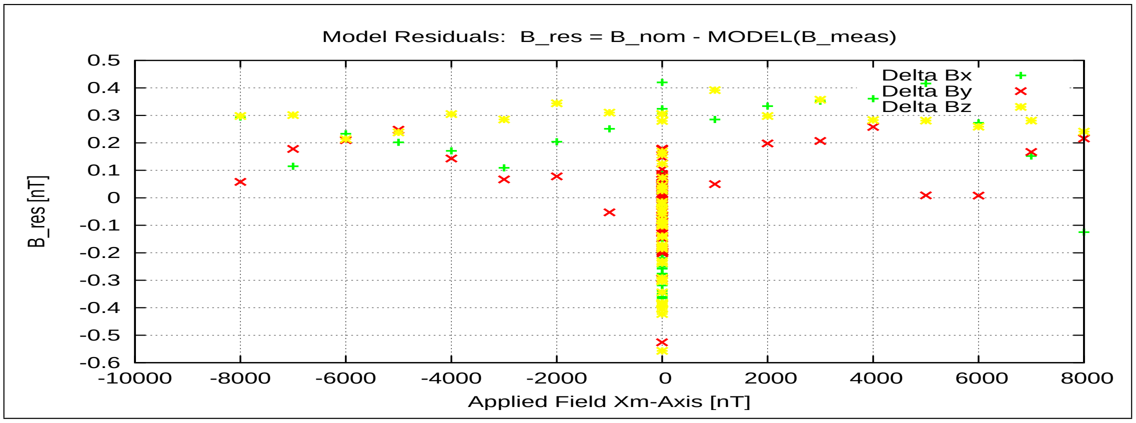

Figure 102: RESIDUALS vs FLD X

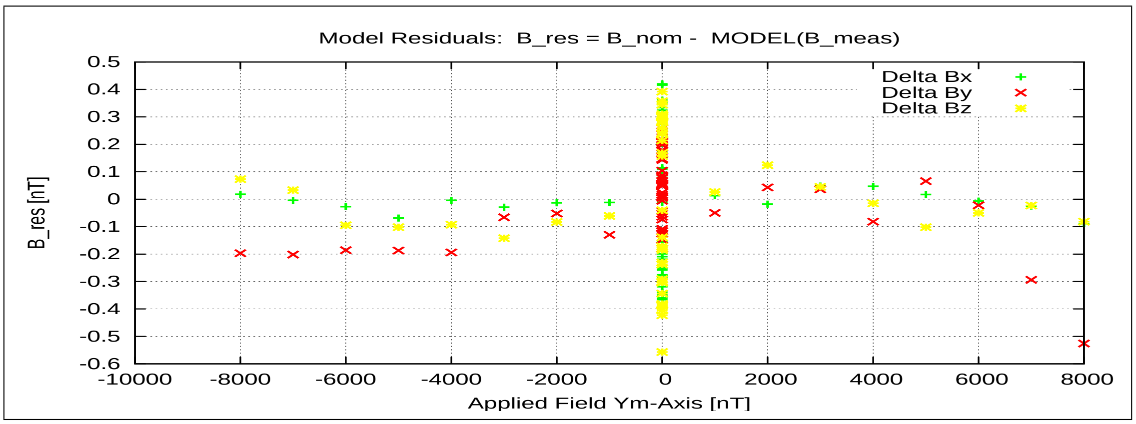

Figure 103: RESIDUALS vs FLD Y

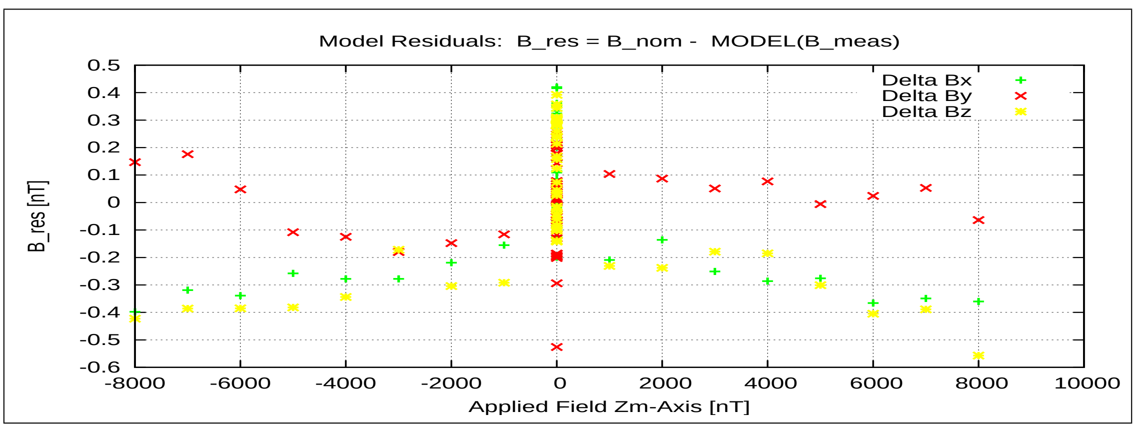

Figure 104: RESIDUALS vs FLD Z

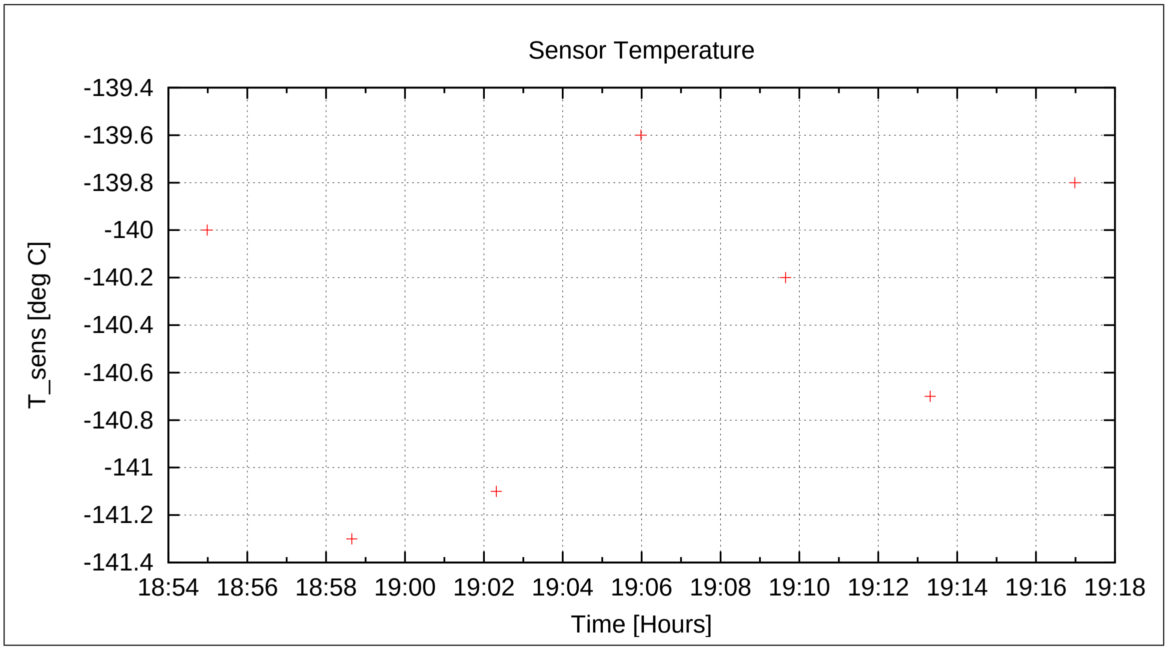

Figure 105: Sensor Temperature

8 Measurements on April 17, 2021

8.1 Data

CCD File	Configuration File	Remark
21-04-17\00-20-44.CCR	NULL.MAG	
21-04-17\00-58-02.CCR	LIN8000XYZ.MAG	
21-04-17\01-21-28.CCR	NULL.MAG	
21-04-17\01-58-45.CCR	LIN8000XYZ.MAG	
21-04-17\02-22-12.CCR	NULL.MAG	
21-04-17\02-59-29.CCR	LIN8000XYZ.MAG	
21-04-17\03-22-56.CCR	NULL.MAG	
21-04-17\04-00-13.CCR	LIN8000XYZ.MAG	
21-04-17\04-23-39.CCR	NULL.MAG	
21-04-17\05-00-57.CCR	LIN8000XYZ.MAG	
21-04-17\05-24-23.CCR	NULL.MAG	
21-04-17\06-01-41.CCR	LIN8000XYZ.MAG	
21-04-17\06-25-08.CCR	NULL.MAG	
21-04-17\07-02-25.CCR	LIN8000XYZ.MAG	
21-04-17\07-25-52.CCR	NULL.MAG	
21-04-17\08-03-10.CCR	LIN8000XYZ.MAG	
21-04-17\08-26-37.CCR	NULL.MAG	
21-04-17\09-03-54.CCR	LIN8000XYZ.MAG	
21-04-17\09-27-22.CCR	NULL.MAG	
21-04-17\10-04-39.CCR	LIN8000XYZ.MAG	
21-04-17\10-28-07.CCR	NULL.MAG	
21-04-17\11-05-24.CCR	LIN8000XYZ.MAG	
21-04-17\11-28-52.CCR	NULL.MAG	
21-04-17\12-06-09.CCR	LIN8000XYZ.MAG	
21-04-17\12-29-37.CCR	NULL.MAG	
21-04-17\13-06-54.CCR	LIN8000XYZ.MAG	
21-04-17\13-30-22.CCR	NULL.MAG	
21-04-17\14-07-39.CCR	LIN8000XYZ.MAG	
21-04-17\14-31-07.CCR	NULL.MAG	
21-04-17\15-08-24.CCR	LIN8000XYZ.MAG	
21-04-17\15-31-52.CCR	NULL.MAG	
21-04-17\16-09-10.CCR	LIN8000XYZ.MAG	
21-04-17\16-32-37.CCR	NULL.MAG	
21-04-17\17-09-55.CCR	LIN8000XYZ.MAG	
21-04-17\17-33-23.CCR	NULL.MAG	
21-04-17\18-10-40.CCR	LIN8000XYZ.MAG	
21-04-17\18-34-08.CCR	NULL.MAG	
21-04-17\19-11-25.CCR	LIN8000XYZ.MAG	
21-04-17\19-34-52.CCR	NULL.MAG	
21-04-17\20-12-10.CCR	LIN8000XYZ.MAG	
21-04-17\20-35-37.CCR	NULL.MAG	
21-04-17\21-12-54.CCR	LIN8000XYZ.MAG	
21-04-17\21-36-21.CCR	NULL.MAG	
21-04-17\22-13-38.CCR	LIN8000XYZ.MAG	
21-04-17\22-37-05.CCR	NULL.MAG	

```
21-04-17\23-14-22.CCR      LIN8000XYZ.MAG
21-04-17\23-37-48.CCR      NULL.MAG
```

No personnel around today. Sensor warmed up autonomously. Linearity measurements and zerofield measurements cont'd.

9 Measurements on April 18, 2021

9.1 Data

CCD File	Configuration File	Remark
21-04-18\00-15-06.CCR	LIN8000XYZ.MAG	
21-04-18\00-38-32.CCR	NULL.MAG	
21-04-18\01-15-49.CCR	LIN8000XYZ.MAG	
21-04-18\01-39-15.CCR	NULL.MAG	
21-04-18\02-16-33.CCR	LIN8000XYZ.MAG	
21-04-18\02-39-59.CCR	NULL.MAG	
21-04-18\03-17-17.CCR	LIN8000XYZ.MAG	
21-04-18\03-40-43.CCR	NULL.MAG	
21-04-18\04-18-00.CCR	LIN8000XYZ.MAG	
21-04-18\04-41-27.CCR	NULL.MAG	
21-04-18\05-18-44.CCR	LIN8000XYZ.MAG	
21-04-18\05-42-11.CCR	NULL.MAG	
21-04-18\06-19-28.CCR	LIN8000XYZ.MAG	
21-04-18\06-42-55.CCR	NULL.MAG	
21-04-18\07-20-13.CCR	LIN8000XYZ.MAG	
21-04-18\07-43-39.CCR	NULL.MAG	
21-04-18\08-20-57.CCR	LIN8000XYZ.MAG	
21-04-18\08-44-24.CCR	NULL.MAG	
21-04-18\09-21-42.CCR	LIN8000XYZ.MAG	
21-04-18\09-45-09.CCR	NULL.MAG	
21-04-18\10-22-26.CCR	LIN8000XYZ.MAG	
21-04-18\10-45-53.CCR	NULL.MAG	
21-04-18\11-23-11.CCR	LIN8000XYZ.MAG	
21-04-18\11-46-38.CCR	NULL.MAG	
21-04-18\12-23-56.CCR	LIN8000XYZ.MAG	
21-04-18\12-47-24.CCR	NULL.MAG	
21-04-18\13-24-41.CCR	LIN8000XYZ.MAG	
21-04-18\13-48-09.CCR	NULL.MAG	
21-04-18\14-25-27.CCR	LIN8000XYZ.MAG	
21-04-18\14-48-54.CCR	NULL.MAG	
21-04-18\15-26-12.CCR	LIN8000XYZ.MAG	
21-04-18\15-49-40.CCR	NULL.MAG	
21-04-18\16-26-57.CCR	LIN8000XYZ.MAG	
21-04-18\16-50-25.CCR	NULL.MAG	
21-04-18\17-27-43.CCR	LIN8000XYZ.MAG	
21-04-18\17-51-10.CCR	NULL.MAG	
21-04-18\18-28-28.CCR	LIN8000XYZ.MAG	
21-04-18\18-51-55.CCR	NULL.MAG	
21-04-18\19-29-12.CCR	LIN8000XYZ.MAG	
21-04-18\19-52-39.CCR	NULL.MAG	
21-04-18\20-29-57.CCR	LIN8000XYZ.MAG	
21-04-18\20-53-24.CCR	NULL.MAG	
21-04-18\21-30-41.CCR	LIN8000XYZ.MAG	
21-04-18\21-54-08.CCR	NULL.MAG	
21-04-18\22-31-25.CCR	LIN8000XYZ.MAG	

```
21-04-18\22-54-52.CCR      NULL.MAG
21-04-18\23-32-10.CCR      LIN8000XYZ.MAG
21-04-18\23-55-36.CCR      NULL.MAG
```

No personnel around today. Sensor warmed up autonomously. Linearity measurements and zerofield measurements cont'd.

10 Measurements on April 19, 2021

10.1 Data

CCD File	Configuration File	Remark
21-04-19\00-32-54.CCR	LIN8000XYZ.MAG	
21-04-19\00-56-20.CCR	NULL.MAG	
21-04-19\01-33-37.CCR	LIN8000XYZ.MAG	
21-04-19\01-57-03.CCR	NULL.MAG	
21-04-19\02-34-21.CCR	LIN8000XYZ.MAG	
21-04-19\02-57-47.CCR	NULL.MAG	
21-04-19\03-35-05.CCR	LIN8000XYZ.MAG	
21-04-19\03-58-31.CCR	NULL.MAG	
21-04-19\04-35-48.CCR	LIN8000XYZ.MAG	
21-04-19\04-59-15.CCR	NULL.MAG	
21-04-19\05-36-32.CCR	LIN8000XYZ.MAG	
21-04-19\05-59-59.CCR	NULL.MAG	
21-04-19\06-37-16.CCR	LIN8000XYZ.MAG	
21-04-19\07-00-43.CCR	NULL.MAG	
21-04-19\07-18-48.CCR	NULL.MAG	
21-04-19\07-52-29.CCR	OFFSET_200.MAG	
21-04-19\08-03-24.CCR	OFFSET_200.MAG	
21-04-19\11-02-30.CCR	LIN2000XYZ.MAG	
21-04-19\11-34-06.CCR	LIN16000XYZ.MAG	
21-04-19\11-59-19.CCR	OFFSET_200.MAG	
21-04-19\13-45-25.CCR	LIN16000XYZ.MAG	
21-04-19\15-52-29.CCR	OFFSET_200.MAG	
21-04-19\16-14-58.CCR	LIN16000XYZ.MAG	
21-04-19\16-37-36.CCR	NULL.MAG	
21-04-19\17-14-53.CCR	LIN16000XYZ.MAG	
21-04-19\17-37-31.CCR	NULL.MAG	
21-04-19\18-14-48.CCR	LIN16000XYZ.MAG	
21-04-19\18-37-26.CCR	NULL.MAG	
21-04-19\19-14-44.CCR	LIN16000XYZ.MAG	
21-04-19\19-37-20.CCR	NULL.MAG	
21-04-19\20-14-38.CCR	LIN16000XYZ.MAG	
21-04-19\20-37-15.CCR	NULL.MAG	
21-04-19\21-14-32.CCR	LIN16000XYZ.MAG	
21-04-19\21-37-09.CCR	NULL.MAG	
21-04-19\22-14-26.CCR	LIN16000XYZ.MAG	
21-04-19\22-37-03.CCR	NULL.MAG	
21-04-19\23-14-20.CCR	LIN16000XYZ.MAG	
21-04-19\23-36-57.CCR	NULL.MAG	

Inspection of the system at 06:50. All looks ok. Box temperature at +5.9 degC. Heating up to 25 degC.

To complete the FGM-OB measurements we perform final offset measurements in both ranges at room temperature.

10.2 DC-Analysis at Room Temperature, Range=R0, FGM-OB

10.2.1 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-19\07-52-29.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER__21-04-19-07-52-29_.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	7.43	0.116
Y	-14.38	0.101
Z	-9.68	0.227

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-0.44
Y	8.27
Z	7.41

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

10.3 DC-Analysis at Room Temperature, Range=R1, FGM-OB

10.3.1 OFFSET Calculation

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\mathrm{off}} = \frac{B_{\mathrm{normal}} + B_{\mathrm{turned}}}{2}$$
$$B_{\mathrm{res}} = \frac{B_{\mathrm{normal}} - B_{\mathrm{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-19\08-03-24.CCR	OFFSET_200.MAG	

Parameter File: `OFF_PARAMETER__21-04-19-08-03-24_.OPF`

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	-1.62	0.088
Y	-5.22	0.094
Z	-4.15	0.110

Residual Field of the Coil System:

Component	B_{res} [enT]
X	-0.62
Y	8.94
Z	7.93

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

10.4 Sensor Mounting: FGM-IB / Orientation / Alignment Check

After the completion of the FGM-OB tests the instrument was switched off and the sensors were disconnected. The FGM-IB sensor was mounted in the thermal test box. See picture 106.

Figure 106: Setup for Standard Calibration of the IB Sensor, Azimuth=180, Elevation=00, view from south.

- POSITION 1: Normal Position for Offset Measurements XY:
  $X_m = -X_c, Y_m = -Y_c, Z_m = +Z_c$
  Elevation=0°. Azimuth= 180°.
- POSITION 2: Turned Position for turned Offset Measurements XY:
  $X_m = +X_c, Y_m = +Y_c, Z_m = +Z_c$
  Elevation=0°. Azimuth= 0°.
- POSITION 3: Normal Position for Offset Measurements Z:
  $X_m = +Z_c, Y_m = +Y_c, Z_m = -X_c$
  Elevation=90°. Azimuth= 00°.
- POSITION 4: Normal Position for turned Offset Measurements Z:
  $X_m = +Z_c, Y_m = -Y_c, Z_m = +Z_c$
  Elevation=0°. Azimuth= 180°.

All linearity measurements are executed in POSITION 1.

Application of separated constant fields in all main axes in standard orientation yielded:

X_m	=	-X_c
Y_m	=	-Y_c
Z_m	=	+Z_c

Thus, this test confirmed that the FM-IB sensor coordinate system is RIGHT-HANDED.

The sensor stayed in Position 1.

10.5 Special Test on the FGM-IB sensor, Room Temperature

At 09:45 a closed loop test at 128 Hz sampling rate was performed. Zerofield conditions. Noise test and Meander test. Analysis to be done by Uli Auster.

At 10:45 an open loop test at 128 Hz sampling rate was performed. Zerofield conditions. Noise test and Meander test. Analysis to be done by Uli Auster.

10.6 DC-Analysis at Room Temp., Range= R2 (4000 nT), FGM-IB, open loop

10.6.1 Calibration on 3 Linear Axes — RANGE R2 — SENSOR IB

Used Files:

CCD File	Configuration File	Remark
21-04-19\11-02-30.CCR	LIN2000XYZ.MAG	

Parameter File: PARAMETER_LIN_R2_IB_21-04-19_11-02-30.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.997521 & -0.068919 & +0.014196 \\ +0.068901 & -0.997622 & -0.001747 \\ +0.014283 & -0.000764 & +0.999898 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m)\text{:} \quad & \lambda_{\mathrm{x}} = && +175^\circ\ 57'54'' \\ \text{Angle } (Y_c, Y_m)\text{:} \quad & \mu_{\mathrm{y}} = && +176^\circ\ 2'52'' \\ \text{Angle } (Z_c, Z_m)\text{:} \quad & \nu_{\mathrm{z}} = && +0^\circ\ 49'10'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 23.0 s
Mean Sensor Temperature: 20.2°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or} = (22.718\ ,\ 10.236\ ,\ 0.340)$ nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.054	0.018	0.016
Mean	0.113	0.052	0.046
Maximum	0.195	0.153	0.107

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +0.962583 & +0.067582 & -0.017604 \\ -0.051471 & +0.977887 & +0.002166 \\ +0.023745 & -0.002608 & +1.239925 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +0.964083 & +0.000000 & +0.000000 \\ +0.014974 & +0.980221 & +0.000000 \\ +0.009988 & -0.001860 & +1.240052 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +0.964083 & +0.000000 & +0.000000 \\ +0.000000 & +0.980103 & +0.000000 \\ +0.000000 & +0.000000 & +1.239983 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ +0.015532 & +1.000121 & +0.000000 \\ +0.010360 & -0.001897 & +1.000056 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\xi_{\text{x,y}} = +90^\circ\ 53'23''$$
$$\xi_{\text{x,z}} = +90^\circ\ 35'43''$$
$$\xi_{\text{y,z}} = +89^\circ\ 52'55''$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	1.572	21.954	0.220
Maximum Error	8.562	120.798	0.465
Minimum Error	-2.315	-35.977	-0.502

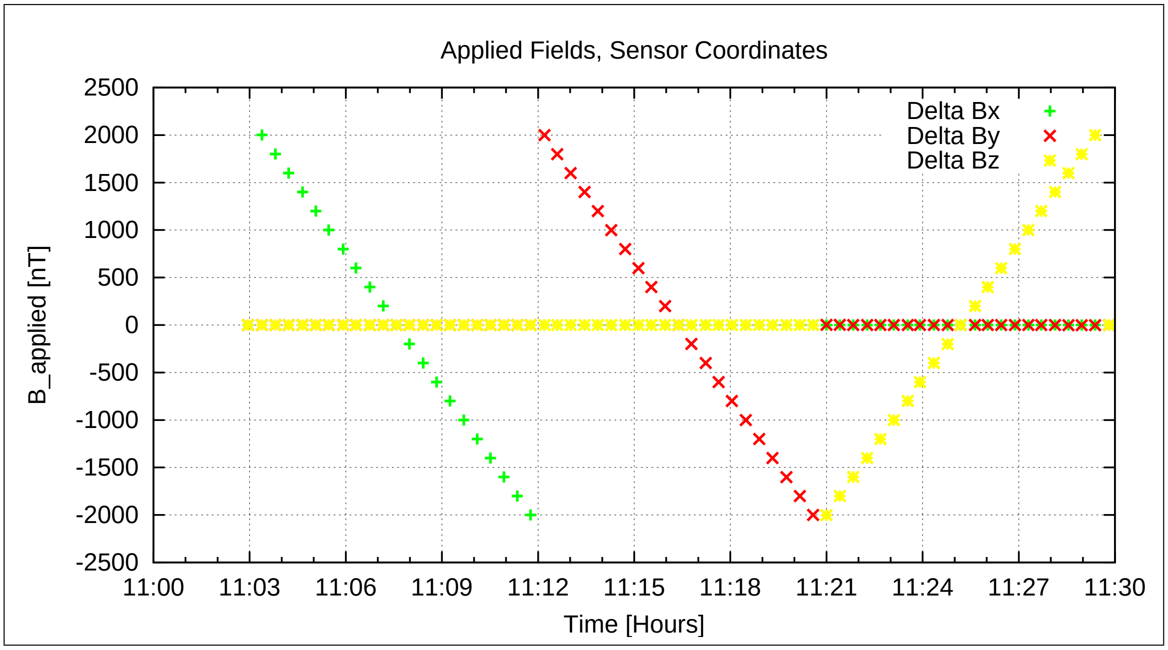

Figure 107: Applied Fields

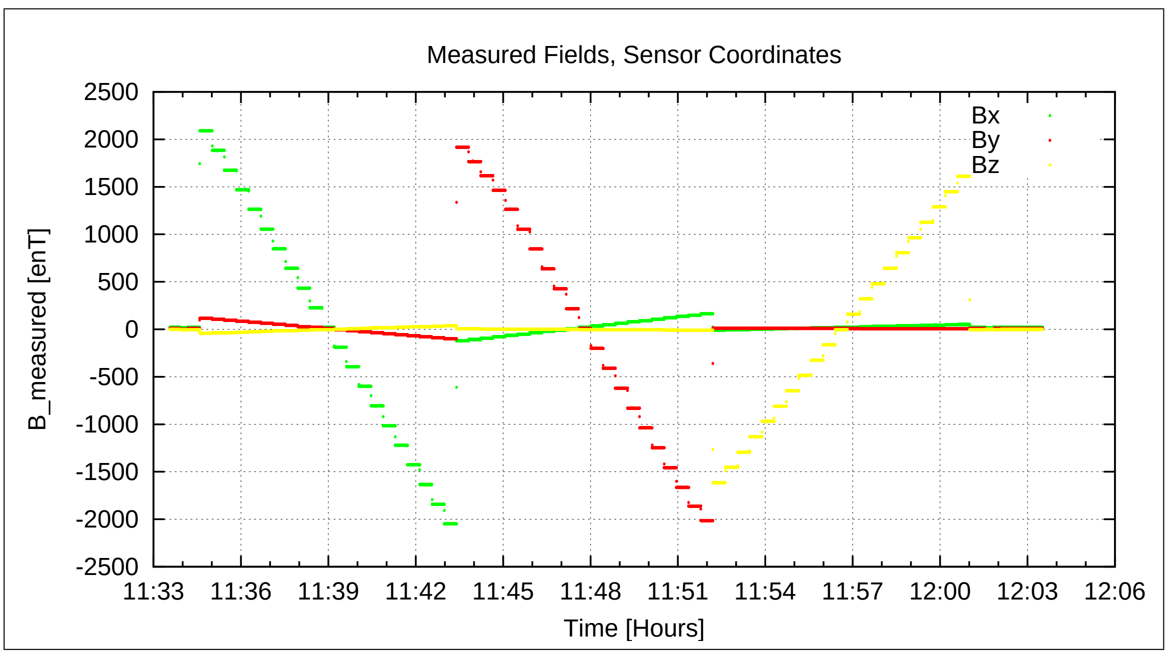

Figure 108: Measured Data

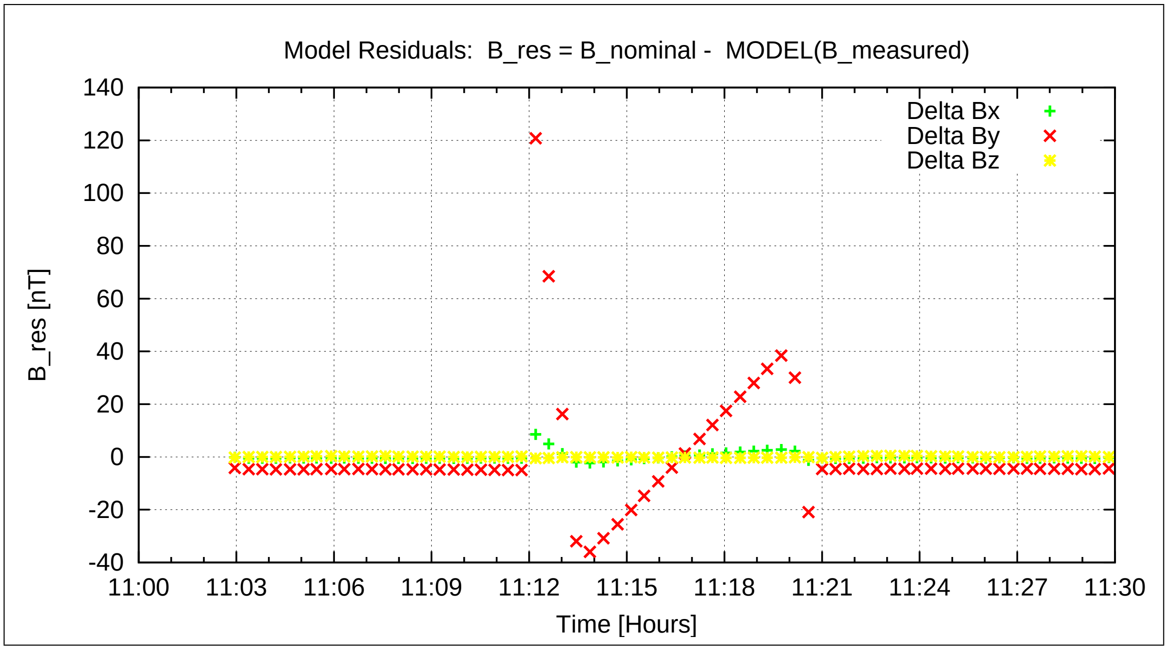

Figure 109: Model Quality - Differences of applied field and modelled measurement data

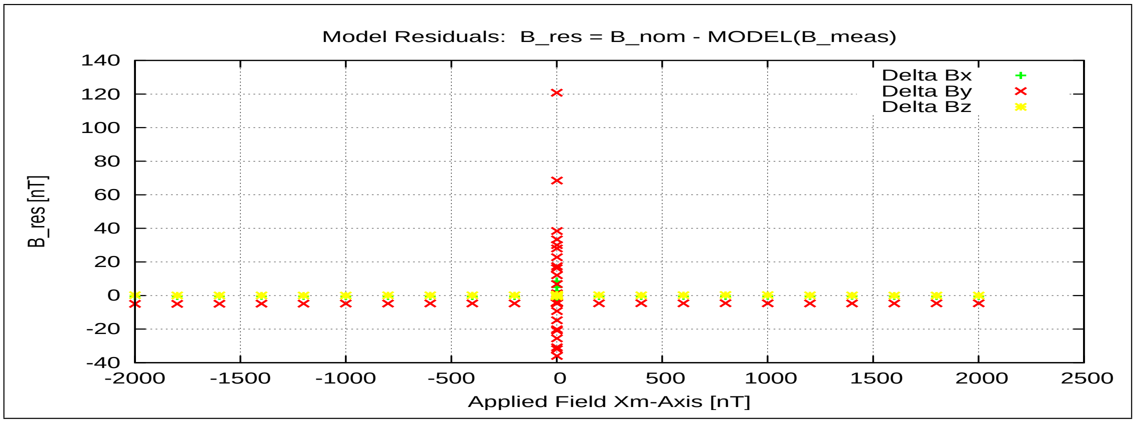

Figure 110: RESIDUALS vs FLD X

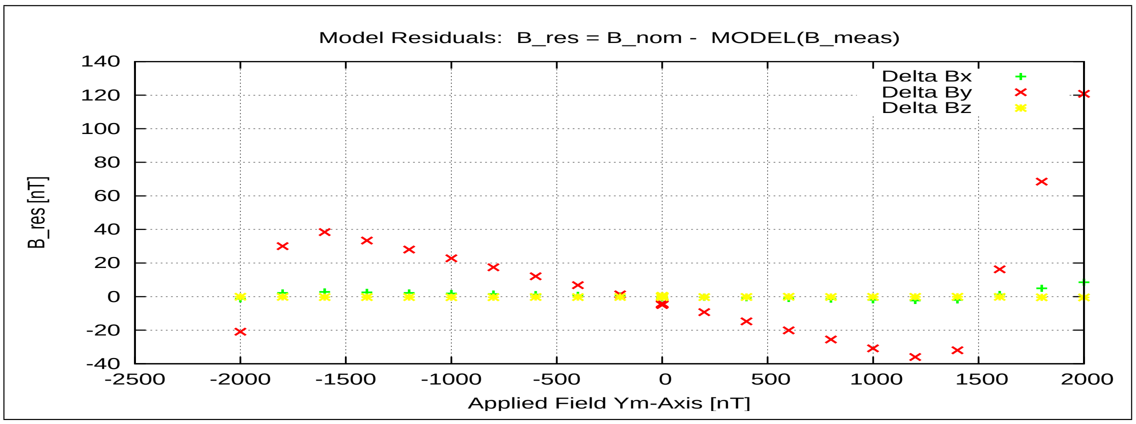

Figure 111: RESIDUALS vs FLD Y

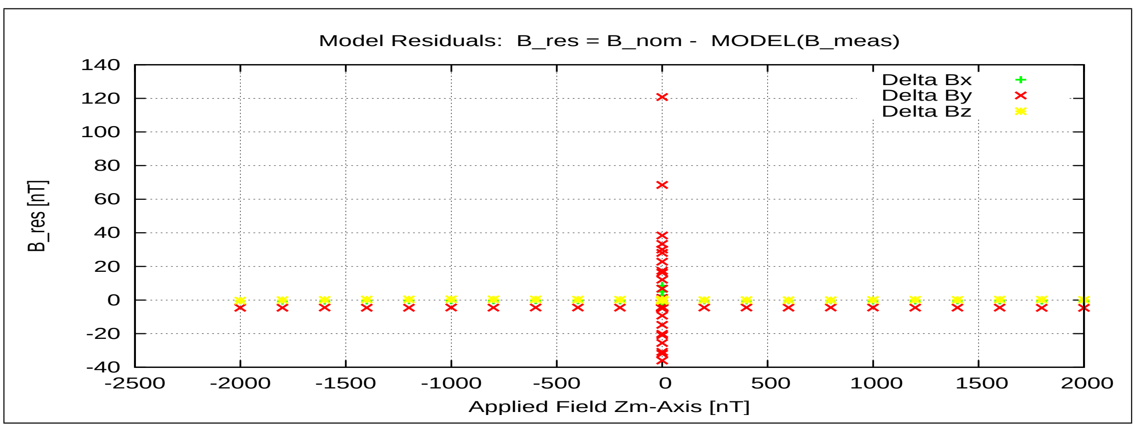

Figure 112: RESIDUALS vs FLD Z

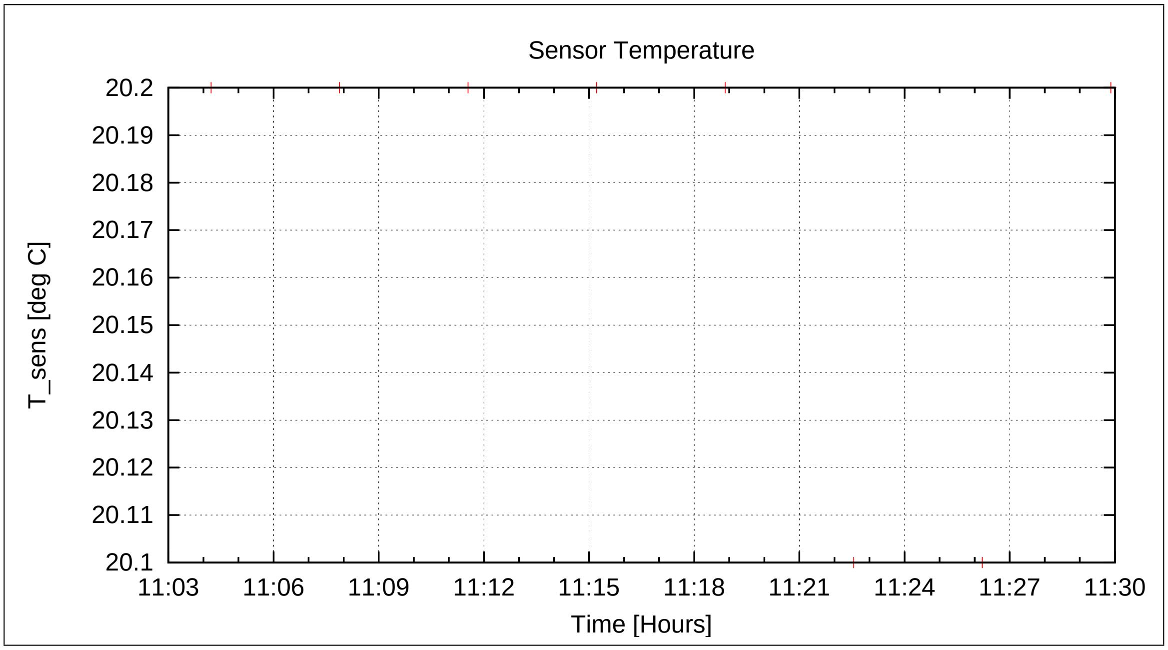

Figure 113: Sensor Temperature

10.7 DC-Analysis at Room Temp. , Range=R4 (16000 nT), FGM-IB, closed loop

10.7.1 Calibration on 3 Linear Axes — RANGE R4 — SENSOR IB

Used Files:

CCD File	Configuration File	Remark
21-04-19\11-34-06.CCR	LIN16000XYZ.MAG	

Parameter File: PARAMETER_LIN_R4_IB_21-04-19_11-34-06.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.999428 & -0.029610 & +0.016313 \\ +0.029420 & -0.999498 & -0.011778 \\ +0.016653 & -0.011291 & +0.999798 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = +178^\circ\ 3'46'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = +178^\circ\ 11'2'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = +1^\circ\ 9'10'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: 20.1°C
Automatic Coil Correction: used

Earthfield Compensation:			
	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or} = (23.650\ ,\ 10.552\ ,\ 0.143)$ nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.058	0.023	0.025
Mean	0.115	0.053	0.057
Maximum	0.191	0.148	0.144

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} \quad = \begin{pmatrix} +0.999557 & +0.029544 & -0.016316 \\ -0.032793 & +0.999484 & +0.011780 \\ +0.032263 & -0.007342 & +0.999986 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \underline{\underline{\tilde{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.000488 & +0.000000 & +0.000000 \\ -0.003544 & +0.999939 & +0.000000 \\ +0.015565 & +0.003949 & +1.000188 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.000488 & +0.000000 & +0.000000 \\ +0.000000 & +0.999933 & +0.000000 \\ +0.000000 & +0.000000 & +1.000059 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} \quad = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ -0.003542 & +1.000006 & +0.000000 \\ +0.015557 & +0.003949 & +1.000129 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\mathrm{x,y}} &= +89^\circ\ 47'49'' \\ \xi_{\mathrm{x,z}} &= +90^\circ\ 53'31'' \\ \xi_{\mathrm{y,z}} &= +90^\circ\ 13'46'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.517	0.557	0.377
Maximum Error	1.460	1.649	1.347
Minimum Error	-0.606	-0.630	-0.384

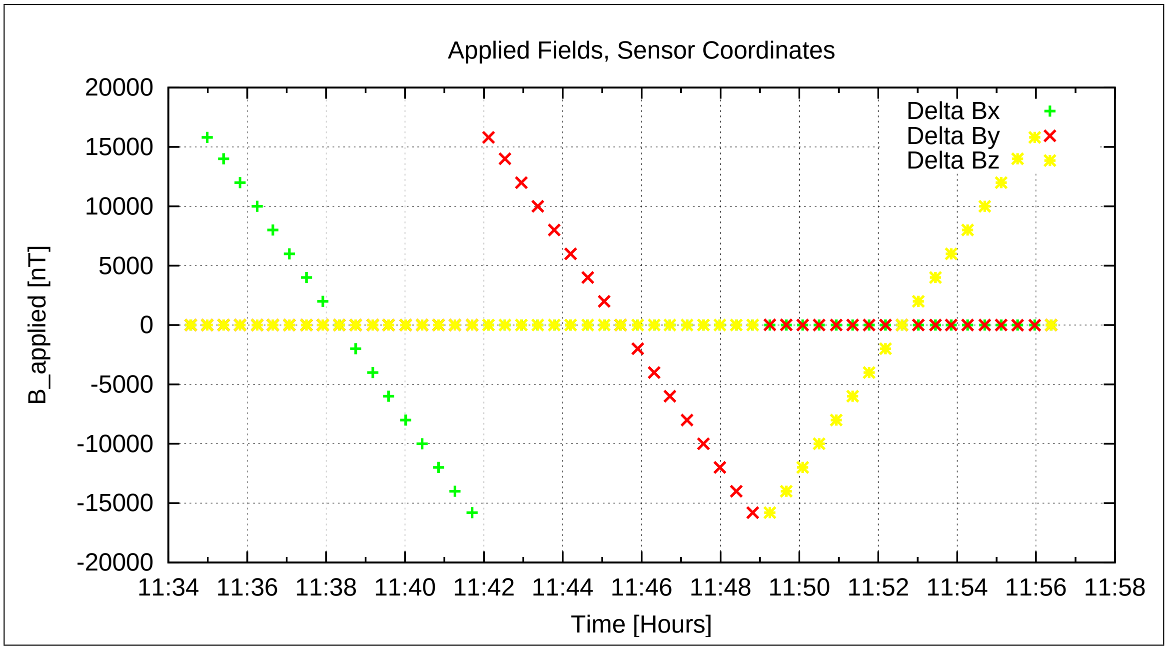

Figure 114: Applied Fields

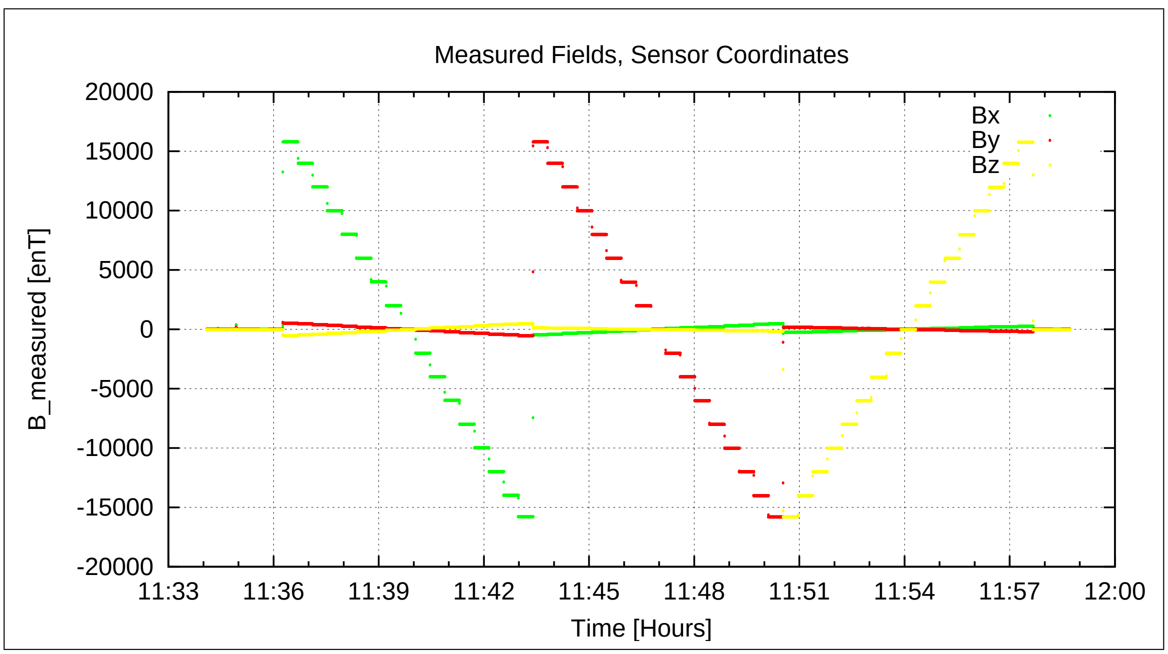

Figure 115: Measured Data

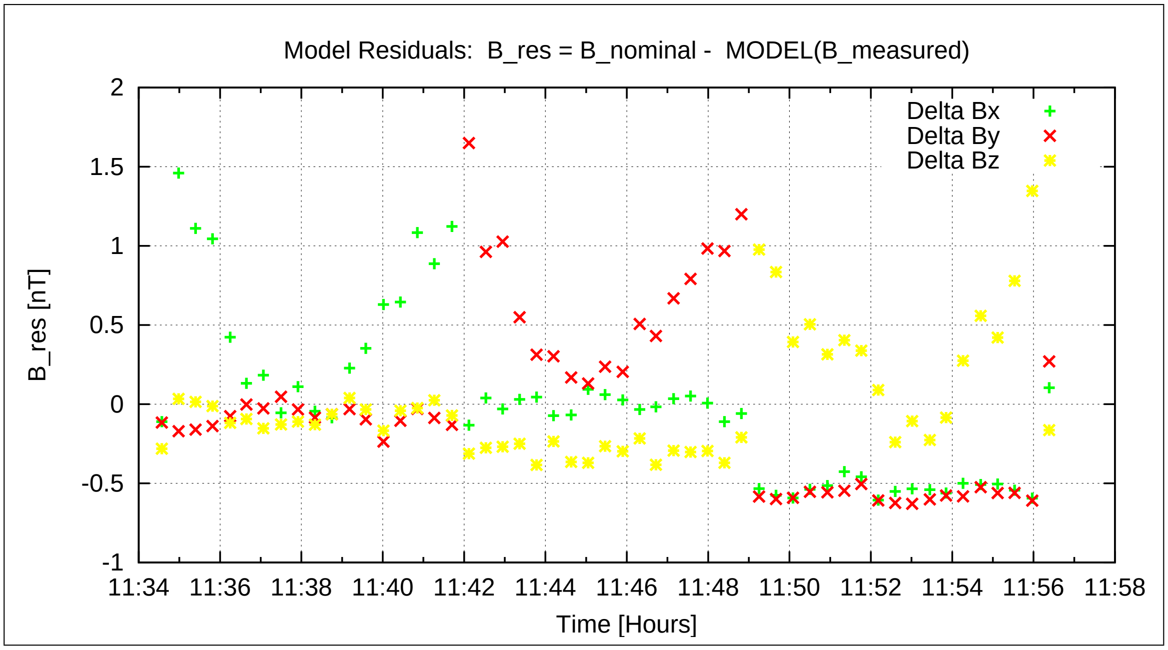

Figure 116: Model Quality - Differences of applied field and modelled measurement data

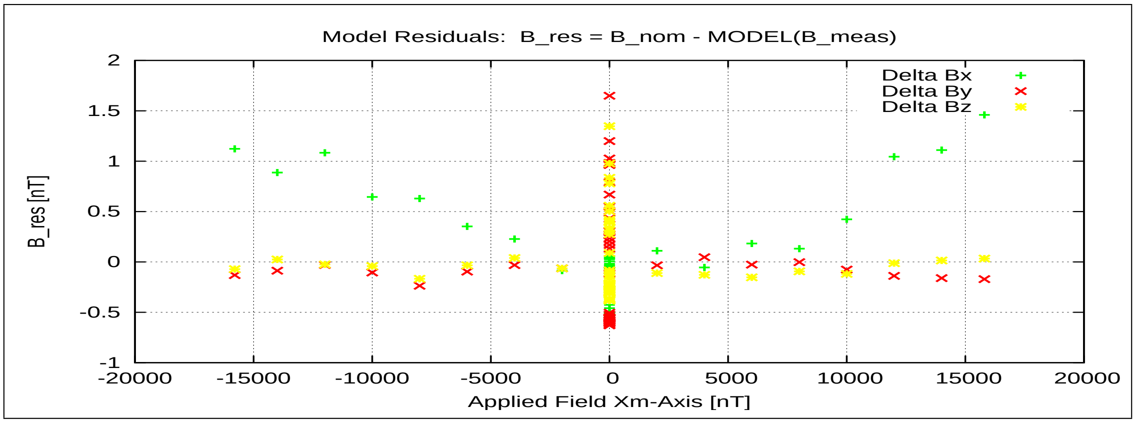

Figure 117: RESIDUALS vs FLD X

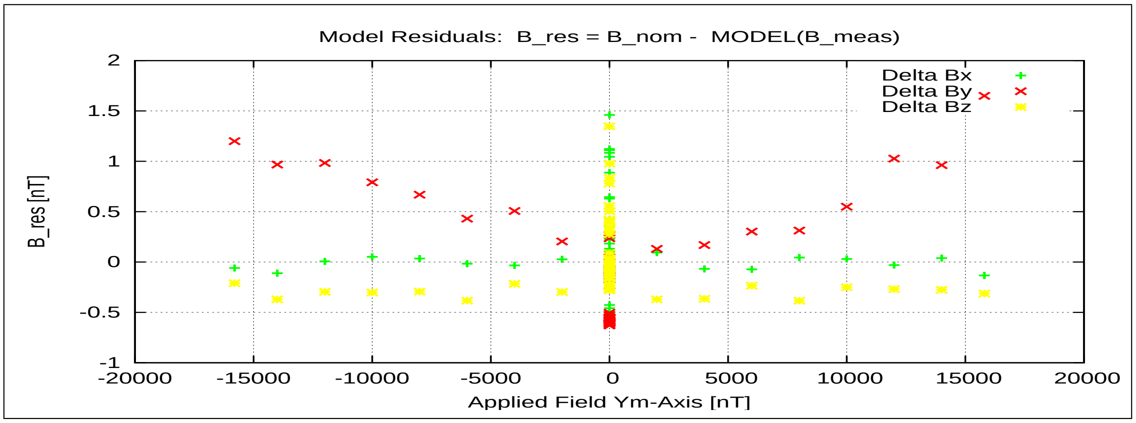

Figure 118: RESIDUALS vs FLD Y

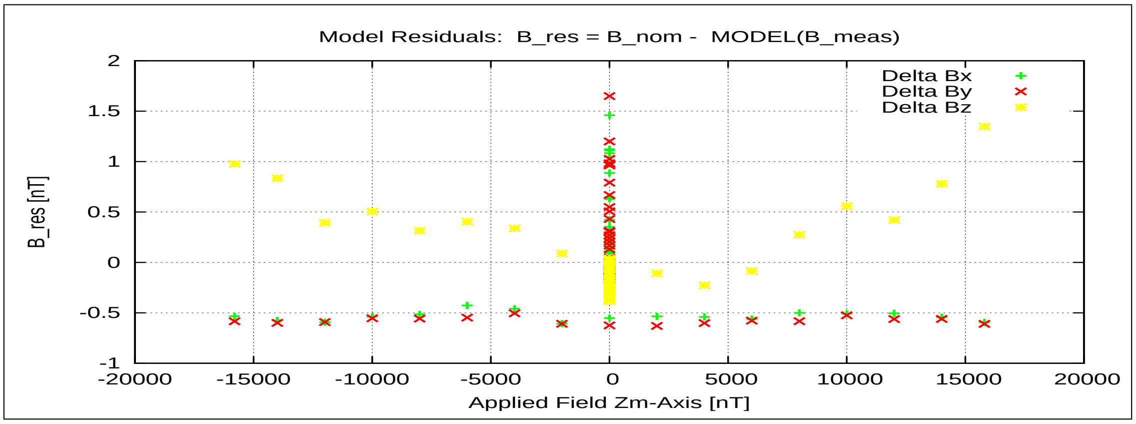

Figure 119: RESIDUALS vs FLD Z

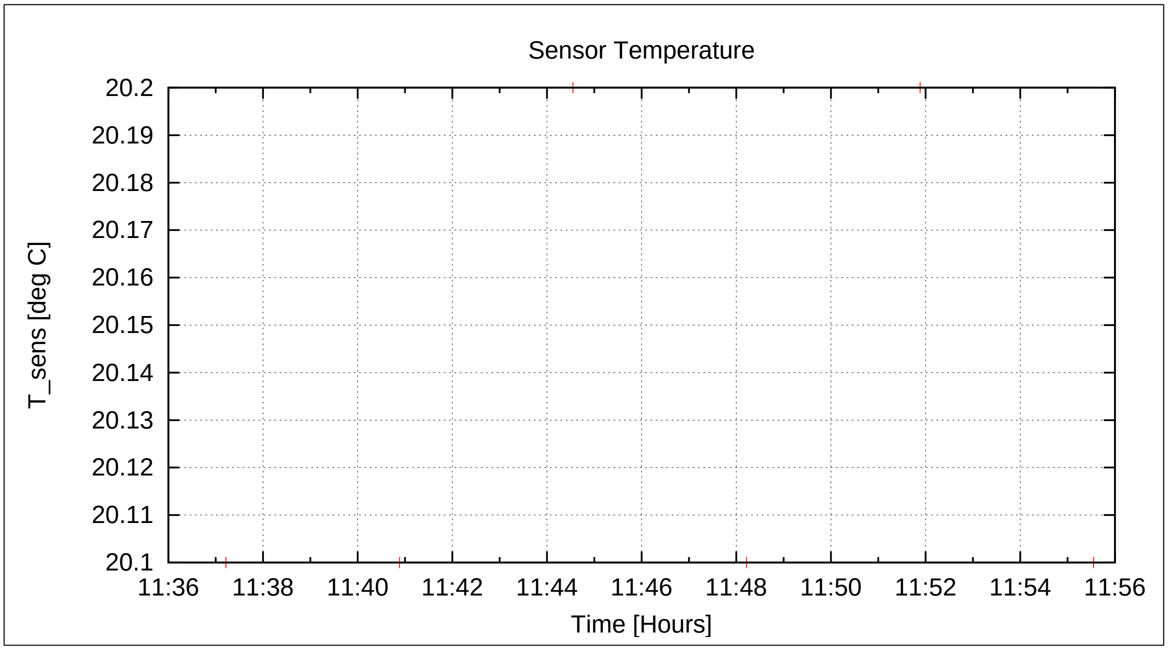

Figure 120: Sensor Temperature

10.8 DC-Analysis at Room Temperature, Range=R0 (1000 nT), FGM-IB

10.8.1 OFFSET Calculation — RANGE R0 — SENSOR IB

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\mathrm{off}} = \frac{B_{\mathrm{normal}} + B_{\mathrm{turned}}}{2}$$
$$B_{\mathrm{res}} = \frac{B_{\mathrm{normal}} - B_{\mathrm{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
`21-04-19\11-59-19.CCR`	`OFFSET_200.MAG`	

Parameter File: `OFF_PARAMETER_R0_21-04-19-11-59-19_IB.OPF`

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	7.20	0.115
Y	9.13	0.091
Z	8.91	0.110

Residual Field of the Coil System:

Component	B_{res} [enT]
X	16.50
Y	2.11
Z	16.33

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

Afterwards we started the cooling cycle. Initially the goal temperature was set to T=-30 degC. Cooling started at 12:12. The system was pressurized to 400 hPa.

10.9 DC-Analysis at T = $-30°$C, Range=R4 (16000 nT), FGM-IB

10.9.1 Calibration on 3 Linear Axes — RANGE R4 — SENSOR IB

Used Files:

CCD File	Configuration File	Remark
21-04-19\13-45-25.CCR	LIN16000XYZ.MAG	

Parameter File: PARAMETER_LIN_R4_IB_21-04-19_13-45-25.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.998733 & -0.032206 & +0.038678 \\ +0.031820 & -0.999438 & -0.010552 \\ +0.038997 & -0.009308 & +0.999196 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +177^\circ\ 6'54'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +178^\circ\ 4'44'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +2^\circ\ 17'52'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -30.3°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (23.153 , 11.538 , -0.671) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.039	0.026	0.032
Mean	0.068	0.053	0.062
Maximum	0.171	0.127	0.144

Calibration Parameter:

$$\text{Transfermatrix:}\quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +0.999391 & +0.032071 & -0.038715 \\ -0.035192 & +1.000023 & +0.010562 \\ +0.054678 & -0.005393 & +1.000146 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:}\quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} = \begin{pmatrix} +1.001377 & +0.000000 & +0.000000 \\ -0.003495 & +1.000544 & +0.000000 \\ +0.015608 & +0.003923 & +1.000950 \end{pmatrix}$$

$$\text{Sensitivity:}\quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.001377 & +0.000000 & +0.000000 \\ +0.000000 & +1.000538 & +0.000000 \\ +0.000000 & +0.000000 & +1.000821 \end{pmatrix}$$

$$\text{Misalignment:}\quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ -0.003490 & +1.000006 & +0.000000 \\ +0.015587 & +0.003921 & +1.000129 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\text{x,y}} &= +89^\circ\ 48'0'' \\ \xi_{\text{x,z}} &= +90^\circ\ 53'38'' \\ \xi_{\text{y,z}} &= +90^\circ\ 13'40'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.441	0.558	0.421
Maximum Error	1.133	1.451	1.438
Minimum Error	-0.550	-0.696	-0.566

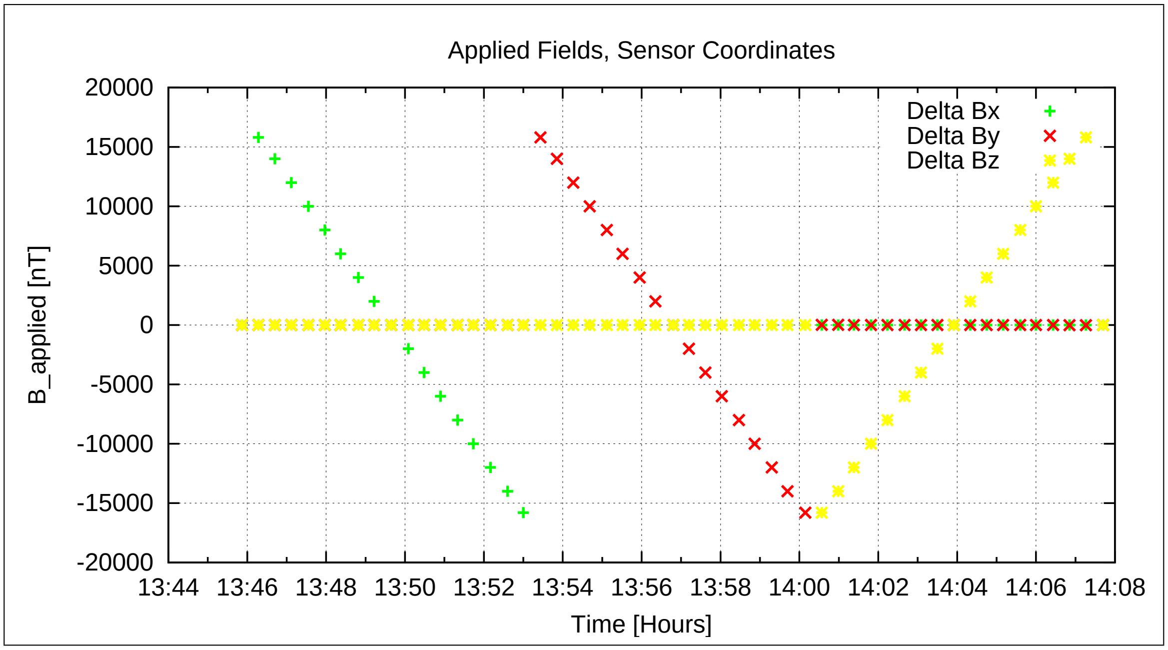

Figure 121: Applied Fields

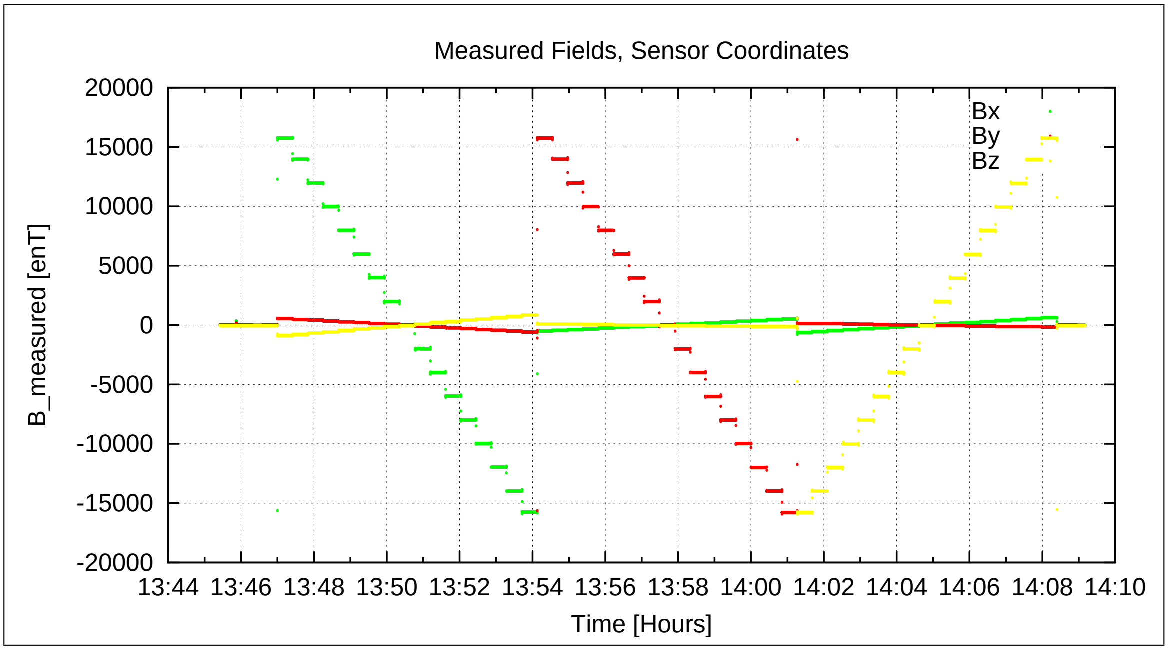

Figure 122: Measured Data

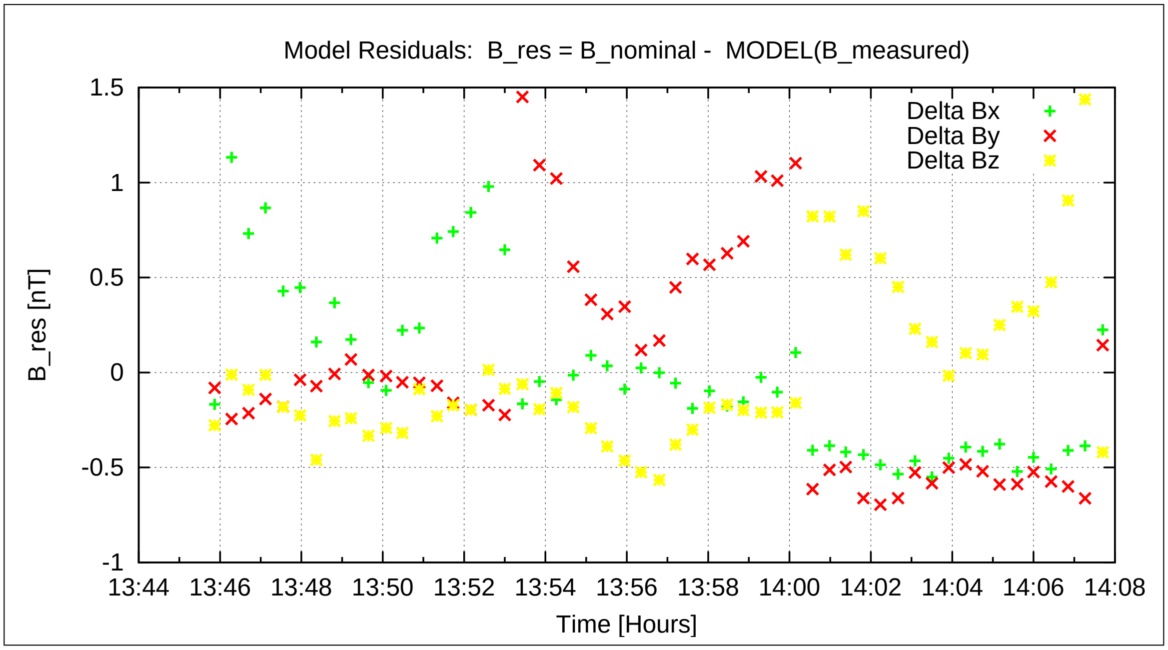

Figure 123: Model Quality - Differences of applied field and modelled measurement data

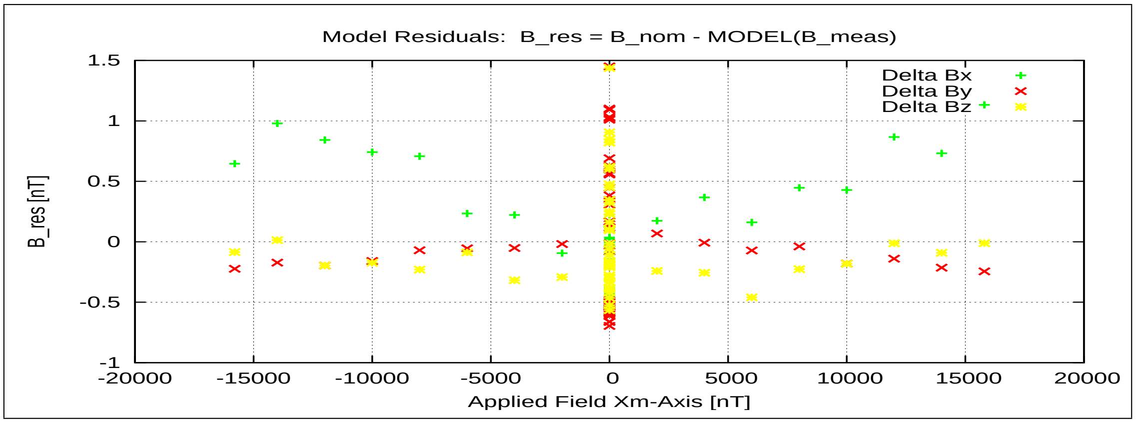

Figure 124: RESIDUALS vs FLD X

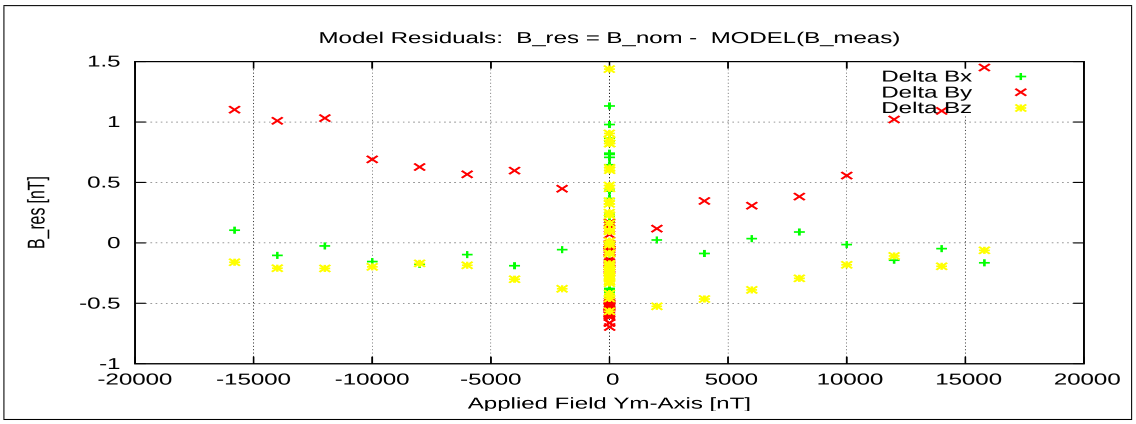

Figure 125: RESIDUALS vs FLD Y

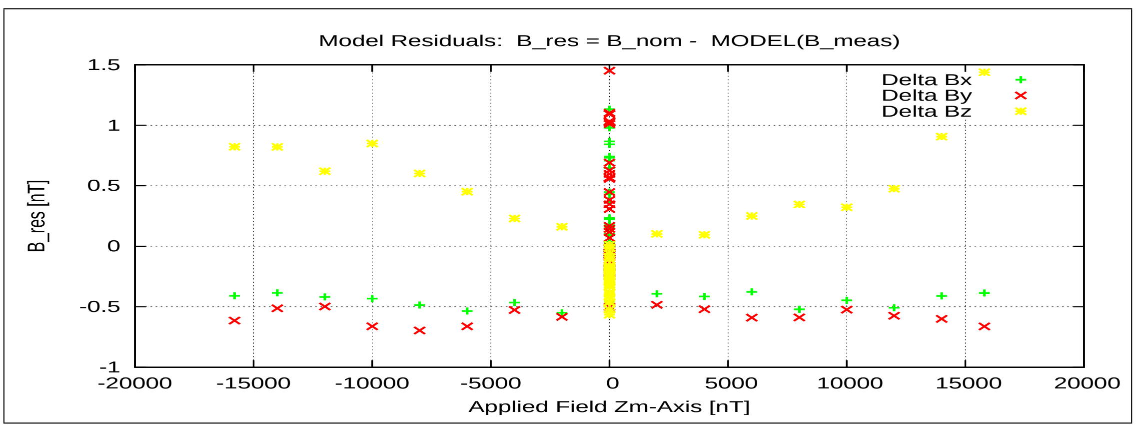

Figure 126: RESIDUALS vs FLD Z

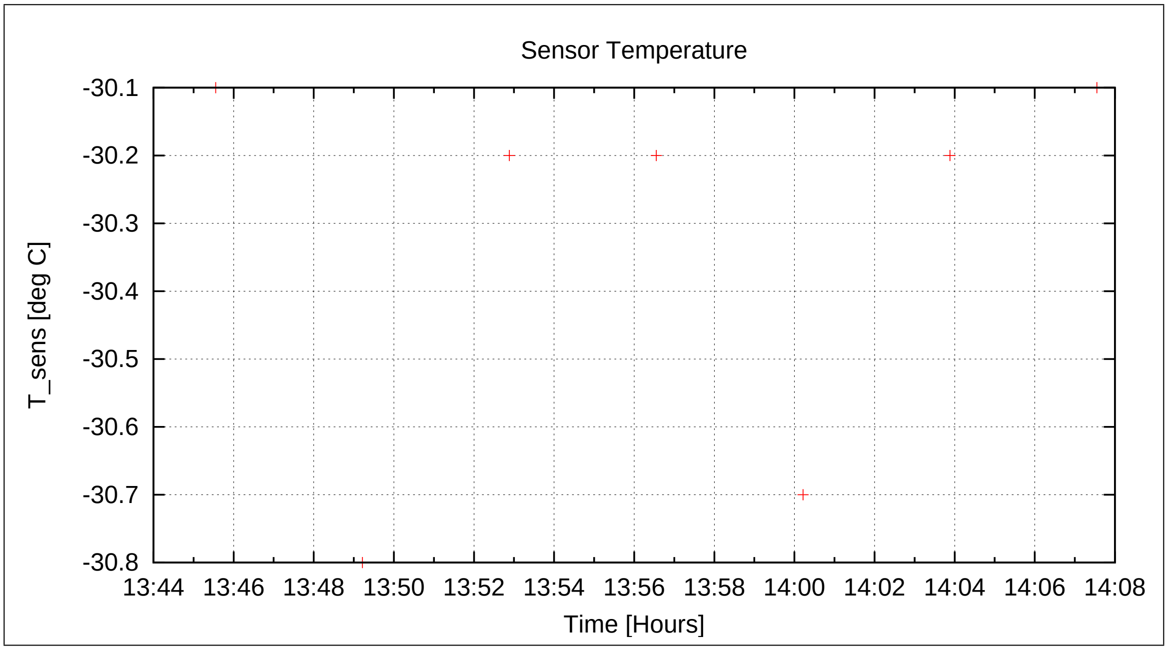

Figure 127: Sensor Temperature

10.10 Special Test on the FGM-IB sensor at T = −30°C

At 14:15 noise measurements, meander tests and find phase tests were were performed in zerofield conditions. Analysis to be done by Uli Auster.

10.11 DC-Analysis at T = −30°C, Range=R0 (1000 nT), FGM-IB

10.11.1 OFFSET Calculation — RANGE R0 — SENSOR IB

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
`21-04-19\15-52-29.CCR`	`OFFSET_200.MAG`	

Parameter File: `OFF_PARAMETER_R0_21-04-19-15-52-29_IB.OPF`

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	7.42	0.099
Y	9.33	0.081
Z	7.21	0.122

Residual Field of the Coil System:

Component	B_{res} [enT]
X	17.13
Y	1.66
Z	16.85

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

10.12 DC-Analysis at T = $-60°$C, Range=R4 (16000 nT), FGM-IB

10.12.1 Calibration on 3 Linear Axes — RANGE R4 — SENSOR IB

Used Files:

CCD File	Configuration File	Remark
21-04-19\19-14-44.CCR	LIN16000XYZ.MAG	

Parameter File: PARAMETER_LIN_R4_IB_21-04-19_19-14-44.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.997800 & -0.033032 & +0.057488 \\ +0.032517 & -0.999422 & -0.009880 \\ +0.057781 & -0.007989 & +0.998297 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = & +176^\circ\ 11'54'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = & +178^\circ\ 3'9'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = & +3^\circ\ 20'38'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -60.0°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (19.599 , 10.708 , -1.466) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.039	0.027	0.034
Mean	0.070	0.054	0.064
Maximum	0.167	0.161	0.160

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\,\underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}} \quad = \begin{pmatrix} +0.998743 & +0.032837 & -0.057571 \\ -0.035871 & +1.000398 & +0.009894 \\ +0.073615 & -0.004060 & +0.999745 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \underline{\underline{\tilde{\phi}}} = \underline{\underline{\omega}}\,\underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.001965 & +0.000000 & +0.000000 \\ -0.003448 & +1.000937 & +0.000000 \\ +0.015719 & +0.003943 & +1.001450 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.001965 & +0.000000 & +0.000000 \\ +0.000000 & +1.000931 & +0.000000 \\ +0.000000 & +0.000000 & +1.001319 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} \quad = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ -0.003441 & +1.000006 & +0.000000 \\ +0.015689 & +0.003939 & +1.000131 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\mathrm{x,y}} &= +89^\circ\ 48'10'' \\ \xi_{\mathrm{x,z}} &= +90^\circ\ 53'59'' \\ \xi_{\mathrm{y,z}} &= +90^\circ\ 13'44'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.315	0.452	0.300
Maximum Error	1.165	1.313	1.216
Minimum Error	-0.294	-0.365	-0.551

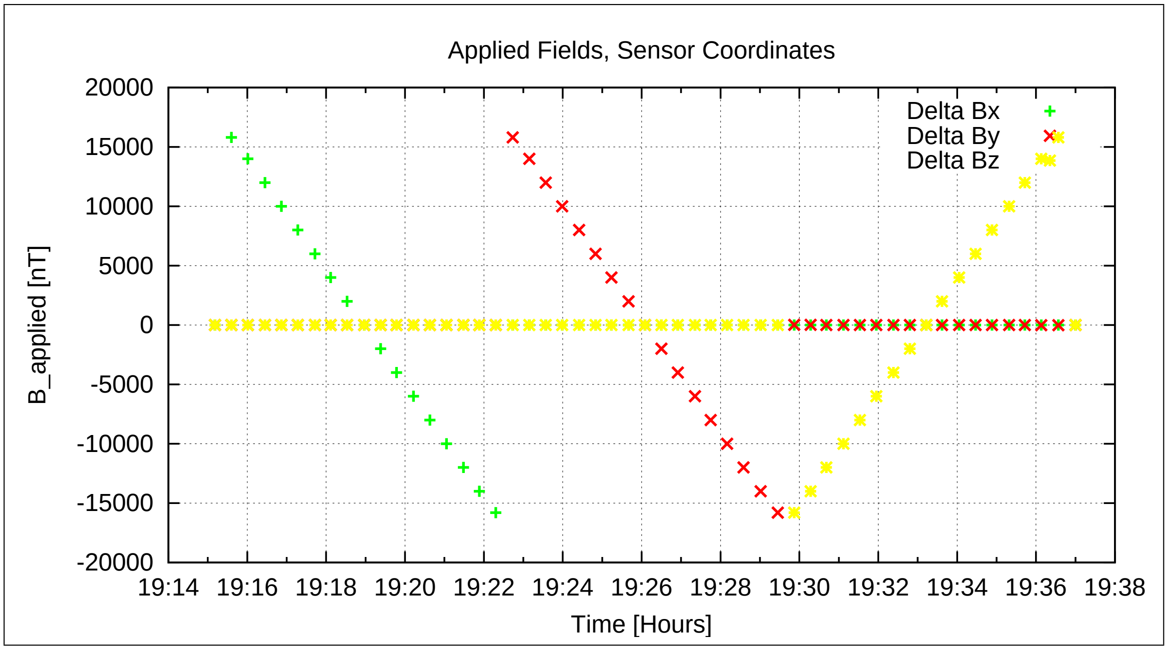

Figure 128: Applied Fields

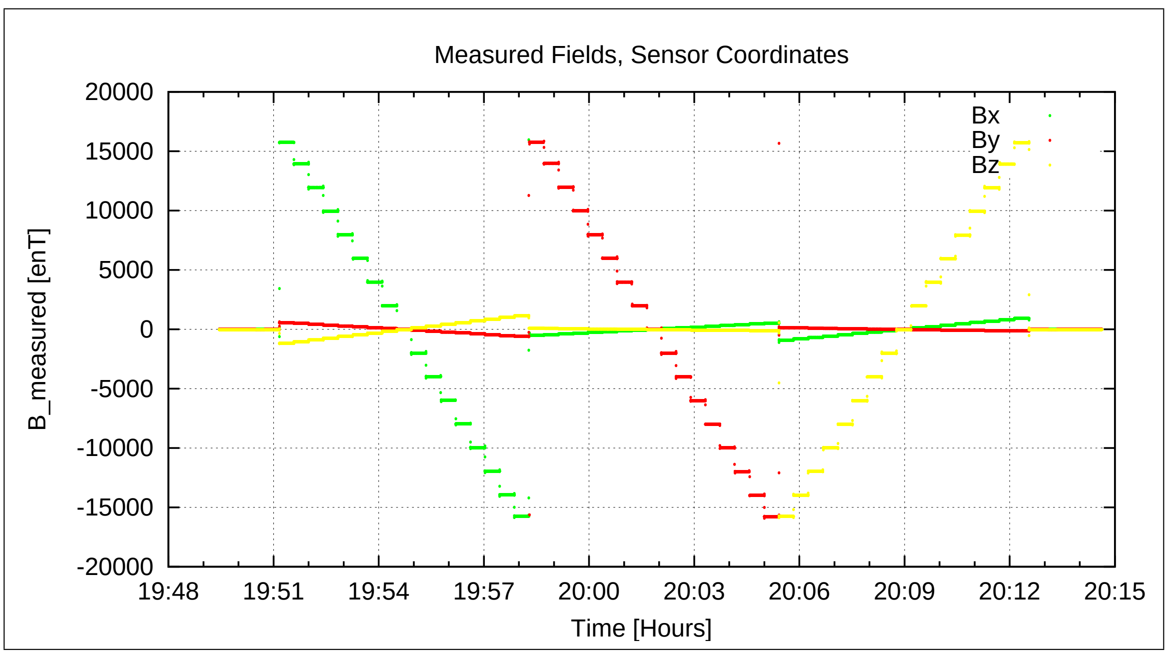

Figure 129: Measured Data

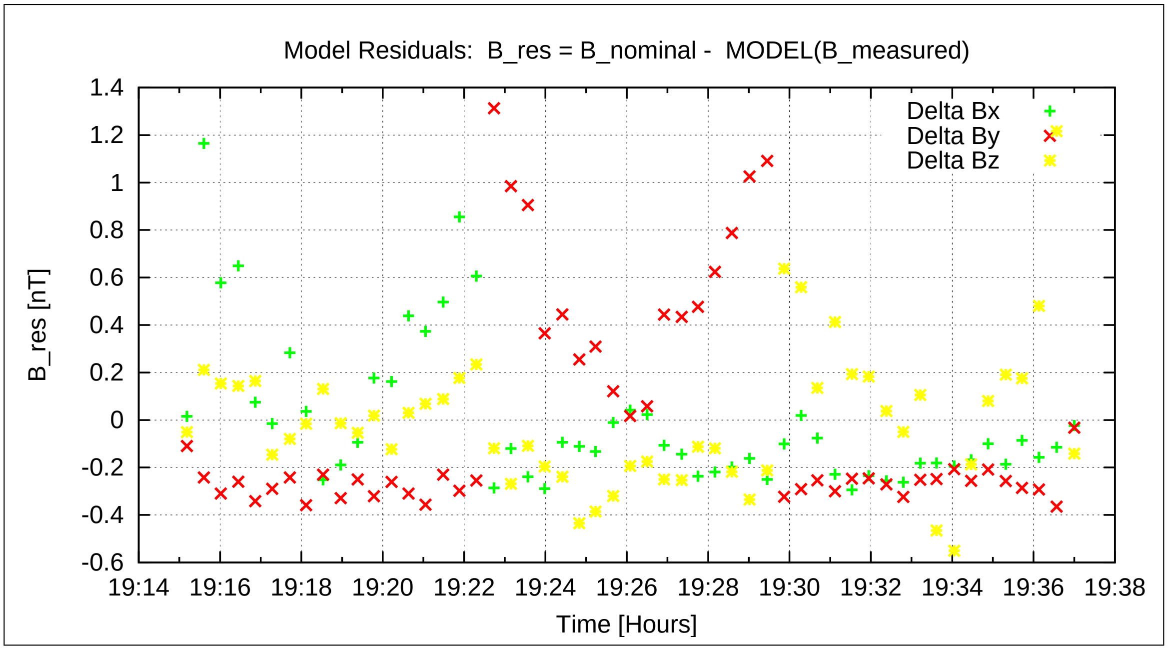

Figure 130: Model Quality - Differences of applied field and modelled measurement data

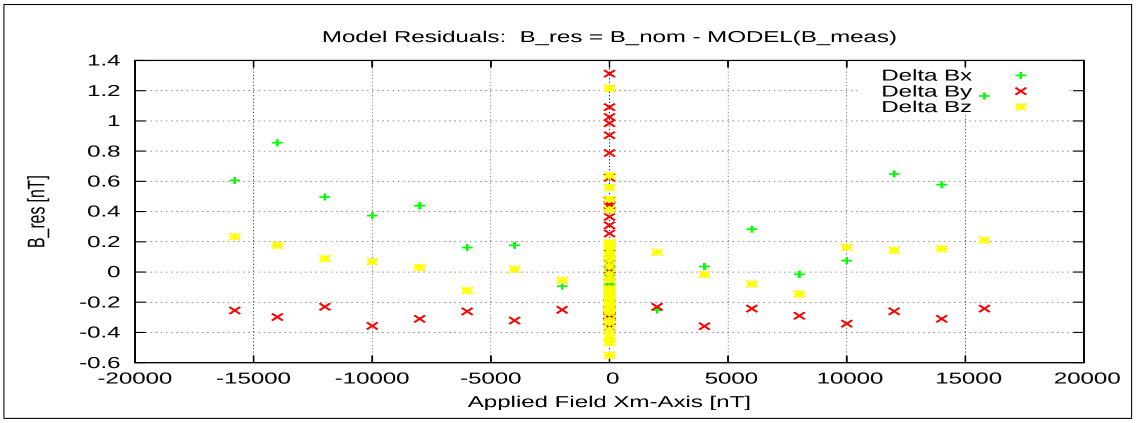

Figure 131: RESIDUALS vs FLD X

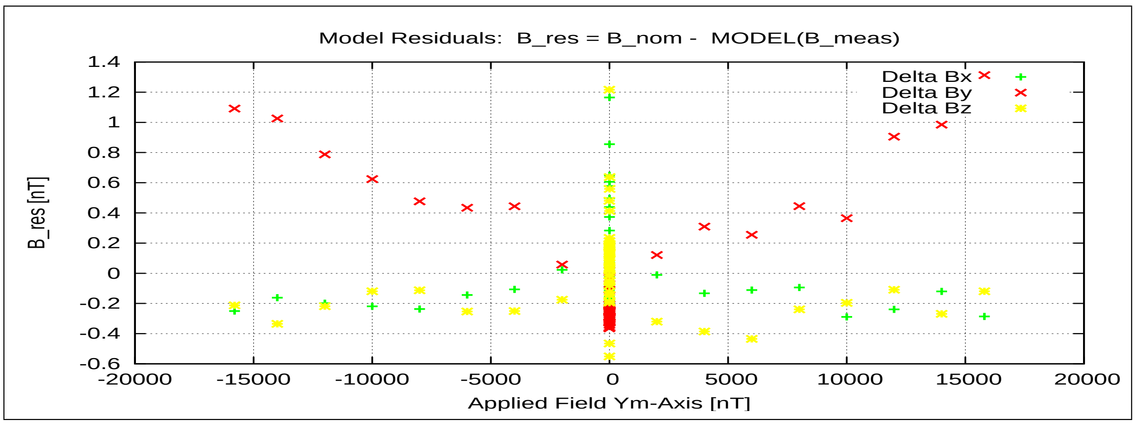

Figure 132: RESIDUALS vs FLD Y

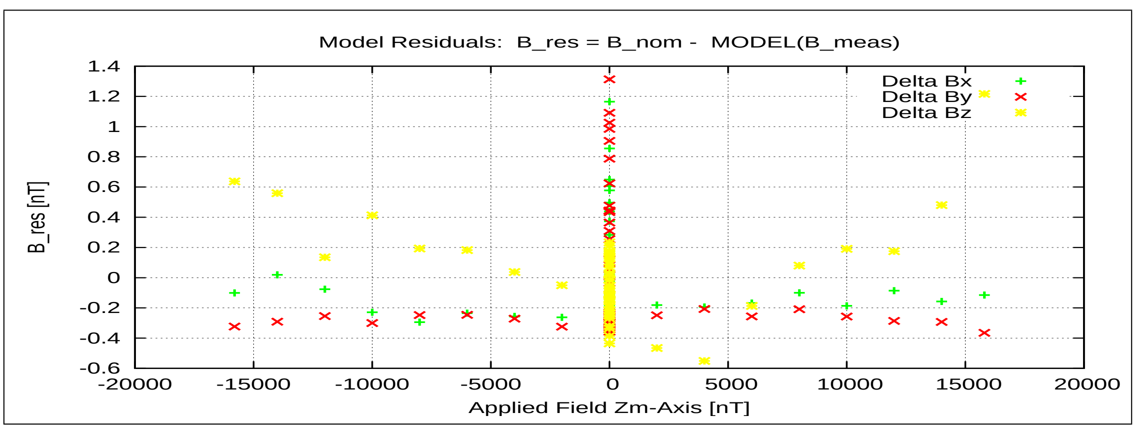

Figure 133: RESIDUALS vs FLD Z

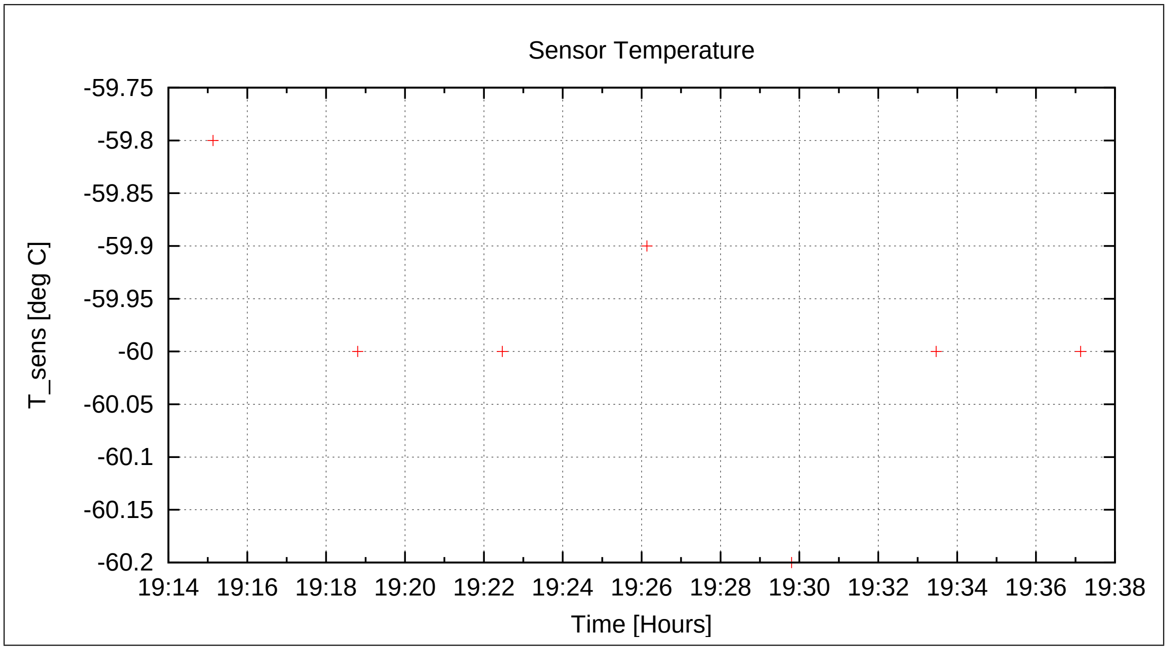

Figure 134: Sensor Temperature

11 Measurements on April 20, 2021

11.1 Data

CCD File	Configuration File	Remark
21-04-20\00-14-14.CCR	LIN16000XYZ.MAG	
21-04-20\00-36-50.CCR	NULL.MAG	
21-04-20\01-14-07.CCR	LIN16000XYZ.MAG	
21-04-20\01-36-43.CCR	NULL.MAG	
21-04-20\02-14-00.CCR	LIN16000XYZ.MAG	
21-04-20\02-36-36.CCR	NULL.MAG	
21-04-20\03-13-54.CCR	LIN16000XYZ.MAG	
21-04-20\03-36-30.CCR	NULL.MAG	
21-04-20\04-13-48.CCR	LIN16000XYZ.MAG	
21-04-20\04-36-24.CCR	NULL.MAG	
21-04-20\05-13-41.CCR	LIN16000XYZ.MAG	
21-04-20\05-36-18.CCR	NULL.MAG	
21-04-20\06-13-35.CCR	LIN16000XYZ.MAG	
21-04-20\06-36-12.CCR	NULL.MAG	
21-04-20\07-23-51.CCR	OFFSET_200.MAG	
21-04-20\09-41-19.CCR	OFFSET_200.MAG	
21-04-20\10-31-18.CCR	LIN16000XYZ.MAG	
21-04-20\11-30-00.CCR	OFFSET_200.MAG	
21-04-20\12-33-52.CCR	LIN16000XYZ.MAG	
21-04-20\13-37-22.CCR	OFFSET_200.MAG	

Inspection of the system at 06:56. All looks ok. Box temperature at -51.2 degC. Sequence stopped at 07:15.

11.2 DC-Analysis at T = $-50°$C, Range=R0 (1000 nT), FGM-IB

11.2.1 OFFSET Calculation — RANGE R0 — SENSOR IB

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$

$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-20\07-23-51.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER_R0_21-04-20-07-23-51_IB.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	6.74	0.123
Y	9.31	0.070
Z	8.72	0.111

Residual Field of the Coil System:

Component	B_{res} [enT]
X	13.11
Y	1.47
Z	12.74

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

11.3 Special Test on the FGM-IB sensor at T = $-50°$C

At 07:45 noise measurements, meander tests and find phase tests were performed in zerofield conditions. Analysis to be done by Uli Auster.

At 08:25 the emptied LN_2 bottle was exchanged by a new, full bottle. The system was pressurized to 400 hPa.

Cooling cont'd. The new temperature goal was set to T= -80 degC.

11.4 Special Test on the FGM-IB sensor at T = $-80°$C

At 09:15 noise measurements, meander tests and find phase tests were performed in zerofield conditions. Analysis to be done by Uli Auster.

11.5 DC-Analysis at T = $-80°$C, Range=R0 (1000 nT), FGM-IB

11.5.1 OFFSET Calculation — RANGE R0 — SENSOR IB

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-20\09-41-19.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER_R0_21-04-20-09-41-19_IB.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	6.25	0.119
Y	10.08	0.078
Z	5.52	0.130

Residual Field of the Coil System:

Component	B_{res} [enT]
X	11.41
Y	-1.26
Z	11.38

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

Cooling cont'd. The new temperature goal was set to T= -110 degC.

11.6 DC-Analysis at T = $-110°$C, Range=R4 (16000 nT), FGM-IB

11.6.1 Calibration on 3 Linear Axes — RANGE R4 — SENSOR IB

Used Files:

CCD File	Configuration File	Remark
21-04-20\10-31-18.CCR	LIN16000XYZ.MAG	

Parameter File: PARAMETER_LIN_R4_IB_21-04-20_10-31-18.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.997782 & -0.041148 & +0.052319 \\ +0.040711 & -0.999127 & -0.009390 \\ +0.052660 & -0.007239 & +0.998586 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): \quad & \lambda_{\mathrm{x}} = && +176^\circ\ 11'0'' \\ \text{Angle } (Y_c, Y_m): \quad & \mu_{\mathrm{y}} = && +177^\circ\ 36'20'' \\ \text{Angle } (Z_c, Z_m): \quad & \nu_{\mathrm{z}} = && +3^\circ\ 2'49'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -109.8°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (16.961 , 9.294 , -5.027) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.041	0.030	0.037
Mean	0.081	0.063	0.075
Maximum	0.239	0.185	0.346

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}} \underline{\underline{\rho}}\, \underline{\underline{\omega}}\, \underline{\underline{\sigma}} = \begin{pmatrix} +0.999517 & +0.041006 & -0.052434 \\ -0.044062 & +1.000680 & +0.009411 \\ +0.068658 & -0.003352 & +1.000774 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\, \underline{\underline{\sigma}} = \begin{pmatrix} +1.002709 & +0.000000 & +0.000000 \\ -0.003392 & +1.001518 & +0.000000 \\ +0.015853 & +0.003904 & +1.002191 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} = \begin{pmatrix} +1.002709 & +0.000000 & +0.000000 \\ +0.000000 & +1.001512 & +0.000000 \\ +0.000000 & +0.000000 & +1.002058 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ -0.003383 & +1.000006 & +0.000000 \\ +0.015810 & +0.003898 & +1.000133 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\begin{aligned} \xi_{\mathrm{x,y}} &= +89^\circ\ 48'22'' \\ \xi_{\mathrm{x,z}} &= +90^\circ\ 54'24'' \\ \xi_{\mathrm{y,z}} &= +90^\circ\ 13'35'' \end{aligned}$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.629	0.657	0.519
Maximum Error	1.297	1.240	1.476
Minimum Error	-0.889	-0.926	-0.715

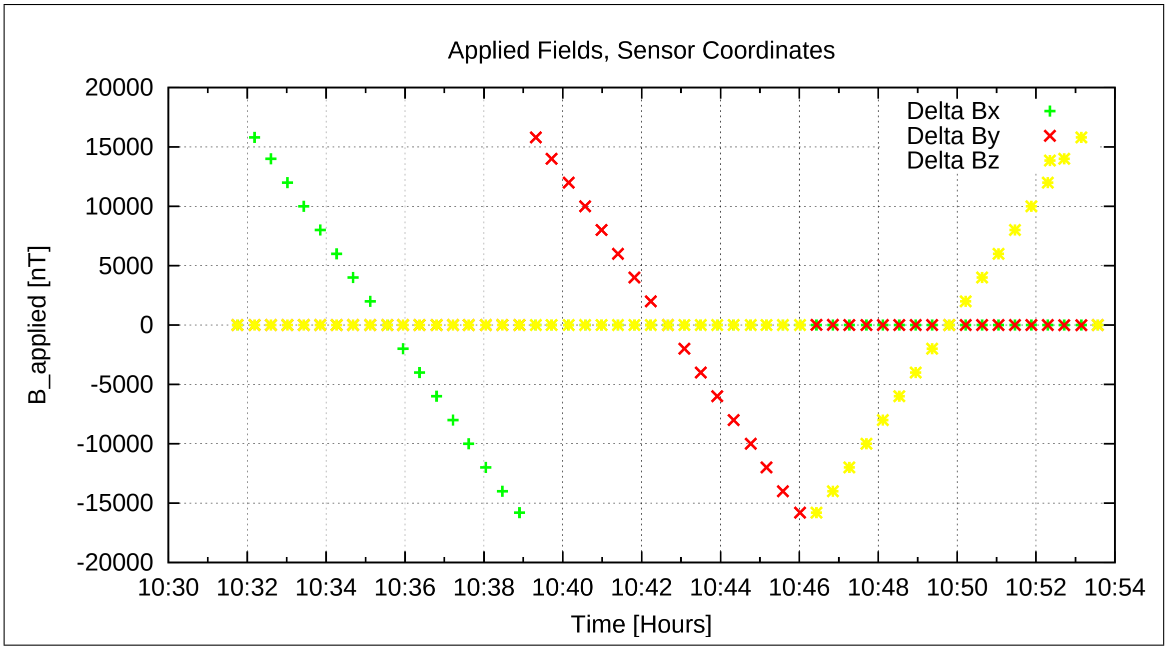

Figure 135: Applied Fields

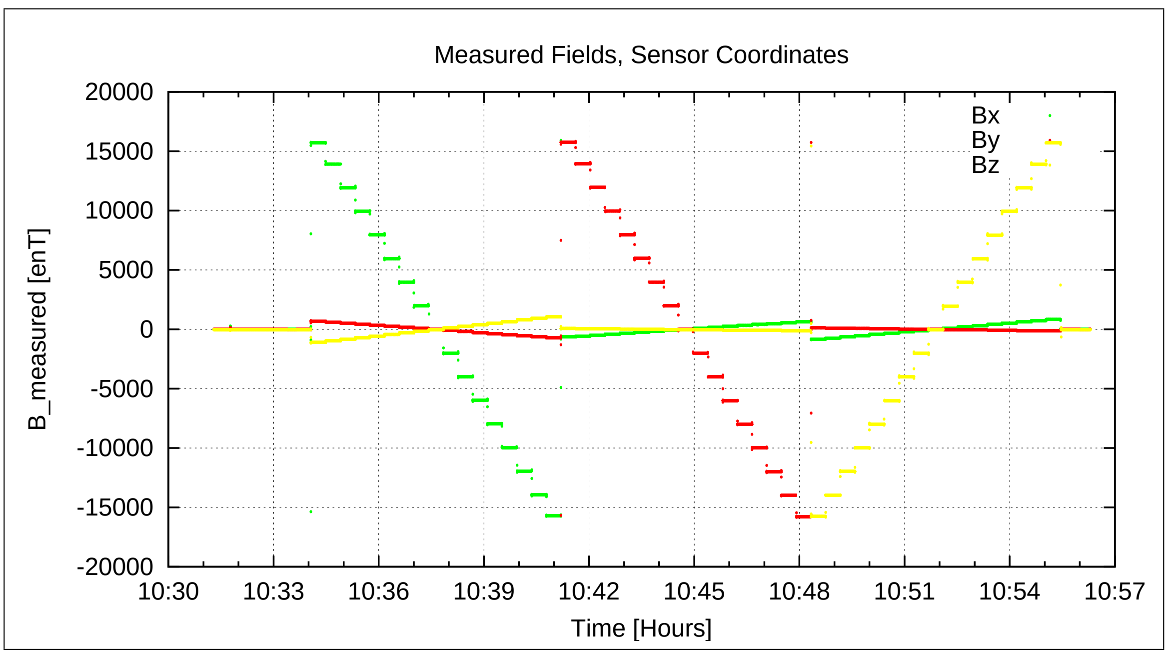

Figure 136: Measured Data

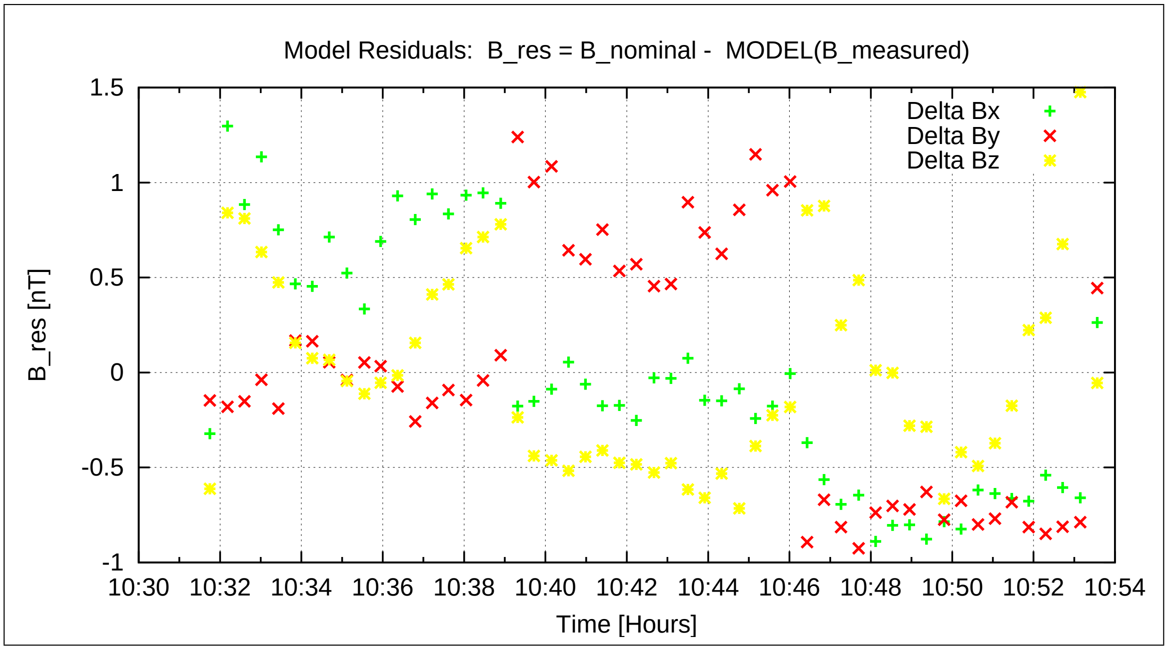

Figure 137: Model Quality - Differences of applied field and modelled measurement data

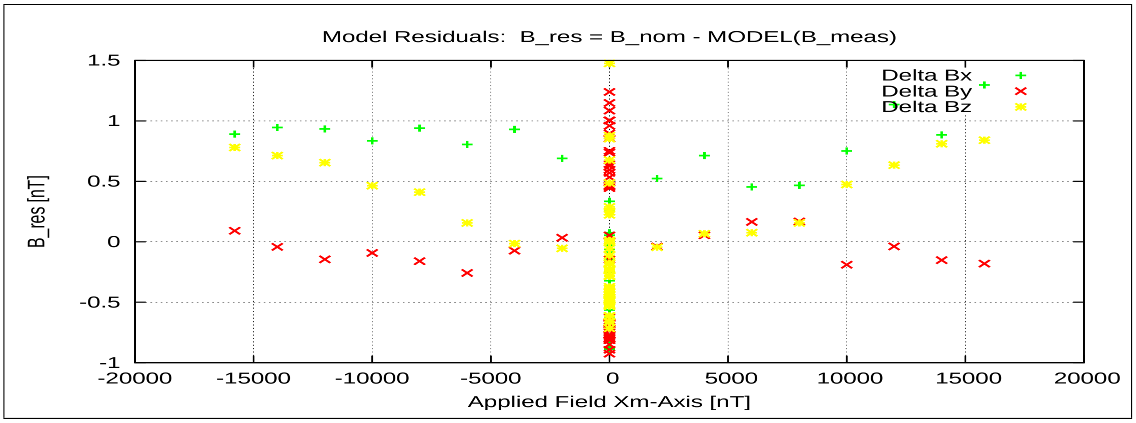

Figure 138: RESIDUALS vs FLD X

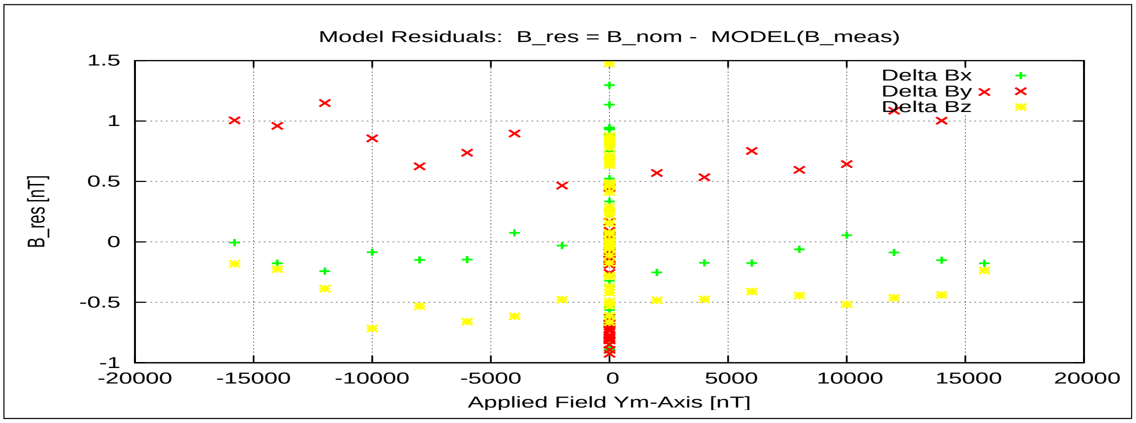

Figure 139: RESIDUALS vs FLD Y

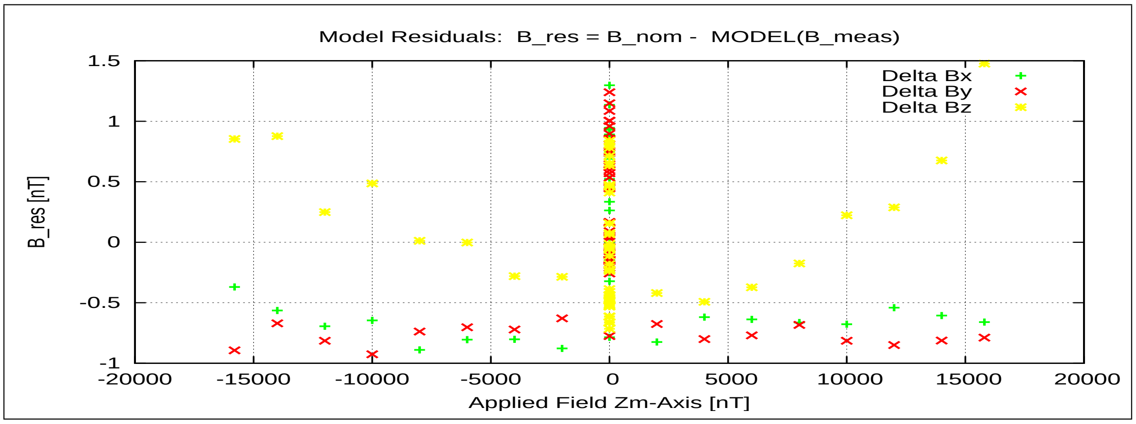

Figure 140: RESIDUALS vs FLD Z

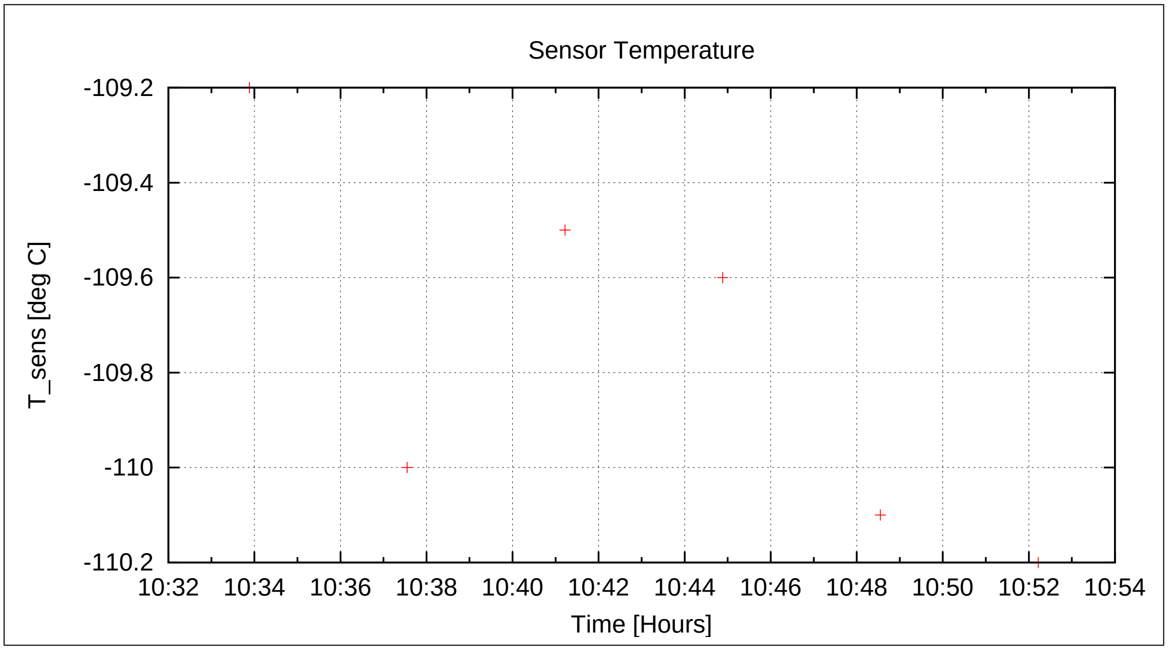

Figure 141: Sensor Temperature

11.7 Special Test on the FGM-IB sensor at T = $-110°$C

At 11:00 noise measurements, meander tests and find phase tests were performed in zerofield conditions. Analysis to be done by Uli Auster.

Cooling cont'd. The new temperature goal was set to T= -140 degC.

11.8 DC-Analysis at T = $-140°$C, Range=R4 (16000 nT), FGM-IB

11.8.1 Calibration on 3 Linear Axes — RANGE R4 — SENSOR IB

Used Files:

CCD File	Configuration File	Remark
21-04-20\12-33-52.CCR	LIN16000XYZ.MAG	

Parameter File: PARAMETER_LIN_R4_IB_21-04-20_12-33-52.CPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Sensor Rotation:

$$\underline{\underline{\rho}} = \begin{pmatrix} -0.996909 & +0.054673 & +0.056424 \\ -0.054982 & -0.998480 & -0.003947 \\ +0.056123 & -0.007038 & +0.998399 \end{pmatrix}$$

Rotation Angles:

$$\begin{aligned} \text{Angle } (X_c, X_m): &\quad \lambda_{\mathrm{x}} = && +175^\circ\ 29'38'' \\ \text{Angle } (Y_c, Y_m): &\quad \mu_{\mathrm{y}} = && +176^\circ\ 50'24'' \\ \text{Angle } (Z_c, Z_m): &\quad \nu_{\mathrm{z}} = && +3^\circ\ 14'33'' \end{aligned}$$

Determinant of Rotation Matrix: 1.00000
Nominal Field Source: SOLARTRON
Fields applied for 24.0 s
Mean Sensor Temperature: -139.9°C
Automatic Coil Correction: used

Earthfield Compensation:	X	=	DYNAMIC
	Y	=	DYNAMIC
	Z	=	DYNAMIC

Offset Treatment:

A polynomial offset trend of order 2 has been fitted and subtracted from the raw data before creating the sensor model.

Mean Coil System Residual + Sensor Offset: $\underline{B}^{or}$ = (18.473 , 7.879 , -4.265) nT

Raw Data Quality:

Standard Deviation of used Raw Data Blocks:

[Values in eng nT]	s_x	s_y	s_z
Minimum	0.054	0.037	0.038
Mean	0.097	0.072	0.077
Maximum	0.269	0.297	0.259

Calibration Parameter:

$$\text{Transfermatrix:} \quad \underline{\underline{\phi}} = \underline{\underline{R}}_{\text{nom}}\, \underline{\underline{\rho}}\, \underline{\underline{\omega}}\, \underline{\underline{\sigma}} \quad = \begin{pmatrix} +0.999372 & -0.054991 & -0.056576 \\ +0.051816 & +1.000327 & +0.003958 \\ +0.072047 & -0.003189 & +1.001075 \end{pmatrix}$$

$$\text{Reduced Transfermatrix:} \quad \tilde{\underline{\underline{\phi}}} = \underline{\underline{\omega}}\, \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.003175 & +0.000000 & +0.000000 \\ -0.003408 & +1.001835 & +0.000000 \\ +0.015747 & +0.003867 & +1.002680 \end{pmatrix}$$

$$\text{Sensitivity:} \quad \underline{\underline{\sigma}} \quad = \begin{pmatrix} +1.003175 & +0.000000 & +0.000000 \\ +0.000000 & +1.001829 & +0.000000 \\ +0.000000 & +0.000000 & +1.002549 \end{pmatrix}$$

$$\text{Misalignment:} \quad \underline{\underline{\omega}} \quad = \begin{pmatrix} +1.000000 & +0.000000 & +0.000000 \\ -0.003397 & +1.000006 & +0.000000 \\ +0.015698 & +0.003860 & +1.000131 \end{pmatrix}$$

Sensor Misalignment Angles:

$$\xi_{\text{x,y}} = +89^\circ\ 48'19''$$
$$\xi_{\text{x,z}} = +90^\circ\ 54'0''$$
$$\xi_{\text{y,z}} = +90^\circ\ 13'27''$$

Model Quality:

Standard Deviation, Maximum and Minimum Error of Calculated Model:

[Values in nT]	X	Y	Z
Standard Deviation	0.352	0.531	0.356
Maximum Error	0.981	1.512	1.434
Minimum Error	-0.590	-0.628	-0.372

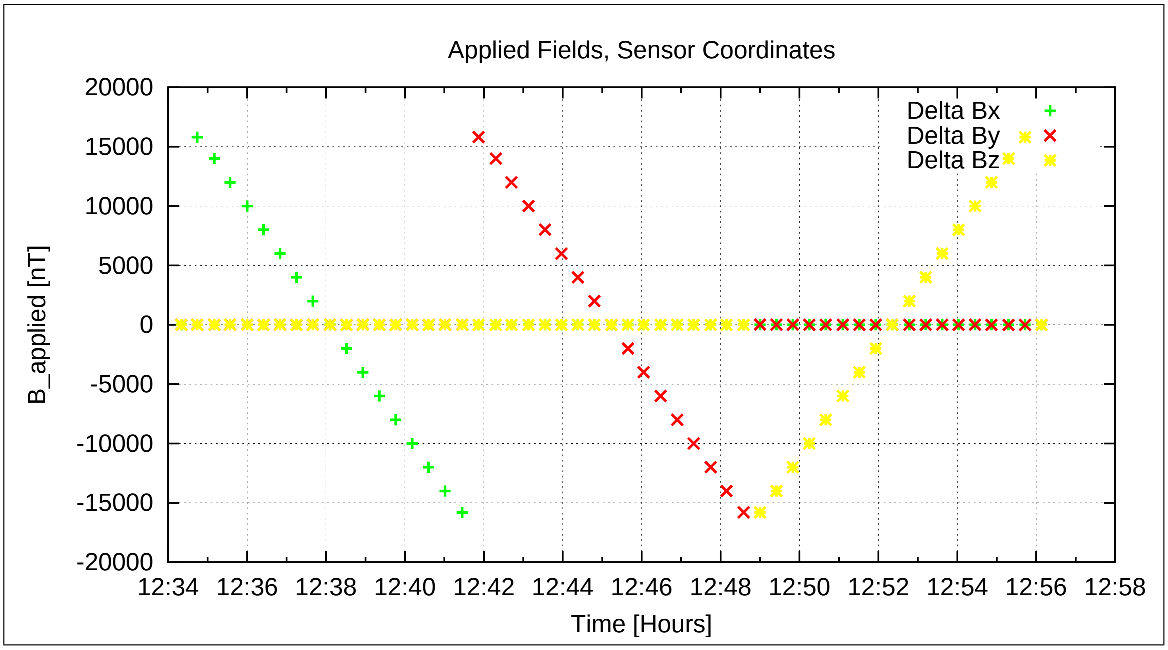

Figure 142: Applied Fields

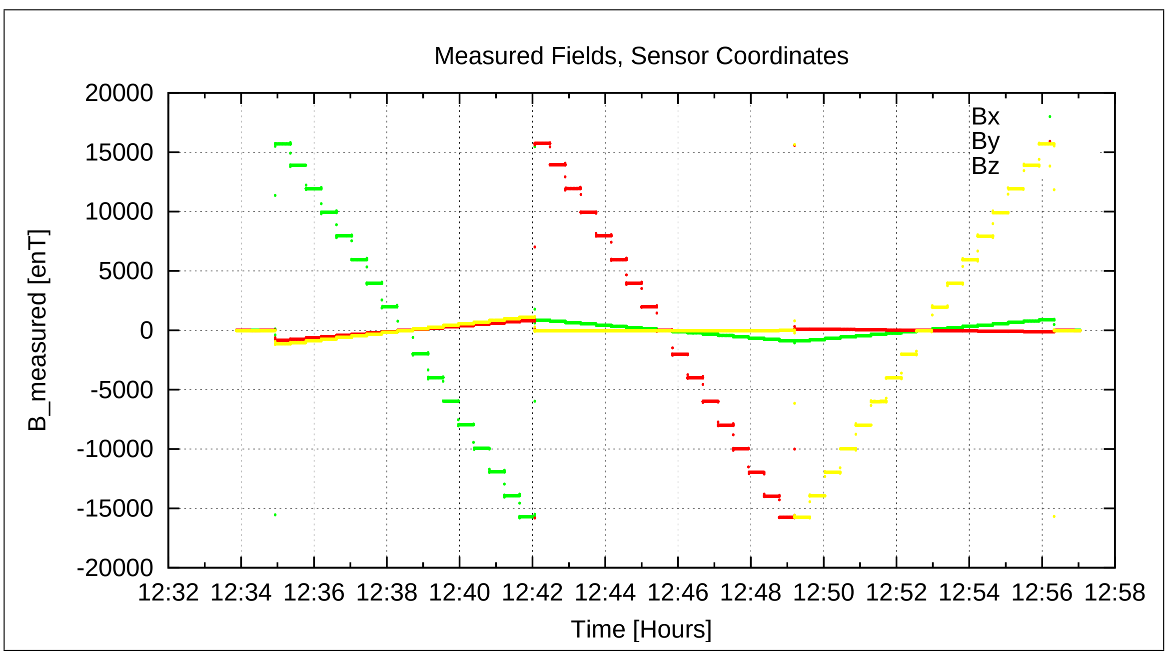

Figure 143: Measured Data

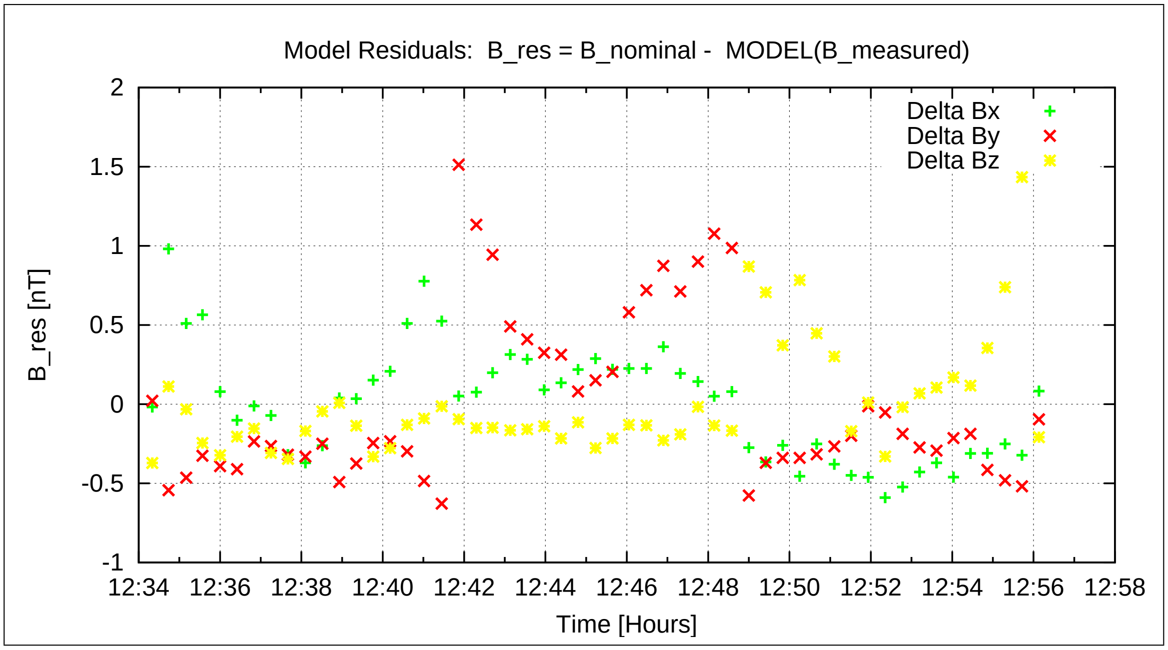

Figure 144: Model Quality - Differences of applied field and modelled measurement data

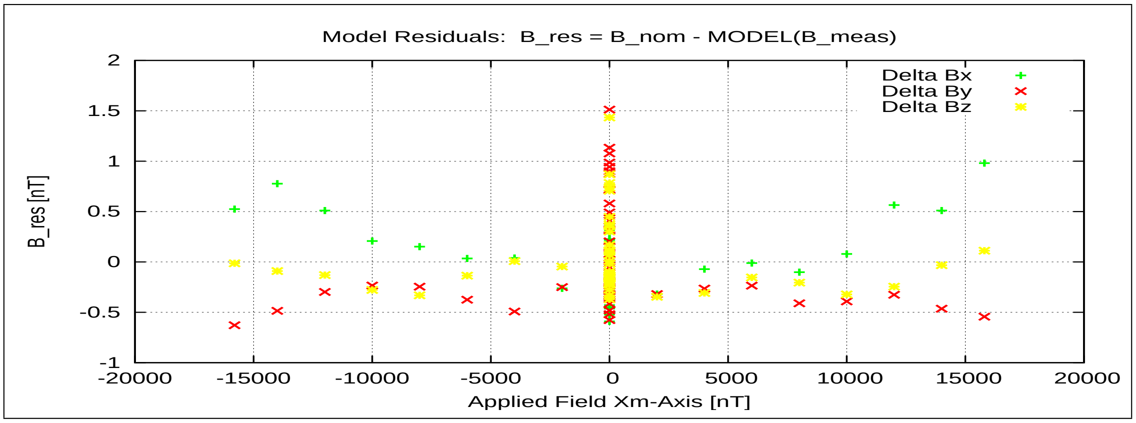

Figure 145: RESIDUALS vs FLD X

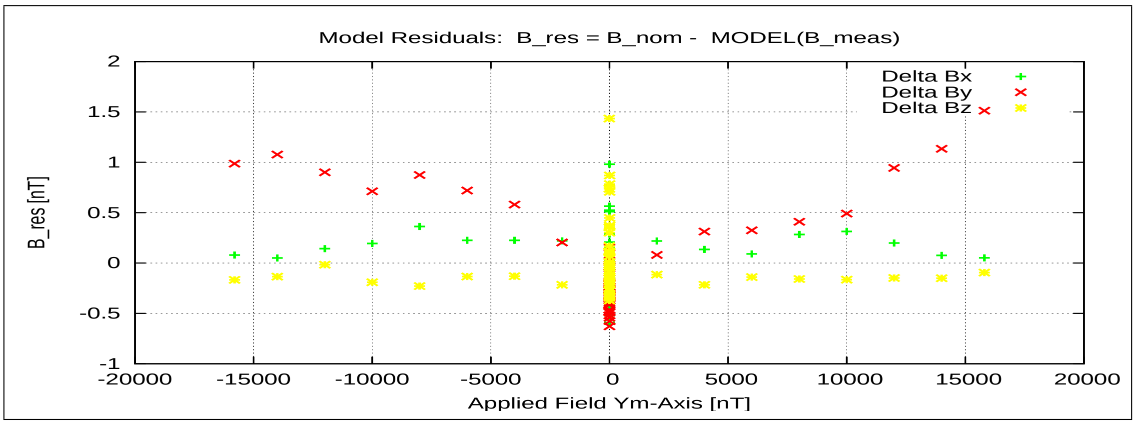

Figure 146: RESIDUALS vs FLD Y

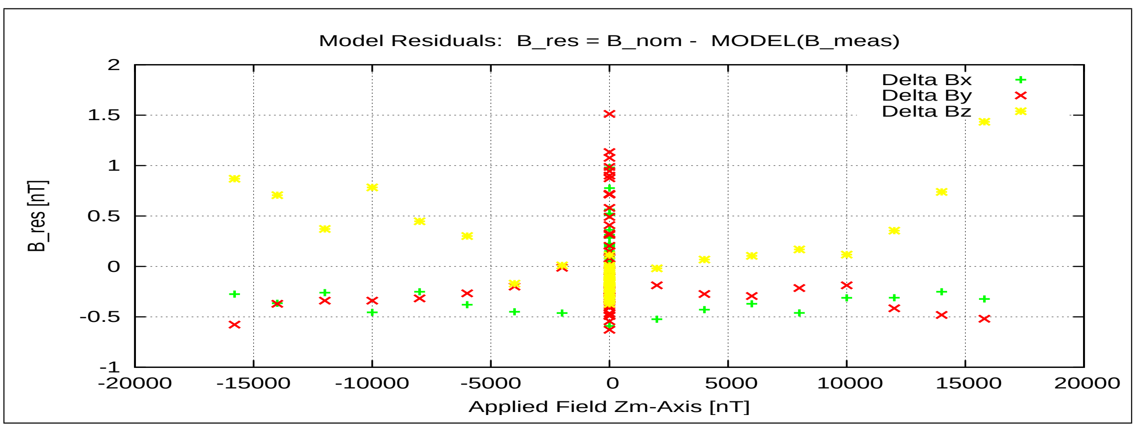

Figure 147: RESIDUALS vs FLD Z

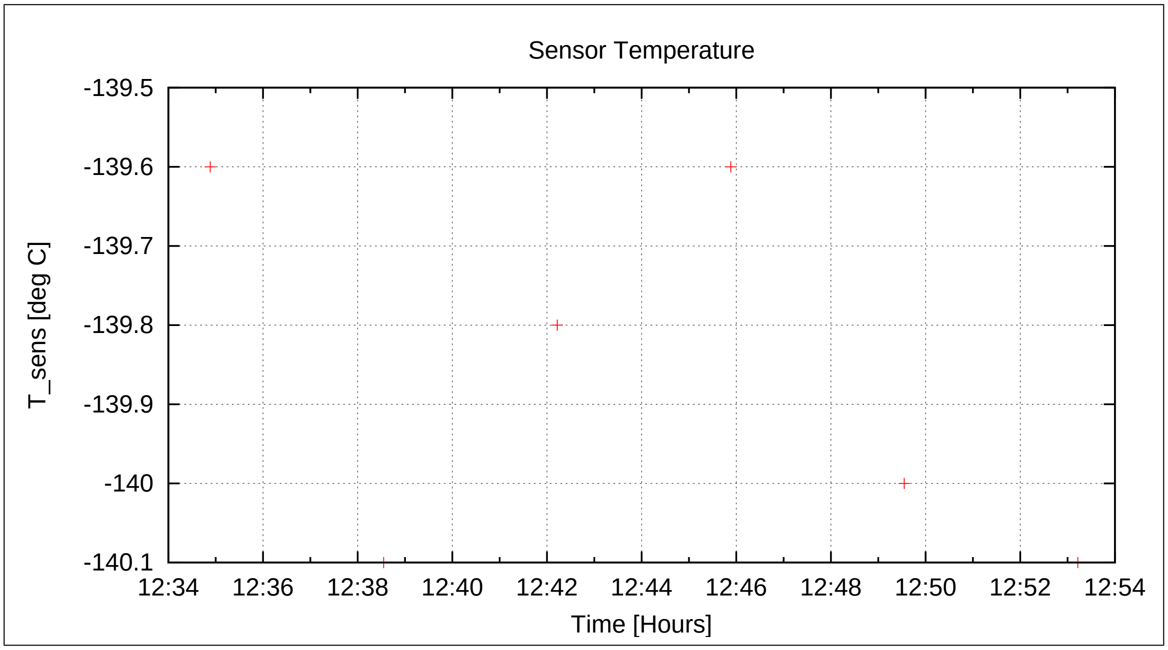

Figure 148: Sensor Temperature

11.9 DC-Analysis at T = $-140°$C, Range=R0 (1000 nT), FGM-IB

11.9.1 OFFSET Calculation — RANGE R0 — SENSOR IB

In this section the instrument offsets and the residual field of the Coil System will be evaluated. Measurements at two orientations for each component act as input for the offset calculation. From the "normal" (0-degrees) and "turned" (180-degrees) orientations the offsets and residual fields can be derived:

$$B_{\text{off}} = \frac{B_{\text{normal}} + B_{\text{turned}}}{2}$$
$$B_{\text{res}} = \frac{B_{\text{normal}} - B_{\text{turned}}}{2}$$

Used Offset Measurements:

CCD File	Configuration File	Remark
21-04-20\13-37-22.CCR	OFFSET_200.MAG	

Parameter File: OFF_PARAMETER_R0_21-04-20-13-37-22_IB.OPF

Calibration Parameter:

Sensor Offsets:

Component	Offset [enT]	Standard deviation [enT]
X	6.77	0.116
Y	10.20	0.118
Z	6.74	0.126

Residual Field of the Coil System:

Component	B_{res} [enT]
X	11.36
Y	-0.40
Z	11.14

Remark: Residual field is given in actual DUT coordinates, not in coil-coordinates!

Cooling cont'd. The temperature goal was remained at T= -140 degC until 17:35. Afterwards to goal was set to T=+20 degC in order to warm up the sensor passively smoothly until Thursday morning (2021-04-22).

The FGM-IB sensor stayed in compensated zerofield conditions over night, as we forgot the start the linearity sequence.

12 Measurements on April 21, 2021

12.1 Data

CCD File	Configuration File	Remark
21-04-21\07-49-05.CCR	LIN16000XYZ.MAG	
21-04-21\08-11-42.CCR	NULL.MAG	
21-04-21\08-48-59.CCR	LIN16000XYZ.MAG	
21-04-21\09-11-36.CCR	NULL.MAG	
21-04-21\09-48-54.CCR	LIN16000XYZ.MAG	
21-04-21\10-11-31.CCR	NULL.MAG	
21-04-21\10-48-48.CCR	LIN16000XYZ.MAG	
21-04-21\11-11-25.CCR	NULL.MAG	
21-04-21\11-48-43.CCR	LIN16000XYZ.MAG	
21-04-21\12-11-20.CCR	NULL.MAG	
21-04-21\12-48-38.CCR	LIN16000XYZ.MAG	
21-04-21\13-11-15.CCR	NULL.MAG	
21-04-21\13-48-33.CCR	LIN16000XYZ.MAG	
21-04-21\14-11-10.CCR	NULL.MAG	
21-04-21\14-48-28.CCR	LIN16000XYZ.MAG	
21-04-21\15-11-05.CCR	NULL.MAG	
21-04-21\15-48-23.CCR	LIN16000XYZ.MAG	
21-04-21\16-11-00.CCR	NULL.MAG	
21-04-21\16-48-18.CCR	LIN16000XYZ.MAG	
21-04-21\17-10-56.CCR	NULL.MAG	
21-04-21\17-48-13.CCR	LIN16000XYZ.MAG	
21-04-21\18-10-50.CCR	NULL.MAG	
21-04-21\18-48-08.CCR	LIN16000XYZ.MAG	
21-04-21\19-10-45.CCR	NULL.MAG	
21-04-21\19-48-02.CCR	LIN16000XYZ.MAG	
21-04-21\20-10-39.CCR	NULL.MAG	
21-04-21\20-47-57.CCR	LIN16000XYZ.MAG	
21-04-21\21-10-33.CCR	NULL.MAG	
21-04-21\21-47-51.CCR	LIN16000XYZ.MAG	
21-04-21\22-10-27.CCR	NULL.MAG	
21-04-21\22-47-44.CCR	LIN16000XYZ.MAG	
21-04-21\23-10-21.CCR	NULL.MAG	
21-04-21\23-47-38.CCR	LIN16000XYZ.MAG	

Inspection of the system at 06:50. All looks ok. Box temperature at -78 degC. Linearity Sequence LIN16000XYZ_NULL.SEQ started at 07:49.

13 Measurements on April 22, 2021

13.1 Data

CCD File	Configuration File	Remark
21-04-22\00-10-14.CCR	NULL.MAG	
21-04-22\00-47-32.CCR	LIN16000XYZ.MAG	
21-04-22\01-10-08.CCR	NULL.MAG	
21-04-22\01-47-25.CCR	LIN16000XYZ.MAG	
21-04-22\02-10-01.CCR	NULL.MAG	
21-04-22\02-47-19.CCR	LIN16000XYZ.MAG	
21-04-22\03-09-55.CCR	NULL.MAG	
21-04-22\03-47-12.CCR	LIN16000XYZ.MAG	
21-04-22\04-09-48.CCR	NULL.MAG	
21-04-22\04-47-06.CCR	LIN16000XYZ.MAG	
21-04-22\05-09-42.CCR	NULL.MAG	
21-04-22\05-46-59.CCR	LIN16000XYZ.MAG	
21-04-22\06-09-36.CCR	NULL.MAG	
21-04-22\06-46-53.CCR	LIN16000XYZ.MAG	
21-04-22\07-09-30.CCR	NULL.MAG	
21-04-22\07-46-48.CCR	LIN16000XYZ.MAG	
21-04-22\08-09-25.CCR	NULL.MAG	

Inspection of the system at 06:50. All looks ok. Box temperature at -7 degC. The system was heated up to T=25 degC actively at 07:10

Linearity Sequence LIN16000XYZ_NULL.SEQ stopped at 08:20. Switching off instrument as well.

13.2 Special Test on the FGM-IB sensor at Room Temperature, Original harness

At 09:30 the FGM-IB sensor was taken out of the box and placed at CoC. It was connected to the electronics with the original harness we used all the days. Noise measurements, meander tests and especially find phase tests were performed in zerofield conditions. Analysis to be done by Uli Auster.

13.3 Special Test on the FGM-IB sensor at Room Temperature, Original harness

After this test the harness was removed and replace by a shorter harness in order to reveal influence of the cable length / capacitor to the phase. Noise measurements, meander tests and especially find phase tests were performed in zerofield conditions. Analysis to be done by Uli Auster.

14 Temperature Analysis of combined measurements: FM-OB

14.1 Linearity & Misalignment, Range=R1, FGM-OB

14.1.1 Temperature Calibration on Linear Axes — RANGE R1 — SENSOR OB

Used Temperature Measurements:

Calibration Parameter File	Remark
PARAMETER_TEMPLIN___21-04-15_20-04-09.CPF	
PARAMETER_TEMPLIN___21-04-16_09-08-53.CPF	
PARAMETER_TEMPLIN___21-04-16_13-23-27.CPF	
PARAMETER_TEMPLIN___21-04-16_15-50-45.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_16-52-07.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_17-52-51.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_18-53-35.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_19-54-20.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_20-55-04.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_21-55-48.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_22-56-33.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-16_23-57-18.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_00-58-02.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_01-58-45.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_02-59-29.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_04-00-13.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_05-00-57.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_06-01-41.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_07-02-25.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_08-03-10.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_09-03-54.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_10-04-39.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_11-05-24.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_12-06-09.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_13-06-54.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_14-07-39.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_15-08-24.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_16-09-10.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_17-09-55.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_18-10-40.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_19-11-25.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_20-12-10.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_21-12-54.CPF	
PARAMETER_TEMPLIN_R1_OB_21-04-17_22-13-38.CPF	

PARAMETER_TEMPLIN_R1_OB_21-04-17_23-14-22.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_00-15-06.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_02-16-33.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_04-18-00.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_06-19-28.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_08-20-57.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_10-22-26.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_12-23-56.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_14-25-27.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_16-26-57.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_18-28-28.CPF
PARAMETER_TEMPLIN_R1_OB_21-04-18_23-32-10.CPF

Thermal Parameter File: THERMAL_PARAMETER_R1_21-04-15-20-04-09_OB.TPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Initial Sensor Rotation:

$$\underline{\underline{R}} = \begin{pmatrix} -0.007403 & +0.053758 & -0.998527 \\ -0.999862 & -0.015262 & +0.006591 \\ -0.014885 & +0.998437 & +0.053864 \end{pmatrix}$$

Initial Rotation Angles:

$$\begin{aligned} \text{Rotation @ X:} \quad \lambda_{\mathrm{x}} &= +90^\circ\ 25'27'' \\ \text{Rotation @ Y:} \quad \mu_{\mathrm{y}} &= +90^\circ\ 52'28'' \\ \text{Rotation @ Z:} \quad \nu_{\mathrm{z}} &= +86^\circ\ 54'44'' \end{aligned}$$

Determinant of Rotation Matrix: 1.0000

Nominal Field Source: FLDS

Automatic Coil correction: used

Used Sensor-Temperature-Channel: T_{59}

Temperature Profile

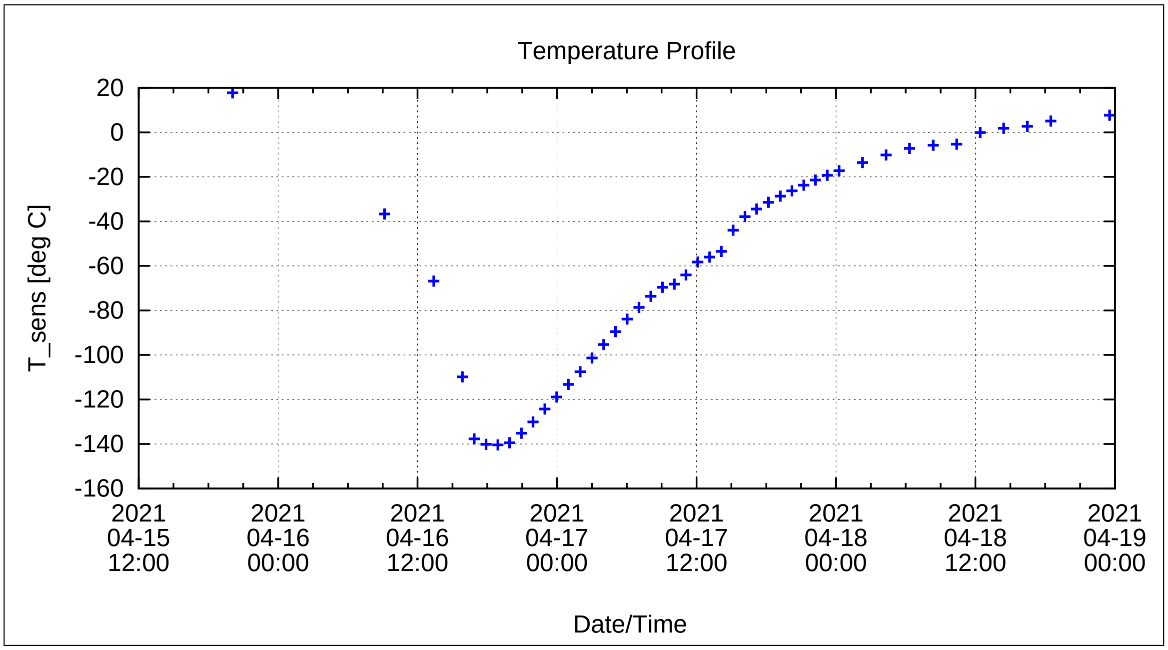

Figure 149: Temperature Profile

Calibration Parameter:

Sensitivity σ_i vs. Temperature:

$$\sigma_i(T) = \sum_{k=0}^{n} \sigma_{k,i} T^k \qquad [1,\ ^\circ\mathrm{C}],\ \mathrm{i=\{x,y,z\}}$$

	$\sigma_{0,i}$	$\sigma_{1,i}$	$\sigma_{2,i}$	$\sigma_{3,i}$	$\sigma_{4,i}$	$\sigma_{5,i}$
σ_x	1.03594E+0	-1.72169E-5				
σ_y	1.08091E+0	-1.54099E-5				
σ_z	1.08127E+0	-1.39002E-5				

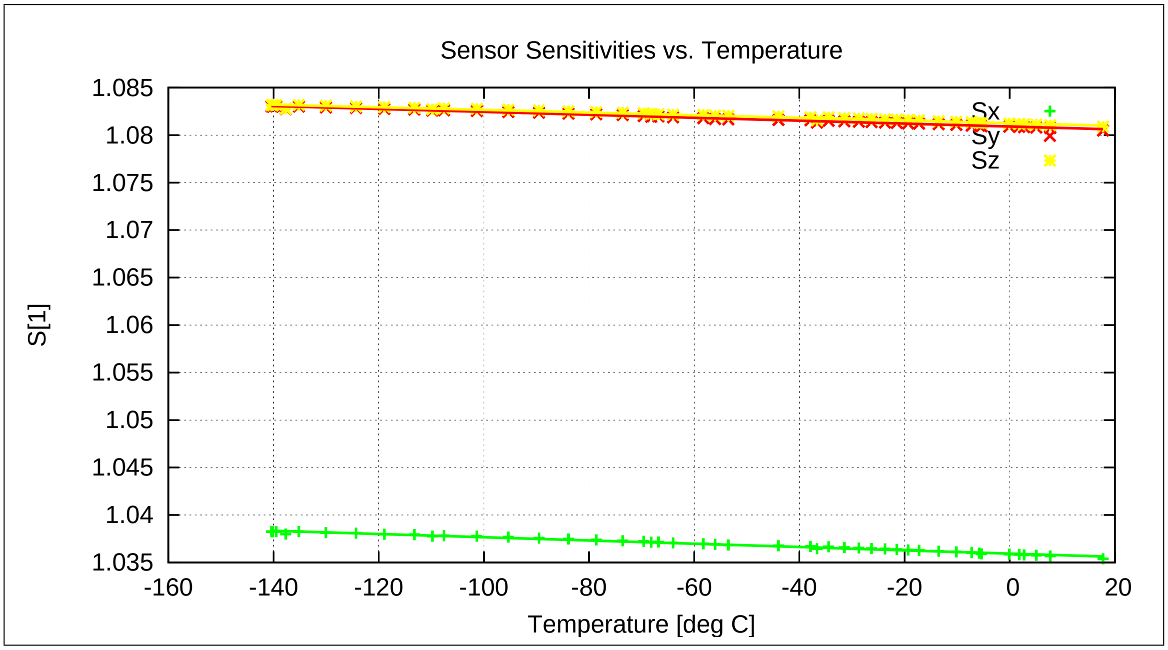

Figure 150: Temperature dependence of Sensitivities

Misalignment Angles ξ_{ij} vs. Temperature:

$$\xi_{ij}(T) = \sum_{k=0}^{n} \xi_{k,ij} T^k \qquad [\text{deg}, \ ^\circ\text{C}], \ \text{ij}= \{\text{xy,xz,yz}\}$$

	$\xi_{0,ij}$	$\xi_{1,ij}$	$\xi_{2,ij}$	$\xi_{3,ij}$	$\xi_{4,ij}$	$\xi_{5,ij}$
ξ_{xy}	9.0374E+1	8.1004E-5				
ξ_{xz}	8.9568E+1	1.0588E-4				
ξ_{yz}	8.9823E+1	1.6671E-4				

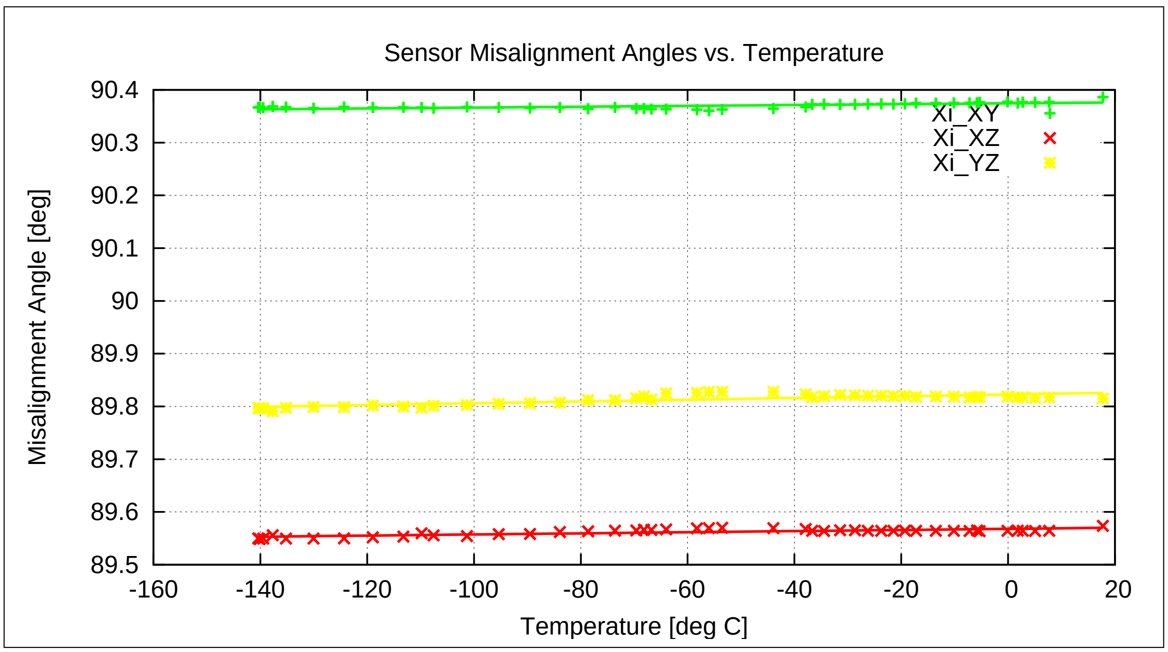

Figure 151: Temperature dependence of Misalignment Angles

Offset & Residual MCF Field $\underline{B}^{or} = \underline{B}^{off} + \underline{B}^{res}$ vs. Temperature:

$$\underline{B}^{or}(T) = \sum_{k=0}^{n} \underline{B}_k^{or} T^k \qquad [\text{enT}, {}^\circ\text{C}]$$

	$\underline{B}_0^{or}$	$\underline{B}_1^{or}$	$\underline{B}_2^{or}$	$\underline{B}_3^{or}$	$\underline{B}_4^{or}$	$\underline{B}_5^{or}$
B_x^{or}	-1.092E+0	-1.065E-2				
B_y^{or}	-1.072E+1	-1.178E-2				
B_z^{or}	5.569E+0	-1.145E-2				

Model Quality:

Minimum and maximum errors of the calculated Model vs. Temperature:

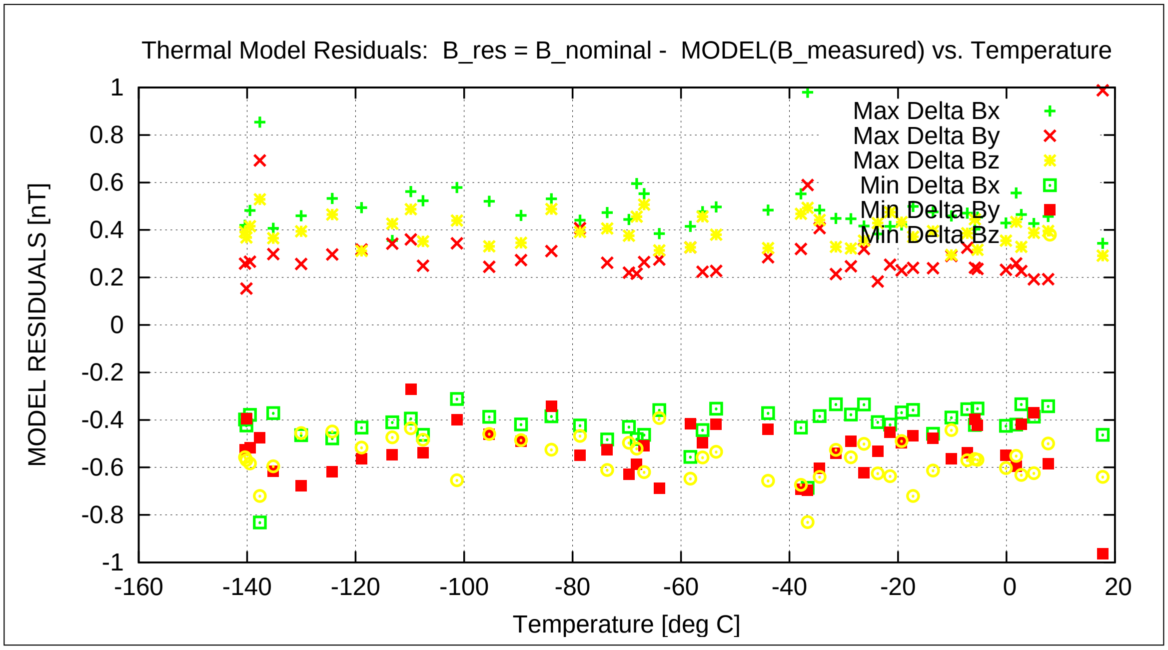

Figure 152: Residuals of Thermal Model

Sensor Rotation during Test:

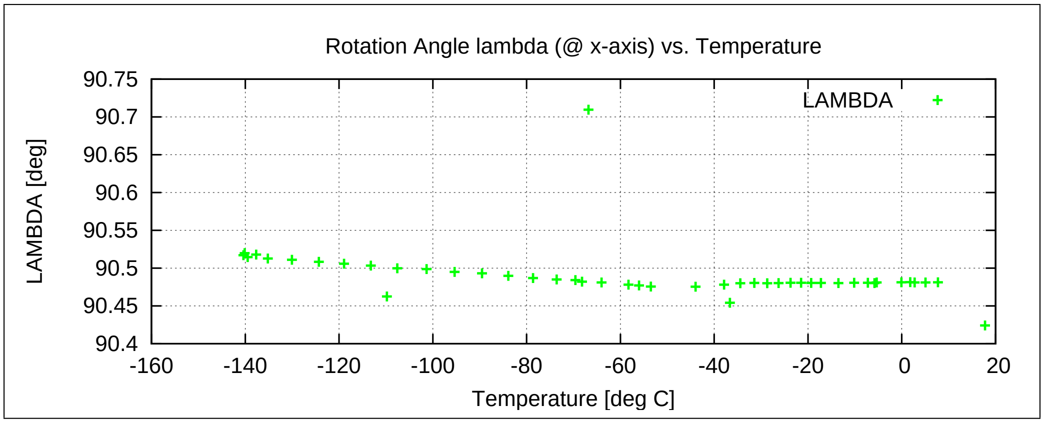

Figure 153: Rotation @ X-Axis

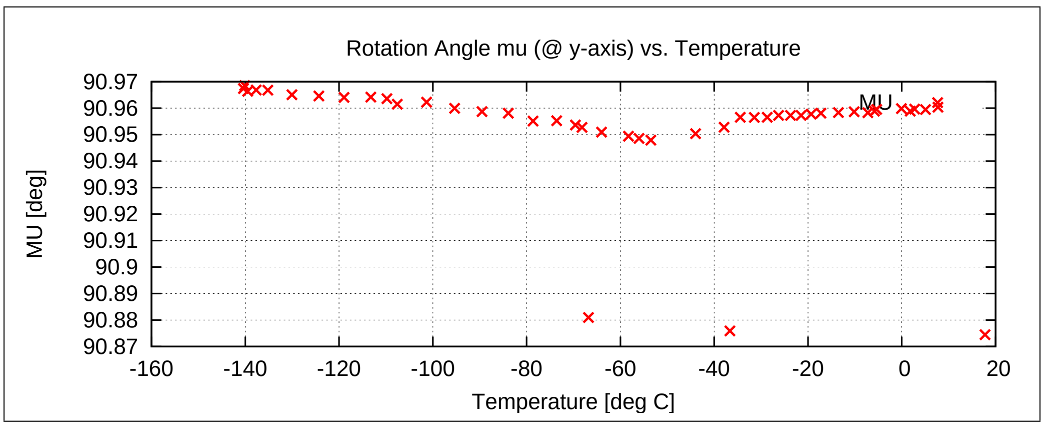

Figure 154: Rotation @ Y-Axis

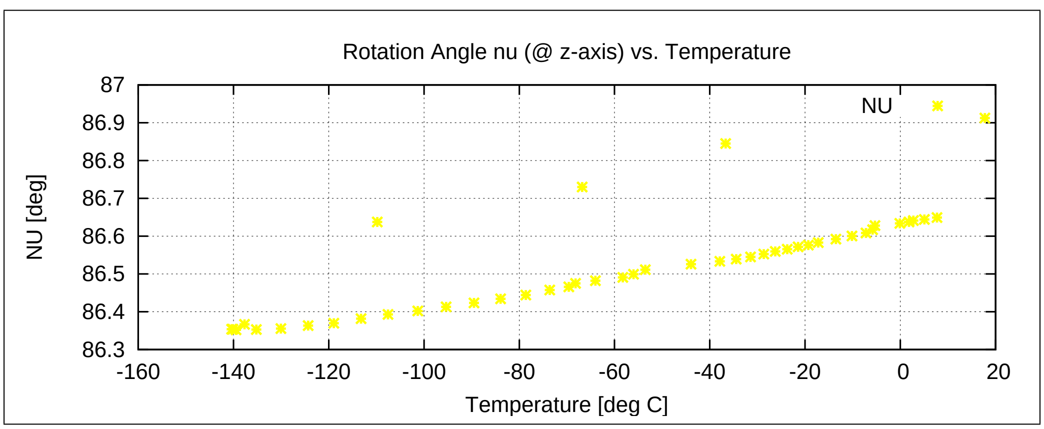

Figure 155: Rotation @ Z-Axis

14.1.2 Temperature Calibration of the Sensor Offset — RANGE R1 — SENSOR OB

Used Temperature Measurements:

Calibration Parameter File	Remark
OFF_PARAMETER__21-04-15-19-33-14_.OPF	
OFF_PARAMETER__21-04-16-09-37-17_.OPF	
OFF_PARAMETER__21-04-16-13-48-45_.OPF	
OFF_PARAMETER__21-04-16-16-17-42_.OPF	
OFF_PARAMETER__21-04-19-08-03-24_.OPF	

Thermal Parameter File: THERMAL_OFF_PARAMETER_R1_21-04-15-19-33-14_OB.TOF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Sensor Temperature Channel: T_{59}

Coil System Temperature Channel: T_{29}

Temperature Profile

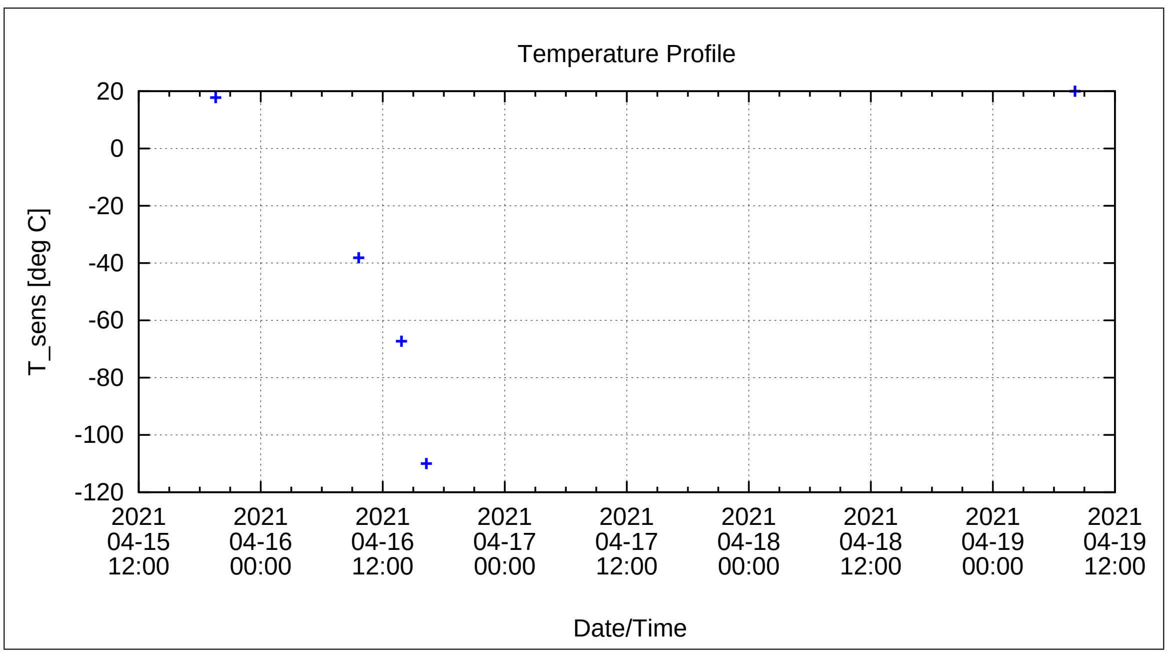

Figure 156: Temperature Profile

Calibration Parameter:

Sensor Offset $\underline{B}_{\mathrm{off}}$ vs. Temperature:

$$B_{\mathrm{off,i}}(T) = \sum_{k=0}^{n} B_{\mathrm{off,k,i}} T^k \qquad [\mathrm{enT},\ ^\circ\mathrm{C}],\ \mathrm{i}=\{\mathrm{x,y,z}\}$$

	$B_{\mathrm{off,0,i}}$	$B_{\mathrm{off,1,i}}$	$B_{\mathrm{off,2,i}}$	$B_{\mathrm{off,3,i}}$	$B_{\mathrm{off,4,i}}$	$B_{\mathrm{off,5,i}}$
$B_{\mathrm{off,x}}$	-1.46286E+0	-2.99227E-2				
$B_{\mathrm{off,y}}$	-5.53053E+0	-9.52054E-4				
$B_{\mathrm{off,z}}$	-3.06769E+0	-2.88930E-2				

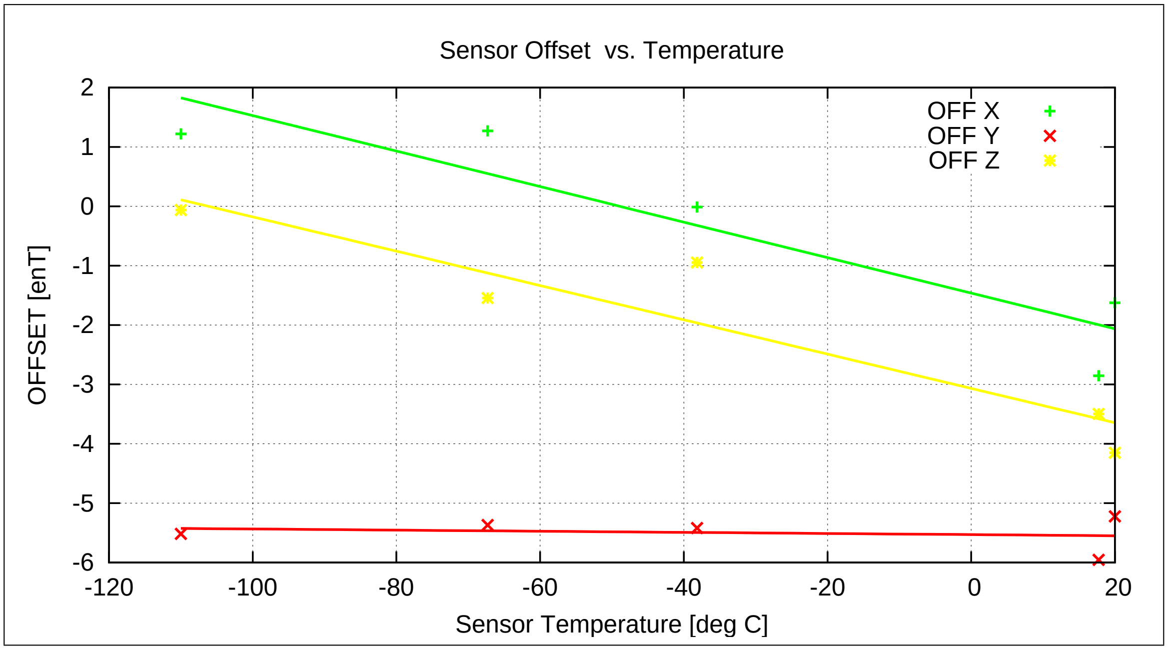

Figure 157: Temperature dependence of Sensor Offsets

Coil System Temperature

The following graph shows the mean Coil System temperature during the complete thermal cycle for the offset measurements.

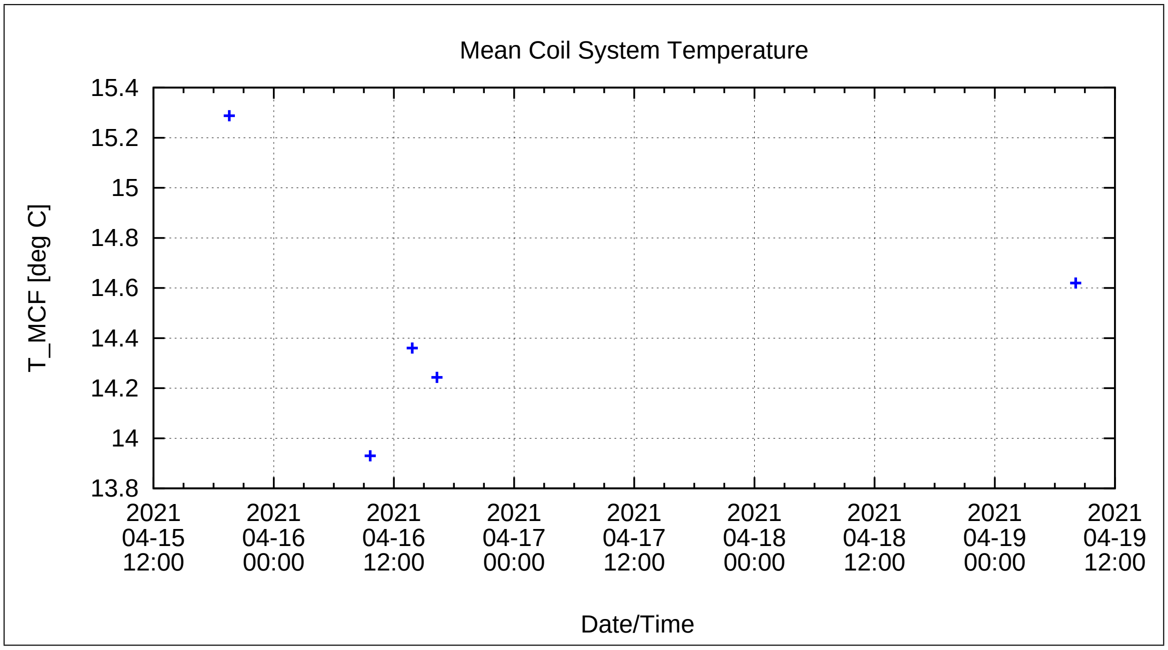

Figure 158: Coil System Temperature during Thermal Measurements

Coil System Residual Field:

The following graph shows the three components of the Coil System Residual field during the whole measurement. Axes designators are related to the actual DUT coordinate system and NOT to Coil System coordinates.

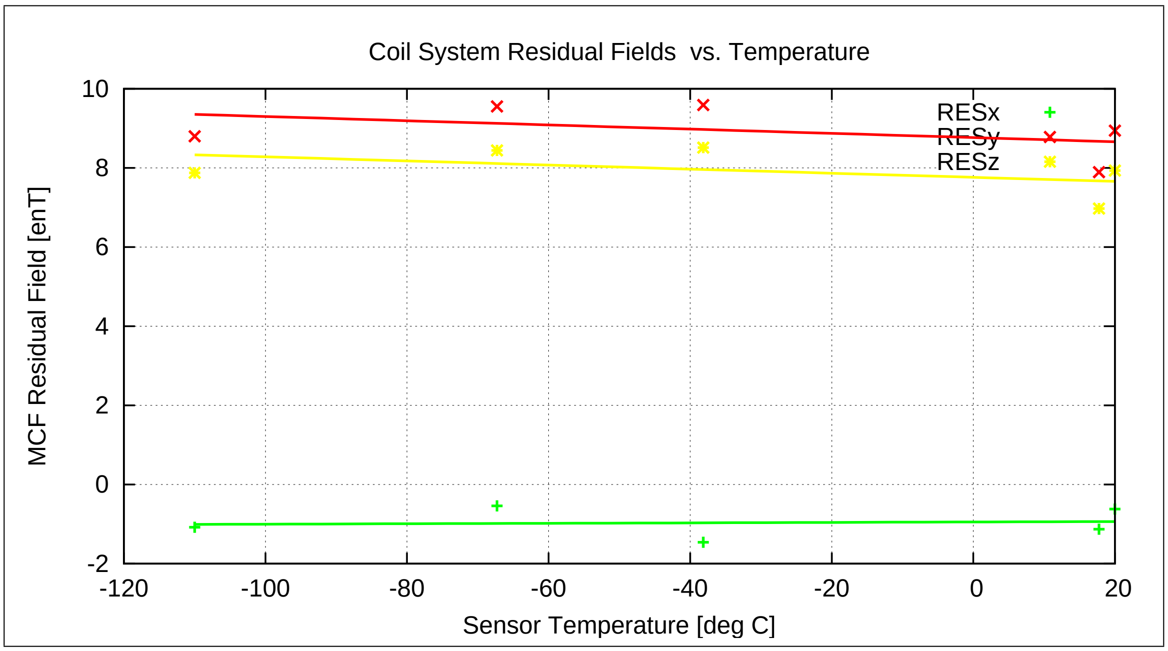

Figure 159: Coil System Residual Fields during Thermal Measurements

14.2 Linearity & Misalignment, Range=R0, FGM-OB

14.2.1 Temperature Calibration on Linear Axes — RANGE R0 — SENSOR OB

Used Temperature Measurements:

Calibration Parameter File	Remark
PARAMETER_TEMPLIN___21-04-15_19-44-03.CPF	
PARAMETER_TEMPLIN___21-04-16_08-11-05.CPF	
PARAMETER_TEMPLIN___21-04-16_13-00-07.CPF	
PARAMETER_TEMPLIN___21-04-16_14-07-15.CPF	

Thermal Parameter File: THERMAL_PARAMETER_R0_21-04-15-19-44-03_OB.TPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Calculated Initial Sensor Rotation:

$$\underline{\underline{R}} = \begin{pmatrix} -0.007521 & +0.053763 & -0.998525 \\ -0.999861 & -0.015247 & +0.006710 \\ -0.014863 & +0.998437 & +0.053870 \end{pmatrix}$$

Initial Rotation Angles:

$$\begin{aligned} \text{Rotation @ X:} \quad & \lambda_{\mathrm{x}} = && +90^\circ\ 25'51'' \\ \text{Rotation @ Y:} \quad & \mu_{\mathrm{y}} = && +90^\circ\ 52'25'' \\ \text{Rotation @ Z:} \quad & \nu_{\mathrm{z}} = && +86^\circ\ 54'43'' \end{aligned}$$

Determinant of Rotation Matrix: 1.0000

Nominal Field Source: FLDS

Automatic Coil correction: used

Used Sensor-Temperature-Channel: T_{59}

Temperature Profile

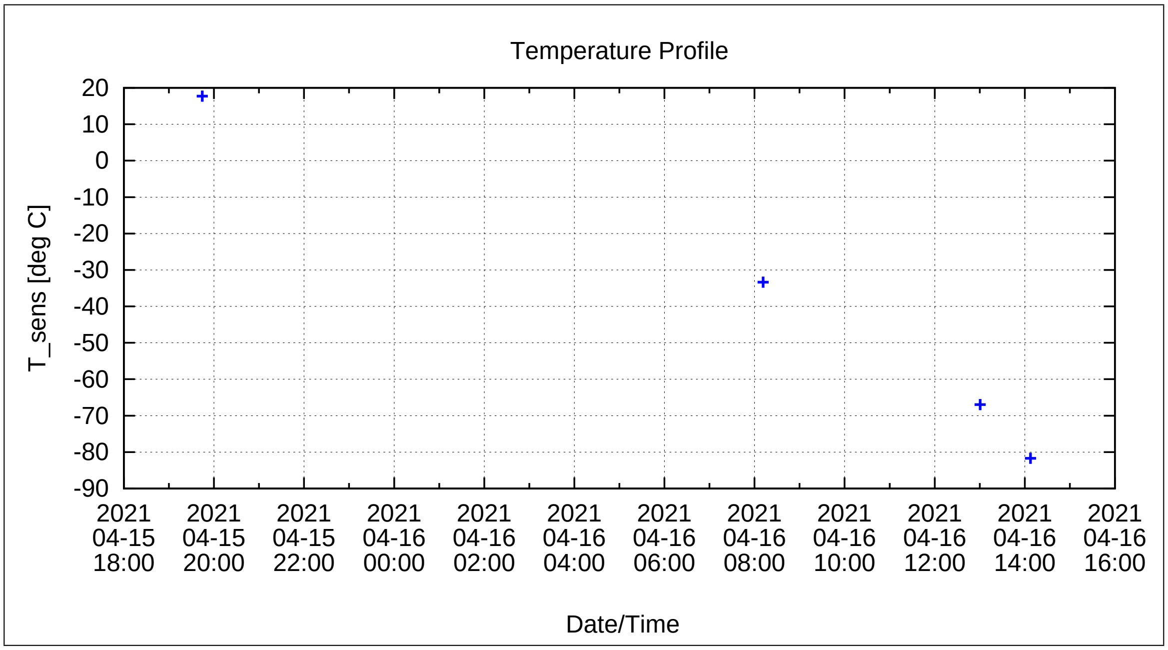

Figure 160: Temperature Profile

Calibration Parameter:

Sensitivity σ_i vs. Temperature:

$$\sigma_i(T) = \sum_{k=0}^{n} \sigma_{k,i} T^k \qquad [1,\ ^\circ\mathrm{C}],\ \mathrm{i}=\{\mathrm{x,y,z}\}$$

	$\sigma_{0,i}$	$\sigma_{1,i}$	$\sigma_{2,i}$	$\sigma_{3,i}$	$\sigma_{4,i}$	$\sigma_{5,i}$
σ_x	1.15318E+0	-5.29975E-5				
σ_y	1.20772E+0	-5.09122E-5				
σ_z	1.20191E+0	-4.80310E-5				

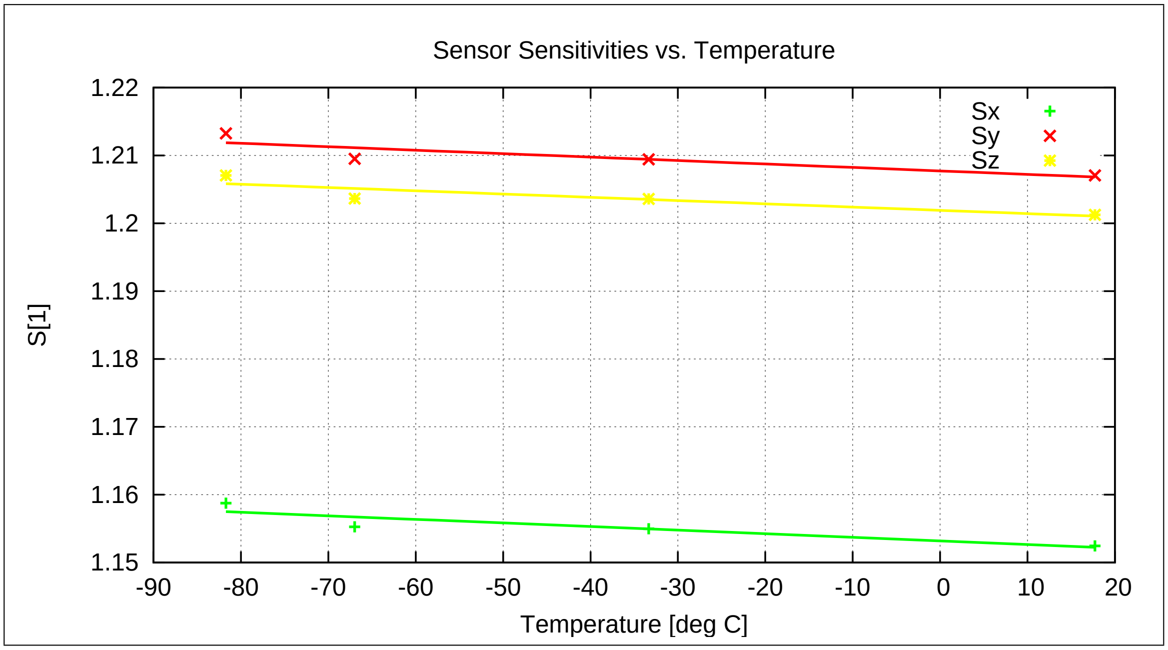

Figure 161: Temperature dependence of Sensitivities

Misalignment Angles ξ_{ij} vs. Temperature:

$$\xi_{ij}(T) = \sum_{k=0}^{n} \xi_{k,ij} T^k \qquad [\text{deg}, \ ^\circ\text{C}], \ \text{ij}= \{\text{xy,xz,yz}\}$$

	$\xi_{0,ij}$	$\xi_{1,ij}$	$\xi_{2,ij}$	$\xi_{3,ij}$	$\xi_{4,ij}$	$\xi_{5,ij}$
ξ_{xy}	9.0379E+1	1.5389E-4				
ξ_{xz}	8.9569E+1	-3.3891E-6				
ξ_{yz}	8.9815E+1	3.3588E-5				

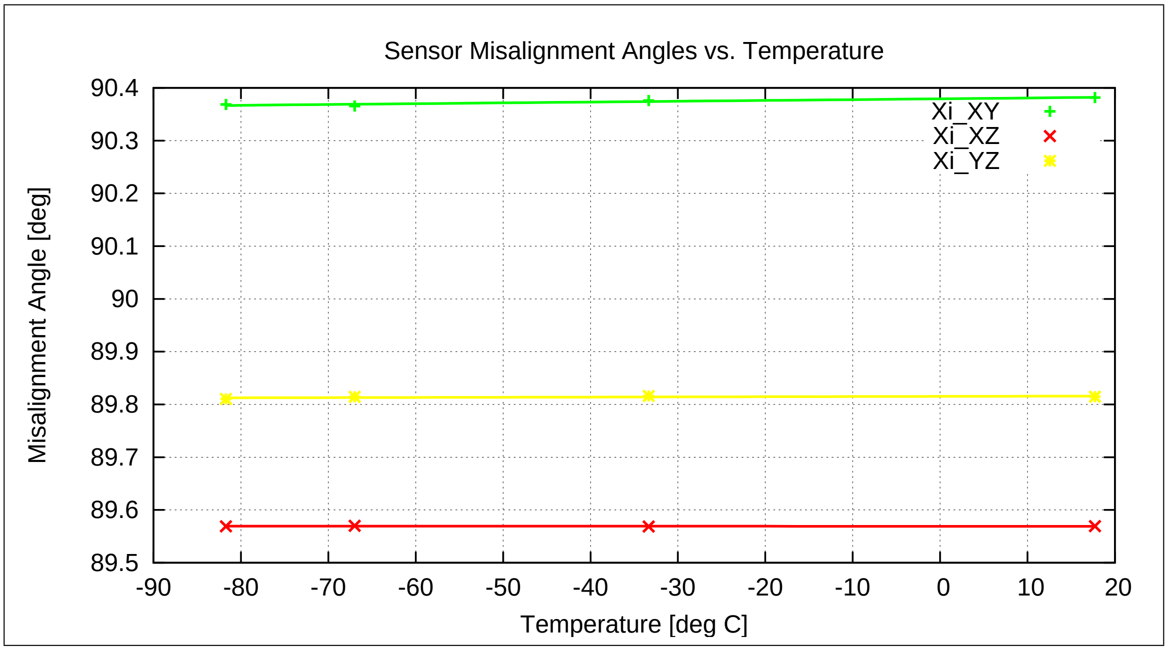

Figure 162: Temperature dependence of Misalignment Angles

Offset & Residual MCF Field $\underline{B}^{or} = \underline{B}^{off} + \underline{B}^{res}$ vs. Temperature:

$$\underline{B}^{or}(T) = \sum_{k=0}^{n} \underline{B}_k^{or} T^k \qquad [\text{enT, °C}]$$

	$\underline{B}_0^{or}$	$\underline{B}_1^{or}$	$\underline{B}_2^{or}$	$\underline{B}_3^{or}$	$\underline{B}_4^{or}$	$\underline{B}_5^{or}$
B_x^{or}	7.625E+0	-3.413E-2				
B_y^{or}	-2.027E+1	1.558E-3				
B_z^{or}	-1.144E+0	-3.767E-2				

Model Quality:

Minimum and maximum errors of the calculated Model vs. Temperature:

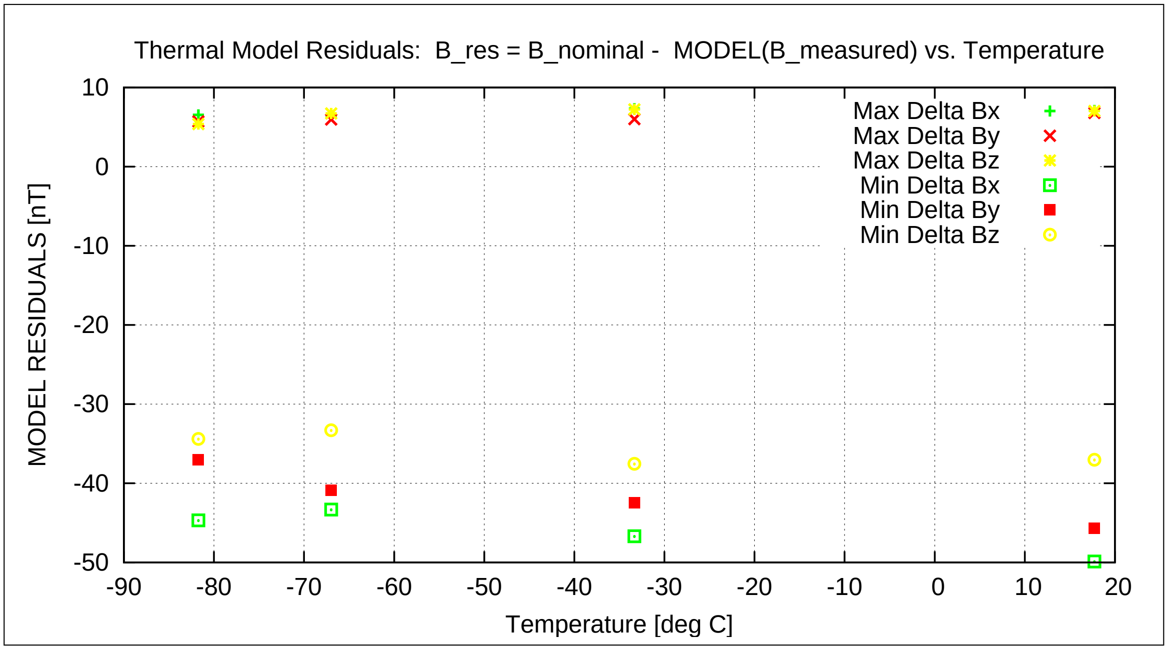

Figure 163: Residuals of Thermal Model

Sensor Rotation during Test:

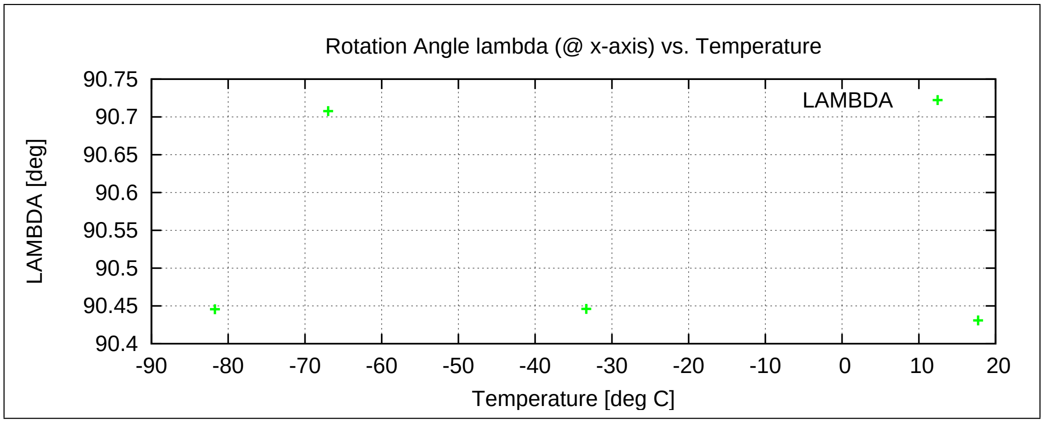

Figure 164: Rotation @ X-Axis

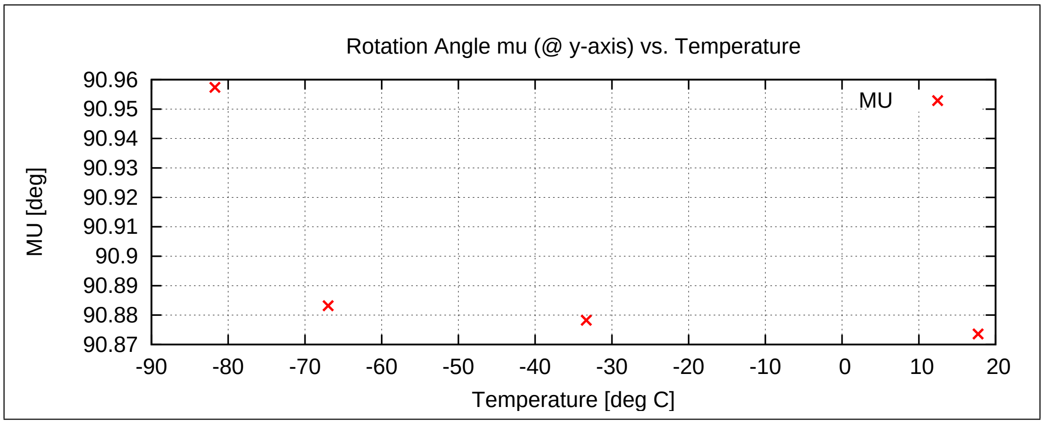

Figure 165: Rotation @ Y-Axis

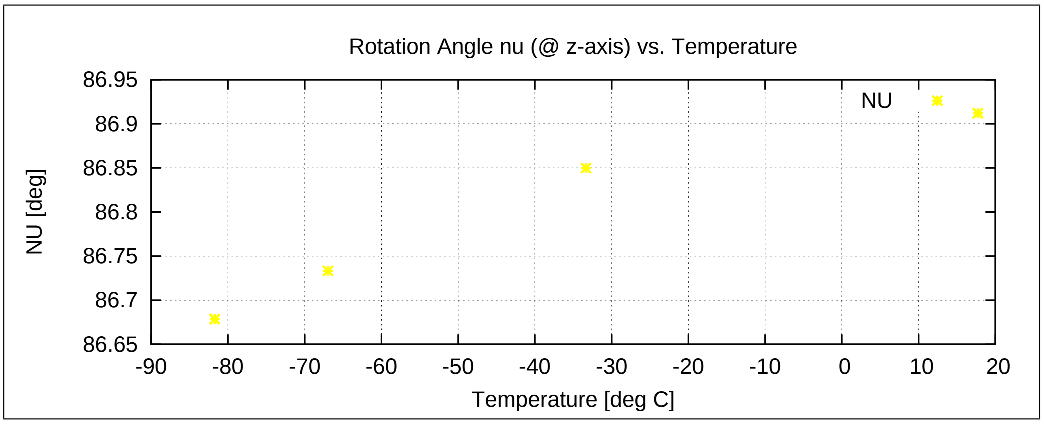

Figure 166: Rotation @ Z-Axis

14.2.2 Temperature Calibration of the Sensor Offset — RANGE R0 — SENSOR OB

Used Temperature Measurements:

Calibration Parameter File	Remark
OFF_PARAMETER__21-04-15-19-23-48_.OPF	
OFF_PARAMETER__21-04-16-13-57-59_.OPF	
OFF_PARAMETER__21-04-16-16-24-15_.OPF	
OFF_PARAMETER__21-04-19-07-52-29_.OPF	

Thermal Parameter File: THERMAL_OFF_PARAMETER_R0_21-04-15-19-23-48_OB.TOF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \\ -1.000000 & +0.000000 & +0.000000 \end{pmatrix}$$

Sensor Temperature Channel: T_{59}

Coil System Temperature Channel: T_{29}

Temperature Profile

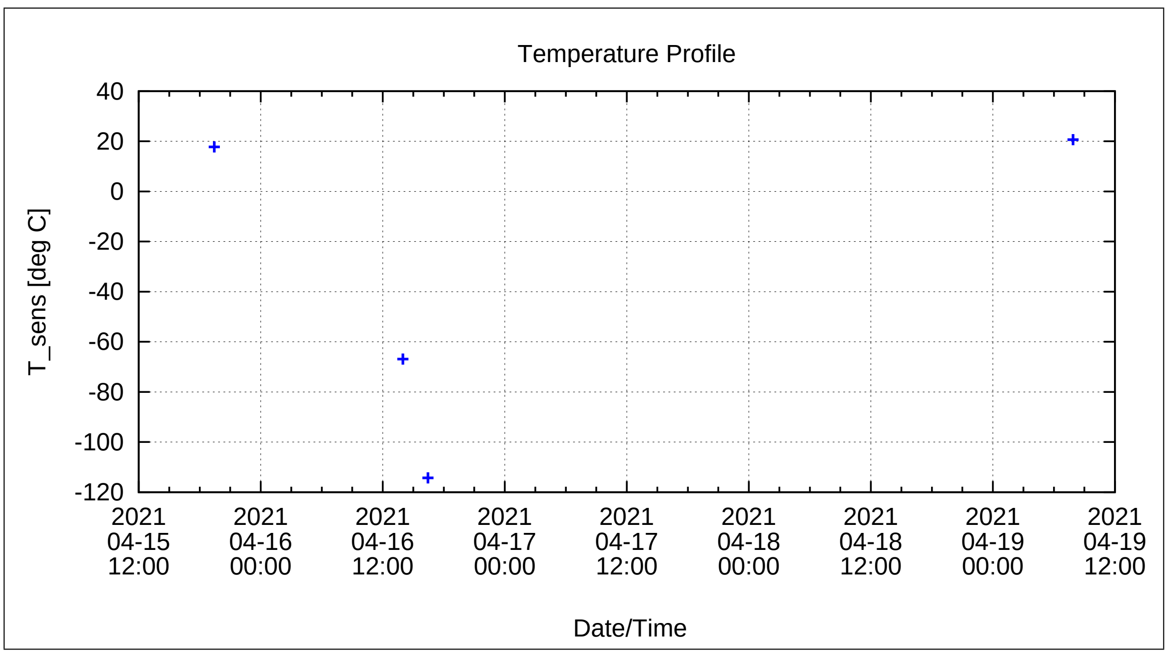

Figure 167: Temperature Profile

Calibration Parameter:

Sensor Offset $\underline{B}_{\text{off}}$ vs. Temperature:

$$B_{\text{off,i}}(T) = \sum_{k=0}^{n} B_{\text{off,k,i}} T^k \qquad [\text{enT},\ ^\circ\text{C}],\ \text{i}=\{\text{x,y,z}\}$$

	$B_{\text{off,0,i}}$	$B_{\text{off,1,i}}$	$B_{\text{off,2,i}}$	$B_{\text{off,3,i}}$	$B_{\text{off,4,i}}$	$B_{\text{off,5,i}}$
$B_{\text{off,x}}$	8.86497E+0	-4.57430E-2				
$B_{\text{off,y}}$	-1.47158E+1	-3.40922E-3				
$B_{\text{off,z}}$	-8.24248E+0	-5.34787E-2				

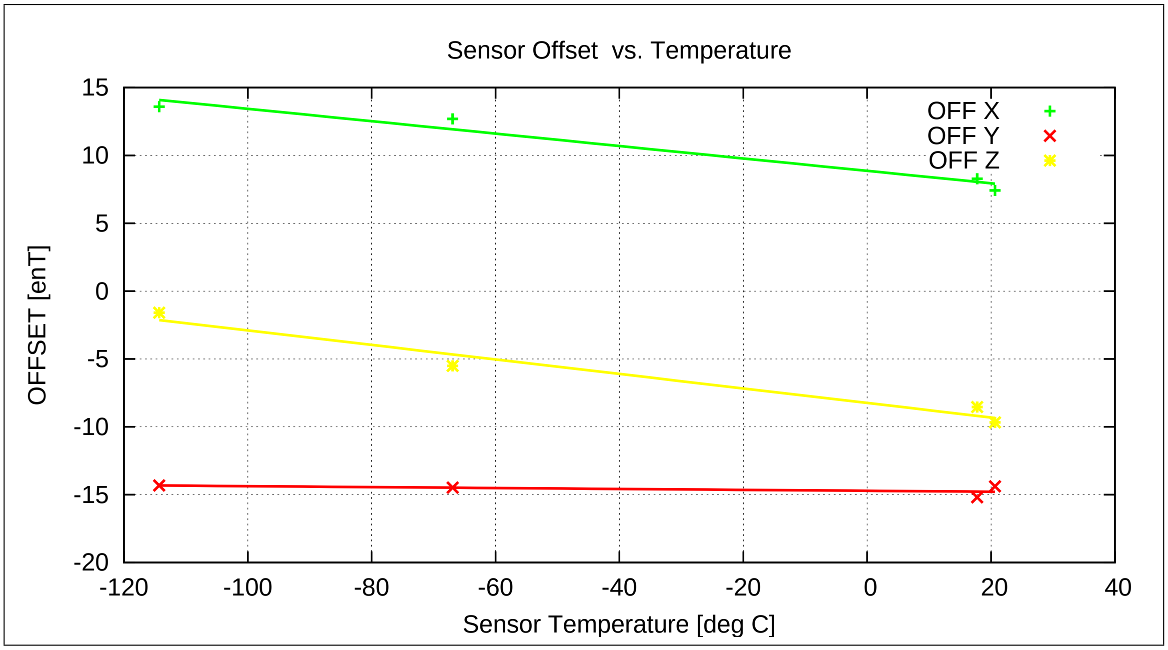

Figure 168: Temperature dependence of Sensor Offsets

Coil System Temperature

The following graph shows the mean Coil System temperature during the complete thermal cycle for the offset measurements.

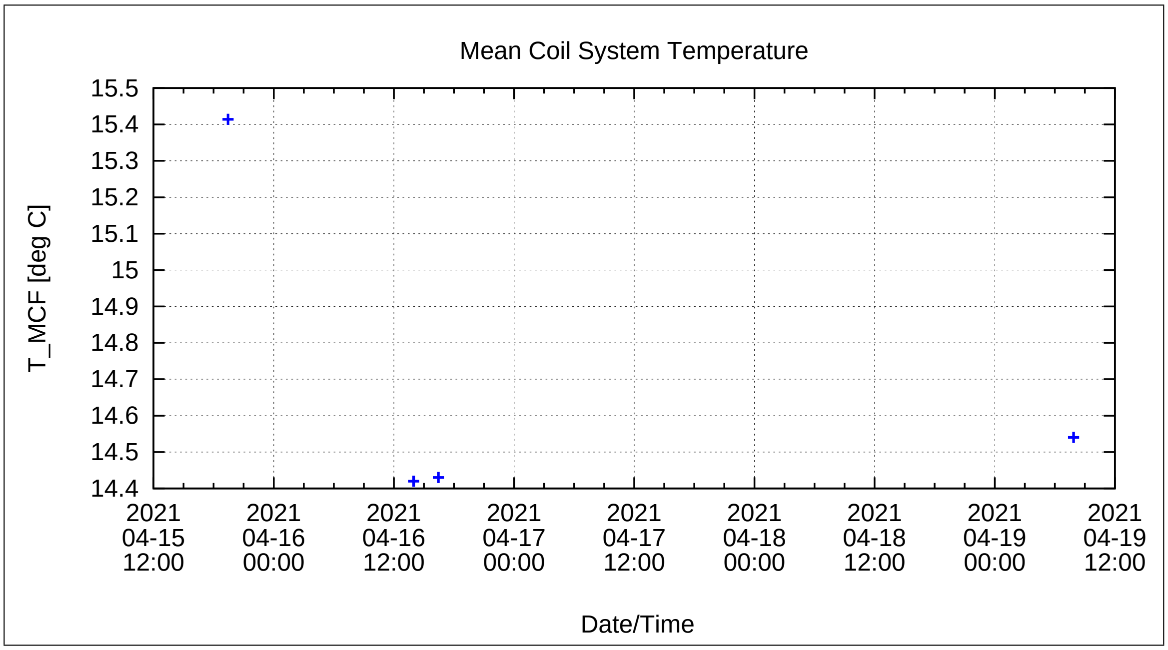

Figure 169: Coil System Temperature during Thermal Measurements

Coil System Residual Field:

The following graph shows the three components of the Coil System Residual field during the whole measurement. Axes designators are related to the actual DUT coordinate system and NOT to Coil System coordinates.

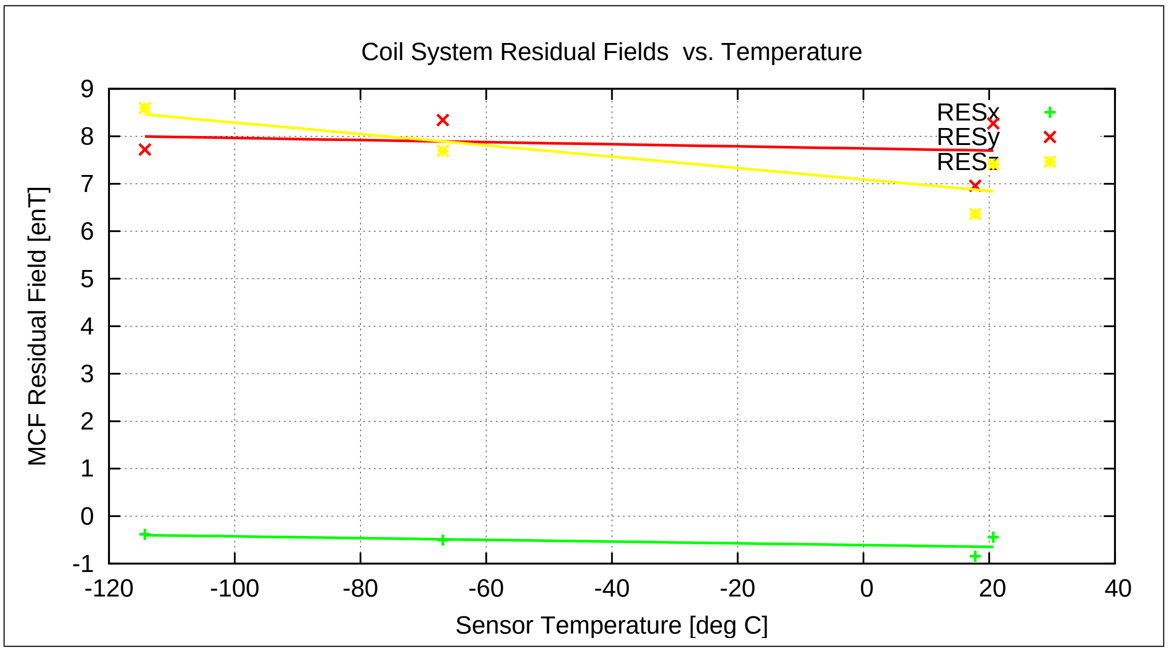

Figure 170: Coil System Residual Fields during Thermal Measurements

15 Temperature Analysis of combined measurements: FM-IB

15.1 Linearity & Misalignment, Range=R4, FGM-IB

15.1.1 Temperature Calibration on Linear Axes — RANGE R4 — SENSOR IB

Used Temperature Measurements:

Calibration Parameter File	Remark
PARAMETER_TEMPLIN_R4_IB_21-04-19_11-34-06.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_13-45-25.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_16-14-58.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_17-14-53.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_18-14-48.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_19-14-44.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_20-14-38.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_21-14-32.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-19_23-14-20.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_00-14-14.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_01-14-07.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_02-14-00.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_03-13-54.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_04-13-48.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_06-13-35.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_10-31-18.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-20_12-33-52.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_08-48-59.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_09-48-54.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_10-48-48.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_13-48-33.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_14-48-28.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_15-48-23.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_16-48-18.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_17-48-13.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_18-48-08.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_20-47-57.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_21-47-51.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-21_22-47-44.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-22_01-47-25.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-22_02-47-19.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-22_03-47-12.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-22_04-47-06.CPF	
PARAMETER_TEMPLIN_R4_IB_21-04-22_05-46-59.CPF	

PARAMETER_TEMPLIN_R4_IB_21-04-22_06-46-53.CPF

Thermal Parameter File: THERMAL_PARAMETER_R4_21-04-19-11-34-06_IB.TPF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Calculated Initial Sensor Rotation:

$$\underline{\underline{R}} = \begin{pmatrix} -0.999428 & -0.029610 & +0.016313 \\ +0.029420 & -0.999498 & -0.011778 \\ +0.016653 & -0.011291 & +0.999798 \end{pmatrix}$$

Initial Rotation Angles:

$$\begin{aligned} \text{Rotation @ X:} \quad \lambda_{\mathrm{x}} &= +178^\circ\ 3'46'' \\ \text{Rotation @ Y:} \quad \mu_{\mathrm{y}} &= +178^\circ\ 11'2'' \\ \text{Rotation @ Z:} \quad \nu_{\mathrm{z}} &= +1^\circ\ 9'10'' \end{aligned}$$

Determinant of Rotation Matrix: 1.0000

Nominal Field Source: SOLARTRON

Automatic Coil correction: used

Used Sensor-Temperature-Channel: T_{59}

Temperature Profile

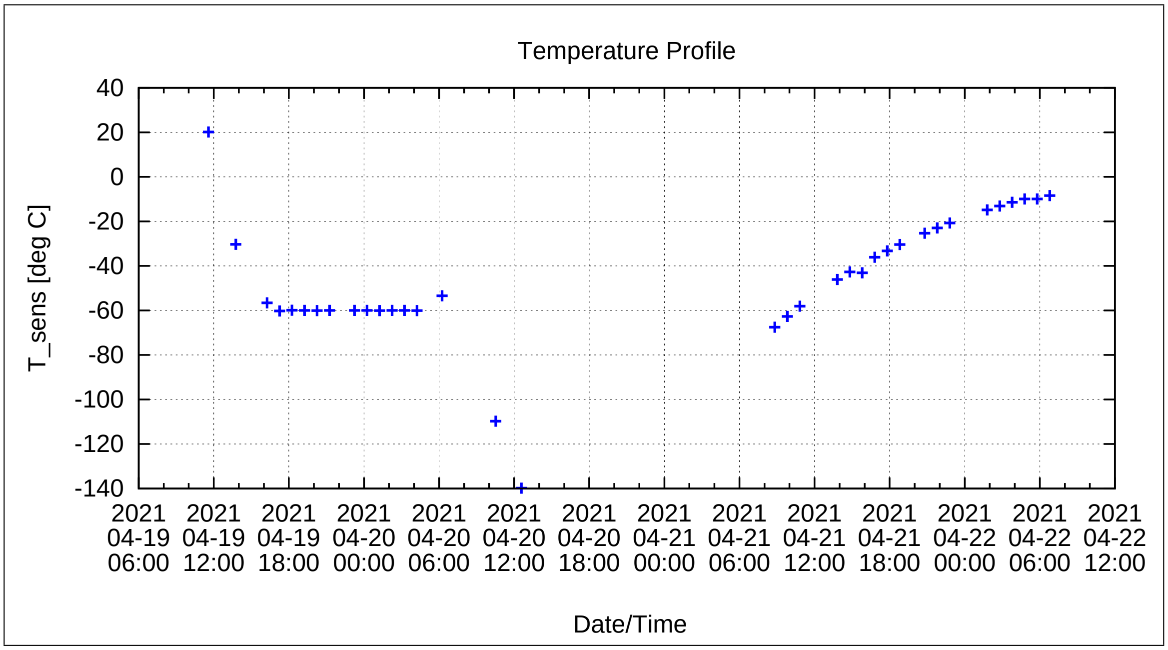

Figure 171: Temperature Profile

Calibration Parameter:

Sensitivity σ_i vs. Temperature:

$$\sigma_i(T) = \sum_{k=0}^{n} \sigma_{k,i} T^k \qquad [1,\ ^\circ\mathrm{C}],\ \mathrm{i=\{x,y,z\}}$$

	$\sigma_{0,i}$	$\sigma_{1,i}$	$\sigma_{2,i}$	$\sigma_{3,i}$	$\sigma_{4,i}$	$\sigma_{5,i}$
σ_x	1.00093E+0	-1.67629E-5				
σ_y	1.00021E+0	-1.19916E-5				
σ_z	1.00036E+0	-1.57435E-5				

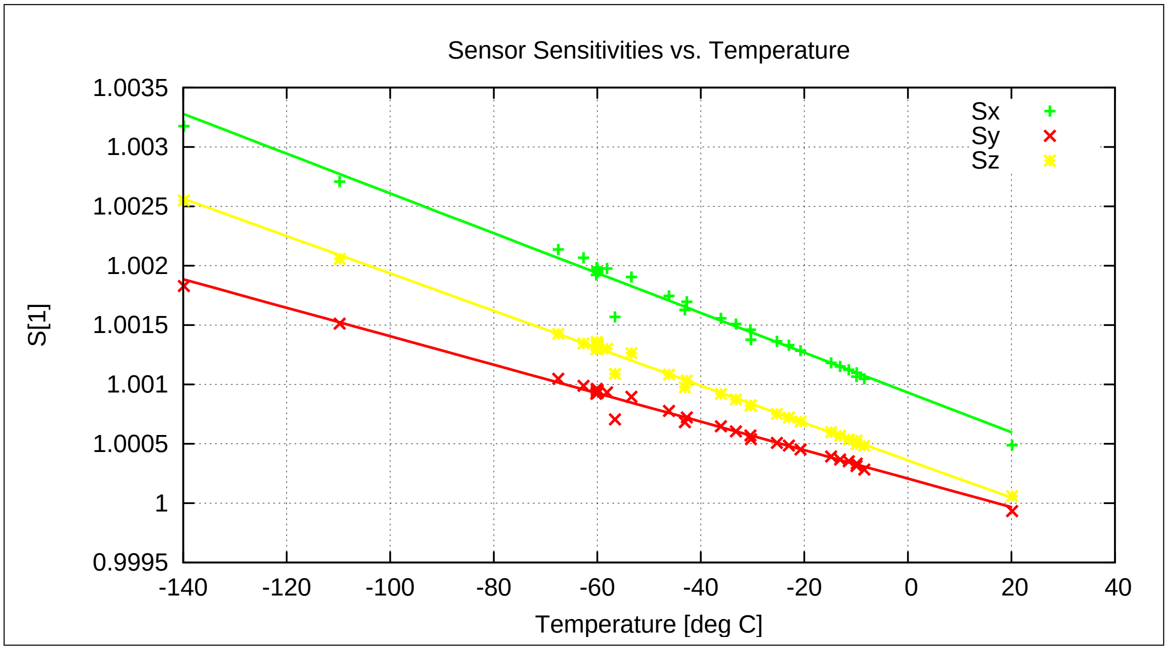

Figure 172: Temperature dependence of Sensitivities

Misalignment Angles ξ_{ij} vs. Temperature:

$$\xi_{ij}(T) = \sum_{k=0}^{n} \xi_{k,ij} T^k \qquad \text{[deg, °C], ij= \{xy,xz,yz\}}$$

	$\xi_{0,ij}$	$\xi_{1,ij}$	$\xi_{2,ij}$	$\xi_{3,ij}$	$\xi_{4,ij}$	$\xi_{5,ij}$
ξ_{xy}	8.9798E+1	-6.1553E-5				
ξ_{xz}	9.0895E+1	-7.8010E-5				
ξ_{yz}	9.0230E+1	3.1229E-5				

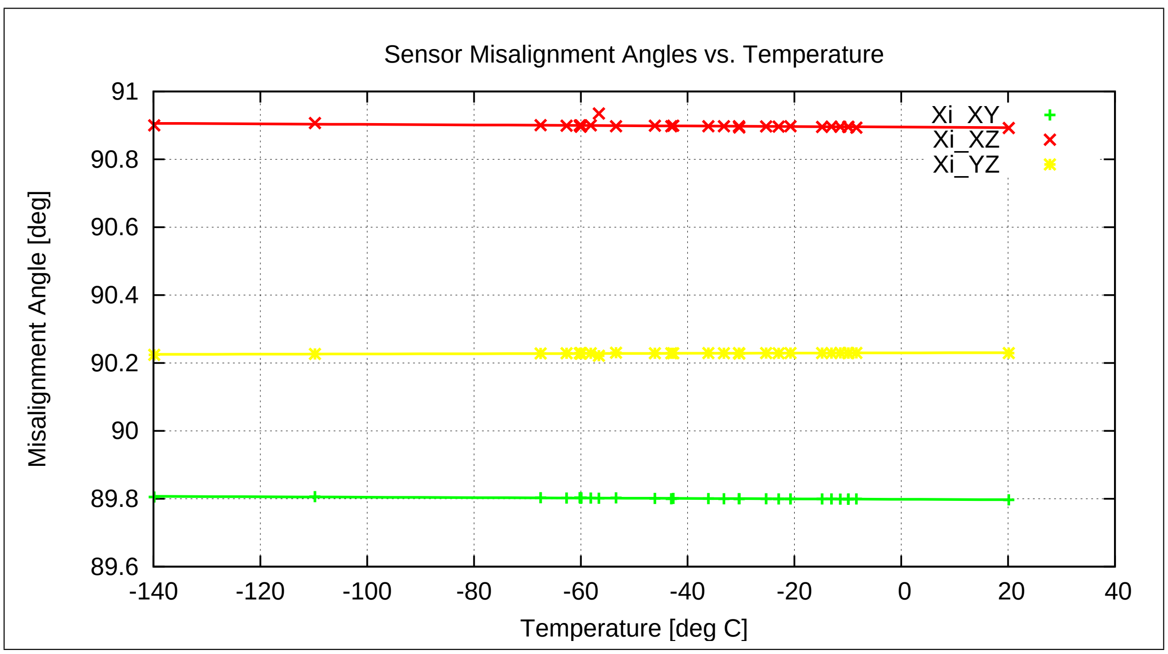

Figure 173: Temperature dependence of Misalignment Angles

Offset & Residual MCF Field $\underline{B}^{or} = \underline{B}^{off} + \underline{B}^{res}$ vs. Temperature:

$$\underline{B}^{or}(T) = \sum_{k=0}^{n} \underline{B}_k^{or} T^k \qquad [\text{enT, °C}]$$

	$\underline{B}_0^{or}$	$\underline{B}_1^{or}$	$\underline{B}_2^{or}$	$\underline{B}_3^{or}$	$\underline{B}_4^{or}$	$\underline{B}_5^{or}$
B_x^{or}	1.877E+1	-2.346E-3				
B_y^{or}	9.105E+0	-8.862E-3				
B_z^{or}	-2.262E+0	7.952E-3				

Model Quality:

Minimum and maximum errors of the calculated Model vs. Temperature:

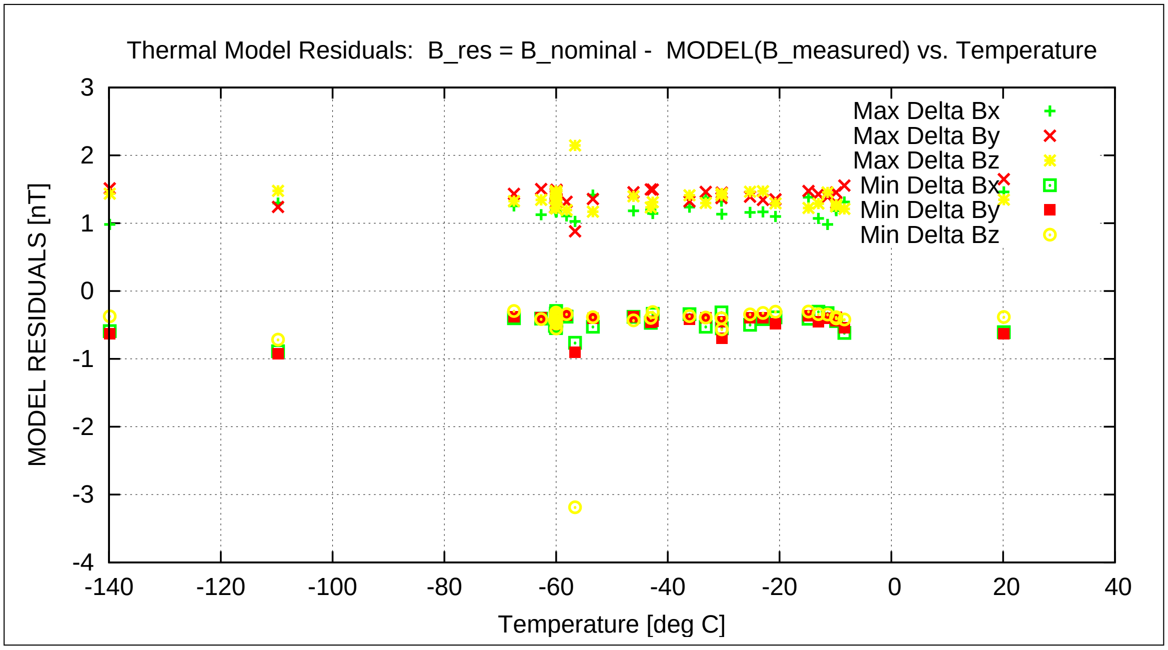

Figure 174: Residuals of Thermal Model

Sensor Rotation during Test:

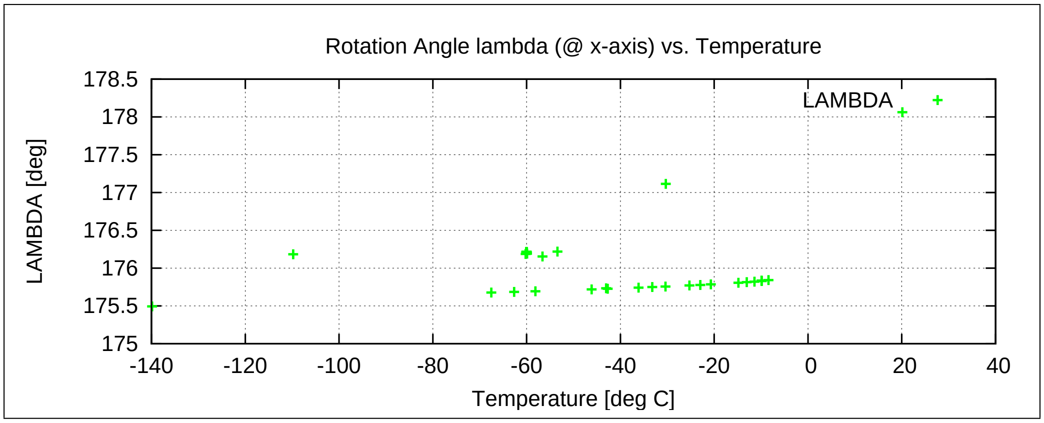

Figure 175: Rotation @ X-Axis

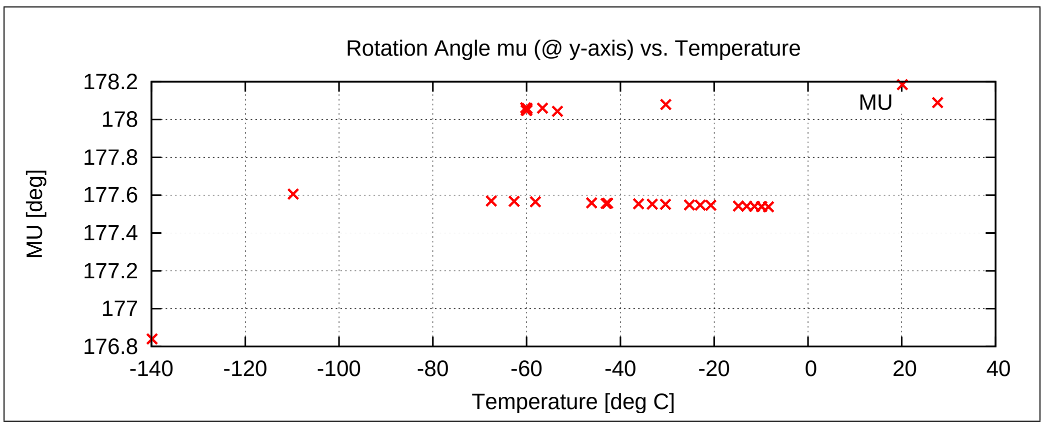

Figure 176: Rotation @ Y-Axis

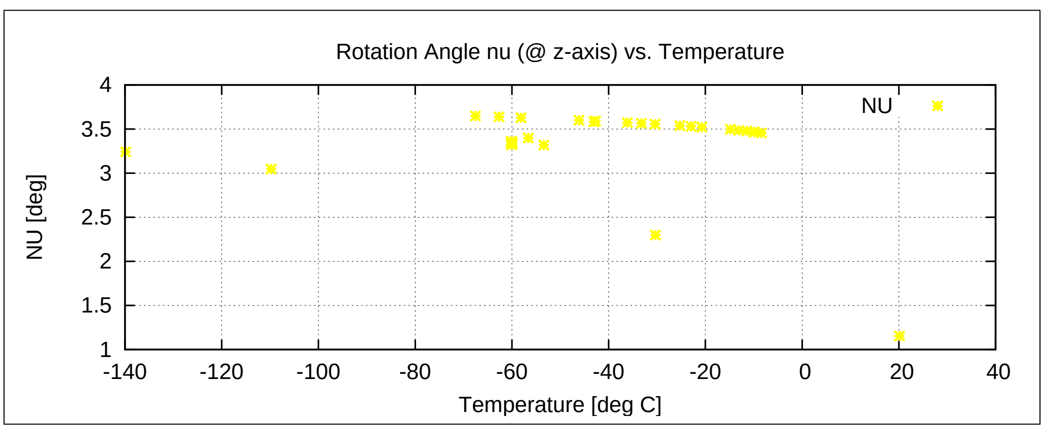

Figure 177: Rotation @ Z-Axis

15.2 Offset, Range=R0, FGM-IB

15.2.1 Temperature Calibration of the Sensor Offset — RANGE R0 — SENSOR IB

Used Temperature Measurements:

Calibration Parameter File	Remark
OFF_PARAMETER_R0_21-04-19-11-59-19_IB.OPF	
OFF_PARAMETER_R0_21-04-19-15-52-29_IB.OPF	
OFF_PARAMETER_R0_21-04-20-07-23-51_IB.OPF	
OFF_PARAMETER_R0_21-04-20-09-41-19_IB.OPF	
OFF_PARAMETER_R0_21-04-20-13-37-22_IB.OPF	

Thermal Parameter File: THERMAL_OFF_PARAMETER_R0_21-04-19-11-59-19_IB.TOF

Facility Parameter:

Nominal Sensor Setup $\underline{B}_{\mathrm{DUT}} = \underline{\underline{R}}_{\mathrm{nom}} \underline{B}_{\mathrm{C}}$

$$\underline{\underline{R}}_{\mathrm{nom}} = \begin{pmatrix} -1.000000 & +0.000000 & +0.000000 \\ +0.000000 & -1.000000 & +0.000000 \\ +0.000000 & +0.000000 & +1.000000 \end{pmatrix}$$

Sensor Temperature Channel: T_{59}

Coil System Temperature Channel: T_{29}

Temperature Profile

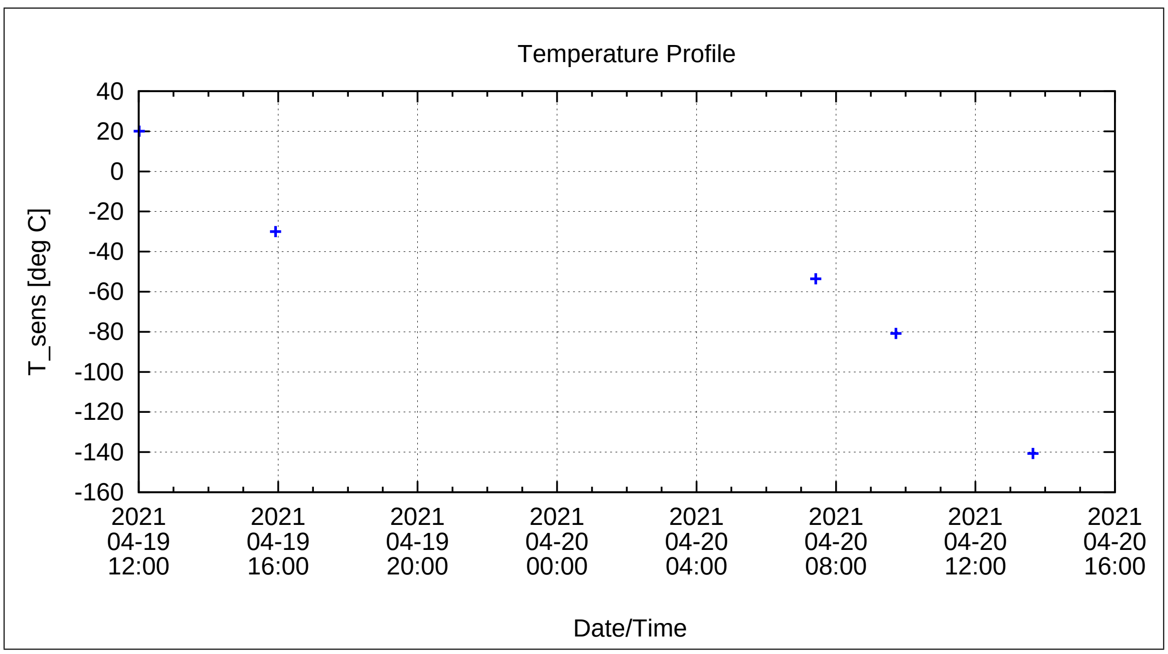

Figure 178: Temperature Profile

Calibration Parameter:

Sensor Offset $\underline{B}_{\mathrm{off}}$ vs. Temperature:

$$B_{\mathrm{off,i}}(T) = \sum_{k=0}^{n} B_{\mathrm{off,k,i}} T^k \qquad [\mathrm{enT},\ ^\circ\mathrm{C}],\ \mathrm{i}=\{\mathrm{x,y,z}\}$$

	$B_{\mathrm{off,0,i}}$	$B_{\mathrm{off,1,i}}$	$B_{\mathrm{off,2,i}}$	$B_{\mathrm{off,3,i}}$	$B_{\mathrm{off,4,i}}$	$B_{\mathrm{off,5,i}}$
$B_{\mathrm{off,x}}$	7.12551E+0	4.37917E-3				
$B_{\mathrm{off,y}}$	9.18695E+0	-7.44677E-3				
$B_{\mathrm{off,z}}$	8.28215E+0	1.51504E-2				

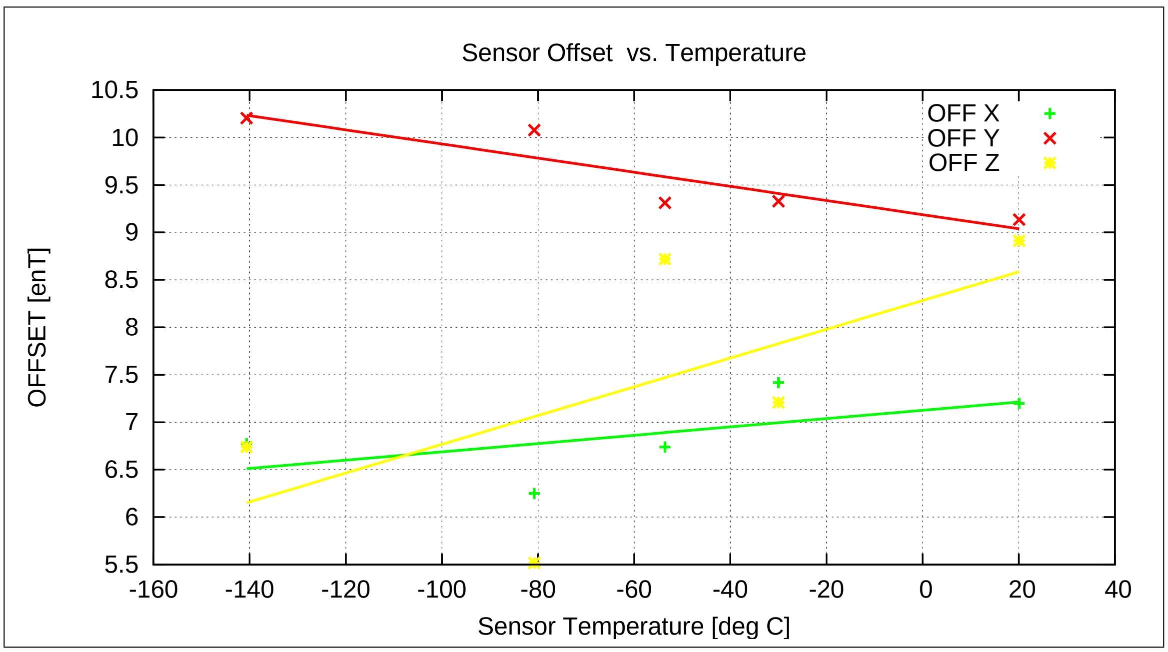

Figure 179: Temperature dependence of Sensor Offsets

Coil System Temperature

The following graph shows the mean Coil System temperature during the complete thermal cycle for the offset measurements.

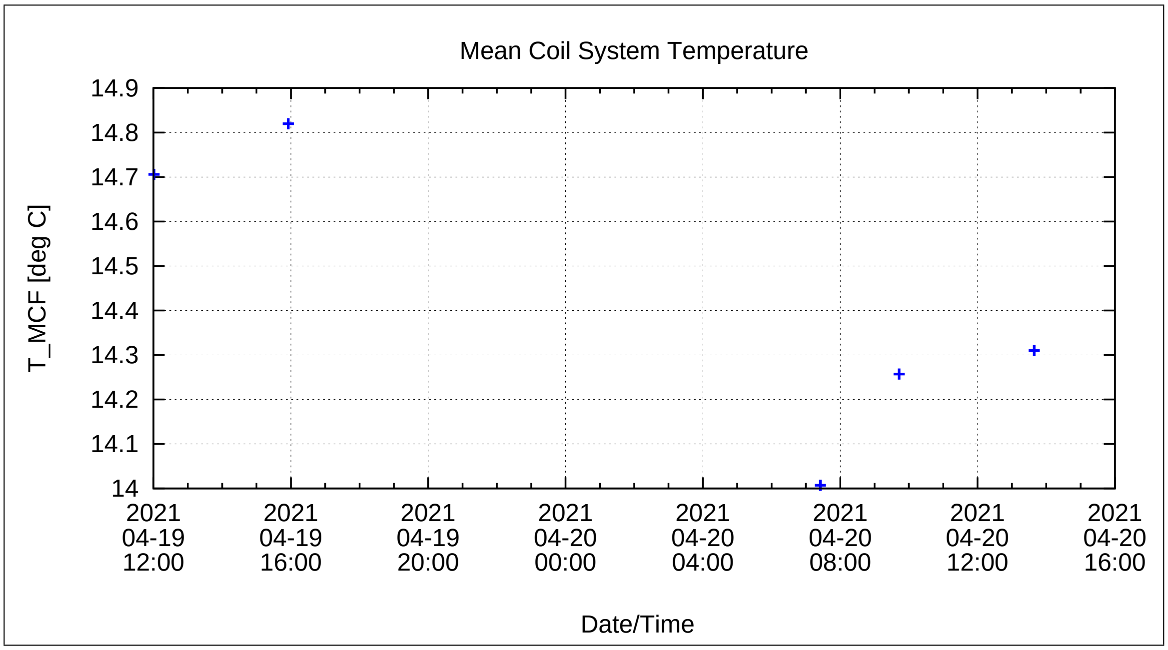

Figure 180: Coil System Temperature during Thermal Measurements

Coil System Residual Field:

The following graph shows the three components of the Coil System Residual field during the whole measurement. Axes designators are related to the actual DUT coordinate system and NOT to Coil System coordinates.

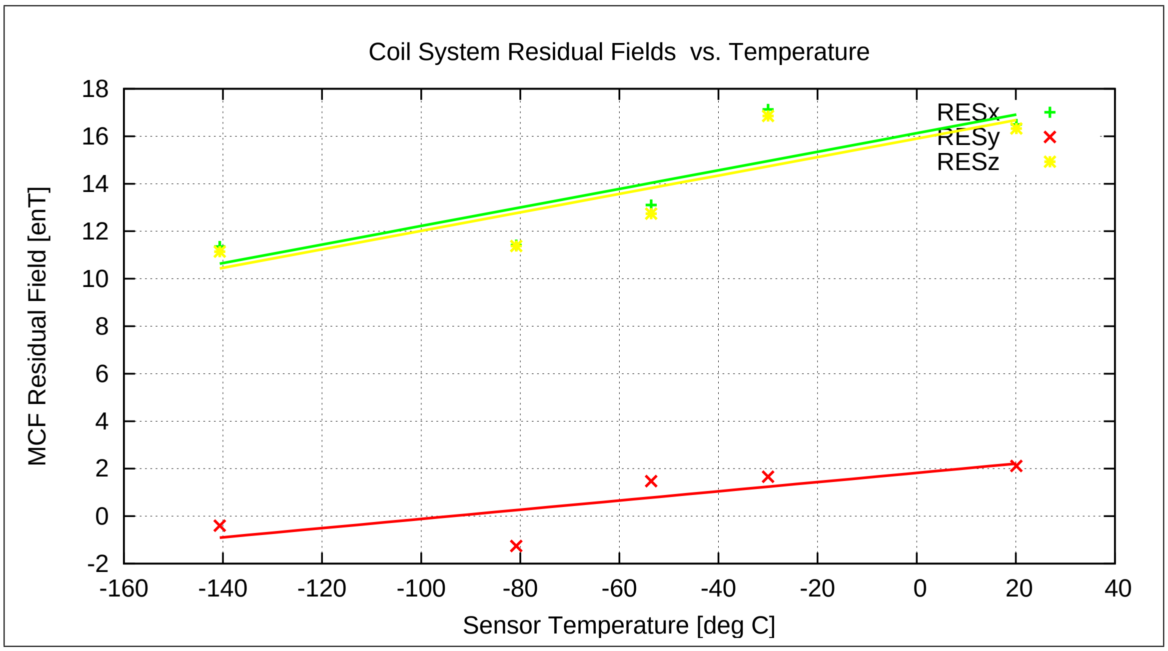

Figure 181: Coil System Residual Fields during Thermal Measurements

16 Complete Temperature Profile for the FM-OB Measurements

The following plot shows the complete thermal cycle of the FGM-OB sensor with all cooling and heating phases. The curves show the desired nominal temperatures in green and the actual measured temperatures in red.

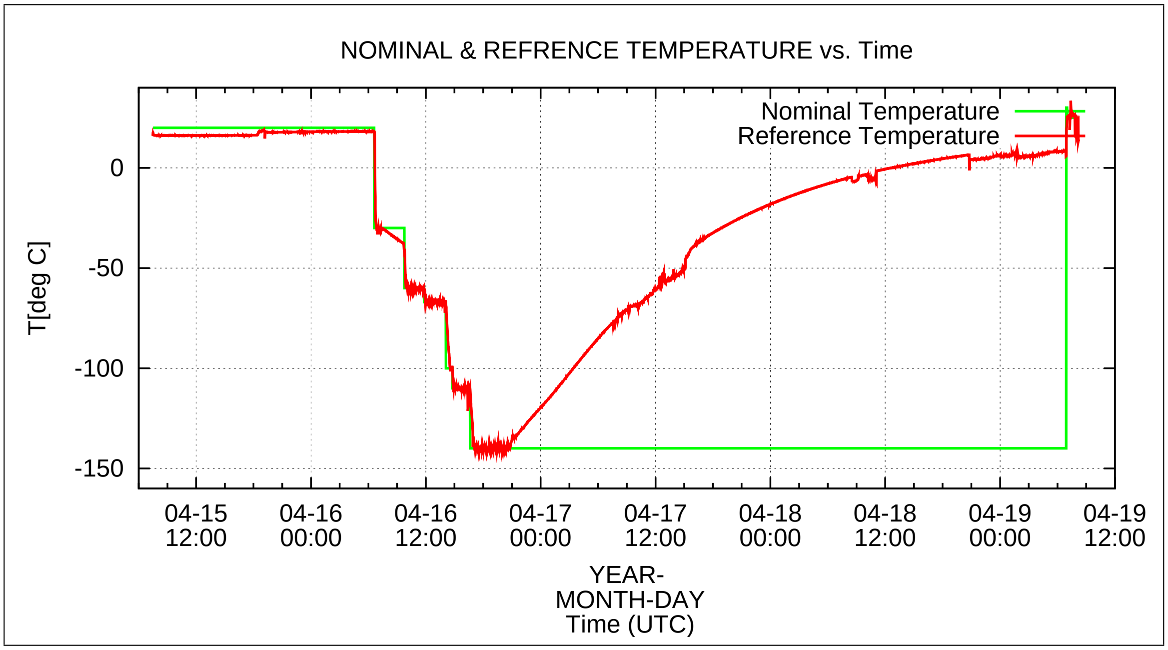

Figure 182: Complete thermal cycle: FM-OB Calibration

17 Complete Temperature Profile for the FM-IB Measurements

The following plot shows the complete thermal cycle of the FGM-IB sensor with all cooling and heating phases. The curves show the desired nominal temperatures in green and the actual measured temperatures in red.

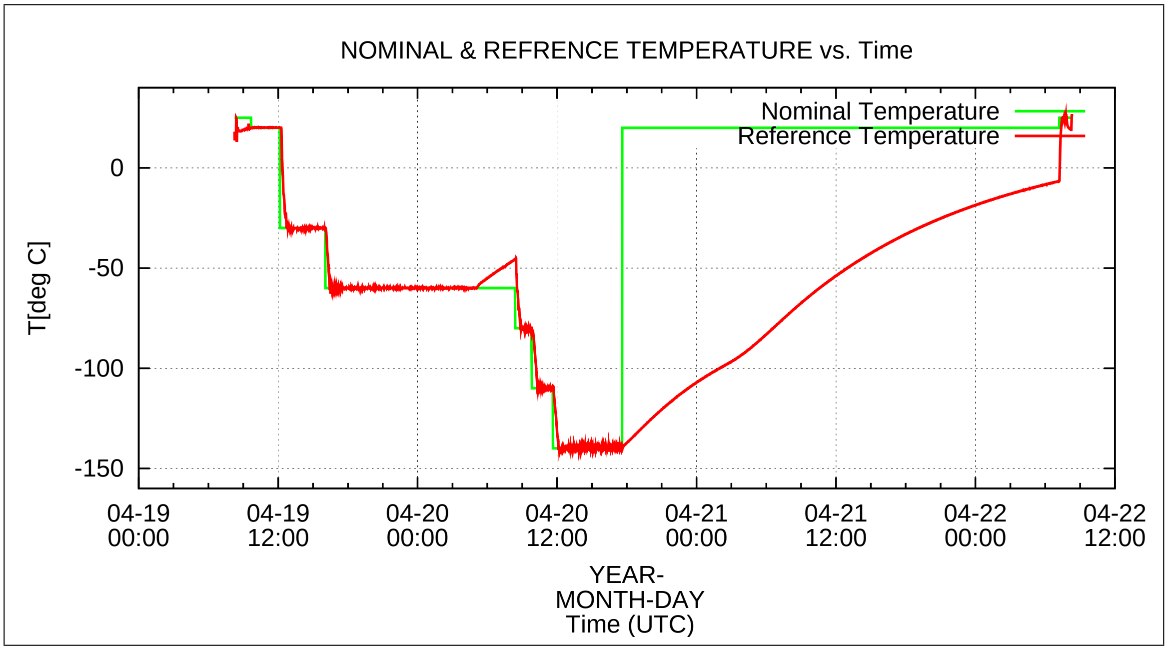

Figure 183: Complete thermal cycle: FM-IB Calibration

18 Mathematical Description of the Calibration

18.1 Basic Principle

The Magnetsrode Coil Facility (MCF) generates an artificial magnetic field $\underline{B}^{\mathrm{FLD}}$ that can be considered as a calibrated, orthogonal magnetic reference field[1] defined in coilsystem coordinates. For the calibration analysis this ideal coilsystem field is rotated to ideal orthogonal sensor coordinates using a nominal Rotation matrix $\underline{\underline{R}}_{\mathrm{nom}}$ in the first step. Nominal means that only rotations of ±90° or ±180° around any axis are considered here to get a coarse alignment of the applied field with the sensor-axes.

$$\underline{B}^{\mathrm{c}} = \underline{\underline{R}}_{\mathrm{nom}}\,\underline{B}^{\mathrm{FLD}}$$

For a standard calibration the matrix $\underline{\underline{R}}_{\mathrm{nom}}$ is just a $\underline{\underline{I}}$ - matrix - sensor coordinates and coilsystem coordinates have roughly the same direction. If the sensor coordinates are left-handed or the sensor is turned by about ±90° or ±180° around any axis $\underline{\underline{R}}_{\mathrm{nom}}$ will contain ±1-elements or 0-elements at any place. The rotated coil system field $\underline{B}^{\mathrm{c}}$ is the used reference field for the following analysis.

The magnetometer under test at the center of the coil system (CoC) generates magnetic raw data $\underline{B}^{\mathrm{r}}$. These measurements may include a residual field caused by the coil system $\underline{B}^{\mathrm{res}}$ and the magnetometer offset $\underline{B}^{\mathrm{off}}$. Both entities are combined in the offset & residual field $\underline{B}^{\mathrm{or}}$:

$$\underline{B}^{\mathrm{or}} = \underline{B}^{\mathrm{off}} + \underline{B}^{\mathrm{res}}$$

Therefore, the second step of the calibration is the generation of offset and residual field corrected measured field data $\underline{B}^{\mathrm{m}}$:

$$\underline{B}^{\mathrm{m}} = \underline{B}^{\mathrm{r}} - \underline{B}^{\mathrm{or}}$$

The actual offset and residual field is automatically taken into account during the calibration analysis. Either a constant field or - if needed - a linear trend of $\underline{B}^{\mathrm{or}}$ is subtracted from the raw data.

The relation between the calibration field and the magnetometer data is then defined by

$$\underline{B}^{\mathrm{c}} = \underline{\underline{\Phi}}\,\underline{B}^{\mathrm{m}}$$

where $\underline{\underline{\Phi}}$ is the complete calibration transfer matrix, defined by

$$\underline{\underline{\phi}} = \underline{\underline{R}}_{\mathrm{nom}}\ \underline{\underline{\rho}}\,\underline{\underline{\omega}}\,\underline{\underline{\sigma}}.$$

[1]During the calibration the temperature dependent sensitivity of the coil system is calculated every 3 minutes and taken into account as well as the static misalignment of the coil system to produce orthogonal, known fields.

$\underline{\underline{\sigma}}(T)$ represents the temperature dependent sensitivity.
$\underline{\underline{\omega}}(T)$ describes the temperature dependent internal sensor misalignment (orthogonalisation matrix).
$\underline{\underline{\rho}}(T)$ describes the real rotation of the sensor against the coil axes.[2]

Thus the calibration algorithms have to solve the following problem:

$$\begin{aligned} \underline{B}^{\mathrm{c}} &= \underline{\underline{\Phi}}\ \underline{B}^{\mathrm{m}} \\ &= \underline{\underline{R}}_{\mathrm{nom}}\ \underline{\underline{\rho}}\ \underline{\underline{\omega}}\ \underline{\underline{\sigma}}\ \underline{B}^{\mathrm{m}} \end{aligned} \tag{1}$$

The separation into these submatrices and evaluation of their elements is done in subsequent steps:

- Calculation of the Sensitivity matrix $\underline{\underline{\sigma}}$
  The sensitivity matrix shall contain the on-axis sensitivity coefficients of the sensors. Therefore, $\underline{\underline{\sigma}}$ has to be a diagonal matrix of the following kind:

  $$\underline{\underline{\sigma}} = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix}$$

  The separation of matrix $\underline{\underline{\sigma}}$ from the transfer function $\underline{\underline{\phi}}$ yields

  $$\underline{\underline{\phi}} = \begin{pmatrix} \frac{\phi_{11}}{\sigma_1} & \frac{\phi_{12}}{\sigma_2} & \frac{\phi_{13}}{\sigma_3} \\ \frac{\phi_{21}}{\sigma_1} & \frac{\phi_{22}}{\sigma_2} & \frac{\phi_{23}}{\sigma_3} \\ \frac{\phi_{31}}{\sigma_1} & \frac{\phi_{32}}{\sigma_2} & \frac{\phi_{33}}{\sigma_3} \end{pmatrix} \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix} :=\hat{\underline{\underline{\phi}}}\ \underline{\underline{\sigma}}$$

  For the computation of the sensitivity coefficients σ_i the transformation of the base-vectors has to be considered. Equation (1) transform the components of the fields, whereas

  $$\underline{e}^{\mathrm{c}} = \left(\underline{\underline{\phi}}^{\mathrm{T}}\right)^{-1} \underline{e}^{\mathrm{m}} := \underline{\underline{\Psi}}\ \underline{e}^{\mathrm{m}}$$

  has to be used for the contragredient base-vector transformation of the skewed sensorsystem Σ^{m} into the orthonormal coilsystem Σ^{c}. The length of the column vectors of $\underline{\underline{\psi}}$ define the sensitivity coefficients σ_i:

  $$\sigma_1 = \frac{1}{\sqrt{\psi_{11}^2 + \psi_{21}^2 + \psi_{31}^2}}$$

[2] Also the rotation matrix is regarded as temperature dependent to be able to consider any thermal setup inadequacies causing fractional rotations.

$$\sigma_2 = \frac{1}{\sqrt{\psi_{12}^2 + \psi_{22}^2 + \psi_{32}^2}}$$
$$\sigma_3 = \frac{1}{\sqrt{\psi_{13}^2 + \psi_{23}^2 + \psi_{33}^2}}$$

- Calculation of the Misalignment matrix $\underline{\underline{\omega}}$

  After the separation of $\underline{\underline{\sigma}}$ the reduced transfer function $\overset{\wedge}{\underline{\underline{\phi}}}$ contains only the misalignment and the real sensor rotation. The misalignment angles $\xi_{xy}, \xi_{xz}, \xi_{yz}$, hence the angles between the base vectors of the affine sensorsystem Σ^{m}, can be evaluated from the scalar products of all these base vectors. The base unit vectors are defined by the inverse transposed matrix of the reduced transfer function:

$$\underline{e}_{\mathrm{X}}^{\mathrm{m}} := \left(\overset{\wedge}{\underline{\underline{\phi}}}^{\mathrm{T}}\right)^{-1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$
$$\underline{e}_{\mathrm{Y}}^{\mathrm{m}} := \left(\overset{\wedge}{\underline{\underline{\phi}}}^{\mathrm{T}}\right)^{-1} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$
$$\underline{e}_{\mathrm{Z}}^{\mathrm{m}} := \left(\overset{\wedge}{\underline{\underline{\phi}}}^{\mathrm{T}}\right)^{-1} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

  The misalignment angles can be derived from the scalar products:

$$\xi_{xy} = \arccos\left(\underline{e}_x^m \cdot \underline{e}_y^m\right)$$
$$\xi_{xz} = \arccos\left(\underline{e}_x^m \cdot \underline{e}_z^m\right)$$
$$\xi_{yz} = \arccos\left(\underline{e}_y^m \cdot \underline{e}_z^m\right)$$

  Let's assume that the x-axis of the reference system $\underline{e}_{\mathrm{X}}^{\mathrm{c}}$ is identical to the $\underline{e}_{\mathrm{X}}^{\mathrm{m}}$-axis of the sensor system. The angle between $\underline{e}_{\mathrm{Y}}^{\mathrm{c}}$ and $\underline{e}_{\mathrm{Y}}^{\mathrm{m}}$ is β (rotation angle around the $\underline{e}_{\mathrm{Z}}^{\mathrm{c}}$ axis). And the sensor $\underline{e}_{\mathrm{Z}}^{\mathrm{m}}$ axis can be constructed by a rotation of η in the $\underline{e}_{\mathrm{X}}^{\mathrm{c}}, \underline{e}_{\mathrm{Y}}^{\mathrm{c}}$-plane and a second rotation of γ out of this plane. Then this misalignment of the base vectors is given by

$$\underline{\underline{O}} = \begin{pmatrix} 1 & \sin\beta & \sin\gamma \\ 0 & \cos\beta & \cos\gamma\sin\eta \\ 0 & 0 & \cos\gamma\cos\eta \end{pmatrix} \tag{2}$$

  By comparison of the angles β, η, γ with misalignment angles $\xi_{xy}, \xi_{xz}, \xi_{yz}$, the misalignment matrix (2) can be written in the following form:

$$\underline{\underline{O}} = \begin{pmatrix} 1 & \cos\xi_{xy} & \cos\xi_{xz} \\ 0 & \sin\xi_{xy} & \dfrac{\cos\xi_{yz} - \cos\xi_{xy}\cos\xi_{xz}}{\sin\xi_{xy}} \\ 0 & 0 & \sqrt{\sin^2\xi_{xz} - \dfrac{(\cos\xi_{yz} - \cos\xi_{xy}\cos\xi_{xz})^2}{\sin^2\xi_{xy}}} \end{pmatrix}$$

$\underline{\underline{O}}$ transforms the base vectors. To achieve the transformation between the field components the transposed, inverse matrix of $\underline{\underline{O}}$ has to be calculated as the final misalignment matrix:

$$\underline{\underline{\omega}} = \left(\underline{\underline{O}}^{\mathrm{T}}\right)^{-1}$$

- Calculation of the Rotation matrix $\underline{\underline{\rho}}$
  With the knowledge of $\underline{\underline{\omega}}$ and the nominal Setup matrix $\underline{\underline{R}}_{\text{nom}}$ the rotation matrix $\underline{\underline{\rho}}$ can be evaluated:

  $$\underline{\underline{\rho}} = (\underline{\underline{R}}_{\text{nom}})^{-1}\ \underline{\underline{\overset{\wedge}{\phi}}}\ (\underline{\underline{\omega}})^{-1}$$

  From this rotation matrix the actual rotation angles of the sensor wrt. the coil system can be calculated using again the scalar product:

  $$\begin{aligned} \lambda &:= \arccos\left(\left[\underline{\underline{\rho}}\begin{bmatrix}1\\0\\0\end{bmatrix}\right]\cdot\begin{bmatrix}1\\0\\0\end{bmatrix}\right) \\ \mu &:= \arccos\left(\left[\underline{\underline{\rho}}\begin{bmatrix}0\\1\\0\end{bmatrix}\right]\cdot\begin{bmatrix}0\\1\\0\end{bmatrix}\right) \\ \nu &:= \arccos\left(\left[\underline{\underline{\rho}}\begin{bmatrix}0\\0\\1\end{bmatrix}\right]\cdot\begin{bmatrix}0\\0\\1\end{bmatrix}\right) \end{aligned}$$

  The rotation matrix $\underline{\underline{\rho}}$ and the rotation angles λ, μ, ν are of interest just for the calibration to determine the right magnetometer parameters.

The transfer function for the normal use of the magnetometer is just given by

$$\underline{\underline{\tilde{\phi}}} = \underline{\underline{\omega}}\ \underline{\underline{\sigma}}$$

19 Nomenclature

Abbreviations in theoretical sections:

Item	Meaning
$\underline{B}^c$	Magnetic calibration field generated by coil system
$\underline{B}^{off}$	Offset of the magnetometer [nT]
$\underline{B}_k^{off}$	Polynomial Fit coefficients for the sensor offset vs. temperature
$\underline{B}^m$	Measured magnetic field raw data, offset & residual field corrected [nT]
$\underline{B}^{or}$	$= \underline{B}^{off} + \underline{B}^{res}$, Offset + Residual field at CoC [enT]
$\underline{B}^r$	Magnetic field raw data
$B^r_{0^\circ}$	Magnetic raw data, measured in normal position (0°)[enT]
$B^r_{180^\circ}$	Magnetic raw data, measured in turned position (180°)[enT]
$\underline{B}^{res}$	Residual field of the coil system
e°C	Engineering degrees centigrade units
enT	Engineering NanoTesla units
c_0, c_1, c_2, c_3	Fit coefficients of the sensor thermistor
i	component x | y | z
CoC	Center of Coil system
DUT	Device under Test
$\underline{\underline{\phi}}$	$= \underline{\underline{R}}_{\text{nom}}\, \underline{\underline{\rho}}\, \underline{\underline{\omega}}\, \underline{\underline{\sigma}}$, Calibration Transfer Matrix
$\tilde{\underline{\underline{\phi}}}$	$= \underline{\underline{\omega}}\, \underline{\underline{\sigma}}$, Transfer Matrix
$\hat{\underline{\underline{\phi}}}$	$= \underline{\underline{\phi}}\, \underline{\underline{\sigma}}^{-1}$, Reduced Transfer Matrix
$\underline{\underline{\psi}}$	$= (\underline{\underline{\phi}}^T)^{-1}$, Auxilliary transfer matrix
$\underline{\underline{O}}$	Base vector Orthogonalisation matrix
$\underline{\underline{\omega}}$	$= (\underline{\underline{O}}^T)^{-1}$, Field components orthogonalisation matrix
$\underline{\underline{R}}_{\text{nom}}$	Nominal Rotation matrix of sensor vs. coil system
$\underline{\underline{\rho}}$	= Actual rotation matrix
$\underline{\underline{\sigma}}$	= Sensitivity matrix
$\sigma_{k,i}$	Polynomial Fit coefficient for sensitivity vs. temperature. Coefficient of order k for the sensor component i
TeMeSys	MRode Temperature Measurement System
T_s^c	Temperature of sensor s, calibrated data [°C]
T_s^r	Temperature of sensor s, raw data [e°C]
s	Sensor IB | OB
$U_{\text{T, IB}}$	IB–Temperature data [V], measured
$U_{\text{T,OB}}$	OB–Temperature data [V], measured
ξ_{ij}	Temperature dependent alignment angle (ij component)
$\xi^0_{k,ij}$	Polynomial Fit coefficient for misalignment angle vs. temperature. Coefficient of order k for the angle ij
λ, μ, ν	Rotation angles wrt. coil system reference coordinates.